Exposé I. Presheaves

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In numbers 0 through 5 of this exposé we present the elementary, and most often well-known, properties of categories of presheaves.[1] These properties are used constantly throughout the rest of the seminar, and knowing them is essential for understanding the subsequent exposés. The proofs are immediate; for the most part they are omitted. In numbers 6 through 9 we take up a few themes that are used several times later on. The reader in a hurry may omit them on a first reading. Number 10 fixes the terminology used. Appendix 11 is due to N. Bourbaki.

1 Universes

A *universe* is a nonempty set[2] $\mathcal{U}$ satisfying the following properties:

($\mathcal{U}$1) If $x \in \mathcal{U}$ and $y \in x$, then $y \in \mathcal{U}$.

($\mathcal{U}$2) If $x, y \in \mathcal{U}$, then $\{x, y\} \in \mathcal{U}$.

($\mathcal{U}$3) If $x \in \mathcal{U}$, then $\mathcal{P}(x) \in \mathcal{U}$.

($\mathcal{U}$4) If $(x_i)_{i\in I}$ is a family of elements of $\mathcal{U}$ and $I \in \mathcal{U}$, then $\bigcup_{i\in I} x_i \in \mathcal{U}$.

From the preceding axioms one easily deduces the following properties:

- if $x \in \mathcal{U}$, then the singleton $\{x\}$ belongs to $\mathcal{U}$;
- if x is a subset of some $y \in \mathcal{U}$, then $x \in \mathcal{U}$;
- if $x, y \in \mathcal{U}$, the ordered pair $(x, y) = \{\{x, y\}, x\}$, in Kuratowski's definition, is an element of $\mathcal{U}$;
- if $x, y \in \mathcal{U}$, the union $x \cup y$ and the product $x \times y$ are elements of $\mathcal{U}$;
- if $(x_i)_{i\in I}$ is a family of elements of $\mathcal{U}$ indexed by an element $I \in \mathcal{U}$, then the product $\prod_{i\in I} x_i$ is an element of $\mathcal{U}$;
- if $x \in \mathcal{U}$, then $\mathrm{card}(x) < \mathrm{card}(\mathcal{U})$ strictly. In particular, the relation $\mathcal{U} \in \mathcal{U}$ is not satisfied.

Thus one may perform all the usual operations of set theory starting from elements of a universe without the final result ceasing to be an element of that universe.

The first use of the notion of universe is that it supplies a definition of the usual categories: the category of sets belonging to the universe $\mathcal{U}$ ($\mathcal{U}$-**Set**), the category of topological spaces belonging to $\mathcal{U}$, the category of commutative groups belonging to $\mathcal{U}$ ($\mathcal{U}$-**Ab**), the category of categories belonging to $\mathcal{U}$, and so on.

However, the only known universe is the set of symbols of the type $\{\{\varnothing\}, \{\{\varnothing\}, \varnothing\}\}$, etc. All the elements of this universe are finite sets, and this universe is countable. In particular, no

[1] The reader may also consult SGA 3, I, §§1–3.

[2] This definition contradicts Bourbaki's appendix below. However, the interest of empty universes seems debatable.

universe is known that contains an element of infinite cardinality. We are therefore led to add to the axioms of set theory the axiom

$(\mathcal{U}A)$ *For every set x there exists a universe $\mathcal{U}$ such that $x \in \mathcal{U}$.*

Since the intersection of a family of universes is a universe, it follows immediately that every set is an element of a smallest universe. One can show that axiom $(\mathcal{U}A)$ is independent of the axioms of set theory.

We shall also add the axiom:

$(\mathcal{U}B)$ *Let $R\{x\}$ be a relation and let $\mathcal{U}$ be a universe. If there exists an element $y \in \mathcal{U}$ such that $R\{y\}$, ther*

The consistency of axioms $(\mathcal{U}A)$ and $(\mathcal{U}B)$ relative to the other axioms of set theory is not proved and, it seems, not provable.

Let $\mathcal{U}$ be a universe and let $c(\mathcal{U})$ be the least upper bound of the cardinals of elements of $\mathcal{U}$; thus $c(\mathcal{U}) \leq \operatorname{card}(\mathcal{U})$. The cardinal $c(\mathcal{U})$ has the following properties:

(FI) If $a < c(\mathcal{U})$, then $2^a < c(\mathcal{U})$.

(FII) If $(a_i)_{i\in I}$ is a family of cardinals strictly smaller than $c(\mathcal{U})$, and if $\operatorname{card}(I)$ is strictly smaller than $c(\mathcal{U})$, then $\sum_{i\in I} a_i < c(\mathcal{U})$.

Cardinals satisfying properties (FI) and (FII) are called *strongly inaccessible cardinals.*

The cardinal 0 and the countably infinite cardinal are strongly inaccessible.

Axiom $(\mathcal{U}A)$ implies:

$(\mathcal{U}A')$ *Every cardinal is strictly bounded above by a strongly inaccessible cardinal.*

One can show (11) that, conversely, the consistency of $(\mathcal{U}A')$ implies the consistency of $(\mathcal{U}A)$, and that the consistency of these axioms implies the consistency of axiom $(\mathcal{U}B)$.

Call a set E *Artinian* if there is no infinite family $(x_n)_{n\in\mathbb{N}}$ such that $x_0 \in E$ and $x_{n+1} \in x_n$. One can then show [*loc. cit.*] that there is a one-to-one correspondence between strongly inaccessible cardinals and Artinian universes, defined as follows: to every strongly inaccessible cardinal c one associates the unique Artinian universe $\mathcal{U}_c$ such that

$$\operatorname{card}(\mathcal{U}_c) = c.$$

If $c < c'$ are two strongly inaccessible cardinals, then

$$\mathcal{U}_c \in \mathcal{U}_{c'}.$$

In particular, the Artinian universes of cardinal smaller than a given cardinal form a well-ordered set for the membership relation. Axiom $(\mathcal{U}A')$ is equivalent to the axiom:

$(\mathcal{U}A'')$ *Every Artinian set is an element of an Artinian universe.*

We note that all the usual sets $(\mathbb{Z}, \mathbb{Q}, \ldots)$ are Artinian sets.

2 $\mathcal{U}$-categories. Presheaves of sets

1.0

In the rest of the seminar, unless the contrary is expressly stated, the universes considered will possess an element of infinite cardinality. Let $\mathcal{U}$ be a universe. A set is said to be *$\mathcal{U}$-small*, or simply *small* when no confusion can result, if it is *isomorphic* to an element of $\mathcal{U}$. We also use the terminology small group, small ring, small category,[3] etc. We shall often suppose, without explicit mention, that the schemes, topological spaces, indexing sets, etc. with which we work are in $\mathcal{U}$, or at least have cardinality in $\mathcal{U}$; however, many of the categories with which we shall work will not themselves be in $\mathcal{U}$.

Definition (1.1)**.** Let $\mathcal{U}$ be a universe and let C be a category. We say that C is a $\mathcal{U}$-category if, for every pair (x, y) of objects of C, the set $\mathrm{Hom}_C(x, y)$ is $\mathcal{U}$-small.

1.1.1

Let C and D be two categories, and let $\mathrm{Fun}(C, D)$ denote the category of functors from C to D. The following assertions are immediate:

(a) If C and D are elements of a universe $\mathcal{U}$ (respectively are $\mathcal{U}$-small), then the category $\mathrm{Fun}(C, D)$ is an element of $\mathcal{U}$ (respectively is $\mathcal{U}$-small).

(b) If C is $\mathcal{U}$-small and D is a $\mathcal{U}$-category, then $\mathrm{Fun}(C, D)$ is a $\mathcal{U}$-category.

Remark (1.1.2)**.** Let D be a category satisfying the following properties:

(C1) the set $\mathrm{Ob}(D)$ is contained in the universe $\mathcal{U}$;

(C2) for every pair (x, y) of objects of D, the set $\mathrm{Hom}_D(x, y)$ is an element of $\mathcal{U}$.

The usual categories constructed from a universe $\mathcal{U}$ have these two properties: $\mathcal{U}$-**Set**, $\mathcal{U}$-**Ab**, and so on. Let C be a category belonging to $\mathcal{U}$. Then the category $\mathrm{Fun}(C, D)$ does not in general have properties (C1) and (C2). For example, the category $\mathrm{Fun}(C, \mathcal{U}\text{-}\mathbf{Set})$ has neither property (C1) nor property (C2). This justifies the definition adopted for a $\mathcal{U}$-category, rather than the more restrictive notion given by conditions (C1) and (C2) above.

Definition (1.2)**.** Let C be a category. The category of presheaves of sets on C relative to the universe $\mathcal{U}$—or, when no confusion can result, the category of presheaves on C—is the category of contravariant functors on C with values in the category of $\mathcal{U}$-sets.

We denote by $\widehat{C}_{\mathcal{U}}$, or more simply by $\widehat{C}$ when no confusion can result, the category of presheaves of sets on C relative to the universe $\mathcal{U}$. The objects of $\widehat{C}_{\mathcal{U}}$ are called $\mathcal{U}$-presheaves, or simply presheaves, on C. When C is $\mathcal{U}$-small, the category $\widehat{C}_{\mathcal{U}}$ is a $\mathcal{U}$-category. When C is a $\mathcal{U}$-category, $\widehat{C}_{\mathcal{U}}$ is not necessarily a $\mathcal{U}$-category.

Construction–Definition (1.3)**.** Let x be an object of a $\mathcal{U}$-category C. The *$\mathcal{U}$-functor represented by x* is the functor

$$h_{\mathcal{U}}(x) : C^{\mathrm{op}} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

constructed as follows.[4] Let y be an object of C.

[3] A category C is viewed as a set of arrows, equipped with the subset of objects, viewed as the set of identity arrows, and with the source and target maps. Thus the expressions 'C is an element of $\mathcal{U}$' and 'C is $\mathcal{U}$-small' make sense.

[4] C^{op} will always denote the category opposite to C.

(a) If $\mathrm{Hom}_C(y,x)$ is an element of $\mathcal{U}$, then

$$h_{\mathcal{U}}(x)(y) = \mathrm{Hom}_C(y,x).$$

(b) Suppose that $\mathrm{Hom}_C(y,x)$ is not an element of $\mathcal{U}$, and let $R(Z,x,y)$ be the relation: 'The set Z is the target of an isomorphism $\mathrm{Hom}_C(y,x) \xrightarrow{\sim} Z$.' We then put

$$h_{\mathcal{U}}(x)(y) = \tau_Z R(Z).$$

By axiom $(\mathcal{U}B)$, $h_{\mathcal{U}}(x)(y)$ is an element of $\mathcal{U}$. Let $R'(u,x,y)$ be the relation: 'u is a bijection

$$u : \mathrm{Hom}_C(y,x) \xrightarrow{\sim} h_{\mathcal{U}}(x)(y)'.$$

We put

$$\varphi(y,x) = \tau_u R'(u).$$

Notice that in both cases (a) and (b) we have a canonical isomorphism

$$\varphi(y,x) : \mathrm{Hom}_C(y,x) \xrightarrow{\sim} h_{\mathcal{U}}(x)(y).$$

In case (a), $\varphi(y,x)$ is the identity. Now let $u : y \to y'$ be an arrow of C. Composition of morphisms defines a map

$$\mathrm{Hom}_C(u,x) : \mathrm{Hom}_C(y',x) \longrightarrow \mathrm{Hom}_C(y,x).$$

We set

$$h_{\mathcal{U}}(x)(u) = \varphi(y,x)\ \mathrm{Hom}_C(u,x)\,\varphi(y',x)^{-1}.$$

One verifies immediately that $h_{\mathcal{U}}(x)$, so defined, is a functor $C^{\mathrm{op}} \to \mathcal{U}$-**Set**.

1.3.1

Similarly, using the isomorphisms φ, for every morphism $v : x \to x'$ one defines a morphism of functors

$$h_{\mathcal{U}}(v) : h_{\mathcal{U}}(x) \longrightarrow h_{\mathcal{U}}(x'),$$

and one verifies immediately that this defines a functor

$$h_{\mathcal{U}} : C \longrightarrow \widehat{C}_{\mathcal{U}}.$$

1.3.2

Let $\mathcal{V}$ now be a universe such that $\mathcal{U} \subset \mathcal{V}$. We then have a canonical fully faithful functor

$$\mathcal{U}\text{-}\mathbf{Set} \hookrightarrow \mathcal{V}\text{-}\mathbf{Set},$$

and hence a fully faithful functor

$$\widehat{C}_{\mathcal{U}} \hookrightarrow \widehat{C}_{\mathcal{V}}.$$

The diagram

$$\begin{tikzcd} C \arrow[r, "h_{\mathcal{U}}"] \arrow[dr, "h_{\mathcal{V}}"'] & \widehat{C}_{\mathcal{U}} \arrow[d] \\ & \widehat{C}_{\mathcal{V}} \end{tikzcd}$$

commutes up to canonical isomorphism. When C is an element of $\mathcal{U}$, this canonical isomorphism is the identity.

1.3.3

In practice, the universe $\mathcal{U}$ is fixed once and for all and is not mentioned. We then use the notation $\widehat{C}$ for the category of $\mathcal{U}$-presheaves of sets and

$$h : C \longrightarrow \widehat{C}.$$

For every object x of C, the presheaf $h(x)$ is called the *presheaf represented by x*, and we shall always identify its value at an object $y \in \mathrm{Ob}(C)$ with $\mathrm{Hom}_C(y, x)$.

Proposition (1.4)**.** Let C be a $\mathcal{U}$-category, let F be a presheaf on C, and let X be an object of C. There is an isomorphism, functorial in X and in F,

$$i : \mathrm{Hom}_{\widehat{C}}(h(X), F) \xleftarrow{\sim} F(X),$$

where, when F is of the form $h(Y)$, the map i is just the map

$$h : \mathrm{Hom}(h(X), h(Y)) \longleftarrow \mathrm{Hom}(X, Y).$$

In particular, the functor h is fully faithful.

1.4.1

This proposition justifies the usual abuse of language that identifies an object of C with the corresponding contravariant functor. A presheaf isomorphic to an object in the image of h, or, using the above abuse of language, isomorphic to an object of C, is called a *representable presheaf*.

1.4.2

Let $\mathcal{V}$ be a universe containing a universe $\mathcal{U}$. The category of $\mathcal{U}$-presheaves is a *full* subcategory of the category of $\mathcal{V}$-presheaves; consequently, a $\mathcal{U}$-presheaf is representable if and only if its image in the category of $\mathcal{V}$-presheaves is a representable $\mathcal{V}$-presheaf.

Continuation anchor. The next untranslated source section is Exposé I, Section 2, 'Projective and inductive limits'.

Exposé I. Presheaves

Sections 2–5

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Translator's status note. This is a working English translation of SGA 4, Exposé I, Sections 2–5, continuing the preceding translated fragment through Section 1.4. Obvious typographical slips in the working French TEX source have been corrected where the mathematics fixes the reading.

2 Projective and inductive limits

Let C be a $\mathcal{U}$-category and let I be a small category. Write $\mathrm{Fun}(I, C)$ for the category of functors from I to C. The category $\mathrm{Fun}(I, C)$ is a $\mathcal{U}$-category. To every object X of C associate the subcategory $\mathcal{X}$ of C having X as its only object and the identity of X as its only arrow. Let $i_X : \mathcal{X} \to C$ be the inclusion functor. There is a unique functor $e_X : I \to \mathcal{X}$, and we shall denote by $k_X : I \to C$ the functor $i_X \circ e_X$. We shall say that k_X is the constant functor with value X. The correspondence $X \mapsto k_X$ is visibly functorial in X, which enables us to define, for every functor $G : I \to C$, the presheaf

$$X \longmapsto \mathrm{Hom}_{\mathrm{Fun}(I,C)}(k_X, G).$$

Definition 2.1. The *projective limit* of G, denoted $\varprojlim_I G$, is the presheaf

$$X \longmapsto \mathrm{Hom}_{\mathrm{Fun}(I,C)}(k_X, G).$$

When the presheaf $\varprojlim_I G$ is representable, we again denote by $\varprojlim_I G$ an object of C representing it. The object $\varprojlim_I G$ is therefore defined only up to isomorphism. When no confusion can result, we use the abbreviated notation $\varprojlim G$.

As G varies, $\varprojlim G$ is a functor from $\mathrm{Fun}(I, C)$ to $\widehat{C}$.

2.1.1

By symmetry, reversing the direction of the arrows in C, one defines similarly the inductive limit of a functor: it is a covariant functor on C with values in the category of $\mathcal{U}$-sets. We shall use the notations $\varinjlim_I G$ and $\varinjlim G$.

Products, fiber products, and kernels are projective limits. Similarly, sums, amalgamated sums, and cokernels are inductive limits.

Definition 2.2. Let I and C be two categories, and let $G : I \to C$ be a functor. We say that the projective limit of G is *representable* if there exists a universe $\mathcal{U}$ such that:

1) the category I is $\mathcal{U}$-small;

2) the category C is a $\mathcal{U}$-category;

3) the presheaf $\varprojlim G$, with values in the category of $\mathcal{U}$-sets, is representable.

It follows from no. 1 that the object $\varprojlim G$ representing the presheaf $\varprojlim G$ does not depend, up to isomorphism, on the universe $\mathcal{U}$. Note also that there is a smallest universe $\mathcal{U}'$ having properties (1) and (2), and that the presheaf $\varprojlim G$ necessarily takes values in the category of $\mathcal{U}'$-sets.

2.2.1

Let I and C be two categories. We say that *I-projective limits in C are representable* if, for every functor $G : I \to C$, the projective limit of G is representable. Finally, let C be a category and let $\mathcal{U}$ be a universe. We say that *$\mathcal{U}$-projective limits in C are representable* if, for every $\mathcal{U}$-small category I and every functor $G : I \to C$, the projective limit of G is representable.

Proposition 2.3. Let C be a category and let $\mathcal{U}$ be a universe. The following assertions are equivalent:

i) $\mathcal{U}$-projective limits in C are representable.

ii) Products indexed by a small set are representable, and kernels of pairs of arrows are representable.

iii) Products indexed by a small set are representable, and fiber products are representable.

Proof. It suffices to observe that there is an isomorphism, functorial in G,

$$\varprojlim G = \operatorname{Ker}\left(\prod_{i\in \operatorname{Ob}(I)} G(i) \rightrightarrows \prod_{u\in \operatorname{Fl}(I)} G(\operatorname{target}(u))\right),$$

where the pair of arrows is defined by the morphisms

$$\prod_{i\in \operatorname{Ob}(I)} G(i) \xrightarrow{\operatorname{pr}_{\operatorname{target}(u)}} G(\operatorname{target}(u)),$$
$$\prod_{i\in \operatorname{Ob}(I)} G(i) \xrightarrow{\operatorname{pr}_{\operatorname{source}(u)}} G(\operatorname{source}(u)) \xrightarrow{G(u)} G(\operatorname{target}(u)).$$

Moreover, it is clear that

$$\operatorname{Ker}\big(X \overset{u}{\underset{v}{\rightrightarrows}} Y\big) = X \times_{X\times Y} X,$$

where the two morphisms from X to $X \times Y$ are $\operatorname{id}_X \times u$ and $\operatorname{id}_X \times v$.

2.3.1

There are, of course, analogous definitions and assertions for inductive limits, which we shall not spell out. The same applies to finite projective and inductive limits, that is, those relative to a finite category I.

Corollary 2.3.2. Let $\mathcal{U}$-**Set** denote the category of $\mathcal{U}$-sets, and let $\mathcal{U}$-**Ab** denote the category of abelian-group objects of $\mathcal{U}$-**Set**. The $\mathcal{U}$-projective and $\mathcal{U}$-inductive limits in $\mathcal{U}$-**Set** and in $\mathcal{U}$-**Ab** are representable.

Proposition 2.3.3. Let I be a small category, let $F : I \to \mathcal{U}$-**Set** be a functor, and let $G \subset \operatorname{Ob}(I)$ be a small set of objects of I such that, for every object X of I, there exists an object $Y \in G$ and a morphism $Y \to X$ (respectively $X \to Y$). Then $\varprojlim_I F$ (respectively $\varinjlim_I F$) is representable, and

$$\operatorname{card}(\varprojlim_I F) \leq \prod_{Y\in G} \operatorname{card}(F(Y))$$

respectively

$$\operatorname{card}(\varinjlim_I F) \leq \sum_{Y \in G} \operatorname{card}(F(Y)).$$

Proof. One checks immediately that $\varprojlim_I F$ (respectively $\varinjlim_I F$) is isomorphic to a subobject of $\prod_{Y \in G} F(Y)$ (respectively to a quotient of $\coprod_{Y \in G} F(Y)$), whence the proposition.

Definition 2.4.1. Let C be a category in which finite projective (respectively inductive) limits are representable, and let $u : C \to C'$ be a functor. We say that u is *left exact* (respectively *right exact*) if it "commutes" with finite projective (respectively finite inductive) limits. A functor that is both left and right exact is called *exact*.

2.4.2

It follows from 2.3 that, for a functor to be left exact, it is necessary and sufficient that it carry the final object, i.e. the empty product, to the final object, the product of two objects to the product of their images, and the kernel of a pair of arrows to the kernel of the image pair; equivalently, that it carry the final object to the final object and fiber products to fiber products. Here it is assumed that finite projective limits in C are representable.

2.5.0

Let I, J, and C be three categories, and let $G : I \times J \to C$ be a functor, i.e. a functor from I with values in the category of functors from J to C. Suppose that the projective limits of the functors

$$\begin{aligned} G_i : J &\longrightarrow C \qquad (i \in \operatorname{Ob}(I)) \\ j &\longmapsto G(i,j) \end{aligned}$$

are representable, and that the functor

$$i \longmapsto \varprojlim_J G_i$$

admits a representable projective limit. Then it is clear that the functor G admits a representable projective limit and that there is a canonical isomorphism

$$\varprojlim_{I \times J} G \xrightarrow{\sim} \varprojlim_I \varprojlim_J G_i,$$

and hence a canonical isomorphism

$$\varprojlim_I \varprojlim_J G_i \xrightarrow{\sim} \varprojlim_J \varprojlim_I G_j.$$

We shall say more briefly that projective limits commute with projective limits. One sees in the same way that inductive limits commute with inductive limits. But it is not true in general that inductive limits commute with projective limits.

Definition 2.5. Let C be a category with representable fiber products, let I be a category, let $G : I \to C$ be a functor, let $g : G \to X$ be a morphism from G to an object of C, i.e. a morphism from G to the constant functor associated with X, and let $m : Y \to X$ be a morphism of C. Let

$$G \times_X Y : I \longrightarrow C$$

be the functor

$$i \longmapsto G(i) \times_X Y.$$

We say that the inductive limit of G is *universal* if, for every object X, every morphism $g : G \to X$, and every morphism $m : Y \to X$,

a) the inductive limit of the functor $G \times_X Y$ is representable;

b) the canonical morphism

$$\varinjlim(G \times_X Y) \longrightarrow (\varinjlim G) \times_X Y$$

is an isomorphism.

Proposition 2.6. Let $\mathcal{U}$ be a universe. Inductive limits in $\mathcal{U}$-**Set** that are representable, in particular $\mathcal{U}$-inductive limits (2.2.1) in $\mathcal{U}$-**Set**, are universal.

We shall also use another commutation result between projective and inductive limits, which we now present.

Definition 2.7. A category I is *pseudo-filtered* if it has the following properties:

PS 1) Every diagram of the form

$$\begin{tikzcd} & j \\ i \arrow[ur] \arrow[dr] & \\ & j' \end{tikzcd}$$

can be inserted into a commutative diagram

$$\begin{tikzcd} & j \arrow[dr] & \\ i \arrow[ur] \arrow[dr] & & k \, . \\ & j' \arrow[ur] & \end{tikzcd}$$

PS 2) Every diagram of the form

$$\begin{tikzcd} i \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & j \end{tikzcd}$$

can be inserted into a diagram

$$\begin{tikzcd} i \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & j \arrow[r, "w"] & k \end{tikzcd}$$

such that $w \circ u = w \circ v$.

A category I is called *filtered* if it is pseudo-filtered, nonempty, and connected, i.e. if any two objects of I can be joined by a finite chain of arrows, with no condition imposed on the directions of the arrows. Equivalently, in the presence of PS 2, this means that $I \neq \varnothing$ and that, for any two objects a, b of I, there is always an object c of I and arrows $a \to c$ and $b \to c$. A category I is called *cofiltered* if I^{op} is filtered.

Example 2.7.1. If, in I, amalgamated sums (respectively sums of two objects) and cokernels of double arrows are representable, then I is pseudo-filtered (respectively filtered).

Proposition 2.8. Let $\mathcal{U}$ be a universe. Filtered $\mathcal{U}$-inductive limits in $\mathcal{U}$-**Set** commute with finite projective limits.

One is immediately reduced to proving that filtered $\mathcal{U}$-inductive limits commute with fiber products. The proof is left to the reader. One may use the following description of the limit.

Lemma 2.8.1. Let I be a small filtered category, and let $i \mapsto X_i$ be a functor from I to $\mathcal{U}$-**Set**. On the sum set $\coprod_{i \in \operatorname{Ob} I} X_i$, let R be the following relation:

(R) Two elements $x_i \in X_i$ and $x_j \in X_j$ are related if there exist an object $k \in \operatorname{Ob} I$ and two morphisms $u : i \to k$ and $v : j \to k$ such that the images in X_k of x_i and x_j under the transition maps u and v, respectively, are equal.

Then:

1) R is an equivalence relation.

2) The quotient $\coprod_{i \in \operatorname{Ob} I} X_i / R$ is canonically isomorphic to $\varinjlim_I X_i$.

3) Two elements $x_i \in X_i$ and $x_j \in X_j$ are respectively equivalent, modulo R, to two elements of one and the same X_k.

4) For every $i \in \operatorname{Ob} I$, two elements α and β of X_i are equivalent modulo R if and only if there exists a morphism $u : i \to j$ such that $u(\alpha) = u(\beta)$.

Assertion 1) follows from PS 1 (2.7). To prove 2), one checks that $\coprod_{i \in \operatorname{Ob} I} X_i / R$ has the universal property of the inductive limit; here only PS 1 is used. Assertion 3) follows from the fact that I is connected. Assertion 4) follows from PS 2.

Corollary 2.9. Let γ be a species of algebraic structure "defined by finite projective limits." The reader is asked to give a mathematical meaning to the preceding phrase; let us merely note that the structures of groups, abelian groups, rings, modules, etc. are structures of this kind. Let $\mathcal{U}$-γ denote the category of γ-objects of $\mathcal{U}$-**Set**. The functor that associates with every object of $\mathcal{U}$-γ its underlying set commutes with filtered $\mathcal{U}$-inductive limits. Consequently filtered $\mathcal{U}$-inductive limits in $\mathcal{U}$-γ commute with finite projective limits.

Corollary 2.10. Pseudo-filtered $\mathcal{U}$-inductive limits in $\mathcal{U}$-**Ab** commute with finite projective limits.

One is reduced to filtered limits by decomposing the indexing category into its connected components.

We now record a result that we shall use constantly. Let C and C' be two $\mathcal{U}$-categories, and let

$$C \underset{v}{\overset{u}{\rightleftarrows}} C'$$

be two functors, with u *left adjoint* to v. Recall that this means that there is an isomorphism of bifunctors with values in $\mathcal{U}$-**Set**,

$$\mathrm{Hom}_{C'}(u(X), X') \xrightarrow{\sim} \mathrm{Hom}_C(X, v(X')).$$

Proposition 2.11. The functor u commutes with representable inductive limits; the functor v commutes with representable projective limits.

This assertion means that, for every category I and every functor $G : I \to C'$ such that the projective limit of G is representable, the functor $v \circ G$ admits a representable projective limit and there is a canonical isomorphism

$$v(\varprojlim G) \longrightarrow \varprojlim(v \circ G).$$

The reader will write out for himself the assertion concerning the functor u.

Finally, let us point out a computation of projective limits in the case where the indexing category admits products.

Proposition 2.12. Let I and C be two categories, and let $\mathcal{V}$ be a universe. Suppose that $\mathcal{V}$-projective limits in C are representable and that there exists in I a family of objects $(i_\alpha)_{\alpha \in A}$, with $A \in \mathcal{V}$, such that:

1) the products $i_\alpha \times i_\beta$ are representable for every pair $(\alpha, \beta) \in A \times A$;

2) every object of I maps to at least one of the i_α.

Then, for every contravariant functor

$$F : I^{\mathrm{op}} \longrightarrow C,$$

the projective limit of F exists and there is an isomorphism, functorial in F,

$$\varprojlim F \xrightarrow{\sim} \mathrm{Ker}\left(\prod_{\alpha \in A} F(i_\alpha) \rightrightarrows \prod_{(\alpha,\beta) \in A \times A} F(i_\alpha \times i_\beta)\right),$$

where the two arrows are defined by the projections of the products $i_\alpha \times i_\beta$ onto the factors.

3 Exactness properties of the category of presheaves

Let C be a $\mathcal{U}$-category, and let $\widehat{C}$ be the category of presheaves on C. The exactness properties of $\widehat{C}$ are all deduced from the following proposition.

Proposition 3.1. The $\mathcal{U}$-projective and $\mathcal{U}$-inductive limits in $\widehat{C}$ are representable. For every object X of C, the functor on $\widehat{C}$

$$F \longmapsto F(X), \qquad F \in \widehat{C},$$

commutes with inductive and projective limits.

In other words, inductive and projective limits in $\widehat{C}$ are computed argument by argument. We list a few corollaries.

Corollary 3.2. Let γ be an algebraic structure defined by finite projective limits. The category of contravariant functors with values in $\mathcal{U}$-γ is equivalent to the category of γ-objects of $\widehat{C}$.

Corollary 3.3. A morphism of $\widehat{C}$ that is both a monomorphism and an epimorphism is an isomorphism. A morphism of $\widehat{C}$ factors uniquely as an epimorphism followed by a monomorphism. Representable inductive limits in $\widehat{C}$ are universal. Filtered $\mathcal{U}$-inductive limits in $\widehat{C}$ commute with finite projective limits. The canonical functor $C \to \widehat{C}$ commutes with representable projective limits.

More generally, one may say that the category $\widehat{C}$ inherits all the properties of the category of $\mathcal{U}$-sets that involve inductive and projective limits.

3.4.0

Let F be an object of $\widehat{C}$. We denote by C/F the following category. Its objects are pairs consisting of an object X of C and a morphism u from X to F. Given two objects (X, u) and (Y, v), a morphism from (X, u) to (Y, v) is a morphism $g : X \to Y$ such that the following diagram is commutative:

$$\begin{tikzcd} X \arrow[rr, "g"] \arrow[dr, "u"'] & & Y \arrow[dl, "v"] \\ & F & \end{tikzcd}$$

Proposition 3.4. With the notation of (3.4.0), the source functor $C/F \to \widehat{C}$ admits a representable inductive limit in $\widehat{C}$. The canonical morphism

$$\varinjlim_{C/F} \mathrm{source}(.) \longrightarrow F$$

is an isomorphism.

Corollary 3.5. Let F and H be two objects of $\widehat{C}$. There is a canonical isomorphism

$$\mathrm{Hom}(F, H) \xrightarrow{\sim} \varprojlim_{(X,u)\in \mathrm{Ob}(C/F)} H(X).$$

Proof. The corollary follows immediately from 3.4 and 1.2.

Proposition 3.6. Let C be a $\mathcal{U}$-category, let $\mathcal{V}$ be a universe containing $\mathcal{U}$, and let

$$i : \widehat{C}_{\mathcal{U}} \longrightarrow \widehat{C}_{\mathcal{V}}$$

be the natural inclusion functor between the corresponding categories of presheaves (1.3). The functor i commutes with inductive and projective limits. Moreover, for every object F of $\widehat{C}_{\mathcal{U}}$, every subobject of $i(F)$ is isomorphic to the image under i of a unique subobject of F; thus one obtains a bijection from the set of subobjects of F to the set of subobjects of $i(F)$.

4 Sieves

Definition 4.1. Let C be a category. A *sieve* of the category C is a full subcategory D of C having the following property: every object of C admitting a morphism to an object of D belongs to D. If X is an object of C, the sieves of the category C/X will be called, by abuse of language, *sieves of X*.

Let $\mathcal{U}$ be a universe such that C is a $\mathcal{U}$-category, and let $\widehat{C}$ be the corresponding category of presheaves. To every sieve of X one associates a subobject of X in $\widehat{C}$ as follows: to every object Y of C one assigns the set of morphisms $f : Y \to X$ such that the object (Y, f) belongs to the sieve.

Proposition 4.2. The map defined above establishes a bijection between the set of sieves of X and the set of subobjects of X in $\widehat{C}$.

Proof. We only indicate the inverse map. To every subfunctor R of X one associates the category C/R of objects of C over R (3.4). One checks immediately that C/R is a sieve of X.

Remark 4.2.1. One sees in the same way that the sieves of C are in canonical one-to-one correspondence with the set of subfunctors of the final functor on C, the final object of $\widehat{C}$.

4.3

Let C be a $\mathcal{U}$-category. By abuse of language, we shall also call the subobjects of X in the category $\widehat{C}$ the sieves of X. This abuse of language allows us, for every presheaf F and every sieve R of X, to define $\mathrm{Hom}_{\widehat{C}}(R, F)$ as the set of morphisms from the functor R to F. There is, moreover, an isomorphism canonical and functorial in F (3.5):

$$\mathrm{Hom}_{\widehat{C}}(R, F) \xrightarrow{\sim} \varprojlim_{C/R} F(.),$$

which also gives a direct definition.[1] Similarly, Proposition 4.2 allows us to transpose the usual operations on functors to sieves. We mention the following.

4.3.1 Base change

Let R be a sieve of X, and let $f : Y \to X$ be a morphism of objects of C. The fiber product $R \times_X Y$ is a sieve of Y, called the *sieve deduced from R by base change*. The corresponding subcategory of C/Y is the inverse image of the subcategory of C/X defined by R, under the canonical functor $C/Y \to C/X$ defined by f.

4.3.2 Order relation, intersection, union

The inclusion relation on subfunctors of X is an order relation. One may define the union and intersection of a family of sieves indexed by an arbitrary set as the least upper bound and greatest lower bound of the corresponding family of subpresheaves.

[1] More simply, C/R identifies with R, so that one has the formula $\mathrm{Hom}_{\widehat{C}}(R, F) \xrightarrow{\sim} \varprojlim_R F$, where F abusively denotes the composite of F with the tautological source functor $R \to C/X \to C$.

4.3.3 Image, generated sieve

Let $(F_\alpha)_{\alpha\in A}$ be a family of presheaves, and for each $\alpha \in A$ let $f_\alpha : F_\alpha \to X$ be a morphism, where X is an object of C. The *image* of this family of morphisms is the union of the images of the f_α. The image of this family is therefore a sieve of X. In particular, if all the F_α are objects of C, the image sieve will be called the sieve generated by the morphisms f_α. The category C/R is the full subcategory of C/X formed by those objects $X' \to X$ over X for which there exists an X-morphism from X' to one of the F_α.

As an exercise, the reader may translate the relations and operations just defined on subfunctors into terms of the categories C/R. He will then see that these relations and operations do not depend on the universe $\mathcal{U}$ such that C is a $\mathcal{U}$-category, and that they are therefore defined for every category, without requiring one to specify the universe to which the sets of morphisms belong. This was in any case already foreseeable *a priori* from 3.6.

5 Functoriality of categories of presheaves

5.0

Let C, C', and D be three categories, and let $u : C \to C'$ be a functor. We denote by u^* the functor obtained by composition with u:

$$u^* : \mathrm{Hom}((C')^{\mathrm{op}}, D) \longrightarrow \mathrm{Hom}(C^{\mathrm{op}}, D), \qquad G \longmapsto G \circ u.$$

The functor u^* commutes with inductive and projective limits.

Proposition 5.1. Suppose that C is small and that, in D, $\mathcal{U}$-inductive limits (respectively projective limits) are representable. The functor u^* admits a left adjoint $u_!$ (respectively a right adjoint u_*). Thus one has an isomorphism

$$\mathrm{Hom}_{\mathrm{Hom}(C^{\mathrm{op}},D)}(F, u^*G) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Hom}((C')^{\mathrm{op}},D)}(u_!F, G),$$
$$\big(\text{respectively } \mathrm{Hom}_{\mathrm{Hom}(C^{\mathrm{op}},D)}(u^*G, F) \simeq \mathrm{Hom}_{\mathrm{Hom}((C')^{\mathrm{op}},D)}(G, u_*F)\big).$$

Proof. We indicate only the proof of the existence of the left adjoint. The corresponding part of the proposition follows formally from the isomorphisms

$$\mathrm{Hom}(C^{\mathrm{op}}, D) \xrightarrow{\sim} \mathrm{Hom}(C, D^{\mathrm{op}})^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Hom}((C^{\mathrm{op}})^{\mathrm{op}}, D^{\mathrm{op}})^{\mathrm{op}}.$$

Let Y be an object of C'. Denote by I_u^Y the following category. Its objects are pairs (X, m), where X is an object of C and m is a morphism $Y \to u(X)$. If (X, m) and (X', m') are two objects of I_u^Y, a morphism from (X, m) to (X', m') is a morphism $\xi : X \to X'$ such that $m' = u(\xi)m$. Composition is defined in the evident way.

Let $f : Y \to Y'$ be a morphism of C'. By composition, f defines a functor $I_u^f : I_u^Y \to I_u^{Y'}$.

We also have a functor $\mathrm{pr}_Y : I_u^Y \to C$ which sends the object (X, m) to the object X. Then the evident diagram

$$\begin{tikzcd} I_u^Y \arrow[r, "I_u^f"] \arrow[d, "\mathrm{pr}_Y"'] & I_u^{Y'} \arrow[dl, "\mathrm{pr}_{Y'}"] \\ C & \end{tikzcd} \tag{*}$$

is commutative.

Now let F be a presheaf on C and set

$$u_!F(Y) = \varinjlim_{I_u^Y} F \circ \mathrm{pr}_Y \,. \tag{5.1.1}$$

The commutativity of diagram $(*)$ and the functoriality of inductive limits make $u_!F$ into a presheaf on C'. Let us show that the functor $u_!$ is left adjoint to u^*. For this it is enough to show that, for every presheaf G on C', there is a functorial isomorphism

$$\mathrm{Hom}(u_!F, G) \xrightarrow{\sim} \mathrm{Hom}(F, u^*G).$$

Let $\xi \in \mathrm{Hom}(u_!F, G)$. For every object X of C, we have a morphism

$$\xi_{u(X)} : u_!F(u(X)) \longrightarrow G(u(X)).$$

But $(X, \mathrm{id}_{u(X)})$ is an object of $I_u^{u(X)}$, and by the definition of the inductive limit there is a canonical morphism

$$F(X) \longrightarrow u_!F(u(X)).$$

We obtain, for every object X of C, a morphism

$$\eta_X : F(X) \longrightarrow G(u(X)),$$

which is visibly functorial in X. Hence a morphism

$$\eta : F \longrightarrow u^*G.$$

Conversely, let $\eta \in \mathrm{Hom}(F, u^*G)$. One obtains, for every object Y of C', a morphism of functors

$$\eta_Y : F \circ \mathrm{pr}_Y \longrightarrow (u^*G) \circ \mathrm{pr}_Y,$$

and hence, by composing with the evident morphism from the functor $(u^*G) \circ \mathrm{pr}_Y$ to the constant functor $G(Y)$, a morphism

$$\xi_Y : u_!F(Y) \longrightarrow G(Y),$$

which is functorial in Y. Hence a morphism

$$\xi : u_!F \longrightarrow G.$$

The reader will check that the two maps thus defined are inverse to one another, completing the proof.

Proposition 5.2. Suppose that in D the $\mathcal{U}$-inductive limits are representable, the finite projective limits are representable, and filtered $\mathcal{U}$-inductive limits commute with finite projective limits. Suppose moreover that, in C, finite projective limits are representable and that the functor u is left exact (2.3.2). Then finite projective limits are representable in $\mathrm{Hom}(C^{\mathrm{op}}, D)$ and in $\mathrm{Hom}((C')^{\mathrm{op}}, D)$, and the functor $u_!$ is left exact.

Proof. The first assertion is trivial. Let us prove the second. By the proof of 5.1, for every presheaf F on C and every object Y of C' one has

$$u_!F(Y) \xrightarrow{\sim} \varinjlim_{I_u^Y} F \circ \mathrm{pr}_Y \,.$$

It is therefore enough to show that the category $(I_u^Y)^{\mathrm{op}}$ satisfies axioms PS 1 and PS 2 (2.7), and that this category is connected. The verification is left to the reader.

5.3

Specializing these results to the case where D is the category of $\mathcal{U}$-sets, one obtains a sequence of three functors

$$u_!, \quad u^*, \quad u_*,$$

which is a "sequence of adjoint functors" in the sense that, for two consecutive functors in the sequence, the one on the right is right adjoint to the other. Their essential properties are summarized in the following proposition.

Proposition 5.4. Let C be a small category, let C' be a $\mathcal{U}$-category, and let $u : C \to C'$ be a functor.

1) The functor $u^* : \widehat{C'} \to \widehat{C}$ commutes with inductive and projective limits.

2) The functor $u_* : \widehat{C} \to \widehat{C'}$ commutes with projective limits. For every presheaf F on C and every object Y of C', one has

$$u_*F(Y) \xrightarrow{\sim} \operatorname{Hom}_{\widehat{C}}(u^*(Y), F).$$

3) The functor $u_! : \widehat{C} \to \widehat{C'}$ commutes with inductive limits. The functor $u_!$ is defined only up to isomorphism, but it can always be chosen so that the diagram

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ {\scriptstyle h}\big\downarrow & & \big\downarrow{\scriptstyle h'} \\ \widehat{C} & \xrightarrow[u_!]{} & \widehat{C'} \end{array}$$

is commutative, where h and h' are the canonical inclusion functors.

For every presheaf F on C, one has

$$u_!F \xrightarrow{\sim} \varinjlim_{C/F} h' \circ u.$$

4) If finite projective limits are representable in C and if u is left exact (2.3.2), then the functor $u_!$ is left exact.

Proof. Assertion (1) is trivial. Assertion (2) follows from the fact that u_* is a right adjoint, by 2.11 and 1.4. The same applies to assertion (3), but one also uses 3.4. Finally, assertion (4) is precisely 5.2, which applies by 2.7.

Proposition 5.5. Let C and C' be two small categories, and let

$$C \underset{v}{\overset{u}{\rightleftarrows}} C'$$

be a pair of functors, where v is left adjoint to u. Then there exist isomorphisms, compatible with the adjunction isomorphisms,

$$v^* \xrightarrow{\sim} u_!, \qquad v_* \xrightarrow{\sim} u^*.$$

Proof. It is enough to exhibit an isomorphism $v^* \xrightarrow{\sim} u_!$; the other isomorphism will follow by adjunction. Let F be a presheaf on C and let Y be an object of C'. Then

$$v^*F(Y) \xrightarrow{\sim} \mathrm{Hom}(v(Y), F).$$

Using 3.4, we get

$$\mathrm{Hom}(v(Y), F) \xrightarrow{\sim} \varinjlim_{C/F} \mathrm{Hom}(v(Y), .).$$

Since v is left adjoint to u, it follows that

$$\varinjlim_{C/F} \mathrm{Hom}(v(Y), .) \xrightarrow{\sim} \varinjlim_{C/F} \mathrm{Hom}(Y, u(.)).$$

Using 5.4.3), we then obtain

$$\varinjlim_{C/F} \mathrm{Hom}(Y, u(.)) \xrightarrow{\sim} \mathrm{Hom}(Y, u_!F) \xrightarrow{\sim} u_!F(Y).$$

Thus, for every object Y of C', we have determined an isomorphism $v^*F(Y) \xrightarrow{\sim} u_!F(Y)$, visibly functorial in Y and in F.

Corollary 5.5.1. Let $u : C \to C'$ be a functor that admits a left adjoint. The functor $u_! : \widehat{C} \to \widehat{C'}$ commutes with projective limits; recall that it commutes with inductive limits by 5.4.3.

Remark 5.5.2. One thus obtains a "sequence of *four* adjoint functors" (cf. 5.3):

$$v_!, \quad v^* = u_!, \quad v_* = u^*, \quad u_*,$$

of which the first three (respectively the last three) commute with inductive limits (respectively projective limits).

Proposition 5.6. The hypotheses are those of 5.4. The following conditions are equivalent:

i) the functor u is fully faithful;

ii) the functor $u_!$ is fully faithful;

iii) the adjunction morphism $\mathrm{id}_{\widehat{C}} \to u^*u_!$ is an isomorphism;

iv) the functor u_* is fully faithful;

v) the adjunction morphism $u^*u_* \to \mathrm{id}_{\widehat{C}}$ is an isomorphism.

Proof. It is clear that ii) $\Leftrightarrow$ iii) and iv) $\Leftrightarrow$ v), by general properties of adjoint functors, and that ii) $\Rightarrow$ i), by 5.4.3). Let us prove that i) $\Rightarrow$ iii). The functors $\mathrm{id}_{\widehat{C}}$, u^*, and $u_!$ commute with inductive limits. By 3.4, it is therefore enough to prove that $H \to u^*u_!H$ is an isomorphism when H is representable, which is evident.

It remains to show that iii) is equivalent to v). For every object H (respectively K) of $\widehat{C}$, denote by $\Phi(H) : H \to u^*u_!H$ (respectively by $\Psi(K) : u^*u_*K \to K$) the adjunction morphism. Then one has a commutative diagram

$$\begin{tikzcd}
\mathrm{Hom}_{\widehat{C}}(H, u^*u_*K) \arrow[r, "\mathrm{Hom}(H,\Psi(K))"] \arrow[d, "\sim"'] & \mathrm{Hom}_{\widehat{C}}(H, K) \\
\mathrm{Hom}_{\widehat{C'}}(u_!H, u_*K) \arrow[d, "\sim"'] & \\
\mathrm{Hom}_{\widehat{C}}(u^*u_!H, K) \arrow[uur, "\mathrm{Hom}(\Phi(H),K)"'] &
\end{tikzcd}$$

Hence $\Phi(H)$ is an isomorphism for every H if and only if $\Psi(K)$ is an isomorphism for every K.

Remark 5.7.

a) The equivalences ii) $\Leftrightarrow$ iii) $\Leftrightarrow$ iv) $\Leftrightarrow$ v) are general facts about adjoint functors.

b) The explicit form of $u_!$, respectively u_*, given in the proof of 5.1 shows immediately that, under the general hypotheses of 5.1, if u is fully faithful, then the adjunction morphism $\mathrm{id} \to u^*u_!$ (respectively $u^*u_* \to \mathrm{id}$) is an isomorphism; equivalently, $u_!$ (respectively u_*) is fully faithful.

5.8.0

Let γ be a species of algebraic structure defined by finite projective limits. Let $\mathcal{U}$-γ-**Set** be the category of γ-objects of $\mathcal{U}$-**Set**, and let

$$\mathrm{und} : \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

be the underlying-set functor. For simplicity, we assume that the species of structure under consideration has a single underlying set. Let C be a category. Composition with und gives a functor denoted

$$\widehat{\mathrm{und}} : \mathrm{Hom}(C^{\mathrm{op}}, \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}) \longrightarrow \widehat{C}.$$

Since in $\widehat{C}$ projective limits are computed argument by argument, the functor $\widehat{\mathrm{und}}$ factors as an equivalence

$$\mathrm{Hom}(C^{\mathrm{op}}, \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}) \xrightarrow{\approx} \widehat{C}_\gamma, \tag{5.8.1}$$

where $\widehat{C}_\gamma$ is the category of γ-objects of $\widehat{C}$, followed by a functor again denoted

$$\widehat{\mathrm{und}} : \widehat{C}_\gamma \longrightarrow \widehat{C},$$

and called the *underlying presheaf of sets* functor.

5.8.2

Suppose that the functor

$$\mathcal{U}\text{-}\gamma\text{-}\mathbf{Set} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

admits a left adjoint

$$\mathrm{Free} : \mathcal{U}\text{-}\mathbf{Set} \longrightarrow \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set},$$

the functor of "freely generated γ-objects." Examples are the free group, free commutative group, free A-module, and so on. In fact one can show that this condition is always satisfied. Composition with Free gives a functor

$$\widehat{\mathrm{Free}} : \widehat{C} \longrightarrow \mathrm{Hom}(C^{\mathrm{op}}, \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}),$$

and, by composing with the equivalence (5.8.1), a functor again denoted

$$\widehat{\mathrm{Free}} : \widehat{C} \longrightarrow \widehat{C}_\gamma,$$

and called the functor "presheaf of freely generated γ-objects." The functor $\widehat{\mathrm{Free}}$ is left adjoint to $\widehat{\mathrm{und}}$.

Proposition 5.8.3. Let γ be a species of algebraic structure defined by finite projective limits such that, in the category of γ-objects of $\mathcal{U}$-**Set**, $\mathcal{U}$-inductive limits are representable.[2] Let C be a category belonging to $\mathcal{U}$, let C' be a $\mathcal{U}$-category, and let $u : C \to C'$ be a functor. Denote by $\widehat{C}_\gamma$ (respectively $\widehat{C'}_\gamma$) the category of γ-objects of $\widehat{C}$ (respectively $\widehat{C'}$), and by $(u^*)^\gamma$ the functor on γ-objects deduced from u^*. It follows from 5.1 and the equivalence 5.8.1 that there exists a functor left adjoint (respectively right adjoint) to $(u^*)^\gamma$. This functor is denoted $u_{!\gamma}$ (respectively $u_{*\gamma}$).

1) The functor $(u^*)^\gamma$ commutes with inductive and projective limits. The diagram

$$\begin{tikzcd} \widehat{C'}_\gamma \arrow[r, "(u^*)^\gamma"] \arrow[d, "\widehat{\mathrm{und}'}"'] & \widehat{C}_\gamma \arrow[d, "\widehat{\mathrm{und}}"] \\ \widehat{C'} \arrow[r, "u^*"'] & \widehat{C} \end{tikzcd} \qquad (*)$$

is commutative; here $\widehat{\mathrm{und}'}$ and $\widehat{\mathrm{und}}$ denote the underlying-set functors.

2) The functor $u_{*\gamma}$ commutes with projective limits. The diagram

$$\begin{tikzcd} \widehat{C}_\gamma \arrow[r, "u_{*\gamma}"] \arrow[d, "\widehat{\mathrm{und}}"'] & \widehat{C'}_\gamma \arrow[d, "\widehat{\mathrm{und}'}"] \\ \widehat{C} \arrow[r, "u_*"'] & \widehat{C'} \end{tikzcd} \qquad (**)$$

is commutative up to isomorphism.

[2] One can show that this condition is always satisfied.

3) The functor $u_{!\gamma}$ commutes with inductive limits. Suppose that $\widehat{\text{und}}$ (respectively $\widehat{\text{und}'}$) admits a left adjoint $\widehat{\text{Free}}$ (respectively $\widehat{\text{Free}'}$), as in (5.8.2). The diagram

$$\begin{tikzcd} \widehat{C} \arrow[r, "u_!"] \arrow[d, "\widehat{\text{Free}}"'] & \widehat{C'} \arrow[d, "\widehat{\text{Free}'}"] \\ \widehat{C}_\gamma \arrow[r, "u_{!\gamma}"'] & \widehat{C'}_\gamma \end{tikzcd} \tag{***}$$

is commutative up to isomorphism.

Suppose that $u_!$ commutes with finite projective limits (5.4 and 5.6). Then the diagram

$$\begin{tikzcd} \widehat{C}_\gamma \arrow[r, "u_{!\gamma}"] \arrow[d, "\widehat{\text{und}}"'] & \widehat{C'}_\gamma \arrow[d, "\widehat{\text{und}'}"] \\ \widehat{C} \arrow[r, "u_!"'] & \widehat{C'} \end{tikzcd} \tag{****}$$

is commutative up to isomorphism, and $u_{!\gamma}$ commutes with finite projective limits.

Proof. Assertion (1) is evident. Assertion (2) is also evident, since u_* is a right adjoint and hence, by 2.11, commutes with projective limits and in particular with finite projective limits; this gives the commutativity of diagram (**). The commutativity of diagram (***) follows from the uniqueness, up to isomorphism, of the left adjoint functor. Finally, the commutativity of diagram (****) follows immediately from the fact that $u_!$ commutes with finite projective limits.

Notation 5.9. By abuse of notation, the functors $(u^*)^\gamma$ and $u_{*\gamma}$ will often be denoted by u^* and u_*; this risks no confusion in view of the commutativity of diagrams $(*)$ and $(**)$. In contrast, when $u_!$ does not commute with finite projective limits, diagram $(****)$ is not commutative up to isomorphism, and the notations $u_!$ and $u_{!\gamma}$ must be used to avoid confusion.

5.10

Let C be a small category. For every object X of $\widehat{C}$, let C/X denote the category of arrows whose target is X and whose source is an object of C. The source functor defines a functor $j_X : C/X \to C$. Let $m : Y \to X$ be a morphism of $\widehat{C}$. Composition of morphisms defines a functor $j_m : C/Y \to C/X$. The diagram

$$\begin{tikzcd} C/Y \arrow[rr, "j_m"] \arrow[dr, "j_Y"'] & & C/X \arrow[dl, "j_X"] \\ & C & \end{tikzcd}$$

is commutative. It follows from 5.1 that, for every object F of $\widehat{C/X}$ and every object Y of C, one has

$$j_{X!}F(Y) = \coprod_{u \in \operatorname{Hom}_{\widehat{C}}(Y,X)} F(u). \tag{5.10.1}$$

Formula (5.10.1) allows one to define $j_{X!}$ when C is a $\mathcal{U}$-category, and one verifies that the functor $j_{X!}$ thus defined is always left adjoint to the functor

$$j_X^* : \widehat{C} \longrightarrow \widehat{C/X}.$$

Proposition 5.11. Let C be a $\mathcal{U}$-category, and let X be a presheaf on C.

1) The functor

$$j_{X!} : \widehat{C/X} \longrightarrow \widehat{C}$$

factors through the category $\widehat{C}/X$:

$$\widehat{C/X} \xrightarrow{e_X} \widehat{C}/X \longrightarrow \widehat{C}.$$

The functor e_X is an equivalence of categories.

2) The functor

$$e_X \circ j_X^* : \widehat{C} \longrightarrow \widehat{C}/X$$

is canonically isomorphic to the functor

$$H \longmapsto (H \times X \xrightarrow{\mathrm{pr}_2} X).$$

Proof.

1) Let f be the final object of $\widehat{C/X}$. One has a canonical isomorphism

$$f \xrightarrow{\sim} \varinjlim_{Y \in \mathrm{Ob}(C/X)} Y,$$

and hence $j_{X!}(f) = \varinjlim_{Y \in \mathrm{Ob}(C/X)} j_X(Y)$ by 5.4. But $j_{X!}(f) \simeq X$, which gives the factorization. To show that e_X is an equivalence, we merely exhibit a quasi-inverse: to every object $H \to X$ of $\widehat{C}/X$ associate the presheaf on C/X

$$(Y \to X) \longmapsto \mathrm{Hom}_{\widehat{C}/X}((Y \to X), (H \to X)).$$

2) The functor $e_X \circ j_X^*$ is right adjoint to the forgetful functor, and therefore $e_X \circ j_X^*$ is canonically isomorphic to the functor $H \mapsto (H \times X \xrightarrow{\mathrm{pr}_2} X)$.

5.12

Let $m : Y \to X$ be a morphism of $\widehat{C}$. By (5.11), the morphism m is canonically isomorphic to the image under e_X of an object of $\widehat{C/X}$, which we shall denote by $[m]$. By restriction to subcategories, the functor e_X defines an equivalence

$$(C/X)/[m] \xrightarrow{e_m} C/Y.$$

The diagram

$$\begin{tikzcd}
(C/X)/[m] \arrow[rr, "e_m"] \arrow[dr, "j_{[m]}"'] & & C/Y \arrow[dl, "j_m"] \\
& C/X &
\end{tikzcd}$$

is commutative up to canonical isomorphism.

5.13

We record a result that will be useful in Exposé VI. Let $u : C \to C'$ be a functor between small categories. For every object H of $\widehat{C}$, denote by

$$u/H : C/H \longrightarrow C'/u_!H$$

the functor which associates to every morphism $m : X \to H$ the morphism

$$u_!X \xrightarrow{u_!m} u_!H$$

(one knows from 5.4 that one may always set $u_!X = uX$). The following diagram is commutative:

$$\begin{tikzcd} C/H \arrow[r, "u/H"] \arrow[d, "j_H"'] & C'/u_!H \arrow[d, "j_{u_!H}"] \\ C \arrow[r, "u"'] & C'. \end{tikzcd} \tag{5.13.1}$$

We therefore have a diagram commutative up to isomorphism; and, since $(u/H)_!$ carries the final object of $\widehat{C/H}$ to the final object of $\widehat{C'/u_!H}$, the diagram

$$\begin{tikzcd} \widehat{C/H} \arrow[r, "(u/H)_!"] \arrow[d, "e_H"'] & \widehat{C'/u_!H} \arrow[d, "e_{u_!H}"] \\ \widehat{C}/H \arrow[r, "u_!/H"'] & \widehat{C'}/u_!H \end{tikzcd} \tag{5.13.3}$$

is commutative up to isomorphism.

Proposition 5.14. Let $u : C \to C'$ be a functor between small categories. Suppose that u has the following property:

(PPF) For every object X of C, the functor

$$u/X : C/X \longrightarrow C'/uX$$

is fully faithful.

Then:

1) Let f be the final object of $\widehat{C}$. The functor u factors as

$$C = C/f \xrightarrow{u/f} C'/u_!f \xrightarrow{j_{u_!f}} C'.$$

The functor u/f is fully faithful.

2) The functor $u_! : \widehat{C} \to \widehat{C'}$ factors as

$$\widehat{C} = \widehat{C/f} \xrightarrow{(u/f)_!} \widehat{C'/u_!f} \xrightarrow{e_{u_!f}} \widehat{C'}/u_!f \longrightarrow \widehat{C'},$$

where the functor $(u/f)_!$ is fully faithful, the functor $e_{u_!f}$ is an equivalence, and $\widehat{C'}/u_!f \to \widehat{C'}$ is the forgetful functor.

3) In particular, the functor $u_!$ is faithful and consequently the adjunction morphism $\mathrm{id} \xrightarrow{\Phi} u^* u_!$ is a monomorphism. Moreover, for every morphism $\alpha : H \to K$ of $\widehat{C}$, the diagram

$$\begin{tikzcd} H \arrow[r, hook, "\Phi(H)"] \arrow[d, "\alpha"'] & u^* u_! H \arrow[d, "u^* u_!(\alpha)"] \\ K \arrow[r, hook, "\Phi(K)"'] & u^* u_! K \end{tikzcd}$$

is cartesian.

Proof.

1) The factorization comes from diagram (5.13.1). The functor u is faithful. Hence u/f is faithful. Let us show that it is fully faithful. Let X and Y be two objects of C, let $\mathrm{can}_X : uX \to u_! f$ and $\mathrm{can}_Y : uY \to u_! f$ be the canonical morphisms, and let

$$\begin{tikzcd} uX \arrow[rr, "m"] \arrow[dr, "\mathrm{can}_X"'] & & uY \arrow[dl, "\mathrm{can}_Y"] \\ & u_! f & \end{tikzcd}$$

be a morphism of $C'/u_! f$. We have $u_! f = \varinjlim_{Z \in \mathrm{Ob}\, C} uZ$, hence by 3.1,

$$\mathrm{Hom}_{\widehat{C'}}(uX, u_! f) = \varinjlim_{Z \in \mathrm{Ob}\, C} \mathrm{Hom}_{C'}(uX, uZ).$$

By the definition of the inductive limit, to say that $\mathrm{can}_Y\, m = \mathrm{can}_X$ is equivalent to saying that there exist

a) a finite sequence of objects X_i of C, $i \in [0, n]$, with $X_0 = X$ and $X_n = Y$;

b) for every i, a morphism $m_i : uX \to uX_i$, with $m_0 = \mathrm{id}_X$ and $m_n = m$;

c) for every pair $(i, i+1)$, a morphism $f_i : X_i \to X_{i+1}$ or a morphism $f_i : X_{i+1} \to X_i$;

such that the corresponding triangular diagrams

$$\begin{tikzcd} & uX \arrow[dl, "m_i"'] \arrow[dr, "m_{i+1}"] & \\ uX_i \arrow[rr, "u(f_i)"'] & & uX_{i+1} \end{tikzcd} \quad \text{or} \quad \begin{tikzcd} & uX \arrow[dl, "m_i"'] \arrow[dr, "m_{i+1}"] & \\ uX_i & & uX_{i+1} \arrow[ll, "u(f_i)"] \end{tikzcd}$$

are commutative. One then proves immediately, by induction on i and using property (PPF), that m_i is of the form $u(p_i)$. In particular $m = u(p)$, and hence u/f is fully faithful.

2) The factorization is immediate. The functor $(u/f)_!$ is fully faithful by 5.6. The other assertions follow from 5.11.

3) The functor $u_!$ is the composite of the forgetful functor, which is faithful, and fully faithful functors. It is therefore faithful. It follows, from the general properties of adjoint functors, that the adjunction morphism $\mathrm{id} \to u^* u_!$ is a monomorphism. By 2), the functor $u_!$ appears

as the composite of a fully faithful functor $v : \widehat{C} \to \widehat{C'}/u_! f$ and the forgetful functor. The functor u^*, right adjoint to $u_!$, is therefore the composite of the "product by $u_! f$" functor, right adjoint to the forgetful functor, and a functor w right adjoint to v. Moreover, since v is fully faithful, the adjunction morphism $\mathrm{id} \to wv$ is an isomorphism. The last assertion follows easily.

Exposé I. Presheaves

Sections 6–7

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Translator's status note. This is a working English translation of SGA 4, Exposé I, Sections 6–7. Obvious typographical slips in the working French TEX source have been corrected where the mathematics fixes the reading; they are recorded in the accompanying worklog.

6 Faithful functors and conservative functors

Definition 6.1. Let E be a category, and let

$$(\varphi_i : E \to F_i)_{i \in I}$$

be a family of functors. We say that the family (φ_i) is *faithful* if, for every pair of objects X, Y of E and every pair of arrows $u, v : X \rightrightarrows Y$, the equalities $\varphi_i(u) = \varphi_i(v)$ for all $i \in I$ imply $u = v$. In other words, for every X, Y the map

$$\mathrm{Hom}_E(X, Y) \longrightarrow \prod_i \mathrm{Hom}_{F_i}(\varphi_i(X), \varphi_i(Y))$$

defined by the φ_i is injective. We say that (φ_i) is *conservative* if every arrow u of E such that $\varphi_i(u)$ is an isomorphism for all $i \in I$ is itself an isomorphism. We say that (φ_i) is conservative for monomorphisms, respectively for epimorphisms, etc., if the preceding condition is verified whenever u is a monomorphism, respectively an epimorphism, etc.

6.1.1

If one introduces the single functor

$$\varphi : E \longrightarrow F = \prod_{i \in I} F_i$$

defined by the family (φ_i), then it is clear that the family is faithful, respectively conservative, respectively conservative for monomorphisms, etc., if and only if φ is faithful, respectively conservative, etc. Here, of course, this means that the one-object family consisting of φ has the corresponding property. Thus there would be no serious loss in subsequently restricting to the case of a family consisting of a single functor. For convenience of later reference, however, we give the following statements for families.

The notions of 6.1 are especially useful when the φ_i satisfy suitable exactness properties; in that case they tend to coincide.

Proposition 6.2. The notation is that of 5.1.

(i) Suppose that kernels of pairs of arrows, or cokernels of pairs of arrows, are representable in E, and that the φ_i commute with them. Then

$$(\varphi_i) \text{ conservative} \implies (\varphi_i) \text{ faithful.}$$

(ii) Suppose that fiber products, respectively amalgamated sums, are representable in E, and that the φ_i commute with them. Suppose that (φ_i) is faithful or conservative. Then, for every arrow u of E, the arrow u is a monomorphism, respectively an epimorphism, if and only if, for every $i \in I$, the arrow $\varphi_i(u)$ has the corresponding property.

(iii) Suppose that in E fiber products and amalgamated sums are representable, that the φ_i commute with them, and that every arrow in E which is a bimorphism, i.e. both a monomorphism and an epimorphism, is an isomorphism. Then

$$(\varphi_i) \text{ faithful} \quad \Longrightarrow \quad (\varphi_i) \text{ conservative.}$$

(iv) Suppose that in E fiber products, respectively amalgamated sums, are representable, and that the φ_i commute with them. If (φ_i) is conservative for monomorphisms, respectively for epimorphisms, then (φ_i) is conservative.

(v) Let D be a diagram type, let $F : d \mapsto F(d)$ be a diagram of type D in E, let X be an object of E, and let $u = (u_d)_{d \in D}$ be a family of arrows

$$X \longrightarrow F(d) \qquad \text{(respectively } F(d) \longrightarrow X\text{).}$$

Suppose that (φ_i) is conservative, that projective limits, respectively inductive limits, of type D are representable in E, and that the φ_i commute with them. Then u makes X a projective limit, respectively an inductive limit, of F in E if and only if, for every $i \in I$, $\varphi_i(u)$ makes $\varphi_i(X)$ a projective limit, respectively an inductive limit, of $\varphi_i(F)$ in F_i.

Proof. For (i), in the non-dual statement, given a pair of arrows $u, v : X \rightrightarrows Y$, it suffices to express the equality $u = v$ by the condition that the inclusion

$$\operatorname{Ker}(u, v) \longrightarrow X$$

is an isomorphism. Here, and below, we do not repeat the dual argument for the dual statement.

For (ii), if (φ_i) is faithful, one expresses the condition that $u : X \to Y$ be a monomorphism by the equality $\mathrm{pr}_1 = \mathrm{pr}_2$ for the fiber product $X \times_Y X$. If (φ_i) is conservative, one expresses it by the condition that the diagonal morphism

$$\delta : X \longrightarrow X \times_Y X$$

be an isomorphism.

For (iv), in the latter argument the morphism δ is a monomorphism, so it would actually have sufficed to assume that (φ_i) is conservative for monomorphisms. This then implies that (φ_i) is conservative tout court. Indeed, if $u \in \mathrm{Fl}(E)$ is such that the $\varphi_i(u)$ are isomorphisms, then by what precedes they are monomorphisms, and hence are isomorphisms by the hypothesis on (φ_i).

Assertion (iii) is a trivial consequence of (ii), and (v) is a trivial consequence of the definitions.

We note the following consequence of (i), (ii), and (iv).

Corollary 6.3. Suppose that in E fiber products and amalgamated sums are representable and that the φ_i commute with them, and that kernels of pairs of arrows or cokernels of pairs of arrows are representable and that the φ_i commute with them. It suffices, for example, that finite projective limits and finite inductive limits be representable in E, and that the φ_i be exact functors. Then the following conditions are equivalent:

a) (φ_i) is faithful;

b) (φ_i) is conservative;

c) (φ_i) is conservative for monomorphisms;

(c') (φ_i) is conservative for epimorphisms.

We also record the following fact.

Proposition 6.4. Let $\varphi : E \to F$ be a functor admitting a right adjoint ψ, so that

$$\mathrm{Hom}_F(\varphi(X), Y) \simeq \mathrm{Hom}_E(X, \psi(Y)).$$

For φ, respectively ψ, to be faithful it is necessary and sufficient that, for every object X of E, respectively every object Y of F, the adjunction morphism

$$X \longrightarrow \psi\varphi(X)$$

be a monomorphism. For φ, respectively ψ, to be fully faithful it is necessary and sufficient that the preceding adjunction morphism be an isomorphism.

Indeed, if X', X are two objects of E, the map

$$\mathrm{Hom}_E(X', X) \longrightarrow \mathrm{Hom}_F(\varphi(X'), \varphi(X)) \tag{$*$}$$

is identified with the map obtained from

$$X \longrightarrow \psi\varphi(X) \tag{$**$}$$

by applying the functor $\mathrm{Hom}_E(X', -)$. Thus, for $(*)$ to be a monomorphism, respectively an isomorphism, for all X', with X fixed, it is necessary and sufficient that the adjunction morphism $(**)$ be a monomorphism, respectively an isomorphism.

Proposition 6.5. Let $p : E' \to E$ be a functor. The following conditions are equivalent:

(i) p is faithful, conservative, and fibred *(SGA 1, VI, 6.1)*.

(ii) p is a fibred functor whose fibers are discrete categories.

(iii) For every $X' \in \mathrm{Ob}(E')$, the functor

$$E'_{/X'} \longrightarrow E_{/p(X')}$$

induced by p is an equivalence of categories, surjective on objects.

(iv) If E is a $\mathcal{U}$-category, there exists an object $F \in \mathrm{Ob}(\widehat{E})$ and an equivalence of categories over E *(SGA 1, VI, 4.3)*

$$E' \xrightarrow{\sim} E_{/F},$$

where $E_{/F}$ is the full subcategory of $\widehat{E}_{/F}$ consisting of the arrows $X \to F$ whose source is in E.

6.5.1

Recall that a category C is said to be *discrete* if it is a *groupoid*, i.e. every arrow is invertible, and if it is *rigid*, i.e. the automorphism group of every object is reduced to the unit group. Equivalently, the category is equivalent to the category C' defined by a set I, with $\mathrm{Ob}(C') = I$ and with identity arrows as its only arrows. When C is already assumed to be a groupoid, saying that C is discrete amounts to saying that, for any two objects X, Y of C, there is at most one arrow from X to Y, i.e. that C is isomorphic to the category defined by a preordered set.

The equivalence of conditions (i) and (ii) in 6.5 is an immediate consequence of these reminders and of the following lemma.

Lemma 6.5.2. Let $p : E' \to E$ be a fibred functor. Then:

(i) for p to be conservative it is necessary and sufficient that its fiber categories be groupoids;

(ii) for p to be faithful it is necessary and sufficient that its fiber categories be ordered categories.

Proof. [Proof of 6.5.2] Suppose first that p is conservative. For every arrow u in a fiber E'_X, one has $p(u) = \mathrm{id}_X$, an isomorphism; hence u is an isomorphism in E', and also in E'_X, since an inverse of u in E' is evidently an inverse in E'_X. Thus E'_X is a groupoid. Conversely, suppose that the E'_X are groupoids, and let u' be an arrow of E' such that $p(u')$ is an isomorphism. We prove that u' is an isomorphism. Since p is fibred, one can factor

$$u' : X' \longrightarrow Y'$$

as a composite

$$X' \longrightarrow u^*(Y') \longrightarrow Y',$$

where the first arrow is an X-morphism, with $X = p(X')$ and $u = p(u')$, and the second is a cartesian morphism over u. The first arrow is an isomorphism because E'_X is a groupoid, and the second is an isomorphism because a cartesian morphism in a fibred category is evidently an isomorphism as soon as its projection is an isomorphism.

Now suppose that p is faithful, and let X', Y' be two objects of a fiber category E'_X. Two arrows from X' to Y' lie over the same arrow id_X of E, hence are identical; therefore E'_X is ordered. Conversely, suppose that the fiber categories are ordered, and prove that p is faithful. Let $u', v' : X' \rightrightarrows Y'$ be arrows of E' over the same arrow $u : X \to Y$ of E. They factor as

$$X' \rightrightarrows u^*(Y') \longrightarrow Y',$$

where the two arrows $X' \rightrightarrows u^*(Y')$ are arrows of E'_X with the same source and the same target. These are therefore equal, and hence $u' = v'$. $\square$

We return to the proof of 6.5. We have proved (i) $\Leftrightarrow$ (ii). On the other hand, (ii) $\Leftrightarrow$ (iv) is fairly clear: indeed, on one hand the category $E_{/F}$ is fibred over E with fibers the discrete categories defined by the sets $F(X)$, as follows at once from the definitions; on the other hand, if p is as in (ii), then by the sorites of SGA 1, VI, 8, the category E' fibred over E is E-equivalent to the split category over E defined by the functor $F : E^{\mathrm{op}} \to \mathbf{Set}$ which sends an object X to the set of isomorphism classes of objects of E'_X. This split category is E-isomorphic to $E_{/F}$.

Since (iv) $\Rightarrow$ (iii) is clear, it remains only to prove (iii) $\Rightarrow$ (i). But it is clear that, for p to be faithful, respectively conservative, it is necessary and sufficient that the induced functors

$$E'_{/X'} \longrightarrow E_{/p(X')}$$

be faithful, respectively conservative; *a fortiori*, it suffices that these functors be fully faithful. It remains only to prove that (iii) implies that p is fibred. But again one sees that a functor p is fibred if and only if the induced functors $E'_{/X'} \to E_{/p(X')}$ are fibred. This is so, in particular, if they are equivalences of categories surjective on objects.

7 Generating and cogenerating subcategories

Definition 7.1. Let E be a category and let C be a full subcategory of E. We say that C is a subcategory of E *generating by strict epimorphisms*, respectively *generating by epimorphisms*, if, for every object X of E, the family of arrows of E with target X and with source $X' \in \mathrm{Ob}(C)$ is strictly epimorphic, respectively epimorphic (1.3). We say that C is a *generating subcategory*, respectively *generating for monomorphisms*, respectively *generating for strict monomorphisms*, if, for every arrow $u : Y \to X$, respectively every monomorphism of E, respectively every strict monomorphism of E (10.5), such that for every $X' \in \mathrm{Ob}(C)$ the corresponding map

$$\mathrm{Hom}_E(X', Y) \longrightarrow \mathrm{Hom}_E(X', X)$$

is bijective, u is an isomorphism. Finally, we say that a family (X_i) of objects of E generates by strict epimorphisms, respectively etc., if the full subcategory C of E generated by this family generates by strict epimorphisms, respectively etc.

7.1.1

In terms of the family $(h_{X'})_{X' \in \mathrm{Ob}(C)}$ of functors represented by the objects $X' \in \mathrm{Ob}(C)$,

$$h_{X'} : E \longrightarrow \mathbf{Set} \qquad (X' \in \mathrm{Ob}(C)),$$

one can express the condition that C be generating, respectively generating for monomorphisms, respectively generating for strict monomorphisms, by saying that the family $(h_{X'})$ is conservative, respectively conservative for monomorphisms, respectively conservative for strict monomorphisms (6.1). It follows immediately from the definitions as well that C generates by epimorphisms if and only if the family $(h_{X'})_{X' \in \mathrm{Ob}(C)}$ is faithful (6.1). We shall also give below, in 7.2(i), an analogous interpretation of the condition that C generate by strict epimorphisms.

7.1.2

As with the notions introduced in 6.1, the notions in 7.1 are especially useful when E has suitable exactness properties; then the various notions introduced have a marked tendency all to be equivalent (7.3). For this reason the question of which of the notions in 7.1 should be considered the most important hardly arises: in the most important cases they coincide, and the term "generating subcategory" may therefore be interpreted indifferently as referring to any of the properties considered in 7.1, for example the first one, which is the strongest of all, as we shall see in 7.2(ii).

7.1.3

Suppose that E is a $\mathcal{U}$-category, and consider the canonical functor

$$\varphi : E \longrightarrow \widehat{C} = \mathrm{Hom}(C, \mathcal{U}\text{-}\mathbf{Set}), \tag{7.1.3.1}$$

obtained by composing $E \to \widehat{E} \to \widehat{C}$, where the first functor is the canonical functor (1.3.3), and the second is the restriction functor to C. It is evident that saying that φ is conservative, respectively faithful, is the same as saying that the family of functors

$$h_{X'} : X \longmapsto \mathrm{Hom}_E(X', X) = \varphi(X)(X')$$

for variable $X' \in \mathrm{Ob}(C)$ is a conservative, respectively faithful, family, i.e. also, by 7.1.2, that C is generating, respectively generating by epimorphisms. Similarly, φ is conservative for monomorphisms, respectively for strict monomorphisms, if and only if the family of the $h_{X'}$ is conservative for monomorphisms, respectively for strict monomorphisms; equivalently, if and only if the subcategory C of E is generating for monomorphisms, respectively for strict monomorphisms.

Proposition 7.2. Let E be a $\mathcal{U}$-category, and let C be a full subcategory.

(i) The following conditions are equivalent:

a) C is a subcategory generating by strict epimorphisms.

b) For every $X \in \mathrm{Ob}(E)$, if $C_{/X}$ denotes the full subcategory of $E_{/X}$ formed by the arrows $X' \to X$ with $X' \in \mathrm{Ob}(C)$, then the natural arrow from the inclusion functor $j : C_{/X} \to E$ to the constant functor on $C_{/X}$ with value X makes X an inductive limit of j:
$$X \xleftarrow{\sim} \varinjlim_{C_{/X}} X'.$$

c) The canonical functor φ of (7.1.3.1) is fully faithful.

(ii) Among the notions of 7.1 one has the following implications:

$$\begin{array}{ccc}
& \text{1) } C \text{ generates by strict epimorphisms } (\varphi \text{ fully faithful}) & \\
& \Downarrow \quad \searrow & \\
\text{2) } C \text{ generates by epimorphisms } (\varphi \text{ faithful}) & & \text{3) } C \text{ is generating } (\varphi \text{ conservative}) \\
& & \Downarrow \\
& \text{4) } C \text{ generates for monomorphisms } (\varphi \text{ conservative for monomorphisms}) & \\
& \Downarrow & \\
& \text{5) } C \text{ generates for strict monomorphisms } (\varphi \text{ conservative for strict monomorphisms}). &
\end{array}$$

(iii) One has the following conditional implications:

a) If in E epimorphic families of arrows are strictly epimorphic, then 2) $\Rightarrow$ 1). If in E monomorphisms are strict, then 5) $\Rightarrow$ 4).

b) If in E kernels of pairs of arrows, respectively fiber products, are representable, then 3) $\Rightarrow$ 2), respectively 4) $\Rightarrow$ 3).

c) If in E every family of morphisms $X_i \to X$ with common target X factors as a strictly epimorphic, respectively epimorphic, family $X_i \to Y$ followed by a monomorphism, respectively by a strict monomorphism, $Y \to X$, then 4) $\Rightarrow$ 1), respectively 5) $\Rightarrow$ 2).

We immediately record the following corollary.

Corollary 7.3. All the notions considered in 7.1, and recalled in the implication diagram of 7.2(ii), are equivalent in each of the following two cases:

(i) In E, kernels of pairs of arrows and fiber products are representable, monomorphisms are strict, and epimorphic families of arrows are strictly epimorphic.

(ii) In E, every family $(X_i \to X)_{i \in I}$ of arrows with common target X factors as an epimorphic family $(X_i \to Y)$ followed by a monomorphism $Y \to X$, every monomorphism of E is strict, and every epimorphic family of arrows of E is strict.

Indeed, in case (i) one has 4) $\Rightarrow$ 3) and 3) $\Rightarrow$ 2) by (b), and 5) $\Rightarrow$ 4) and 2) $\Rightarrow$ 1) by (a). In case (ii) one has, by (c), the implications 4) $\Rightarrow$ 1) and 5) $\Rightarrow$ 2). The conclusion follows from the implication diagram 7.2(ii).

Proof. [Proof of 7.2] For (i), the implication b) $\Rightarrow$ a) follows at once from the definitions. We prove a) $\Rightarrow$ b). Under hypothesis a), one must prove that, for every $X, Y \in \mathrm{Ob}(E)$, every system of arrows

$$u_{X'} : X' \longrightarrow Y$$

indexed by the $X' \in \mathrm{Ob}(C_{/X})$ such that $u_{X'} f = u_{X''}$ for every arrow $f : X'' \to X'$ in $C_{/X}$ factors through an arrow, necessarily unique by hypothesis a), $X \to Y$. By a), it suffices to check that, for every object Z of $E_{/X}$ and every pair of morphisms $v' : Z \to X'$ and $v'' : Z \to X''$ in $E_{/X}$, with X' and X'' in $C_{/X}$, one has $u_{X'} v' = u_{X''} v''$. But by hypothesis a), the family of arrows $w : X''' \to Z$, with $X''' \in \mathrm{Ob}(C)$, is epimorphic; hence it suffices to check that for every such w one has

$$(u_{X'} v') w = (u_{X''} v'') w,$$

which is also written $u_{X'}(v' w) = u_{X''}(v'' w)$ and follows immediately from the hypothesis on the family u.

We now prove the equivalence of b) and c). For every $X \in \mathrm{Ob}(E)$, the object $\varphi(X)$ of $\widehat{C}$ is the inductive limit in $\widehat{C}$ of the canonical functor $\widehat{C}_{/\varphi(X)} \to \widehat{C}$ (3.4). But $\widehat{C}_{/\varphi(X)}$ is canonically isomorphic to $C_{/X}$, so that in $\widehat{C}$ one has

$$\varphi(X) = \varinjlim_{C_{/X}} X',$$

where the object X' of C is identified with the functor in $\widehat{C}$ that it represents. Consequently, for a second object Y of E, there is a canonical isomorphism

$$\begin{aligned} \mathrm{Hom}_{\widehat{C}}(\varphi(X), \varphi(Y)) &\simeq \varprojlim_{C_{/X}} \mathrm{Hom}_{\widehat{C}}(X', \varphi(Y)) \\ &\simeq \varprojlim_{C_{/X}} \varphi(Y)(X') \\ &\simeq \varprojlim_{C_{/X}} \mathrm{Hom}_E(X', Y). \end{aligned} \qquad (*)$$

Thus the map
$$\mathrm{Hom}_E(X,Y) \longrightarrow \mathrm{Hom}_{\widehat{C}}(\varphi(X),\varphi(Y))$$
defined by φ is, through the isomorphism between the extreme terms of $(*)$, exactly the map
$$\mathrm{Hom}_E(X,Y) \longrightarrow \varprojlim_{C_{/X}} \mathrm{Hom}_E(X',Y)$$
induced by the inductive system of arrows $X' \to X$ indexed by $C_{/X}$ considered in 7.2(i)b). Therefore the first map is bijective for every Y, with X fixed, if and only if X is an inductive limit of the inclusion functor $j : C_{/X} \to E$. This proves the equivalence of b) and c).

For (ii), the implications 1) $\Rightarrow$ 2) and 3) $\Rightarrow$ 4) $\Rightarrow$ 5) are trivial by the definitions. The implication 1) $\Rightarrow$ 3) is obtained by interpreting property 1) as the full faithfulness of φ, by (i), and observing that a fully faithful functor is conservative. We have already observed in 7.1.1 that 3) means that φ is conservative.

For (iii), assertion (a) is a tautology. Assertion (b) follows from 6.2(i), respectively 6.2(iv), applied to the functor φ, noting that φ is left exact. Finally we prove (c). For $X \in \mathrm{Ob}(E)$, consider the family of all morphisms $X_i \to X$ with $X_i \in \mathrm{Ob}(C)$. By the hypothesis on E, it factors as a strictly epimorphic, respectively epimorphic, family $X_i \to Y$ followed by a monomorphism, respectively by a strict monomorphism, $Y \to X$. Then hypothesis 4), respectively 5), implies that $Y \to X$ is an isomorphism. Thus the family in question is strictly epimorphic, respectively epimorphic. $\square$

Proposition 7.4. Let E be a category, let C be a full subcategory of E, and let X be an object of E. Suppose that C is generating in E, respectively that C generates for monomorphisms and that the fiber product over X of two subobjects of X is representable in E. Then a strict subobject (10.11), respectively a subobject, X' of X is known once, for every $T \in \mathrm{Ob}(C)$, one knows the subset of $\mathrm{Hom}_E(T,X)$ which is the image of $\mathrm{Hom}_E(T,X')$. Consequently, the cardinal of the set of strict subobjects, respectively of the set of subobjects, of X is bounded above by
$$\prod_{T \in \mathrm{Ob}(C)} 2^{\mathrm{card}\,\mathrm{Hom}_E(T,X)},$$
and if $X \in \mathrm{Ob}(C)$, it is bounded above by $2^{\mathrm{card}\,\mathrm{Fl}(C)}$.

We first prove the statement with "respectively". Let X' and X'' be two subobjects of X such that, for every $T \in \mathrm{Ob}(C)$, the images of $\mathrm{Hom}_E(T,X')$ and $\mathrm{Hom}_E(T,X'')$ in $\mathrm{Hom}_E(T,X)$ are equal. They are therefore also equal to the image of $\mathrm{Hom}_E(T,X''')$, where X''' is the fiber product of X' and X'' over X. Since C generates for monomorphisms, it follows that the monomorphisms $X''' \to X'$ and $X''' \to X''$ are isomorphisms. Hence X' and X'' are equal, each being equal to the subobject X''' of X.

We now prove the non-respective assertion. By the definition of strict subobject, it suffices to check that knowing the subset $\mathrm{Hom}_E(T,X')$ of $\mathrm{Hom}_E(T,X)$ for every $T \in \mathrm{Ob}(C)$ determines the set of pairs of arrows $u, v : X \rightrightarrows T$ such that $ui = vi$, where $i : X' \to X$ is the canonical injection. Since C is generating, the relation $ui = vi$ is equivalent to the relation $(ui)f = (vi)f$ for every $f \in \mathrm{Hom}_E(T,X')$, i.e. to $ug = vg$ for every $g \in \mathrm{Hom}_E(T,X)$ coming from $\mathrm{Hom}_E(T,X')$ (that is, of the form if with $f \in \mathrm{Hom}_E(T,X')$). This proves the assertion.

Corollary 7.5. Let E be a category, let C be a full generating subcategory, and let X be an object of C. Then a strict quotient (10.8) X' of X is known once, for every $T \in \mathrm{Ob}(C)$, one knows the subset of $\mathrm{Hom}_E(T,X)^2$ formed by the pairs (u,v) such that $qu = qv$, where $q : X \to X'$ is the canonical morphism. Thus the cardinal of the set of strict quotients of X is bounded above by

$$\prod_{T\in\mathrm{Ob}(C)} 2^{\mathrm{card}\,\mathrm{Hom}_E(T,X)^2}.$$

Indeed, by definition, a strict quotient X' of X is known once, for every object Y of E, one knows the subset of $\mathrm{Hom}_E(Y,X) \times \mathrm{Hom}_E(Y,X)$ formed by the pairs (u,v) such that $qu = qv$, where $q : X \to X'$ is the canonical morphism. But the relation $qu = qv$ is equivalent to $(qu)f = (qv)f$ for every morphism $f : T \to Y$ whose source T lies in C, since C is generating. This relation may also be written $q(uf) = q(vf)$, which proves the first assertion of 7.5. The second follows immediately.

7.5.1

One can generalize 7.5 by introducing, for a family of objects $(X_i)_{i\in I}$ of E, the notion of a *strict quotient in E of the family.* By this we mean a strictly epimorphic family $(p_i : X_i \to X')_{i\in I}$ of morphisms of E, with the convention, as for ordinary quotients, that two such families $(p_i : X_i \to X')$ and $(q_i : X_i \to X'')$ are identified if there exists an isomorphism $v : X' \to X''$ (necessarily unique) such that $vp_i = q_i$ for every $i \in I$. With this terminology, the proof of 7.5 applies *mutatis mutandis* to give the following variant.

Variant 7.5.2. Let E and C be as in 7.5, and let $(X_i)_{i\in I}$ be a family of objects of E. Then a strict quotient X' of $(X_i)_{i\in I}$ in E (7.5.1) is known once, for every $T \in \mathrm{Ob}(C)$ and every pair $(i,j) \in I \times I$, one knows the subset of $\mathrm{Hom}_E(T,X_i) \times \mathrm{Hom}_E(T,X_j)$ formed by the pairs (u,v) such that $p_iu = p_jv$, where $p_i : X_i \to X'$ denotes the canonical morphism for each $i \in I$. Consequently, the cardinal of the set of strict quotients of $(X_i)_{i\in I}$ in E is bounded above by

$$\prod_{T\in\mathrm{Ob}(C)} \prod_{i,j\in I} 2^{\mathrm{card}\,\mathrm{Hom}_E(T,X_i)\times\mathrm{card}\,\mathrm{Hom}_E(T,X_j)}.$$

7.5.3

One sees at once that the analogous conclusion is true if one assumes only that C generates for strict monomorphisms, provided one assumes that the products $X_i \times X_j$ are representable and restricts to *effective* quotients of the family $(X_i)_{i\in I}$, i.e. to strict quotients X' such that the fiber products $X_i \times_{X'} X_j$ are representable in E; this is no restriction if E is stable under fiber products.

Proposition 7.6. Let E be a category, let C be a full subcategory of E generating by epimorphisms (7.1), let Π_0 be an infinite cardinal $\geq \mathrm{card}\,\mathrm{Fl}(C)$, let $\Pi \geq \Pi_0$ be a cardinal, and let $(u_i : X_i \to X)_{i\in I}$ be a universal epimorphic family (10.3) in E formed by squareable morphisms (10.7), with sources $X_i \in \mathrm{Ob}(C)$, and such that $\mathrm{card}(I) \leq \Pi$. Then

$$\mathrm{card}\,\mathrm{Fl}(C_{/X}) \leq \Pi^{\Pi_0}.$$

Let $T \in \mathrm{Ob}(C)$. It will be enough to prove that

$$\mathrm{card}\,\mathrm{Hom}_E(T,X) \leq \Pi^{\Pi_0}. \tag{7.6.1}$$

for then one successively obtains

$$\operatorname{card}\operatorname{Ob}(C_{/X}) \leq (\Pi^{\Pi_0}) \times \operatorname{card}\operatorname{Ob}(C) \leq (\Pi^{\Pi_0}) \times \Pi_0 = \Pi^{\Pi_0},$$

and finally

$$\operatorname{card}\operatorname{Fl}(C_{/X}) \leq ((\Pi^{\Pi_0})^2) \times \Pi_0 = \Pi^{\Pi_0},$$

since for two objects S, T of $C_{/X}$ one has

$$\operatorname{card}\operatorname{Hom}_{C_{/X}}(S,T) \leq \operatorname{card}\operatorname{Hom}_C(S,T) \leq \Pi_0.$$

To prove (7.6.1), first note the following lemma.

Lemma 7.6.2. Let J be a set such that $\operatorname{card}(J) = \Pi_0$. For every object T of C and every morphism $f : T \to X$, there exists an epimorphic family $(v_j : S_j \to T)_{j\in J}$, with sources $S_j \in \operatorname{Ob}(C)$, and for every $j \in J$ an element $i(j) \in I$ and a morphism $g_j : S_j \to X_{i(j)}$, such that

$$u_{i(j)}g_j = fv_j.$$

Indeed, the family of projections

$$X_i \times_X T \longrightarrow T$$

is epimorphic by hypothesis. On the other hand, since C generates by epimorphisms, for every i the family of arrows $S \to X_i \times_X T$ with source $S \in \operatorname{Ob}(C)$ is epimorphic. Thus, by transitivity, the family of arrows $v : S \to T$ with source in C which factor through one of the $X_i \times_X T$ is epimorphic. These are precisely the arrows $v : S \to T$ with source in C for which there exist $i \in I$ and $g : S \to X_i$ such that $u_i g = fv$. Since the set of these arrows $v : S \to T$ is contained in $\operatorname{Fl}(C)$ and therefore has cardinal $\leq \Pi_0$, it can be indexed by the index set J. This proves the lemma.

Now a morphism $f : T \to X$ is known once the fv_j are known, and these are known once the g_j are known. Thus $\operatorname{card}\operatorname{Hom}_E(T,X)$ is bounded above by the cardinal of the set of families $(i(j), S_j, v_j, g_j)_{j\in J}$. The cardinal of the set of maps $j \mapsto i(j)$ from J to I is $\leq \Pi^{\Pi_0}$; and for such a map fixed, the cardinal of the set of the corresponding families S_j, g_j, v_j is bounded above by the cardinal of the set of maps from J to $\operatorname{Fl}(C) \times \operatorname{Fl}(C)$, which is $\leq \Pi_0^{\Pi_0}$, since $\operatorname{card}(\operatorname{Fl}(C) \times \operatorname{Fl}(C)) = \Pi_0^2 = \Pi_0$. Thus the left hand side of (7.6.1) is bounded above by

$$\Pi^{\Pi_0} \times \Pi_0^{\Pi_0} = \Pi^{\Pi_0},$$

which completes the proof of 7.6.

Proposition 7.7. Let E be a category in which fiber products are representable, let $(X_\alpha)_{\alpha\in A}$ be a generating family of objects of E, and let $(\varphi_i)_{i\in I}$ be a family of functors $\varphi_i : E \to E_i$ commuting with fiber products. For (φ_i) to be conservative (6.1), it is necessary and sufficient that for every $\alpha \in A$ and every subobject X' of X_α distinct from X_α, there exist $i \in I$ such that the morphism

$$\varphi_i(X') \longrightarrow \varphi_i(X_\alpha)$$

is not an isomorphism. In this case, if E is a $\mathcal{U}$-category and A is $\mathcal{U}$-small, there exists a $\mathcal{U}$-small subset J of I such that $(\varphi_j)_{j\in J}$ is already a conservative family of functors.

The necessity of the condition of conservativity under consideration is evident; we prove its sufficiency. By 6.2(iv), it suffices to prove that (φ_i) is conservative for monomorphisms. Let $u : Y' \to Y$ be a monomorphism in E which is not an isomorphism. We must prove that there is an $i \in I$ such that $\varphi_i(u)$ is not an isomorphism. By the hypothesis on (X_α), there exists an α and a morphism $X_\alpha \to Y$ which does not factor through Y'. In other words, the inverse image X' of Y' is a subobject of X_α distinct from X_α. By hypothesis, there exists an $i \in I$ such that

$$\varphi_i(X') \longrightarrow \varphi_i(X_\alpha)$$

is not an isomorphism. Since

$$\varphi_i(X') \simeq \varphi_i(X_\alpha) \times_{\varphi_i(Y)} \varphi_i(Y'),$$

it follows that $\varphi_i(Y') \to \varphi_i(Y)$ is not an isomorphism.

The second assertion of 7.7 follows at once from the first, taking 7.4 into account.

Corollary 7.7.1. Let E be a $\mathcal{U}$-category in which fiber products are representable, and suppose that E admits a generating family of objects which is $\mathcal{U}$-small. Then, for every generating family $(Y_i)_{i \in I}$ of E, there exists a $\mathcal{U}$-small generating subfamily $(Y_j)_{j \in J}$, with $\mathrm{card}(J) \in \mathcal{U}$.

It suffices to apply 7.7 to a $\mathcal{U}$-small generating family $(X_\alpha)_{\alpha \in A}$ of E and to the family of functors φ_i represented by the Y_i.

Proposition 7.8. Let E be a category, let C be a full subcategory generating by strict epimorphisms, let D be a category, and let $\mathrm{Hom}'(E, D)$ be the full subcategory of $\mathrm{Hom}(E, D)$ consisting of the functors which commute with inductive limits of type $C_{/X}$, where X ranges over the objects of E. Then the functor $F \mapsto F|_C$ induces a fully faithful functor

$$\mathrm{Hom}'(E, D) \longrightarrow \mathrm{Hom}(C, D).$$

Consequently, if $\mathrm{Hom}(C, D)$ is a $\mathcal{U}$-category (1.1), for example by (1.1.1 b)) if C is $\mathcal{U}$-small and D is a $\mathcal{U}$-category, then $\mathrm{Hom}'(E, D)$ is also a $\mathcal{U}$-category.

Let $F, G : E \to D$ be two functors in the source category, and let $u : F \to G$ be a morphism. If $X = \varinjlim X_i$ in E and if F and G commute with the inductive limit in question, then $u(X) : F(X) \to G(X)$ is identified with the limit of the morphisms $u(X_i) : F(X_i) \to G(X_i)$, and hence is known once the $u(X_i)$ are known. This shows that the functor considered in 7.8 is faithful, by the implication a) $\Rightarrow$ b) in 7.2(i). Conversely, let $v : F|_C \to G|_C$ be a morphism; we prove that it comes from a morphism $u : F \to G$. For every $X \in \mathrm{Ob}(E)$ define

$$u(X) : F(X) = \varinjlim_{C_{/X}} F(X_i) \longrightarrow G(X) = \varinjlim_{C_{/X}} G(X_i)$$

as the inductive limit of the $v(X_i)$. It is immediate that this gives a morphism functorial in X, hence a morphism $u : F \to G$, and finally that the morphism induced by u from $F|_C$ to $G|_C$ is v. This completes the proof.

7.1 Families and subcategories cogenerating

Let E be a category, and let C be a full subcategory of E. We say that C is cogenerating by strict monomorphisms, respectively cogenerating by monomorphisms, respectively cogenerating, respectively cogenerating for epimorphisms, respectively cogenerating for strict epimorphisms, if the full subcategory C^{op} of E^{op} generates by strict epimorphisms, respectively etc. The terminology for families is analogous. Thus all the results in this section concerning the notion of generating subcategory and its variants (7.1) give corresponding results for the dual notions, which we leave to the reader to formulate for personal satisfaction.

Proposition 7.10. Let E be a category, let C be a full generating subcategory (7.1), and let D be a full subcategory of E. For D to be cogenerating (7.9), it is necessary and sufficient that, for every pair of arrows

$$T \underset{v}{\overset{u}{\rightrightarrows}} X$$

in E with source $T \in \mathrm{Ob}(C)$ and with $u \neq v$, there exist an arrow $w : X \to I$ with target $I \in \mathrm{Ob}(D)$ such that $wu \neq wv$.

By definition, to say that D is cogenerating means that for every pair of arrows $Y \rightrightarrows X$ in E such that $u \neq v$, there exists an arrow $w : X \to I$, with $I \in \mathrm{Ob}(D)$, such that $wu \neq wv$. Thus 7.10 simply says that it suffices to test this property when $Y \in \mathrm{Ob}(C)$. Since C is generating, the hypothesis $u \neq v$ implies that there exists $f : T \to Y$ such that $uf \neq vf$. By the hypothesis there exists $w : X \to I$, with target $I \in \mathrm{Ob}(D)$, such that $w(uf) \neq w(vf)$, and hence $wu \neq wv$. □

Corollary 7.11. With the notation of 7.10, suppose that the objects I of D are injective objects of E, i.e. that for every monomorphism $X \to Y$ in E, the corresponding map

$$\mathrm{Hom}_E(Y, I) \longrightarrow \mathrm{Hom}_E(X, I)$$

is surjective. Suppose moreover that every pair of arrows

$$T \underset{v}{\overset{u}{\rightrightarrows}} X$$

in E factors as an epimorphic, respectively effective epimorphic, pair $T \rightrightarrows X'$ followed by a monomorphism $i : X' \to X$. Then in the criterion 7.10 for D to be cogenerating, one may restrict to pairs (u, v) which are epimorphic, respectively effective epimorphic.

We conclude:

Corollary 7.12. Let E be a $\mathcal{U}$-category admitting a small generating subcategory C, and such that every object of E is the source of a monomorphism into an injective object of E. Suppose moreover that, for every $T \in \mathrm{Ob}(C)$, the sum $T \amalg T$ in E is representable, and that every morphism with source $T \amalg T$ factors as an effective epimorphism followed by a monomorphism. Then E admits a small full cogenerating subcategory D. More precisely, one may take D such that

$$\operatorname{card} \mathrm{Ob}(D) \leq \prod_{T, T' \in \mathrm{Ob}(C)} 2^{\operatorname{card}(\mathrm{Hom}_E(T', T \amalg T))^2}.$$

Indeed, by 7.11 it is enough, for every $T \in \mathrm{Ob}(C)$ and every effective quotient X of $T \amalg T$, to choose an embedding of X into an injective object I of E, and to take for D the full subcategory of E generated by these I. The conclusion then follows from 7.5.

Examples 7.13. To construct small cogenerating subcategories in terms of small generating subcategories, one is therefore led to look for conditions ensuring that a $\mathcal{U}$-category E has “enough injectives”, i.e. that every object embeds into an injective object by a monomorphism. It is well known [Tohoku] that this condition is satisfied in a $\mathcal{U}$-abelian category with exact small filtered inductive limits (axiom AB5 of *loc. cit.*) admitting a small generating family. The construction of *loc. cit.* is not, moreover, essentially tied to abelian categories, and also works in the category of sheaves of $\mathcal{U}$-sets on a topological space $X \in \mathcal{U}$. We shall not state here the exactness properties which make the construction in question work, and shall limit ourselves to observing that in the special case of the category of sheaves of sets on X, one immediately reduces to the case of *loc. cit.* as follows.

Note that if $\mathcal{O}_X$ is a ring on X, then every injective object I of the category of $\mathcal{O}_X$-modules is also injective as an object of the category of sheaves of sets. Indeed, if F is a sheaf of sets and $\mathcal{O}_X[F]$ is the “free $\mathcal{O}_X$-module generated by F”, then by definition there is a homomorphism of sheaves of sets

$$F \longrightarrow \mathcal{O}_X[F] \tag{$*$}$$

giving an isomorphism, functorial in F,

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X[F], I) \xrightarrow{\sim} \mathrm{Hom}(F, I).$$

Since the functor $F \mapsto \mathcal{O}_X[F]$ plainly carries monomorphisms to monomorphisms, it follows at once that, if I is an injective $\mathcal{O}_X$-module, then it is also an injective sheaf of sets. It follows that if the morphism $(*)$ is a monomorphism, as happens if one takes for $\mathcal{O}_X$ a constant ring with value a ring $A \neq 0$, then an embedding of $\mathcal{O}_X[F]$ into an injective $\mathcal{O}_X$-module I gives an embedding of F into the underlying injective sheaf of sets of I.

The same argument applies, without change, to the category of sheaves of $\mathcal{U}$-sets on a $\mathcal{U}$-site, which will be introduced in the next exposé.

The interest of the existence of a small cogenerating subcategory lies chiefly in the representability criterion 8.12.7 below.

Exposé I. Presheaves

Section 8.1–8.6

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 8.1 through Section 8.6. The source has one visibly truncated sentence in the proof of Proposition 8.6.4; the translation marks the mathematically forced completion as referring to the immediately following Corollary 8.6.5.

8 Ind-objects and pro-objects

8.1 Cofinal functors and cofinal subcategories

Definition 8.1.1. Let

$$\varphi : I \longrightarrow I' \tag{8.1.1.1}$$

be a functor. We say that φ is a *cofinal functor* if, for every category C and every functor $u : I' \to C$, considering $\varinjlim u$ and $\varinjlim u \circ \varphi$ as objects of $\mathrm{Hom}(G(\mathcal{V}\text{-}\mathbf{Set}))^{\mathrm{op}}$ (2.1, 2.3.1), where $\mathcal{V}$ is a universe such that $I, I' \in \mathcal{V}$ and C is a $\mathcal{V}$-category, the canonical morphism

$$\varinjlim u \circ \varphi \longrightarrow \varinjlim u \tag{8.1.1.2}$$

is an isomorphism. When φ is the inclusion functor of a subcategory of I', we say that I is a *cofinal subcategory* if φ is cofinal.

8.1.2

Notice that, for fixed φ and u, the bijectivity of (8.1.1.2) does not depend on the choice of the universe $\mathcal{V}$.

Returning to the definition of the terms appearing in (8.1.1.2), one sees that to say that φ is cofinal is also to say that, for every universe $\mathcal{V}$ such that I and I' are $\mathcal{V}$-small, and for every functor

$$v : I'^{\mathrm{op}} \longrightarrow (\mathcal{V}\text{-}\mathbf{Set}),$$

the canonical homomorphism

$$\varprojlim v \longrightarrow \varprojlim v \circ \varphi \tag{8.1.2.1}$$

is an isomorphism, that is, a bijection. To see that this condition is indeed necessary, observe, putting $v = v^{\mathrm{op}}$, that it also says that the condition of Definition 8.1.1 is fulfilled when one takes $C = (\mathcal{V}\text{-}\mathbf{Set})^{\mathrm{op}}$; this implies that the inductive limits considered in (8.1.1.2) are representable in C.

8.1.2.2

It is immediate from the definition that *the composite of two cofinal functors is cofinal.*

Proposition 8.1.3. Let $\varphi : I \to I'$ be a functor.

a) For φ to be cofinal, it is necessary that φ satisfy the condition:

F 1) For every object i' of I', there exists an object i of I such that $\varphi(i)$ dominates i', i.e. such that $\mathrm{Hom}(i', \varphi(i)) \neq \emptyset$.

b) Suppose that I is filtered. For φ to be cofinal, it is necessary and sufficient that φ satisfy condition (F 1) and the following condition:

F 2) For every object i of I and every double arrow

$$i' \overset{f'}{\underset{g'}{\rightrightarrows}} \varphi(i)$$

in I' with target $\varphi(i)$, there exists an arrow $h : i \to j$ in I such that $\varphi(h)f' = \varphi(h)g'$.

Moreover, if φ is cofinal, then I' is filtered.

c) Suppose that I' is filtered and that φ is fully faithful. For φ to be cofinal, it is necessary and sufficient that it satisfy condition F 1) of a). This implies that I is filtered.

Proof. For necessity in a) and b), one uses Definition 8.1.1 only in the case where u is the canonical inclusion functor

$$u : I' \hookrightarrow \widehat{I'}$$

(choosing a universe $\mathcal{U}$ such that I and I' are $\mathcal{U}$-small). Then (8.1.1.2) may be interpreted as an arrow of $\widehat{I'}$, whose target is the *final* functor on I'. One sees immediately that condition F 1) expresses that this arrow is an epimorphism of $\widehat{I'}$, i.e. that it is surjective on each argument, since inductive limits in $\widehat{I'}$ are calculated argument by argument. This proves a).

Now suppose that I is filtered and that φ is cofinal. Then I' is filtered: indeed, the condition that two objects of I' be dominated by a third follows at once from the corresponding condition on I and from F 1). It remains only to prove condition PS 2) of 2.7, i.e. that for every double arrow

$$f', g' : i' \rightrightarrows j'$$

in I', there exists an arrow $h' : j' \to k'$ in I' such that $h'f' = h'g'$. By F 1) we may suppose that k' is of the form $\varphi(i)$. But by the hypothesis that φ is cofinal, for every object i' of I' the set $\varinjlim_i \mathrm{Hom}(i', \varphi(i))$ has one element. It follows from the standard description of filtered inductive limits in **Set** (2.8.1) that there is an arrow $h : i \to j$ in I such that $\varphi(h)f' = \varphi(h)g'$. This finishes the proof that I' is filtered, and proves F 2) at the same time.

To prove that the conditions stated in b) are sufficient for φ to be cofinal, use the form 8.1.2 of the definition. One immediately checks that F 1) implies that (8.1.2.1) is a monomorphism, i.e. injective on each argument, while F 2), together with F 1), ensures that it is bijective, since projective limits in $\widehat{I'^{\mathrm{op}}}$ are calculated argument by argument. This proves b).

Finally, if I' is filtered and φ is fully faithful, it is immediate that F 1) implies that I is filtered and implies F 2). Thus c) follows from a) and b).

8.1.4

When I' is a filtered category and I is a full subcategory of I', Proposition 8.1.3 c) shows that the condition that I be cofinal in I' depends only on the subset $\mathrm{Ob}\, I$ of the preordered set $\mathrm{Ob}\, I'$ (for the preorder relation "$x \leq y$" iff "$\mathrm{Hom}(x, y) \neq \emptyset$"). If J' is a preordered set and J is a subset of J', we shall sometimes say that J is a *cofinal subset of* J' when every element of J' is dominated by an element of J. When J' is filtered, this therefore means that the inclusion functor $\mathcal{J} \hookrightarrow \mathcal{J}'$ of the associated categories is cofinal.

8.1.5

In what follows, we shall use the notion of cofinal functor only in cases where the categories I and I' are filtered. Classically, one even restricted oneself to categories associated with preordered sets, i.e. categories in which there is at most one arrow with given source and target. It turns out, however, that this restriction is inconvenient in applications, because the "natural" filtered categories which arise in many applications are *not* ordered categories. The following result, due to P. Deligne, nevertheless shows that there is no essential difference between the two points of view.

Proposition 8.1.6. Let I be a small filtered category. Then there exists a small ordered set E, and a cofinal functor

$$\varphi : \mathcal{E} \longrightarrow I,$$

where $\mathcal{E}$ denotes the category associated with E.

First suppose that the preordered set $\mathrm{Ob}\, I$ has no greatest element. By a subdiagram of I we mean a pair $D = (\mathcal{O}, F)$ consisting of a subset F of $\mathrm{Fl}(I)$ and a subset $\mathcal{O}$ of $\mathrm{Ob}\, I$, such that for every $f \in F$, the source and target of f belong to $\mathcal{O}$. An element e of $\mathcal{O}$ is called a *final object* of the subdiagram D if, for every $x \in D$, the set $\mathrm{Hom}(x, e) \cap F$ of arrows of D with source x and target e has exactly one element f_x, if for every arrow $g : x \to y$ of D one has $f_x = f_y \circ g$, and if $f_e = \mathrm{id}_e$.

Let E be the set of *finite* subdiagrams D of I having a *unique* final element $\varphi(D)$, and order E by inclusion. If D, D' are elements of E and $D \subset D'$, then there is a unique arrow $\varphi(D', D)$ of D' with source $\varphi(D)$ and target $\varphi(D')$; if $D \subset D' \subset D''$ are inclusions in E, then of course

$$\varphi(D'', D) = \varphi(D'', D')\varphi(D', D),$$

and also $\varphi(D, D) = \mathrm{id}_{\varphi(D)}$. Thus one obtains a functor $\varphi : \mathcal{E} \to I$, and it remains to prove that E is filtered and that φ is cofinal. By Proposition 8.1.3 b), three verifications are required.

1) Condition F 1) of 8.1.3. For every $i \in I$, take D to be the subdiagram of I reduced to the object i and its identity arrow. Then $i \leq \varphi(D) = i$. This condition F 1) also shows that $E \neq \emptyset$.

2) Condition F 2). For every $i \in \mathrm{Ob}\, I$, every $D \in E$, and every double arrow $i \rightrightarrows \varphi(D)$, one must find a diagram $D' \in E$, containing D, such that

$$\varphi(D', D) : \varphi(D) \longrightarrow \varphi(D')$$

equalizes the double arrow. Since I is filtered, there exists an object j of I and an arrow $f : \varphi(D) \to j$ equalizing the double arrow. Since I has no greatest element, we may suppose that j is a *strict* upper bound of $\varphi(D)$, which implies that it is distinct from all the other objects of D. Let D' be the subdiagram of I whose set of objects is $\mathcal{O} \cup \{j\}$, and whose set of arrows consists of the arrows of D, together with the composites

$$x \xrightarrow{f_x} \varphi(D) \xrightarrow{f} j,$$

where x is an object of D and f_x is the unique arrow of D with source x and target $\varphi(D)$, and together with id_j. It is then clear that D' is a finite subdiagram of I, that it has j as its unique final object, hence that $D' \in E$, and that D' satisfies the required condition.

3) Let D and D' be two subdiagrams in E. They are contained in some common $D'' \in E$. Indeed, one can find a strict upper bound j of $\varphi(D)$ and $\varphi(D')$,

$$\begin{tikzcd} \varphi(D) \arrow[dr, "f"] & \\ & j \\ \varphi(D') \arrow[ur, "f'"'] & \end{tikzcd}$$

and take D'' to be the subdiagram whose set of objects is the union of the sets of objects of D and D' and of $\{j\}$, and whose set of arrows is the union of the sets of arrows of D and D', the composites $f \circ f_x$ with x an object of D, the composites $f' \circ f'_{x'}$ with x' an object of D', and $\{\mathrm{id}_j\}$. This is indeed a finite subdiagram of I. We show that, after replacing j, f, f' by j', gf, gf' with a suitable $g : j \to j'$, we may arrange for j to be a final object of D''. This is the same as saying that, for every x which is an object of both D and D', one has

$$f f_x = f' f'_{x'}.$$

But this is only a *finite* set of double arrows to be equalized by some $g : j \to j'$, and this is possible because I is filtered.

This completes the proof in the case considered. The general case is reduced to it by introducing the filtered category $\mathbb{N}$ associated with the ordered set $\mathbb{N}$ of natural numbers, and observing that the category $\mathbb{N} \times I$ is filtered, has no greatest element, and that the projection $\mathbb{N} \times I \to I$ is a cofinal functor.

Corollary 8.1.7. Let I be a $\mathcal{U}$-category. For there to exist a small filtered ordered set E and a cofinal functor $\mathcal{E} \to I$, it is necessary and sufficient that I be filtered and that $\mathrm{Ob}\, I$ admit a small cofinal subset I' (8.1.4).

This is necessary by 8.1.3 a), and sufficient by 8.1.6 applied to the full subcategory of I defined by I'.

Definition 8.1.8. Let I be a filtered category. We say that I is *essentially small* if I is a $\mathcal{U}$-category and if it satisfies the equivalent conditions of 8.1.7.

8.2 Ind-objects and ind-representable functors

8.2.1

In what follows we fix a $\mathcal{U}$-category C, which we always regard as embedded in the category $\widehat{C}$ of $\mathcal{U}$-presheaves by means of the canonical functor (1.3.1)

$$h : C \hookrightarrow \widehat{C}. \tag{8.2.1.1}$$

We shall call an *ind-object of* C (or an *inductive system* of C) any functor

$$\varphi : I \longrightarrow C, \tag{8.2.1.2}$$

where I is a *filtered* category (2.7), called the *index category* of the ind-object. Apart from this, I is arbitrary; in particular we do not require I to be a $\mathcal{U}$-category. It will often be convenient to denote an ind-object φ by the indexed notation $(X_i)_{i\in \operatorname{Ob} I}$, or even, by a further abuse of notation, by $(X_i)_{i\in I}$, $(X_i)_i$, or (X_i), where $X_i = \varphi(i)$ for $i \in \operatorname{Ob} I$. This notation is no more and no less abusive than the classical notation for inductive systems indexed by filtered ordered sets, where the transition morphisms are likewise not mentioned. The X_i are called the *component objects* of the ind-object $\mathcal{X}$. Notice that, in the general case considered here, the "*transition morphisms*" $\varphi(f)$ of the inductive system are indexed by the set $\operatorname{Fl}(I)$ of arrows of I, which in general is not identified with a subset of $\operatorname{Ob} I \times \operatorname{Ob} I$. In other words, for given i and j, with j dominating i, there may exist more than one transition morphism from X_i to X_j.

8.2.2

The most useful ind-objects $\mathcal{X} = (X_i)_{i\in I}$ of C are those for which the index category I is essentially small (8.1.8); such an ind-object is called *essentially small.* If $(X_i)_{i\in I}$ is such an object, using the fact that small inductive limits in $\widehat{C}$ are representable (3.1), one may consider

$$L(\mathcal{X}) := \varinjlim h \cdot \mathcal{X} = \varinjlim_i X_i \in \operatorname{Ob} \widehat{C}, \tag{8.2.2.1}$$

where the last limit is taken *in* $\widehat{C}$. Thus $L(\mathcal{X})$ is the presheaf

$$L(\mathcal{X}) : Y \longmapsto \varinjlim_i \operatorname{Hom}(Y, X_i) \tag{8.2.2.2}$$

on C. We say that this presheaf is the *presheaf ind-represented by the ind-object* $\mathcal{X} = (X_i)_{i\in I}$ *of* C. A presheaf F on C is called an *ind-representable presheaf* if it is isomorphic to a presheaf $L(\mathcal{X})$ ind-represented by an essentially small ind-object. It follows from 8.1.6 that, in this definition, one may suppose that I is a small category, and even that I is associated with an ordered set belonging to $\mathcal{U}$.

8.2.3

Consider an ind-object $\mathcal{X} = (X_i)_{i\in I}$, given by a functor $\varphi : I \to C$, and let

$$u : I' \longrightarrow I$$

be a functor, where I' is a second filtered category. Then the functor $\varphi \circ u$ is an ind-object $\mathcal{X}'$ of C with index category I', written in indexed notation as $(X_{u(i')})_{i'\in I'}$. It is called the ind-object of C *deduced from* $(X_i)_{i\in I}$ *by change of index categories* using the functor u. If I and I', equivalently $\mathcal{X}$ and $\mathcal{X}'$, are essentially small, one obtains a canonical morphism

$$L(\mathcal{X}') = \varinjlim_{i'} X_{u(i')} \longrightarrow L(\mathcal{X}) = \varinjlim_i X_i \tag{8.2.3.1}$$

between the presheaves ind-represented by $\mathcal{X}'$ and $\mathcal{X}$. When u is a cofinal functor (8.1.1), $\mathcal{X}$ is essentially small if and only if $\mathcal{X}'$ is so (8.1.7), and the preceding homomorphism (8.2.3.1) is an isomorphism

$$L(\mathcal{X}') \simeq L(\mathcal{X}). \tag{8.2.3.2}$$

8.2.4

Let

$$\mathcal{X} = (X_i)_{i \in I}, \qquad \mathcal{Y} = (Y_j)_{j \in J} \tag{8.2.4.1}$$

be two essentially small ind-objects of C, indexed by filtered essentially small categories I and J. A *morphism* from the ind-object $\mathcal{X}$ to the ind-object $\mathcal{Y}$ is by definition a morphism

$$L(\mathcal{X}) \longrightarrow L(\mathcal{Y}) \tag{8.2.4.2}$$

between the presheaves they ind-represent. We put

$$\mathrm{Hom}_{\text{ind-ob}}(\mathcal{X}, \mathcal{Y}) = \mathrm{Hom}_{\widehat{C}}(L(\mathcal{X}), L(\mathcal{Y})). \tag{8.2.4.3}$$

We suppress the subscript "ind-ob" on Hom if no confusion is likely. Composition of morphisms of ind-objects of C is defined as composition of morphisms of the presheaves they ind-represent. If $\mathcal{V} \supset \mathcal{U}$ is a universe containing $\mathcal{U}$, and if one restricts to essentially small ind-objects which belong to $\mathcal{V}$, these form the set of objects of a category whose arrows are triples $(\mathcal{X}, \mathcal{Y}, u)$, where $\mathcal{X}$ and $\mathcal{Y}$ are essentially small ind-objects belonging to $\mathcal{V}$ and $u : \mathcal{X} \to \mathcal{Y}$ is a morphism of ind-objects, i.e. a morphism $L(\mathcal{X}) \to L(\mathcal{Y})$. The category so obtained is denoted

$$\mathrm{Ind}_{\mathcal{V}}(C, \mathcal{U}). \tag{8.2.4.4}$$

When $\mathcal{V} = \mathcal{U}$, it is denoted simply

$$\mathrm{Ind}_{\mathcal{U}}(C) \quad \text{or} \quad \mathrm{Ind}(C), \tag{8.2.4.5}$$

and is called the *category of ind-objects of C* relative to $\mathcal{U}$, the mention of $\mathcal{U}$ being suppressed when no confusion is likely.

Of course, if $\mathcal{V} \subset \mathcal{V}'$ are two universes containing $\mathcal{U}$, then $\mathrm{Ind}_{\mathcal{V}}(C, \mathcal{U})$ is a full subcategory of $\mathrm{Ind}_{\mathcal{V}'}(C, \mathcal{U})$. It follows from 8.2.3 that the inclusion functor is an *equivalence of categories*

$$\mathrm{Ind}_{\mathcal{V}}(C, \mathcal{U}) \xrightarrow{\approx} \mathrm{Ind}_{\mathcal{V}'}(C, \mathcal{U}). \tag{8.2.4.6}$$

This shows in particular that *every essentially small ind-object of C defines an element of* $\mathrm{Ind}(C)$*, uniquely up to isomorphism.* This observation justifies the fairly common practical abuse of language which consists in identifying any essentially small ind-object of C with an object of $\mathrm{Ind}(C)$.

It follows at once from the definitions that for every universe $\mathcal{V}$ we have a canonical functor

$$L : \mathrm{Ind}_{\mathcal{V}}(C, \mathcal{U}) \longrightarrow \widehat{C} \tag{8.2.4.7}$$

which is fully faithful. These functors are of course known up to unique isomorphism, by (8.2.4.6), once one of them is known, in particular once one knows the canonical functor

$$L : \mathrm{Ind}(C) \longrightarrow \widehat{C}. \tag{8.2.4.8}$$

Since this functor is fully faithful, it is often used to identify an object of the first category with the functor it pro-represents, or even to identify $\mathrm{Ind}(C)$ with its essential image in $\widehat{C}$, consisting of the ind-representable presheaves. Notice, however, that this identification has distinctly more drawbacks than the analogous identification of C with a full subcategory of $\widehat{C}$ via the functor h (8.2.1.1), because unlike the latter functor, the functor (8.2.4.8) is not in general injective on objects.

8.2.5

Let us spell out definition (8.2.4.3) in terms of the indexed expressions (8.2.4.1) of the ind-objects under consideration. Taking account of the definition (8.2.2.1), one obtains

$$\mathrm{Hom}(\mathcal{X},\mathcal{Y}) \simeq \varprojlim_i \mathrm{Hom}(X_i,\mathcal{Y}) := \varprojlim_i \mathrm{Hom}(X_i, L(\mathcal{Y})) = \varprojlim_i \varinjlim_j \mathrm{Hom}_C(X_i, Y_j),$$

the last equality coming from (8.2.2.2) applied to $\mathcal{Y}$. Thus there is a canonical bijection

$$\mathrm{Hom}((X_i)_{i\in I}, (Y_j)_{j\in J}) \simeq \varprojlim_{i\in I} \varinjlim_{j\in J} \mathrm{Hom}_C(X_i, Y_j). \tag{8.2.5.1}$$

We leave to the reader the task of making the composition of morphisms of ind-objects explicit using this formula. Notice that it follows at once from this formula that the set $\mathrm{Hom}(\mathcal{X},\mathcal{Y})$ of homomorphisms of ind-objects is $\mathcal{U}$-*small*; equivalently, the categories (8.2.4.4) are $\mathcal{U}$-categories. Equivalently again, by (8.2.4.6), the *category* $\mathrm{Ind}(C)$ *is a* $\mathcal{U}$-*category*; that is, the right hand side of (8.2.5.1) is small when $I, J \in \mathcal{U}$, which follows immediately from the fact that C is a $\mathcal{U}$-category, hence the $\mathrm{Hom}_C(X_i, Y_j)$ are small.

8.2.6

Let I be an essentially small filtered category. Then one sees either from (8.2.5.1), or from the functoriality of the inductive limit of a functor $I \to \widehat{C}$ with respect to that functor, that there is a canonical functor

$$\mathrm{Hom}(I, C) \longrightarrow \mathrm{Ind}(C). \tag{8.2.6.1}$$

Notice that this functor is *not* in general fully faithful, or even faithful.

Remark 8.2.7. The definitions of 8.2.4 make clear the notion of *isomorphism* of two essentially small ind-objects, which also means the isomorphism of the presheaves they ind-represent. Notice that this is a very weak notion of isomorphism when compared with the notion of isomorphism in categories of the form $\mathrm{Hom}(I, C)$. Thus many fairly natural relations one can impose on an ind-object, such as having its components in a given strictly full subcategory, or being strict (8.12 below), are *not* stable under isomorphism. Therefore, if one wants to work with notions stable under isomorphism of ind-objects, one must "saturate" the notions under consideration by passing to the strictly full subcategory of $\mathrm{Ind}(C)$ generated by the subcategory of $\mathrm{Ind}(C)$ consisting of objects satisfying the condition in question. See, for example, the notion of essentially constant ind-object introduced below (8.4).

Exercise 8.2.8. Let $\mathcal{U} \subset \mathcal{V}$ be two universes and let C be a $\mathcal{U}$-category belonging to $\mathcal{V}$. Let $\mathrm{SysInd}(C)$ denote the following category.

a) The objects of $\mathrm{SysInd}(C)$ are functors $\varphi : I \to C$, where I is an essentially $\mathcal{U}$-small filtered category belonging to $\mathcal{V}$.

b) If (I,φ) and (J,ψ) are two objects of $\mathrm{SysInd}(C)$, a morphism in $\mathrm{SysInd}(C)$ from (I,φ) to (J,ψ) is a pair (m,u), where $m : I \to J$ is a functor and $u : \varphi \to \psi \circ m$ is a morphism of functors. The composition of morphisms in $\mathrm{SysInd}(C)$ is defined in the evident way.

Let S be the set of morphisms $(m,u) : (I,\varphi) \to (J,\Psi)$ of $\mathrm{SysInd}(C)$ such that m is cofinal and u is an isomorphism. Let (I,φ) be an object of $\mathrm{SysInd}(C)$. The category $\mathrm{Fl}(I)$ of arrows of I maps to I by two natural functors: source and target. Moreover these two functors are related by the canonical morphism of functors $v : \mathrm{source} \to \mathrm{target}$. Hence there are two morphisms in $\mathrm{SysInd}(C)$ from $(\mathrm{Fl}(I), \varphi \circ \mathrm{source})$ to (I,φ):

$$p_1(I,\varphi) = (\mathrm{source}, \mathrm{id}),$$

and

$$p_2(I,\varphi) = (\mathrm{target}, \varphi^* v : \varphi \circ \mathrm{source} \to \varphi \circ \mathrm{target}).$$

Let $p : \mathrm{SysInd}(C) \to \mathrm{Ind}(C)$ be the evident functor. Let B be a category. Show that the functor $F \mapsto F \circ p$,

$$\mathrm{Hom}(\mathrm{Ind}(C), B) \longrightarrow \mathrm{Hom}(\mathrm{SysInd}(C), B),$$

is fully faithful, and that a functor $G : \mathrm{SysInd}(C) \to B$ belongs to its essential image if and only if it has the following two properties:

1) For every $s \in S$, $F(s)$ is an isomorphism of B.

2) For every object (I,φ) of $\mathrm{SysInd}(C)$, $F(p_1(I,\varphi)) = F(p_2(I,\varphi))$.

8.3 Characterization of ind-representable functors

Proposition 8.3.1. Let F be an ind-representable presheaf on C. Then F is left exact, i.e. (2.3.2), for every finite category J and every functor $\varphi : J \to C$ such that $\varinjlim \varphi$ is representable (i.e. $\varprojlim \varphi^{\mathrm{op}}$ is representable), the canonical map

$$F(\varinjlim \varphi) \longrightarrow \varprojlim F \circ \varphi$$

is bijective.

Indeed, representable presheaves are plainly left exact; hence by 2.8 so is every filtered inductive limit of such functors. $\square$

8.3.2

Given a presheaf F on C, we shall often have to work with the category

$$C_{/F} \hookrightarrow \widehat{C}_{/F}, \tag{8.3.2.1}$$

which denotes the full subcategory of the second category consisting of arrows $X \to F$ *whose source lies in* C. It follows from 1.4 that the objects of this category are identified with pairs (X,u), where X is an object of C and $u \in F(X)$. A morphism from (X,u) to (X',u') is then interpreted as an arrow $f : X \to X'$ in C such that $F(f)(u') = u$. The "forget-the-target" functor from $\widehat{C}_{/F}$ to $\widehat{C}$ induces a functor, called the *canonical* functor,

$$C_{/F} \longrightarrow C, \tag{8.3.2.2}$$

already considered in 3.4, where it is proved that the inductive limit of this functor exists in $\widehat{C}$ (with no smallness condition on $C_{/F}$) and is canonically isomorphic to F.

8.3.2.3

Notice that if finite inductive limits are representable in C, and if F is left exact, i.e. carries them to finite projective limits in **Set**, then the same is true in $C_{/F}$, and *a fortiori* (2.7.1) $C_{/F}$ is filtered. More generally, if sums of two objects and cokernels of double arrows are representable in C, and if F carries them respectively to products and to kernels, then $C_{/F}$ is stable under the same types of finite inductive limits. Hence it is filtered if and only if it is nonempty, i.e. if and only if the functor F is not the constant functor with value $\emptyset$.

Theorem 8.3.3. Let C be a $\mathcal{U}$-category and let $F \in \operatorname{Ob} \widehat{C}$, say $F : C^{\mathrm{op}} \to (\mathcal{U}\text{-}\mathbf{Set})$ is a $\mathcal{U}$-presheaf on C. The following conditions are equivalent:

(i) F is ind-representable (8.2.2).

(ii) The category $C_{/F}$ (8.3.2) is filtered and essentially small.

(iii) If finite inductive limits are representable in C: the functor F is left exact, and $\operatorname{Ob} C_{/F}$ admits a small cofinal subset (8.1.4).

(iii bis) If sums of two objects and cokernels of double arrows are representable in C: the functor F carries sums of two objects of C to products and cokernels of double arrows of C to kernels, the category $C_{/F}$ is nonempty, i.e. the functor F is not the constant functor with value $\emptyset$, and finally there exists a small family of objects of $C_{/F}$ such that every object X of $C_{/F}$ is dominated by some X_i, i.e. admits an F-morphism $X_i \to X$.

(iv) If the category C is equivalent to a small category: the category $C_{/F}$ is filtered.

(v) If the category C is equivalent to a small category and if filtered inductive limits in C are representable: the functor F is left exact.

Proof. (i) $\Rightarrow$ (ii). Suppose that F is ind-represented by $(X_i)_{i\in I}$, with I small. We prove that $C_{/F}$ is filtered. Let X, X' be two objects of $C_{/F}$, i.e. objects of C equipped with morphisms $X \to F$ and $X' \to F$. We prove that they are dominated by a third object of $C_{/F}$. The morphisms $X \to F$ and $X' \to F$ come from morphisms $X \to X_i$ and $X' \to X'_{i'}$, and after replacing i, i' by a common upper bound in $\operatorname{Ob} I$, we may suppose $i = i'$. We then take as common upper bound of X and X' the object X_i equipped with the canonical morphism $X_i \to F$.

Now let

$$X \overset{f}{\underset{g}{\rightrightarrows}} X' \xrightarrow{h} F$$

be a double arrow in $C_{/F}$. We prove that it is equalized by an arrow $X' \to X''$ of $C_{/F}$. The morphism $h : X' \to F$ is given by a morphism $h_i : X' \to X_i$, and the condition $hf = hg$ says that there exists $\alpha : i \to j$ in I such that

$$\varphi(\alpha)(h_i f) = \varphi(\alpha)(h_i g).$$

Thus, after replacing h_i by $\varphi(\alpha)h_i$, we equalize f and g. This proves that $C_{/F}$ is filtered. It is then immediate that it is essentially small, since the objects $(X_i \to F)_{i\in \operatorname{Ob} I}$ form a small cofinal family in $\operatorname{Ob} C_{/F}$.

(ii) $\Rightarrow$ (i). Put $I = C_{/F}$ and consider the canonical functor (8.3.2.1)

$$\varphi : I = C_{/F} \longrightarrow C.$$

The presheaf represented by this ind-object of C is known to be F (3.4), so F is ind-representable by definition (8.2.2).

(ii) $\Leftrightarrow$ (iii). Indeed, (ii) $\Rightarrow$ (iii) because an ind-representable functor is left exact (8.3.1), and (iii) $\Rightarrow$ (ii) because we have observed (8.3.2.3) that left exactness of F implies that $C_{/F}$ is filtered, and one applies Definition 8.1.8 to conclude that this category is essentially small.

The equivalence (ii) $\Leftrightarrow$ (iii bis) is proved in the same way as (ii) $\Leftrightarrow$ (iii).

The equivalences (ii) $\Leftrightarrow$ (iv) and (iii) $\Leftrightarrow$ (v) are trivial, since if C is equivalent to a small category, then $C_{/F}$ is evidently also equivalent to a small category and *a fortiori* is automatically essentially small as soon as it is filtered. This completes the proof of 8.3.3.

Remark 8.3.4. Let $F : C' \to C''$ be a functor between $\mathcal{U}$-categories. It appears that the notion of left exactness (2.3.2) is of practical interest only when finite projective limits are representable in C'. In the case where $C'' = (\mathcal{U}\text{-}\mathbf{Set})$ and C' is $\mathcal{U}$-small, writing C' in the form C^{op} (so $C = C'^{\mathrm{op}}$), the "right notion" which in the general case replaces left exactness is that of ind-representability of F when it is considered as a presheaf on C^{op}; equivalently, it is the pro-representability of F when it is considered as a functor on C' (cf. 8.10). This notion does coincide with left exactness when finite projective limits are representable in C', i.e. when finite inductive limits are representable in C (8.3.3). When one no longer assumes that C' is $\mathcal{U}$-small, or at least equivalent to a $\mathcal{U}$-small category, the two notions for a functor F, left exactness and pro-representability, no longer necessarily coincide, even if finite projective limits are representable in C' (cf. 8.12.9). In fact, it seems that in this case left exact functors which are not ind-representable should be regarded as pathological; the "good" objects remain the ind-representable functors. Compare 8.13.3. For the case of functors $F : C' \to C''$, with C' and C'' again arbitrary $\mathcal{U}$-categories, we shall develop below (8.11.5), more generally, a notion which "improves" that of left exactness, namely the notion of a functor admitting a pro-adjoint functor.

8.4 Constant and essentially constant ind-objects

Choose a final category e, i.e. one such that $\mathrm{Ob}\, e$ and $\mathrm{Fl}\, e$ are each reduced to a single element; for example, one may take for e the full subcategory of **Set** consisting of the empty set, if one wants a canonical choice. This is plainly a filtered and small category. If X is an object of C, we associate with it the constant ind-object indexed by e, with value X, denoted $c(X)$. It is clear that, as X varies, one obtains a fully faithful functor

$$c : C \longrightarrow \mathrm{Ind}(C), \tag{8.4.1}$$

which is moreover injective on objects, and through which we shall identify C with a full subcategory of $\mathrm{Ind}(C)$. Notice that the composite functor

$$C \xrightarrow{c} \mathrm{Ind}(C) \xrightarrow{L} \widehat{C} \tag{8.4.2}$$

is the canonical functor h (8.2.1.1).

8.4.3

More generally, let I be a filtered category. The constant functor on I with value an object X of C is also called *the constant ind-object of value X indexed by I*. If I is essentially small, it is clear that this ind-object ind-represents the functor $h(X)$ represented by X, hence this ind-object is isomorphic to $c(X)$.

8.4.4

An ind-object $\mathcal{X}$ of C is called *essentially constant* if it is isomorphic (in a category $\mathrm{Ind}_{\mathcal{V}}(C, \mathcal{V}')$ for suitable $\mathcal{V}, \mathcal{V}'$) to an ind-object of the form $c(X)$, with $X \in \mathrm{Ob}\, C$. The object X, which is then determined up to canonical isomorphism, is called the *value* of the essentially constant ind-object in question. Thus, by definition, the functor c of (8.4.1) induces an equivalence of C with the full subcategory of $\mathrm{Ind}(C)$ consisting of essentially constant ind-objects; this subcategory is the essential image of the functor c.

Obviously a constant ind-object is essentially constant; the converse is true only in the trivial case where C is empty or a one-point category.

8.5 Filtered inductive limits in $\mathrm{Ind}(C)$

Proposition 8.5.1. Let C be a $\mathcal{U}$-category. In $\mathrm{Ind}(C)$, small filtered inductive limits are representable, and the canonical functor (8.2.4.8)

$$L : \mathrm{Ind}(C) \longrightarrow \widehat{C}$$

commutes with them.

Taking account of the fact that L is fully faithful, the assertions of the proposition are equivalent to the following one.

Corollary 8.5.2. Every small filtered inductive limit, in $\widehat{C}$, of ind-representable presheaves is ind-representable.

This follows easily from criterion 8.3.3(ii). The details of the verification are left to the reader.

8.5.3

The "tautological" case of filtered inductive limits in $\mathrm{Ind}(C)$ is the case where one starts from an essentially small ind-object $\mathcal{X} = (X_i)_{i \in I}$ of C. One then has, in $\widehat{C}$ and hence also in $\mathrm{Ind}(C)$,

$$\mathcal{X} = \varinjlim_{I} X_i. \tag{8.5.3.1}$$

In writing this formula, one should take care that this is not an inductive limit in C, and that even when the latter exists, it is not isomorphic in $\mathrm{Ind}(C)$ to $\mathcal{X}$: indeed, the canonical functor (8.4.1) $c : C \to \mathrm{Ind}(C)$ *does not in general commute with filtered inductive limits.* See Exercise 8.5.5 below.

To avoid this possible confusion in writing (8.5.3.1), "certain authors" (= P. Deligne) prefer to write it

$$\mathcal{X} = \text{“}\varinjlim_{I}\text{”}\, X_i, \tag{8.5.3.2}$$

the role of the quotation marks being to indicate that the inductive limit is taken in a category of ind-objects $\mathrm{Ind}(C)$. By extension, one should then denote by "$\varinjlim$" every inductive-limit operation in $\mathrm{Ind}(C)$, without requiring the components of the inductive system considered in $\mathrm{Ind}(C)$ to lie in the image, or in the essential image, of $c : C \to \mathrm{Ind}(C)$.

8.5.4

To compute the inductive or projective limit of a functor

$$\varphi : J \longrightarrow \operatorname{Ind}(C), \tag{8.5.4.1}$$

where J is now an arbitrary category, it is often convenient to make φ explicit by means of a functor

$$\Phi : J \times I \longrightarrow C, \tag{8.5.4.2}$$

where I is an essentially small, or preferably small, filtered category. Such a Φ indeed defines a functor

$$j \longmapsto (i \longmapsto \Phi(j,i)) : J \longrightarrow \operatorname{Hom}(I, C),$$

hence, after composing with the canonical arrow (8.2.6.1) $\operatorname{Hom}(I, C) \to \operatorname{Ind}(C)$, a functor

$$j \longmapsto (i \longmapsto \Phi(j,i)) : J \longrightarrow \operatorname{Ind}(C). \tag{8.5.4.3}$$

We may say that Φ is an *indexed expression of φ, indexed by the filtered category I*, if the corresponding functor (8.5.4.3) is isomorphic to φ. We shall study below (8.8) general existence conditions for an indexed expression of a given functor φ. Here we shall merely start from a functor φ given in indexed form (8.5.4.3), in the case where J is a small filtered category, in order to indicate the calculation of

$$\text{“}\varinjlim\text{”}\,\varphi = \text{“}\varinjlim_{j}\text{”}\,\varphi(j).$$

Formula (8.5.3.2), together with the associativity formula for inductive limits (2.5.0), then immediately gives the *tautological calculation* of “$\varinjlim$” φ:

$$\text{“}\varinjlim_{J}\text{”}\,\varphi = \text{“}\varinjlim_{J\times I}\text{”}\,\Phi, \tag{8.5.4.4}$$

i.e.

$$\text{“}\varinjlim_{j}\text{”}\,(i \longmapsto \Phi(j,i)) = \text{“}\varinjlim_{j,i}\text{”}\,\Phi(j,i). \tag{8.5.4.5}$$

Thus the inductive system sought is none other than Φ itself, the index category being $J \times I$, which is indeed filtered since J and I are so.

Exercise 8.5.5. Let C be a $\mathcal{U}$-category.

a) Prove that the following conditions are equivalent:

   (i) Small inductive limits are representable in C, and the functor $c : C \to \operatorname{Ind}(C)$ commutes with them.

   (ii) Small inductive limits are representable in C, and for every $X \in \operatorname{Ob} C$, the functor $Y \mapsto \operatorname{Hom}(X, Y)$ commutes with these limits.

   (iii) The functor c is an equivalence of categories.

   (iv) Small filtered inductive limits are representable in C, and the functor $\lim c : \operatorname{Ind} C \to C$ (cf. 8.7.1.5) is fully faithful, or equivalently an equivalence.

b) If C is a finite category, prove that the preceding conditions are equivalent to the following one: for every projector in C (10.6), its image is representable in C.

8.6 Extension of a functor to ind-objects

8.6.1

Let

$$f : C \longrightarrow C' \tag{8.6.1.1}$$

be a functor between $\mathcal{U}$-categories. For every ind-object

$$\varphi : I \longrightarrow C$$

of C, where I is a filtered category, the composite functor $f \circ \varphi$ is an ind-object of C', also denoted $\mathrm{Ind}(f)(\varphi)$. In indexed notation, if φ is written as $\mathcal{X} = (X_i)_{i \in I}$, one obtains

$$\mathrm{Ind}(f)(X_i)_{i\in I} = (f(X_i))_{i\in I}. \tag{8.6.1.2}$$

Let $\mathcal{Y} = (Y_j)_{j\in J}$ be a second inductive system of C. Then one obtains an evident map, also denoted $u \mapsto \mathrm{Ind}(f)(u)$:

$$\begin{aligned}\mathrm{Hom}(\mathcal{X},\mathcal{Y}) &= \varprojlim_i \varinjlim_j \mathrm{Hom}(X_i, Y_j)\\ &\longrightarrow \mathrm{Hom}(\mathrm{Ind}(f)(\mathcal{X}), \mathrm{Ind}(f)(\mathcal{Y})) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}(f(X_i), f(Y_j)).\end{aligned} \tag{8.6.1.3}$$

One immediately checks that these maps are compatible with the composition of morphisms of ind-objects, referring to the unwritten formula for composition of morphisms of ind-objects (8.2.5). We have thus obtained, for every universe $\mathcal{V}$, a functor

$$\mathrm{Ind}(f) : \mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U}) \longrightarrow \mathrm{Ind}_{\mathcal{V}}(C',\mathcal{U}) \tag{8.6.1.4}$$

(notation of (8.2.4.5)), and in particular a functor

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C'). \tag{8.6.1.5}$$

If one has a second functor

$$g : C' \longrightarrow C'',$$

then of course there is an identity of functors between categories of ind-objects, or of $\mathrm{Ind}_{\mathcal{V}}$-objects if desired,

$$\mathrm{Ind}(gf) = \mathrm{Ind}(g) \circ \mathrm{Ind}(f), \tag{8.6.1.6}$$

and for $C' = C$ one has the perennial formula

$$\mathrm{Ind}(\mathrm{id}_C) = \mathrm{id}_{\mathrm{Ind}(C)}\,. \tag{8.6.1.7}$$

Thus, figuratively speaking, one may say that the category $\mathrm{Ind}(C)$ *depends functorially on* C, covariantly.[1] We leave to the reader the task of making explicit even a 2-*functorial* dependence, by defining, for every pair of $\mathcal{U}$-categories C, C', a canonical functor

$$\mathsf{Ind} : \mathrm{Hom}(C, C') \longrightarrow \mathrm{Hom}(\mathrm{Ind}(C), \mathrm{Ind}(C')), \tag{8.6.1.8}$$

and by specifying the functorial nature of the identity isomorphisms (8.6.1.6) and (8.6.1.7).

[1] For a contravariant dependence, under certain conditions, cf. 8.11.

8.6.2

Return to the functor (8.6.1.1), and consider the corresponding functor (5.1)

$$f_! : \widehat{C} \longrightarrow \widehat{C'}, \tag{8.6.2.1}$$

and the diagram of functors

$$\begin{tikzcd} \mathrm{Ind}(C) \arrow[r, "\mathrm{Ind}(f)"] \arrow[d, "L_C"'] & \mathrm{Ind}(C') \arrow[d, "L_{C'}"] \\ \widehat{C} \arrow[r, "f_!"'] & \widehat{C'}. \end{tikzcd} \tag{8.6.2.2}$$

Here the vertical arrows denote the canonical functors L of (8.2.4.8). Since the functor $f_!$ "extends f" and commutes with inductive limits (5.4.3), and since the functors L_C and $L_{C'}$ commute with small filtered inductive limits (8.5.1), it follows from (8.5.3.2) that for every ind-object $(X_i)_{i\in I}$ of C there is, in $\widehat{C'}$, a canonical isomorphism

$$f_!(L_C((X_i)_{i\in I})) \simeq \varinjlim_i f_! L_C(X_i) \simeq \varinjlim_i L_{C'} f(X_i) = L_{C'}(f(X_i)_{i\in I}),$$

i.e. a canonical isomorphism

$$f_! L_C(\mathcal{X}) \simeq L_{C'}(\mathrm{Ind}(f)(\mathcal{X})).$$

We leave to the reader the verification that the latter is functorial, i.e. that it corresponds to a canonical isomorphism

$$f_! L_C \simeq L_{C'} \circ \mathrm{Ind}(f). \tag{8.6.2.3}$$

In other words, *the square (8.6.2.2) is commutative up to canonical isomorphism.* Taking account of the fact that $f_!$ commutes with small inductive limits, and of 8.5.1, we conclude as follows.

Proposition 8.6.3. Let $f : C \to C'$ be a functor between $\mathcal{U}$-categories. Then the functor

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C')$$

commutes with filtered inductive limits. Moreover, it makes the following diagram commutative:

$$\begin{tikzcd} C \arrow[r, "f"] \arrow[d, "c_C"'] & C' \arrow[d, "c_{C'}"] \\ \mathrm{Ind}(C) \arrow[r, "\mathrm{Ind}(f)"'] & \mathrm{Ind}(C'), \end{tikzcd}$$

where the vertical arrows $c_C, c_{C'}$ are the canonical functors (8.4.1).

The last assertion is trivial from the definitions and has been included for ease of reference. Note moreover that the properties stated in 8.6.3 *characterize the functor* $\mathrm{Ind}(f)$ up to unique isomorphism as being induced by the functor $f_!$ (8.6.2.1), as follows from the proof just given of (8.6.2.3).

Proposition 8.6.4. The notation is that of 8.6.3.

a) For $\mathrm{Ind}(f)$ to be faithful, respectively fully faithful, it is necessary and sufficient that f be so.

b) For $\mathrm{Ind}(f)$ to be an equivalence of categories, it is necessary and sufficient that f be fully faithful and that every object of C' be isomorphic to a direct factor (10.6) of an object in the image of f.

Proof.

a) Necessity follows immediately from the fact that f is induced by $\mathrm{Ind}(f)$. For sufficiency, it is enough to use the form (8.6.1.3) of $\mathrm{Ind}(f)$ on the sets $\mathrm{Hom}(\mathcal{X}, \mathcal{Y})$, remembering that filtered inductive limits of sets, and arbitrary projective limits, carry monomorphisms to monomorphisms and isomorphisms to isomorphisms.

b) We may already suppose f, hence $\mathrm{Ind}(f)$, fully faithful. Since every object of $\mathrm{Ind}(C')$ is a small filtered inductive limit of objects of C', full faithfulness of $\mathrm{Ind}(f)$ implies that for this functor to be essentially surjective it is equivalent that every object X' of C' lie in its essential image. If one has an isomorphism

$$X' \xrightarrow{\sim} \text{“}\varinjlim\text{”}\, f(X_i),$$

then this isomorphism factors through one of the $f(X_i)$; this shows that X' is a direct factor of $f(X_i)$, proving necessity in b). For sufficiency, using the full faithfulness of $\mathrm{Ind}(f)$, the assertion follows at once from the following corollary.

Corollary 8.6.5. In $\mathrm{Ind}(C)$ the images of projectors (10.6) are representable, and the functor $\mathrm{Ind}(f) : \mathrm{Ind}(C) \to \mathrm{Ind}(C')$ commutes with the formation of these images.

Indeed, this follows from the fact that in $\mathrm{Ind}(C)$ small filtered inductive limits are representable (8.5.1) and that $\mathrm{Ind}(f)$ commutes with them (8.6.3), taking account of the fact that the image of a projector $p : X \to X$ may be interpreted as the inductive limit of the filtered inductive system indexed by $\mathbb{N}$

$$X \xrightarrow{p} X \xrightarrow{p} X \longrightarrow \cdots,$$

or, equivalently, as the inductive limit of the evident functor on the filtered category P having a single object and a nonidentity arrow Π such that $\Pi^2 = \Pi$.

Exposé I. Presheaves

Section 8.7–8.8

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 8.7 through Section 8.8. A few evident typographical slips in the source notation are corrected where the mathematical context forces the correction.

8 Ind-objects and pro-objects

8.7 The functor $\varinjlim_C : \mathrm{Ind}(C) \to C$. Universal characterizations of $\mathrm{Ind}(C)$

8.7.1

Let C again be a $\mathcal{U}$-category, and let

$$c : C \longrightarrow \mathrm{Ind}(C) \tag{8.7.1.1}$$

be the canonical functor. Let $\mathcal{X} = (X_i)_{i\in I}$ be an object of $\mathrm{Ind}(C)$. It follows immediately from the definition of representable inductive limits (2.1, 2.1.1) that $\varinjlim_i X_i$ is *representable in C if and only if the left adjoint of c is defined at $\mathcal{X}$*; in that case $\varinjlim_i X_i$, computed in C, is precisely the value of that left adjoint at $\mathcal{X}$:

$$\mathrm{Hom}_C(\varinjlim_i X_i, Y) \simeq \mathrm{Hom}_{\mathrm{Ind}(C)}((X_i)_{i\in I}, c(Y)). \tag{8.7.1.2}$$

Let

$$\mathrm{Ind}(C)' \subset \mathrm{Ind}(C) \tag{8.7.1.3}$$

denote the full subcategory of $\mathrm{Ind}(C)$ formed by the ind-objects of C which admit an inductive limit *in* C. The preceding observation implies that this subcategory is *strictly full*, and that one has a canonical functor

$$\varinjlim_C : \mathrm{Ind}(C)' \longrightarrow C, \tag{8.7.1.4}$$

whose value on an object $(X_i)_{i\in I}$ of $\mathrm{Ind}(C)'$ is its inductive limit in C. In the special case in which small filtered inductive limits are representable in C, one therefore obtains a natural functor

$$\varinjlim_C : \mathrm{Ind}(C) \longrightarrow C. \tag{8.7.1.5}$$

8.7.1.6

Of course, the preceding functors may also be defined for categories of the form $\mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U})'$ and $\mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U})$; but, in view of (8.2.4.6), they are already determined, up to unique isomorphism, by the corresponding functors in the case $\mathcal{V} = \mathcal{U}$.

8.7.1.7

From the preceding construction of $\varinjlim_C$ as a left adjoint, it follows immediately that this functor *commutes with arbitrary inductive limits*, and in particular with small filtered inductive limits, the latter being representable in $\mathrm{Ind}(C)$ by 8.5.1. Notice also that $\mathrm{Ind}(C)'$ always contains the essential image of c, formed by the essentially constant ind-objects, and that one has a canonical isomorphism, functorial in $X \in \mathrm{Ob}\,\mathrm{Ind}(C)'$,

$$\varinjlim_{C} c(X) \simeq X. \tag{8.7.1.8}$$

Equivalently, in the favorable case where C is stable under small inductive limits, one has a canonical isomorphism

$$\varinjlim_{C} \circ c \simeq \mathrm{id}_C\,. \tag{8.7.1.9}$$

8.7.2

Now consider a functor

$$f : C \longrightarrow E, \tag{8.7.2.1}$$

where E is a $\mathcal{U}$-category in which small inductive limits are representable, so that there is a functor

$$\varinjlim_{E} : \mathrm{Ind}(E) \longrightarrow E.$$

Since (8.6) also defined

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(E),$$

one may consider the composite

$$\bar{f} = \varinjlim_{E} \circ \mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow E, \tag{8.7.2.2}$$

which is sometimes called the *canonical extension of f to ind-objects*, although it should not be confused with $\mathrm{Ind}(f)$. Since it is a composite of two functors commuting with small filtered inductive limits (8.6.3, 8.7.1.7), it also commutes with small filtered inductive limits. Moreover, from the isomorphism (8.7.1.9), applied to E, and from (8.6.2.3), it follows that $\bar{f}$ *extends* f up to isomorphism; that is, there is a canonical isomorphism

$$\bar{f} \circ c_C \simeq f. \tag{8.7.2.3}$$

It is also rather clear, taking into account the fact that every object of $\mathrm{Ind}(C)$ is a small filtered inductive limit of objects of C, that the two preceding properties still *characterize* $\bar{f}$ up to unique isomorphism. This can be made precise so as to obtain, at the same time, a universal characterization, up to equivalence, of $\mathrm{Ind}(C)$ among the $\mathcal{U}$-categories E in which small filtered inductive limits are representable. If E and F are two such $\mathcal{U}$-categories, let

$$\mathrm{Hom}(E,F)' \subset \mathrm{Hom}(E,F) \tag{8.7.2.4}$$

denote the full subcategory of $\mathrm{Hom}(E,F)$ formed by the functors which commute with small filtered inductive limits. Then one has the following result.

Proposition 8.7.3. Let C be a $\mathcal{U}$-category, and use the notation above (8.7.2.4). Then the canonical functor

$$c_C : C \longrightarrow \mathrm{Ind}(C)$$

is 2-universal among functors with source C and target a $\mathcal{U}$-category in which small filtered inductive limits are representable. In other words, $\mathrm{Ind}(C)$ is such a category (8.2.5, 8.5.1), and if E is such a category, the functor

$$g \longmapsto g \circ c_C : \mathrm{Hom}(\mathrm{Ind}(C), E)' \longrightarrow \mathrm{Hom}(C, E) \tag{8.7.3.1}$$

is an equivalence of categories.

To prove the last assertion, it is enough to verify that the assignment $f \mapsto \bar{f}$ defined by (8.7.2.2) can be made into a *functor*

$$f \longmapsto \bar{f} = \varinjlim_{E} \cdot \mathrm{Ind}(f) : \mathrm{Hom}(C, E) \longrightarrow \mathrm{Hom}(\mathrm{Ind}(C), E)', \tag{8.7.3.2}$$

and that this functor is a quasi-inverse of (8.7.3.1). The details of the verification, which are essentially trivial, are left to the reader. One may also invoke 7.8 to conclude first that (8.7.3.1) is fully faithful, and then use (8.7.2.3) to conclude that it is essentially surjective.

8.7.4

Let us return to a functor between $\mathcal{U}$-categories

$$f : C \longrightarrow E, \tag{8.7.4.1}$$

where E is a $\mathcal{U}$-category in which small filtered inductive limits are representable. This gives a functor

$$\bar{f} : (X_i)_{i \in I} \longmapsto \varinjlim_{i} f(X_i) : \mathrm{Ind}(C) \longrightarrow E. \tag{8.7.4.2}$$

We shall study conditions on f which ensure that $\bar{f}$ is fully faithful, respectively an equivalence of categories. Since f is isomorphic to the composite $\bar{f} \circ c_C$, and since $c_C : C \to \mathrm{Ind}(C)$ is fully faithful, it is *necessary*, for $\bar{f}$ to be fully faithful, that f be so. Replacing C by its essential image in E, we therefore lose no essential generality by assuming, as we shall do below in order to simplify notation, that f is the inclusion functor of a subcategory C of E.

Proposition 8.7.5. The notation is that of 8.7.4.

a) For the functor $\bar{f}$ to be fully faithful, it is necessary and sufficient that the following condition hold:

(PF) Every object X of C satisfies the following property: for every small filtered inductive system $(Y_i)_{i \in I}$ in C, with inductive limit $\varinjlim_i Y_i$ in E, the canonical map

$$\varinjlim_{i} \mathrm{Hom}(X, Y_i) \longrightarrow \mathrm{Hom}(X, \varinjlim_{i} Y_i) \tag{8.7.5.1}$$

is bijective.

b) For the functor $\bar{f}$ to be an equivalence of categories, it is necessary and sufficient that C satisfy condition (PF) and the following two conditions:

(ii) C is a subcategory of E generating by strict epimorphisms (7.1).

(iii) For every object X of E, the full subcategory $C_{/X}$ of $E_{/X}$ formed by objects X' over X whose source is in $\operatorname{Ob} C$ is filtered and essentially small.

Condition (iii) holds in particular if C is equivalent to a small category, if finite inductive limits are representable in C, and if the inclusion functor $f : C \to E$ commutes with them; that is, for every finite category J and every functor $\varphi : J \to C$, the inductive limit of $f\varphi$ is representable and is isomorphic in E to an object of C.

Proof.

a) With the notation of condition (PF), if $\mathcal{Y}$ denotes the ind-object (Y_i) and $\mathcal{X}$ the constant ind-object defined by X, then (8.7.5.1) is precisely the canonical map

$$\operatorname{Hom}(\mathcal{X}, \mathcal{Y}) \longrightarrow \operatorname{Hom}(\bar{f}(\mathcal{X}), \bar{f}(\mathcal{Y})).$$

Thus, if $\bar{f}$ is fully faithful, this map is bijective. Conversely, if $\mathcal{X} = (X_j)_{j\in J}$ and $\mathcal{Y} = (Y_i)_{i\in I}$ are arbitrary ind-objects of C, then the map

$$u : \operatorname{Hom}(\mathcal{X}, \mathcal{Y}) \longrightarrow \operatorname{Hom}(\bar{f}(\mathcal{X}), \bar{f}(\mathcal{Y}))$$

is the projective limit over j of the maps

$$u_j : \operatorname{Hom}(\mathcal{X}_j, \mathcal{Y}) \longrightarrow \operatorname{Hom}(\bar{f}(\mathcal{X}_j), \bar{f}(\mathcal{Y})),$$

where $\mathcal{X}_j$ is the constant ind-object with value X_j. Hence u is bijective if the u_j are, and condition (PF) implies that $\bar{f}$ is fully faithful.

b) Suppose that (PF), (ii), and (iii) hold, and prove that $\bar{f}$ is an equivalence. It remains only to prove that it is essentially surjective. Let $X \in \operatorname{Ob} E$. Hypothesis (ii) means precisely that

$$\varinjlim_{C_{/X}} Z \longrightarrow X$$

is an isomorphism, while (iii) says that the category $C_{/X}$ is filtered and essentially small. Thus X is the image of the ind-object of C defined by the natural functor $C_{/X} \to E$.

Conversely, suppose that $\bar{f}$ is an equivalence. In view of a), it remains to check (ii) and (iii). We may suppose that $E = \operatorname{Ind}(C)$, with C identified with the subcategory $c(C)$ of E. But it is known (8.3.3(ii)) that, for every $X \in \operatorname{Ind}(C)$, identified if desired with the functor F on C^{op} which it ind-represents, $C_{/X} = C_{/F}$ is a filtered essentially small category. This proves (iii). Moreover, the inductive limit in $\widehat{C}$ of the functor $C_{/F} \to \widehat{C}$ is F. A fortiori the same is true in the full subcategory E of $\widehat{C}$, since $F = X$ lies in E. This proves (ii).

It remains to prove the last assertion of b). But the hypothesis made on C plainly implies that finite inductive limits are representable in the category $C_{/X}$; a fortiori $C_{/X}$ is filtered.

Remark 8.7.5.2. The preceding argument shows more generally that, when condition (PF) holds, the essential image of the fully faithful functor $\bar{f}$ (8.7.4.2) consists of the objects $X \in \operatorname{Ob} E$ such that the category $C_{/X}$ is filtered and essentially small–a condition automatically satisfied if C satisfies the conditions stated at the end of b)–and such that the inductive limit of the canonical functor $C_{/X} \to E$ is X.

Corollary 8.7.6. Let E_{PF} be the full subcategory of E formed by the objects satisfying condition (PF) of Proposition 8.7.5 a). Then, for every functor $J \to E_{PF}$, with J a finite category, which admits an inductive limit X in E, one has $X \in \operatorname{Ob} E_{PF}$. The canonical functor

$$(X_i)_{i \in I} \longmapsto \varinjlim_{i,E} X_i,$$

obtained from the inclusion $g : E_{PF} \hookrightarrow E$,

$$\bar{g} : \operatorname{Ind}(E_{PF}) \longrightarrow E \tag{8.7.6.1}$$

is fully faithful. If finite inductive limits are representable in E and E_{PF} is equivalent to a small category, then the essential image of $\bar{g}$ is formed by the objects X of E for which the canonical morphism

$$\varinjlim_{(E_{PF})_{/X}} Y \longrightarrow X$$

is an isomorphism. If, moreover, the subcategory E_{PF} of E generates by strict epimorphisms (7.1), then the functor (8.7.6.1) is an equivalence of categories.

All the assertions are evident, in the order in which they are stated, from 8.7.5, using 2.8 for the first assertion.

Corollary 8.7.7. Suppose that finite inductive limits are representable in E. Let C be a full subcategory of E, equivalent to a small category, and consider the functor

$$\bar{f} : \operatorname{Ind}(C) \longrightarrow E, \qquad (X_i)_{i \in I} \longmapsto \varinjlim_{i,E} X_i. \tag{8.7.7.1}$$

Let E_{PF} be as in 8.7.6. The following conditions are equivalent:

(i) The functor $\bar{f} : \operatorname{Ind}(C) \to E$ is an equivalence.

(ii) The category C is a subcategory of E generating by strict epimorphisms, is contained in E_{PF}, and every object of E_{PF} is isomorphic in E (or in E_{PF}, which is the same thing by 8.7.6) to a direct factor of an object of C.

When the subcategory C of E is stable under direct factors (10.6), the preceding conditions are also equivalent to the following:

(ii bis) $C = E_{PF}$, and C generates in E by strict epimorphisms.

(iii) The subcategory C of E generates by strict epimorphisms, is contained in E_{PF}, and finite inductive limits are representable in C.

When, moreover, finite inductive limits are representable in E, these conditions are also equivalent to

(iii bis) The subcategory C of E generates by strict epimorphisms, is contained in E_{PF}, and is stable in E under finite inductive limits; equivalently, under finite sums and cokernels of double arrows.

We already know (8.7.5) that condition (i) implies that $C \subset E_{PF}$ and that C generates by strict epimorphisms. Let us also prove that then every object X of E_{PF} is isomorphic to a direct factor of an object of C. Indeed, X is a filtered inductive limit $\varinjlim_i X_i$ in E of objects of C; by the definition of E_{PF}, for a suitable i the canonical homomorphism $X_i \to X$ admits a left inverse, so that X is indeed a direct factor of the object X_i of C. This proves (i) $\Rightarrow$ (ii), without any further hypothesis on C or E.

Let us prove (ii) $\Rightarrow$ (i). Since C is equivalent to a small category and every object of E_{PF} is a direct factor of an object of C, it follows that E_{PF} is also equivalent to a small category. Hence, by 8.7.6, the functor (8.7.6.1) is an equivalence of categories, and one concludes by applying 8.6.4 b) to the inclusion $C \hookrightarrow E_{PF}$.

When every direct factor in E of an object of C lies in C, it is clear that (ii) is equivalent to (ii bis). On the other hand, (ii bis) implies (iii), respectively (iii bis), by 8.7.6. Finally, (iii), and *a fortiori* (iii bis), implies (i) by 8.7.5. This completes the proof.

Exercise 8.7.8 (Karoubi envelopes). Let C be a $\mathcal{U}$-category, let $E = \mathrm{Ind}(C)$, and let $E_{PF} \supset C$ be the full subcategory of E defined in 8.7.6.

a) Prove that images of projectors are representable in E_{PF}, and that every object of E_{PF} is a direct factor of an object of C.

b) Call a category F *Karoubian* if images of projectors are representable in it. Let

$$\varphi : C \longrightarrow K$$

be a fully faithful functor, with K Karoubian, such that every object of K is a direct factor of an object in the image of C. Prove that φ is 2-*universal* among functors from C to Karoubian categories, more precisely: for every Karoubian category F, the functor

$$g \longmapsto g \circ \varphi : \mathrm{Hom}(K, F) \longrightarrow \mathrm{Hom}(C, F)$$

is an equivalence of categories. Use the fact that every functor commutes with images of projectors. The category K equipped with φ, determined up to equivalence, itself determined up to unique isomorphism, by the preceding properties, is called the *Karoubi envelope* $\widetilde{C}$ of C. Compare also IV, 7.5.

c) Deduce from a) and b) that $\mathrm{Ind}(C)_{PF}$, equipped with the functor $C \to \mathrm{Ind}(C)_{PF}$ induced by $c : C \to \mathrm{Ind}(C)$, makes $\mathrm{Ind}(C)_{PF}$ a Karoubi envelope of C.

d) Show that every functor $f : C \to C'$ extends, essentially uniquely, to a functor

$$\widetilde{f} : \widetilde{C} \longrightarrow \widetilde{C'}$$

between the Karoubi envelopes, and that if one takes these Karoubi envelopes as in c), then $\widetilde{f}$ is the functor induced by $\mathrm{Ind}(f) : \mathrm{Ind}(C) \to \mathrm{Ind}(C')$. Show that the conditions of 8.5.4 b) are also equivalent to the following condition: $\widetilde{f}$ is an equivalence of categories.

Exercise 8.7.9. Let E be a $\mathcal{U}$-category.

a) Show that the following conditions are equivalent:

(i) E is equivalent to a category of the form $\mathrm{Ind}(C)$, with C a $\mathcal{U}$-category.

(i bis) As in (i), with C moreover Karoubian (8.7.8 b)).

(ii) The subcategory E_{PF} of E (8.7.6) generates by strict epimorphisms, and for every object X of E, $(E_{PF})_{/X}$ is a filtered essentially small category.

Show that, if C is a *Karoubian* $\mathcal{U}$-category such that E is equivalent to $\mathrm{Ind}(C)$, then C is equivalent to E_{PF}, by an equivalence determined up to unique isomorphism. Establish a 2-equivalence between the 2-category formed by the categories E satisfying the preceding conditions, with functors between them *commuting with small filtered inductive limits*, and the 2-category formed by the *Karoubian* $\mathcal{U}$-categories and arbitrary functors between them.

b) Show that the following conditions are equivalent:

(i) E is equivalent to a category of the form $\mathrm{Ind}(C)$, where C is a $\mathcal{U}$-category in which finite inductive limits are representable.

(i bis) As in (i), but with C moreover Karoubian.

(ii) Finite inductive limits are representable in E, the subcategory E_{PF} of E generates by strict epimorphisms, and for every object X of E there exists a small subset I of $\mathrm{Ob}\, E_{PF/X}$ such that every object of $E_{PF/X}$ is dominated by an object of I. Use 8.9.5 b).

8.8 Indexed representation of a functor $J \to \mathrm{Ind}(C)$

8.8.1

Let

$$\varphi : J \longrightarrow \mathrm{Ind}(C), \qquad \varphi \in \mathrm{Ob}\,\mathrm{Hom}(J, \mathrm{Ind}(C)),$$

be a functor, where C is a $\mathcal{U}$-category. Recall (8.5.4) that an *indexed representation of* φ is, by definition, an ind-object of $\mathrm{Hom}(J, C)$,

$$\psi : I \longrightarrow \mathrm{Hom}(J, C),$$

where I is a filtered essentially small category, whose inductive limit in $\mathrm{Hom}(J, \mathrm{Ind}(C))$ is isomorphic to φ, by a specified isomorphism. Thus φ admits an indexed representation if and only if it is isomorphic to an essentially small filtered inductive limit in $\mathrm{Hom}(J, \mathrm{Ind}(C))$ of objects of the full subcategory $\mathrm{Hom}(J, C)$. We shall say that the category J is *admissible for* C if every functor $\varphi : J \to \mathrm{Ind}(C)$ admits an indexed representation.

Since small filtered inductive limits are representable in $\mathrm{Ind}(C)$ (8.5.1), the same is true in $\mathrm{Hom}(J, \mathrm{Ind}(C))$, and they are computed argument by argument. Consequently, the inclusion functor

$$\mathrm{Hom}(J, C) \hookrightarrow \mathrm{Hom}(J, \mathrm{Ind}(C)) \tag{8.8.1.1}$$

extends canonically to a functor

$$\mathrm{Ind}(\mathrm{Hom}(J, C)) \longrightarrow \mathrm{Hom}(J, \mathrm{Ind}(C)) \tag{8.8.1.2}$$

by 8.7.2. The elements of the essential image of this functor are then precisely the φ which admit an indexed representation. Thus saying that J is admissible for C is the same as saying that the functor (8.8.1.2) is essentially surjective. In this connection we record the following result.

Proposition 8.8.2. If the category J is equivalent to a finite category, then the canonical functor (8.8.1.2) is fully faithful. Hence J is admissible for C if and only if this functor is an equivalence of categories.

By 8.7.5 a), everything reduces to proving that, for every small filtered inductive system $(\varphi_i)_{i\in I}$ in $\mathrm{Hom}(J,C)$ and every object φ of $\mathrm{Hom}(J,C)$, the canonical map

$$\varinjlim_i \mathrm{Hom}(\varphi,\varphi_i) \longrightarrow \mathrm{Hom}(\varphi, \varinjlim_i \varphi_i) \tag{8.8.2.1}$$

is bijective, where $\varinjlim_i \varphi_i$ denotes the inductive limit taken in $\mathrm{Hom}(J,\mathrm{Ind}(C))$. But, if φ and ψ are two functors $J \to C$, one has an exact diagram of sets, functorial in φ and ψ,

$$\mathrm{Hom}(\varphi,\psi) \longrightarrow \prod_{X\in \mathrm{Ob}\, J} \mathrm{Hom}(\varphi(X),\psi(X)) \rightrightarrows \prod_{\substack{f\in \mathrm{Fl}\, J\\ f:X\to Y}} \mathrm{Hom}(\varphi(X),\psi(Y)).$$

Since filtered inductive limits commute with kernels of double arrows and with finite products, the bijectivity of (8.8.2.1) follows when J is finite. The case in which J is equivalent to a finite category immediately reduces to this one.

Proposition 8.8.3. Let $\varphi : J \to \mathrm{Ind}(C)$ be a functor, with J equivalent to a small category. For φ to admit an indexed representation, it is sufficient that the following two conditions hold; these conditions are also necessary when J is equivalent to a finite category.

a) The category $\mathrm{Hom}(J,C)_{/\varphi}$, formed by the arrows of $\mathrm{Hom}(J,\mathrm{Ind}(C))$ with target φ and source in $\mathrm{Hom}(J,C)$, is filtered.

b) For every object j of J, the functor

$$\psi \longmapsto \psi(j) : \mathrm{Hom}(J,C)_{/\varphi} \longrightarrow C_{/\varphi(j)} \tag{8.8.3.1}$$

is cofinal, i.e. satisfies conditions F 1) and F 2) of 8.1.3.

Proof. Suppose a) and b) hold, and prove that φ admits an indexed representation. We may of course suppose that J is small. Put

$$I = \mathrm{Hom}(J,C)_{/\varphi}, \tag{8.8.3.2}$$

which is a filtered category by hypothesis, and first suppose that I is essentially small. Then the "inclusion" of I into $\mathrm{Hom}(J,C)$ defines an ind-object of $\mathrm{Hom}(J,C)$, say $i \mapsto \psi_i$. Moreover, one has a canonical homomorphism

$$\varinjlim_i \psi_i \longrightarrow \varphi, \tag{8.8.3.3}$$

given argument by argument by the homomorphism

$$\varinjlim_I \psi_i(j) \longrightarrow \varphi(j) = \varinjlim_{C_{/\varphi(j)}} X$$

deduced from the functor (8.8.3.1). Since the latter functor is cofinal, it follows that (8.8.3.3) is an isomorphism, and hence the conclusion follows.

In the case where I is not assumed essentially small, it is enough to construct an essentially small full subcategory I' such that (8.8.3.3) remains an isomorphism when $\varinjlim_{I'}$ is used instead of $\varinjlim_{I}$. For this, it is enough that the composite functors

$$I' \longrightarrow I \longrightarrow C_{/\varphi(j)}$$

induced by the functors (8.8.3.1) all be cofinal. One then concludes by the following lemma, whose proof is immediate from criterion (8.1.3 b)) and is left to the reader.

Lemma 8.8.3.4. Let I be a filtered $\mathcal{U}$-category, and let

$$f_j : I \longrightarrow I_j \qquad (j \in J)$$

be a small family of cofinal functors from I to essentially small filtered categories. Then there exists a full subcategory I' of I which is filtered and essentially small, and such that the functors induced by the f_j are cofinal.

Let us finally prove the necessity in 8.8.3 when J is assumed equivalent to a finite category. An indexed representation of φ by means of an essentially small filtered index category I defines a functor ψ and, for each j, a functor ψ_j:

$$\begin{tikzcd} I \arrow[r, "\psi"] \arrow[dr, "\psi_j"'] & \mathrm{Hom}(J,C)_{/\varphi} \arrow[d] \\ & C_{/\varphi(j)}. \end{tikzcd} \tag{8.8.3.5}$$

It is enough to prove that ψ and each of the resulting functors ψ_j are cofinal. By 8.1.3 b), it will then follow that $\mathrm{Hom}(J,C)_{/\varphi}$ is filtered, and by Definition 8.1.1 that the vertical arrow in (8.8.3.5) is also cofinal. By 8.8.2 one may identify φ with an object of $\mathrm{Ind}(E)$, where $E = \mathrm{Hom}(J,C)$, and the fact that the functor

$$\psi : I \longrightarrow E_{/\varphi}$$

deduced from the ind-object φ of E indexed by I is cofinal is a general fact, verified immediately by means of the criteria F 1) and F 2) of 8.1.3. The same argument, with E and φ replaced by C and $\varphi(j)$, shows that ψ_j is cofinal. This completes the proof.

Remark 8.8.4.

a) In the case where J is equivalent to a finite category, if φ admits an indexed representation, we have seen that (8.8.3.5) is a cofinal functor. This implies that the category $\mathrm{Hom}(J,C)_{/\varphi}$ itself is essentially small.

b) Suppose that finite inductive limits are representable in C. Then the same is true in $E = \mathrm{Hom}(J,C)$, where these limits are computed argument by argument; they are also inductive limits in $\mathrm{Hom}(J,\widehat{C})$ and *a fortiori* in $\mathrm{Hom}(J,\mathrm{Ind}(C))$. It follows immediately that condition a) of 8.8.3 is then automatically satisfied, and everything reduces to condition b). I do not know whether this condition is automatically satisfied when, in addition, J is assumed small.

We now come to the principal result of the present subsection.

Proposition 8.8.5. Let J be a category equivalent to a finite category, and suppose moreover that J is rigid, i.e. that for every $j \in \operatorname{Ob} J$, every endomorphism of j is the identity. Then J is admissible for C, whatever the $\mathcal{U}$-category C may be; equivalently, every functor $J \to \operatorname{Ind}(C)$ admits an indexed representation.

Replacing J by an equivalent category, we may suppose that J is *reduced*, i.e. that two isomorphic objects of J are identical. Then J is finite. We proceed by induction on $\operatorname{card} \operatorname{Ob} J$, the case where this cardinal is ≤ 0 being trivial; hence we suppose it is at least 1. Let j_0 be a *maximal* object of J, i.e. such that for every arrow $j_0 \to j$ there exists an arrow $j \to j_0$. Since J is rigid, this implies that $j_0 \to j$ is an isomorphism, and since J is reduced, this implies $j_0 = j$. Let J' be the full subcategory obtained from J by removing the object j_0, and let J'' be the full subcategory reduced to the object j_0. The proposition then follows from the following lemma.

Lemma 8.8.5.1. Let J be a finite category, and let J' and J'' be two full subcategories such that

$$\operatorname{Ob} J = \operatorname{Ob} J' \cup \operatorname{Ob} J'',$$

and such that, for every $j' \in \operatorname{Ob} J'$ and $j'' \in \operatorname{Ob} J''$, one has

$$\operatorname{Hom}(j'', j') = \varnothing.$$

If J' and J'' are admissible for C, then so is J.

Everything reduces to proving criteria a) and b) of 8.8.3. This amounts to six elementary verifications: the last two conditions for filtered categories for $\operatorname{Hom}(J, C)_{/\varphi}$, and the conditions F 1) and F 2) for the functors (8.8.3.1), successively in the case $j \in \operatorname{Ob} J'$ and in the case $j \in \operatorname{Ob} J''$; these latter conditions also imply that $\operatorname{Hom}(J, C)_{/\varphi}$ is nonempty. The rather tedious verification presents no difficulty and is left to the reader. The editor searched in vain for an elegant proof bypassing these unpleasant verifications.

Exercise 8.8.6.

a) Let J and J' be two categories admissible for C, one of them finite. Prove that $J \times J'$ is admissible for C. Use 8.8.2.

b) Prove that a small discrete category is admissible for every category C.

c) Let J be a category satisfying the following conditions: (i) J is rigid; (ii) $\operatorname{Ob} J$ is countable; (iii) for any two objects j, j' of J, $\operatorname{Hom}(j, j')$ is finite; (iv) for every object j of J, the set of objects of J which are dominated by j, i.e. which are sources of an arrow with target j, is finite. Show that J is admissible for every $\mathcal{U}$-category C. First show that J can be written as a filtered union of finite full subcategories J_n such that every object of J dominated by an object of J_n lies in J_n; then verify criteria a) and b) of 8.8.3.

Exercise 8.8.7. In the notation, we identify an ordered set with the category it defines.

a) Let C be an ordered set. Let $\widetilde{C}$ be the set of subsets A of C which are filtered and such that $x \in A$ and $y \leq x$ imply $y \in A$. Let $L_0 : C \to \widetilde{C}$ be the map which sends $x \in C$ to the set $L_0(x)$ of all $y \in C$ such that $y \leq x$. Show that L_0 is injective and that the order on C

is induced by that on $\widetilde{C}$. For every $A \in \widetilde{C}$, consider the inclusion $f(A) : A \to C$, which is an ind-object of C. For every ind-object $\varphi : I \to C$ of C, let $g(\varphi) \in \widetilde{C}$ be the subset of C formed by the elements of C dominated by an element of the form $\varphi(i)$. Show that in this way one obtains two quasi-inverse equivalences

$$\mathrm{Ind}(C) \underset{g}{\overset{f}{\rightleftarrows}} \widetilde{C}$$

which transform the functor L_0 into the canonical functor $L : C \to \mathrm{Ind}(C)$.

b) Suppose that, for every $x \in C$, the set of elements majorizing, respectively minorizing, x is finite. Show that every ind-object, respectively pro-object, of C is essentially constant; more precisely, for every such $\varphi : I \to C$, respectively $\varphi : I^{\mathrm{op}} \to C$, there exists $i_0 \in \mathrm{Ob}\, I$ such that the induced functor on $I_{/i_0}$ is constant, i.e. sends every arrow to an isomorphism.

c) Let $C \subset \mathbb{N}^{\mathrm{op}} \times \mathbb{N}$ be the ordered set formed by the pairs of natural numbers (j, i) such that $i \geq j$. Show that $\widetilde{\mathbb{N}}$ is isomorphic to the ordered set obtained from $\mathbb{N}$, identified with a subset of $\widetilde{\mathbb{N}}$ as in a), by adjoining a greatest element ∞. Show that $\widetilde{C}$ is isomorphic to $\mathbb{N}^{\mathrm{op}} \times \widetilde{\mathbb{N}}$. Consider the functor

$$\varphi : \mathbb{N}^{\mathrm{op}} \longrightarrow \widetilde{C}, \qquad \varphi(j) = (j, \infty).$$

Show that:

1) the projective limit of φ is not representable in $\widetilde{C}$, although filtered projective limits in C are representable, and are essentially constant by b);

2) for every functor $\psi : \mathbb{N}^{\mathrm{op}} \to C$, one has $\mathrm{Hom}(\psi, \varphi) = \varnothing$, and *a fortiori* the functor φ admits no representation in indexed form.

Exercise 8.8.8. Let $X = \mathbb{N} \amalg \mathbb{N}$ be the disjoint union of two copies of $\mathbb{N}$, let s be the symmetry of X, and for every $i \in \mathbb{N}$, let X_i be the subset $\Delta_{i+1} \amalg \Delta_i$ of X, where Δ_i denotes the subset of $\mathbb{N}$ formed by the j such that $j \leq i$. Let C be the subcategory of the category of subsets of X whose objects are the X_i and whose arrows are the maps between the X_i induced either by the identity of X or by its symmetry s. Let $\widetilde{C}$ be the category defined analogously, but in which the object X is also allowed.

a) Show that $\mathrm{Ind}(C)$ is equivalent to $\widetilde{C}$, with the canonical functor $C \to \mathrm{Ind}(C)$ corresponding to the inclusion $C \to \widetilde{C}$. Show that there is no pair (Y, f), consisting of an object Y of C and an endomorphism f of Y, together with a morphism of objects-with-endomorphism in $\mathrm{Ind}(C)$ from (Y, f) to (X, s).

b) Conclude that, if J is the category with one object and, besides the identity arrow, a single arrow of order 2, then the functor $J \to \mathrm{Ind}(C)$ defined by (X, s) admits no indexed representation.

Exposé I. Presheaves

Sections 8.9–8.13

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Translator's status note. This is a working English translation of SGA 4, Exposé I, Sections 8.9 through 8.13. A few evident typographical slips in the source notation are corrected where the mathematical context forces the correction.

8 Ind-objects and pro-objects

8.9 Exactness properties of $\mathrm{Ind}(C)$

Proposition 8.9.1. Let C be a $\mathcal{U}$-category.

a) The canonical functors L (8.2.4.8) and c (8.4.1)

$$C \xrightarrow{c} \mathrm{Ind}(C) \xrightarrow{L} \widehat{C}$$

commute with projective limits.

b) Suppose that finite inductive limits are representable in C, and that C is equivalent to a small category. Then small projective limits are representable in $\mathrm{Ind}(C)$.

c) If small projective limits (respectively finite projective limits) are representable in C, then the same is true in $\mathrm{Ind}(C)$. Let J be a category which is finite and rigid, or discrete. If projective limits of type J are representable in C, the same type of projective limits is representable in $\mathrm{Ind}(C)$.

d) Small filtered inductive limits in $\mathrm{Ind}(C)$, which are representable by 8.5.1, are left exact, that is, they commute with finite projective limits.

Proof.

a) The fact that L commutes with projective limits follows from the fact that it is fully faithful and from the computation of projective limits in $\widehat{C}$ argument by argument. Thus, in order to verify that a projective system of morphisms $F \to F_i$ $(i \in I)$ in $\widehat{C}$ makes F the projective limit of the F_i, it is enough to verify the analogous assertion for the projective systems of sets

$$\mathrm{Hom}(X, F) \longrightarrow \mathrm{Hom}(X, F_i)$$

for all $X \in \mathrm{Ob}\, C$. But $C \subset \mathrm{Ind}(C)$.

The fact that c commutes with projective limits follows formally from the fact that L does so, together with the fact that the composite $L \circ c : C \to \widehat{C}$ does so.

b) In view of a), the assertion says that every projective limit of ind-representable presheaves is ind-representable. This follows immediately from criterion 8.3.3(v), since a projective limit of left exact functors is left exact; this is just the fact that projective limits commute with projective limits (2.5.0).

c) This follows formally from the analogous property of $\widehat{C}$ (3.3), together with the facts that L is conservative, being fully faithful, and that it commutes with the limits of the type in question, by a) and 8.5.1.

d) It is well known (cf. 2.3) that representability of finite projective limits is equivalent to representability of projective limits of the special finite ordered types consisting of the empty type and the elementary fiber-product type. Representability of small projective limits is equivalent to representability of products and finite projective limits. Hence the second assertion made in c) implies the first.

First suppose that I is finite. Using 8.8.5 on the representability of functors $\varphi : J \to \mathrm{Ind}(C)$ in indexed form (8.5.4)

$$\Phi : J \times I \longrightarrow C, \tag{8.9.1.1}$$

the existence of $\varprojlim \varphi$ is a special case of the following more precise and more general result.

Corollary 8.9.2. Let $\varphi : J \to \mathrm{Ind}(C)$ be a functor given in indexed form Φ as in (8.9.1.1), where J is a finite category. If, for every $i \in \mathrm{Ob}\, I$, the partial functor $j \mapsto \Phi(j, i)$ has a projective limit (respectively an inductive limit) representable in C, then φ has a projective limit (respectively an inductive limit) representable in $\mathrm{Ind}(C)$, and there is a canonical isomorphism

$$\varprojlim \varphi \simeq \text{“}\varinjlim_i\text{”} \varprojlim_j \Phi(j, i), \tag{8.9.2.1}$$

respectively

$$\varinjlim \varphi \simeq \text{“}\varinjlim_i\text{”} \varinjlim_j \Phi(j, i), \tag{8.9.2.2}$$

where $\varprojlim_j$ (respectively $\varinjlim_j$) is computed in C.

For the first formula, one uses only that in $\mathrm{Ind}(C)$ filtered inductive limits commute with $\varprojlim_j$, and that c commutes with $\varprojlim_J$ by 8.9.1(d) and a):

$$\begin{aligned}\varprojlim \varphi &= \varprojlim_j \text{“}\varinjlim_i\text{”}\, \Phi(j, i) \\ &\xleftarrow{\sim} \text{“}\varinjlim_i\text{”} \varprojlim_j{}^{\mathrm{Ind}(C)} \Phi(j, i) \simeq \text{“}\varinjlim_i\text{”} \varprojlim_j^{C} \Phi(j, i).\end{aligned}$$

The second is proved in the same way, using the commutation of inductive limits of type I with inductive limits of type J (2.5.0) and the fact that c commutes with inductive limits of type J, proved in 8.9.4(a) below.

The case where J is discrete. This case is contained in the following more precise assertion.

Corollary 8.9.3. Let I_α be essentially small filtered categories indexed by a small set A, and, for each $\alpha \in A$, let

$$X(\alpha) = (X(\alpha)_{i_\alpha})_{i_\alpha \in I_\alpha}$$

be an ind-object of C indexed by I_α. Let I be the product category of the I_α, and suppose that, for every $\mathbf{i} = (i_\alpha)_{\alpha \in A} \in I$, the product

$$\prod_{\alpha \in A} X(\alpha)_{i_\alpha}$$

is representable in C. Then the product $\prod_{\alpha\in A} X(\alpha)$ is representable in $\mathrm{Ind}(C)$, and is canonically isomorphic to the ind-object indexed by I given by the formula

$$\prod_{\alpha\in A} \text{“}\varinjlim_{i_\alpha\in I_\alpha}\text{”}\, X(\alpha)_{i_\alpha} \simeq \text{“}\varinjlim_{\mathbf{i}=(i_\alpha)_{\alpha\in A}\in I}\text{”} \left(\prod_{\alpha\in A} X(\alpha)_{i_\alpha}\right), \tag{8.9.3.1}$$

where the product on the right is computed in C. Notice that I is filtered and essentially small because the I_α are.

Indeed, it is well known, and immediate after reducing to the category of sets, that the displayed formula is valid when the limits are computed in $\widehat{C}$. The conclusion then follows from the fact that L commutes with the limits under consideration (8.9.1(a) and 8.5.1).

Remarks 8.9.4. The proof just given for 8.9.2 and 8.9.3 shows, more generally, that if, for every $i \in \mathrm{Ob}\, I$, the limit $\varprojlim_j \Phi(j,i)$ (respectively $\varinjlim_j \Phi(j,i)$), computed in $\mathrm{Ind}(C)$, is representable, then the corresponding limit of φ is representable. This and the argument in c) show that, for a given category J arising from a finite ordered set or a discrete category, or more generally one which is C-admissible in the sense of 8.3.1, projective limits (respectively inductive limits) of type J are representable in $\mathrm{Ind}(C)$ if and only if, for every functor $\varphi : J \to C$, the projective limit (respectively inductive limit) of φ, computed in $\mathrm{Ind}(C)$, is representable. In the non-parenthesized case this also means, by a), that every projective limit of type J of representable presheaves on C is ind-representable.

Proposition 8.9.5. Let C be a $\mathcal{U}$-category.

a) The canonical functor
$$c : C \longrightarrow \mathrm{Ind}(C)$$
is right exact, hence exact in view of 8.9.1(a).

b) If finite inductive limits (respectively finite sums) are representable in C, then small inductive limits (respectively small sums) are representable in $\mathrm{Ind}(C)$. Let J be a finite or discrete preordered set. If inductive limits of type J are representable in C, then the same is true in $\mathrm{Ind}(C)$.

Proof.

a) Suppose that in C one has $X \simeq \varinjlim_J X_j$, with J finite and with X and the X_j in C. Then, for every ind-object $\mathcal{Y} = (Y_i)_{i\in I}$ of C,

$$\begin{aligned}\mathrm{Hom}(X,\mathcal{Y}) &\simeq \varinjlim_i \mathrm{Hom}(X, Y_i) \simeq \varinjlim_i \varprojlim_j \mathrm{Hom}(X_j, Y_i) \\ &\simeq \varprojlim_j \varinjlim_i \mathrm{Hom}(X_j, Y_i) \simeq \varprojlim_j \mathrm{Hom}(X_j, \mathcal{Y}),\end{aligned}$$

where the middle isomorphism is 2.8. The desired conclusion follows by comparing the extreme terms.

b) It was already observed in 8.8.4 that the assertions follow from a) and 8.9.2, at least for finite inductive limits. To prove the conclusions in the infinite cases, one is reduced to the case of sums, which may be interpreted as a filtered inductive limit of finite sums; the conclusion follows from 8.5.1.

Remark 8.9.6. We have already observed that the functor c does not in general commute with filtered inductive limits, and therefore not with infinite sums either, in contrast with what happens for $L : \mathrm{Ind}(C) \to \widehat{C}$. Conversely, apart from commutation with filtered inductive limits (8.5.1), L practically never has commutation properties with any other type of inductive limit: initial object, sum of two objects, or cokernels of double arrows. Thus, if C has an initial object $\varnothing_C$, which is then an initial object of $\mathrm{Ind}(C)$ by a), $\varnothing_C$ is never an initial object of $\widehat{C}$, that is, never identical with the constant presheaf of value $\varnothing$, since $\mathrm{Hom}(\varnothing_C, \varnothing_C) \neq \varnothing$.

Proposition 8.9.7. Let

$$f : C \longrightarrow C'$$

be a functor between $\mathcal{U}$-categories, and let J be a finite and rigid category (8.8.5). If inductive limits (respectively projective limits) of type J are representable in C and if f commutes with them, then

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C')$$

also commutes with this type of limit.

This follows immediately from 8.8.5 and from the calculation, in 8.9.2, of finite limits in a category of ind-objects, for a functor represented in indexed form.

Corollary 8.9.8. If finite inductive limits (respectively finite projective limits) are representable in C, and if f is right exact (respectively left exact), then the same is true of $\mathrm{Ind}(f)$. In the non-parenthesized case, $\mathrm{Ind}(f)$ even commutes with arbitrary small inductive limits.

The first assertion follows from 8.9.7. The second follows from the first, taking into account that $\mathrm{Ind}(f)$ commutes with filtered inductive limits by 8.6.3.

Exercise 8.9.9. Let C be a $\mathcal{U}$-category.

a) Suppose finite sums are representable in C. Show that if finite sums in C are disjoint (respectively universal; cf. II 4.5), then small sums in $\mathrm{Ind}(C)$ are disjoint (respectively universal).

b) Suppose every morphism in C factors as an epimorphism followed by a monomorphism (respectively as an effective epimorphism followed by a monomorphism, respectively as an epimorphism followed by an effective monomorphism, respectively as an effective epimorphism followed by an effective monomorphism). Show that $\mathrm{Ind}(C)$ satisfies the same property. In the last parenthesized case one concludes that every epimorphism of $\mathrm{Ind}(C)$ is effective, every monomorphism of $\mathrm{Ind}(C)$ is effective, and every bimorphism of $\mathrm{Ind}(C)$ is an isomorphism. Show that, in this case, every epimorphism (respectively monomorphism) of $\mathrm{Ind}(C)$ can be represented by an inductive system of epimorphisms (respectively monomorphisms) of C. If, moreover, every epimorphism (respectively every monomorphism) in C is universal, then the same property holds in $\mathrm{Ind}(C)$.

c) Suppose C is additive (respectively abelian). Then $\mathrm{Ind}(C)$ is additive (respectively abelian).

d) Let $X = \text{“}\varinjlim\text{”}_I X_i$ be an ind-object of C, and let $R \rightrightarrows X$ be an equivalence relation in X. For every $i \in \mathrm{Ob}\, I$, let $R_i \rightrightarrows X_i$ be the equivalence relation in X_i, considered as an object of $\mathrm{Ind}(C)$, induced by R via $X_i \to X$. Here one assumes that the fiber product

$$R_i = R \times_{X \times X} (X_i \times X_i)$$

in $\mathrm{Ind}(C)$ is representable by an object of $\mathrm{Ind}(C)$, which is the case if finite projective limits are representable in C. Show that, for R to be effective, it is enough that the R_i be effective.

e) Let X be an object of C, and let R be an equivalence relation in $c(X)$, that is, in X considered as an object of $\mathrm{Ind}(C)$. Suppose fiber products are representable in C. For R to be effective, it is necessary that R be of the form "$\varinjlim$"$_i R_i$, where the R_i are equivalence relations in X, viewed as an object of C; this condition is sufficient if equivalence relations are effective in C.

f) Suppose every morphism in C factors as an effective epimorphism followed by an effective monomorphism, and suppose finite projective limits are representable. Let X be an object of C, and let R be a subobject of $X \times X$, viewed as an object of $\mathrm{Ind}(C)$. Write R in the form "$\varinjlim$"$_i Z_i$, where $(Z_i)_{i \in I}$ is a filtered increasing family of subobjects of $X \times X$ in C; this is possible by b). Show that, for R to be an equivalence relation in X, it is necessary and sufficient that for every $i \in \mathrm{Ob}\, I$ there exist $j \in \mathrm{Ob}\, I$ containing $s(Z_i)$ and $Z_i \circ Z_i$, where s is the symmetry of $X \times X$.

g) Suppose finite projective limits and finite sums are representable in C, every equivalence relation in C is effective, and every morphism in C factors as an effective epimorphism followed by an effective monomorphism. Show that equivalence relations in $\mathrm{Ind}(C)$ are universal if and only if C satisfies the following condition:

ST) For every object X of C and every subobject Z of $X \times X$ in C, if one defines recursively a sequence of subobjects Z_n $(n \geq 0)$ of $X \times X$ in C by $Z_0 = Z$ and

$$Z_{n+1} = \mathrm{Sup}(Z_n, s(Z_n), Z_n \circ Z_n),$$

where the supremum is taken in the set of subobjects of $X \times X$, which exists by the hypotheses on C, then the sequence $(Z_n)_{n \geq 0}$ is stationary.

k) Show that in $\mathrm{Ind}(\mathbf{Set})$ equivalence relations are not necessarily effective.

*NB. For criteria ensuring that $\mathrm{Ind}(C)$ is a topos, see VI 8.9.9.$_*$

Exercise 8.9.10. Let C be a $\mathcal{U}$-category. Put $\mathrm{Pro}(C) = (\mathrm{Ind}(C^{\mathrm{op}}))^{\mathrm{op}}$ (cf. 8.11).

a) Suppose finite sums (respectively small sums) are representable in C. Show that if they are disjoint (cf. II 4.5), then $\mathrm{Pro}(C)$ satisfies the same condition.

b) Let J be a small set such that sums indexed by J are representable in C, hence also in $\mathrm{Pro}(C)$. Let $(X(\alpha))_{\alpha \in J}$ be a family of elements of $\mathrm{Pro}(C)$, with

$$X(\alpha) = \text{“}\varprojlim_{I_\alpha}\text{”} X_{i_\alpha}.$$

Show that, for the sum of the $X(\alpha)$ in $\mathrm{Pro}(C)$ to be universal, it is enough that the same be true for each of the families $\varprojlim_J X_{i_\alpha}$, where $(i_\alpha)_{\alpha \in J} \in \prod_{\alpha \in J} I_\alpha$. Deduce that, if sums of type J are universal in C, then for the same to be true in $\mathrm{Pro}(C)$ it is necessary and sufficient that, for every family $(X(\alpha))_{\alpha\in J}$ as above with $I_\alpha = I$ for all $\alpha \in J$, the canonical homomorphism in $\mathrm{Pro}(C)$

$$\text{“}\varprojlim_{I}\text{”}\ \amalg_\alpha X(\alpha)_i \longleftarrow \text{“}\varprojlim_{I^J}\text{”}\ \amalg_{\alpha\in J} X(\alpha)_{i_\alpha}$$

be an isomorphism. Deduce that in $\mathrm{Pro}(\mathbf{Set})$ sums of type J are universal if and only if J is finite.

Exercise 8.9.11. Let C be a $\mathcal{U}$-category.

a) Let $(X_i \to X)_{i\in I}$ be a family of morphisms in C. Show that for it to be epimorphic in $\mathrm{Ind}(C)$, it is enough that it admit a finite subfamily which is epimorphic in C. Prove that this condition is also necessary if one assumes that every finite family of morphisms with target X in C factors as an epimorphic family followed by an effective monomorphism (10.5). For the latter assertion, let P be the set of finite subsets of I, ordered by inclusion, let $X' = (X'_J)_{J\in P}$ be the ind-object formed by the images of the finite subfamilies of the given family, and finally let

$$X'' = X \amalg_{X'} X = \text{“}\varinjlim\text{”}_J\, X \amalg_{X'_J} X.$$

Show that the two canonical morphisms $X \rightrightarrows X''$ are distinct but coincide on the X'_J.

b) Suppose that in C every finite family of morphisms with the same target factors as an epimorphic family followed by an effective monomorphism. Prove that $\mathrm{Ind}(C)$ has a small subcategory generating by epimorphisms (7.1) if and only if C has a small subcategory C' such that every object of C is the target of a finite epimorphic family with source in C'. If one assumes that C has a small generating subcategory (7.1), then $\mathrm{Ind}(C)$ has a small subcategory generating by epimorphisms if and only if C is equivalent to a small category. For this last statement, use 7.5.2.

c) $\mathrm{Ind}(\mathbf{Set})$ has no small generating subcategory.

Exercise 8.9.12. Let C be a $\mathcal{U}$-category, and consider

$$\mathrm{Pro}(C) = \mathrm{Ind}(C^{\mathrm{op}})^{\mathrm{op}}.$$

a) Show that if $\mathrm{Pro}(C)$ has a small subcategory P' generating by epimorphisms (7.1), then there is a small subcategory C' of C which generates by epimorphisms in $\mathrm{Pro}(C)$. Take the category of components of the objects of P'.

b) Show that the category $\mathrm{Pro}(\mathbf{Set})$ has no small subcategory generating by epimorphisms. If C' is as in a), choose a set X whose cardinal strictly dominates the cardinals of the elements of C', and consider the pro-set $\mathcal{X}$ formed by the complements in X of the subsets of cardinal $< \mathrm{card}(X)$. Show that, for every nonempty set Y of cardinal $< \mathrm{card}(X)$, one has $\mathrm{Hom}(Y, \mathcal{X}) = \varnothing$, but that $\mathcal{X}$ is not isomorphic to the constant pro-set $\varnothing$.

8.10 Dual notions: pro-objects and pro-representable functors

Let C be a $\mathcal{U}$-category. A *pro-object* of C is a functor

$$\varphi : I^{\mathrm{op}} \longrightarrow C, \tag{8.10.1}$$

where I is an essentially small filtered category, called the index category, and where, as usual, I^{op} denotes the opposite category. Let $\mathcal{V}$ be a universe such that $\mathcal{V} \supset \mathcal{U}$. Notice that pro-objects of C indexed by $I \in \mathcal{V}$ are in bijective correspondence with ind-objects of C^{op} indexed by $I \in \mathcal{V}$: to such an ind-object $\psi : I \to C^{\mathrm{op}}$ one associates the pro-object $\varphi = \psi^{\mathrm{op}} : I^{\mathrm{op}} \to C$. Following this correspondence, one defines, by transport of structure and reversal of arrows, the notion of morphism between pro-objects of C from the analogous notion (8.2.4.2, 8.2.5.1) for ind-objects. One accordingly obtains the category of pro-objects of C indexed by $I \in \mathcal{V}$, together with a canonical isomorphism

$$\mathrm{Pro}_{\mathcal{V}}(C,\mathcal{U}) \simeq \mathrm{Ind}_{\mathcal{V}}(C^{\mathrm{op}},\mathcal{U})^{\mathrm{op}}. \tag{8.10.2}$$

For $\mathcal{V} = \mathcal{U}$ one simply speaks of the *category of pro-objects of* C, denoted $\mathrm{Pro}_{\mathcal{U}}(C)$ or simply $\mathrm{Pro}(C)$; it is therefore defined by

$$\mathrm{Pro}(C) = \mathrm{Pro}_{\mathcal{U}}(C,\mathcal{U}) \simeq \mathrm{Ind}(C^{\mathrm{op}})^{\mathrm{op}}. \tag{8.10.3}$$

If two pro-objects are given in indexed form

$$\mathcal{X} = (X_i)_{i\in I}, \qquad \mathcal{Y} = (Y_j)_{j\in J}, \tag{8.10.4}$$

then formula (8.2.5.1) takes here the form

$$\mathrm{Hom}(\mathcal{X},\mathcal{Y}) \simeq \varprojlim_j \varinjlim_i \mathrm{Hom}(X_i, Y_j). \tag{8.10.5}$$

The functor (8.4.1), applied to C^{op}, gives, after passage to opposite categories, a canonical functor, denoted by the same letter c when no confusion is possible, which allows one to identify C with a full subcategory of $\mathrm{Pro}(C)$:

$$c : C \longrightarrow \mathrm{Pro}(C). \tag{8.10.6}$$

The arrows were reversed in the definition of morphisms of pro-objects precisely in order to obtain such a functor, rather than a functor $C \to \mathrm{Pro}(C)^{\mathrm{op}}$. Pro-objects in the essential image of (8.10.6) are again called *essentially constant pro-objects*.

Put

$$\check{C} = (C^{\mathrm{op}})\widehat{\ } = \mathrm{Hom}(C, \mathbf{Set}). \tag{8.10.7}$$

Then the functor (8.2.4.7) may be regarded as a canonical fully faithful functor

$$L : \mathrm{Pro}_{\mathcal{V}}(C,\mathcal{U}) \longrightarrow \check{C}^{\mathrm{op}}, \tag{8.10.8}$$

and in particular one obtains

$$L : \mathrm{Pro}(C) \hookrightarrow \check{C}^{\mathrm{op}}. \tag{8.10.9}$$

We shall leave to the reader the task of translating, as needed, the results of the preceding sections and of the following ones from the language of ind-objects into that of pro-objects. Depending on the mathematical context, one or the other language is the more useful, not to mention the cases in which both notions occur simultaneously, for instance when one has to consider complex categories such as $\mathrm{Pro}(\mathrm{Ind}(C))$ or $\mathrm{Ind}(\mathrm{Pro}(C))$. We content ourselves with giving a few additional indications, in order to fix terminology and notation and to clarify the yoga.

8.10.10

A *covariant functor* $F : C \to \mathbf{Set}$ is called a *pro-representable functor* if it lies in the essential image of (8.10.9), or equivalently of 8.10.8. The full subcategory of $\check{C}$ consisting of such functors is therefore equivalent to $\mathrm{Pro}(C)^{\mathrm{op}}$. One generally prefers to work in the opposite category, as the essential image as a subcategory of (8.10.9), hence equivalent to $\mathrm{Pro}(C)$ itself. In accord with this usage, it is often preferable to regard covariant functors

$$F : C \longrightarrow \mathbf{Set}$$

as objects of $\mathrm{Hom}(C, \mathbf{Set})^{\mathrm{op}} = \check{C}^{\mathrm{op}}$, and consequently to write the category of morphisms

$$X \longrightarrow F \quad \text{in } \check{C}^{\mathrm{op}},$$

that is, the pairs (X, u) consisting of an object X of C and an element $u \in F(X)$, as

$${}_F\backslash C. \tag{8.10.11}$$

With this convention one obtains a covariant functor, the source functor,

$${}_F\backslash C \longrightarrow C, \tag{8.10.12}$$

and 3.4 is rewritten in the form

$$F \xrightarrow{\sim} \text{“}\varprojlim_{({}_F\backslash C)^{\mathrm{op}}}\text{”} X. \tag{8.10.13}$$

8.10.14

Criterion 8.3.3, applied to C^{op}, becomes a criterion of pro-representability for F: it is necessary and sufficient that $({}_F\backslash C)^{\mathrm{op}}$ be filtered and essentially small. The latter condition is superfluous if C is equivalent to a small category. If, moreover, finite projective limits are representable in C, this is equivalent to saying that F commutes with them, that is, that F is *left exact*.

8.10.15

In $\mathrm{Pro}(C)$, small filtered projective limits are representable, and the functor (8.10.9) commutes with them. In other words, this functor transforms projective limits in $\mathrm{Pro}(C)$ into inductive limits in $\check{C} = \mathrm{Hom}(C, \mathbf{Set})$, just as does its composite $C \to \check{C}^{\mathrm{op}}$ with (8.10.6), which shares with it the unfortunate property of being contravariant when it is regarded as taking values in $\check{C}$. Notice carefully that pro-representable functors from C to $\mathbf{Set}$ are functors which are small filtered inductive limits of representable functors, and not projective limits, as the terminology might perhaps suggest.

8.11 Ind-adjoints and pro-adjoints

8.11.1

Let

$$f : C \longrightarrow C' \tag{8.11.1.1}$$

be a functor between $\mathcal{U}$-categories. It induces the functor $F \mapsto F \circ f$

$$f^* : \widehat{C'} \longrightarrow \widehat{C}. \tag{8.11.1.2}$$

We say that f *admits an ind-adjoint* if this functor sends ind-representable functors to ind-representable functors. Since f^* commutes with inductive limits, and since the full subcategory of $\widehat{C'}$ formed by the ind-representable functors is stable under small filtered inductive limits, it is equivalent to say that f admits an ind-adjoint, or that f^* sends the representable objects of $\widehat{C'}$ to ind-representable functors on C, that is,

$$X \longmapsto \operatorname{Hom}(f(X), Y') \quad \text{is ind-representable for every } Y' \in \operatorname{Ob} C'. \tag{8.11.1.3}$$

Another way to express the condition that f admit an ind-adjoint is to say that there is a functor

$$g : \operatorname{Ind}(C') \longrightarrow \operatorname{Ind}(C) \tag{8.11.1.4}$$

which is essentially induced by f^*, that is, such that there is an isomorphism of bifunctors

$$\operatorname{Hom}_{\operatorname{Ind}(C)}(X, g(\mathcal{Y}')) \simeq \operatorname{Hom}_{\operatorname{Ind}(C')}(f(X), \mathcal{Y}'), \tag{8.11.1.5}$$

where $X \in \operatorname{Ob} C$ and $\mathcal{Y}' \in \operatorname{Ob} \operatorname{Ind}(C')$. In fact it is enough to have a functor

$$g_0 : C' \longrightarrow \operatorname{Ind}(C) \tag{8.11.1.6}$$

and an isomorphism of bifunctors

$$\operatorname{Hom}_{\operatorname{Ind}(C)}(X, g_0(Y')) \simeq \operatorname{Hom}_{C'}(f(X), Y') \tag{8.11.1.7}$$

for $X \in \operatorname{Ob} C$ and $Y' \in \operatorname{Ob} C'$. The functor g, respectively g_0, is plainly determined by f up to unique isomorphism, and conversely f can be reconstructed up to canonical isomorphism from g, respectively from g_0. Formula (8.11.1.5) makes it clear that g commutes with inductive limits and extends g_0; this determines it up to unique isomorphism in terms of g_0 by 8.7.2. The functor g, and sometimes also the functor g_0 that it extends, is called the *ind-adjoint functor of* f. There is no need here to specify "right", since the other one, if it exists, will be called the pro-adjoint (cf. 8.11.5 below).

Of course, if f admits a right adjoint

$$f^{\mathrm{ad}} : C' \longrightarrow C,$$

then it admits an ind-adjoint g, and this is canonically isomorphic to the canonical extension of f^{ad} to ind-objects:

$$g \simeq \operatorname{Ind}(f^{\mathrm{ad}}). \tag{8.11.1.8}$$

The notion of ind-adjoint is therefore a natural generalization of the notion of right adjoint.

Now consider the canonical extension

$$\operatorname{Ind}(f) : \operatorname{Ind}(C) \longrightarrow \operatorname{Ind}(C'). \tag{8.11.1.9}$$

We then have the following proposition.

Proposition 8.11.2. For the functor $f : C \to C'$ to admit an ind-adjoint, it is necessary and sufficient that the functor $\operatorname{Ind}(f)$ in (8.11.1.9) admit a right adjoint. This right adjoint is canonically isomorphic to the ind-adjoint (8.11.1.4).

The sufficiency and the last assertion are trivial from the ind-adjunction formula (8.11.1.5). For necessity, note that, by passage to the projective limit in this formula over the X_i, one deduces an adjunction isomorphism for the ind-objects $\mathcal{X} = (X_i)_{i\in I}$ and $\mathcal{Y}'$:

$$\mathrm{Hom}_{\mathrm{Ind}(C)}(\mathcal{X}, g(\mathcal{Y}')) \simeq \mathrm{Hom}_{\mathrm{Ind}(C')}(\mathrm{Ind}(f)(\mathcal{X}), \mathcal{Y}'). \tag{8.11.2.1}$$

□

Corollary 8.11.3. If f admits an ind-adjoint, then $\mathrm{Ind}(f)$ commutes with inductive limits, and the ind-adjoint g commutes with projective limits.

Proposition 8.11.4. For the functor $f : C \to C'$ to admit an ind-adjoint, it is necessary that f be right exact, and this condition is sufficient when C is equivalent to a small category.

This follows from criterion (8.11.1.3), and from 8.3.1 and 8.3.3(iv). For another criterion in terms of the notion of accessible functor, see 8.13.3.

8.11.5

Now consider the functor $F \mapsto F \circ f$

$$f^{\mathrm{op}*} : \check{C}' \longrightarrow \check{C} \tag{8.11.5.1}$$

induced by f. We shall say that f *admits a pro-adjoint* if the preceding functor sends pro-representable functors to pro-representable functors, that is, if there exists a functor, called the *pro-adjoint functor of* f,

$$g : \mathrm{Pro}(C') \longrightarrow \mathrm{Pro}(C), \tag{8.11.5.2}$$

and an isomorphism of bifunctors

$$\mathrm{Hom}_{\mathrm{Pro}(C)}(g(\mathcal{Y}'), \mathcal{X}) \simeq \mathrm{Hom}_{\mathrm{Pro}(C')}(\mathcal{Y}', \mathrm{Pro}(f)(\mathcal{X})). \tag{8.11.5.3}$$

Of course, saying that f admits a pro-adjoint means that f^{op} admits an ind-adjoint, so the notions and results for ind-adjoints translate trivially into terms of pro-adjoints. Let us merely point out that f admits a pro-adjoint if and only if

$$\mathrm{Pro}(f) : \mathrm{Pro}(C) \longrightarrow \mathrm{Pro}(C')$$

admits a left adjoint, and that in this case the preceding functor g is such a left adjoint of $\mathrm{Pro}(f)$; and that for this it is necessary that f be left exact, this condition also being sufficient when C is equivalent to a small category. In this case, f is exact if and only if it admits both an ind-adjoint and a pro-adjoint.

Example 8.11.6. Consider the case of a functor

$$f : C \longrightarrow \mathbf{Set}.$$

If this functor admits a pro-adjoint, then it is pro-representable. The converse is true if and only if the full subcategory of $\check{C}$ formed by pro-representable functors is stable under small products; this is the case in particular if C is equivalent to a small category (8.10.14), or if small products are representable in C (8.9.5(b) applied to C^{op}). Thus, up to minor qualifications, one may say that, for a functor $f : C \to C'$, the existence of a pro-adjoint is the *natural generalization of pro-representability of* f, which is defined when $C' = \mathbf{Set}$.

8.12 Strict ind-objects and pro-objects. Application to a representability criterion

8.12.1

Let

$$\mathcal{X} = (X_i)_{i\in I}$$

be an ind-object of the $\mathcal{U}$-category C, and let F be the presheaf which it ind-represents. It is immediate that the following two conditions are equivalent: the canonical morphisms

$$X_i \longrightarrow F = \text{“}\varinjlim_i\text{”} X_i$$

are monomorphisms of $\mathrm{Ind}(C)$, or equivalently of $\widehat{C}$, that is, monomorphisms of functors argument by argument; and, for every arrow $i \to j$ of I, the corresponding transition morphism

$$X_i \longrightarrow X_j$$

is a monomorphism. When these conditions hold, and when moreover I is an ordered category, $\mathcal{X}$ is called a *strict ind-object*. Notice that this condition is not invariant under isomorphism of ind-objects. An ind-object will be called *essentially strict* if it is isomorphic to a strict ind-object. A presheaf will be called *strictly ind-representable* if it is ind-representable by a strict ind-object; thus $F = \text{“}\varinjlim_i\text{”} X_i$ is strictly ind-representable if and only if $\mathcal{X} = (X_i)_{i\in I}$ is essentially strict.

8.12.1.1. Let F be a presheaf on C, and consider the full subcategory of $C_{/F}$ formed by the arrows $X \to F$ with source in C which are monomorphisms. We shall call it the *category of representable subfunctors of* F; it is the category associated with the ordered set of representable subfunctors of F, ordered by the order induced from the set of subobjects of F. The following then follows immediately from the definitions.

Proposition 8.12.2. For a presheaf F on C to be strictly ind-representable, it is necessary and sufficient that the ordered category I of representable subfunctors of F be filtered and essentially small and that one have

$$\varinjlim_I X_i \xrightarrow{\sim} F. \qquad (8.12.2.1)$$

When, for every object X of C, the set of subobjects of X in C is small, for example when C admits a small generating subcategory (7.4), this implies that the category of representable subfunctors is even small.

8.12.2.2. If F is strictly ind-representable, there is therefore a preferred way to ind-represent it by an ind-object, and that ind-object is even strict: take the representation (8.12.2.1). A strict ind-object $\mathcal{Y}$ is called *saturated* if it is isomorphic to an ind-object of the form (X_i) appearing in (8.12.2.1), for a suitable F. Thus there exists, up to unique isomorphism, only one saturated strict ind-object isomorphic to the given strict ind-object, namely the one considered in 8.12.2, taking $F = \varinjlim \mathcal{Y}$ as a limit in $\widehat{C}$.

8.12.2.3. Suppose one already knows that a small cofinal subset can be found in $\mathrm{Ob}\, C_{/F}$, which is the case in particular if F is ind-representable. Then the same condition follows in the full subcategory I considered in 8.12.2, so in the criterion 8.12.2 one may omit the condition that I be essentially small.

8.12.3

Let F be a presheaf on C, and consider an object

$$u : X \longrightarrow F$$

of $C_{/F}$. We say that u, or the pair (X, u), is *minimal* if, for every factorization of u as

$$X \xrightarrow{p} X' \xrightarrow{u'} F, \tag{8.12.3.1}$$

with p a strict epimorphism (10.2), the morphism p is an isomorphism. Considering u as an element of $F(X)$ (1.4), to say that u is minimal therefore means that every strict epimorphism $p : X \to X'$ such that

$$u \in \operatorname{Im}\bigl(F(p) : F(X') \to F(X)\bigr)$$

is an isomorphism. This notion is clarified by part a) of the following lemma.

Lemma 8.12.4. Let F be a presheaf on C.

a) Suppose that F transforms cokernels into kernels, and consider a morphism

$$u : X \longrightarrow F$$

with $X \in \operatorname{Ob} C$. For u to be a monomorphism, it is sufficient, when cokernels of double arrows are representable in C, that u be minimal; this condition is also necessary if fiber products are representable in C.

b) Suppose finite inductive limits are representable in C. For the subcategory of representable subfunctors of F (8.12.1.1) to be filtered, not necessarily small, and to have inductive limit F, it is sufficient that F be left exact and that every morphism $u : X \to F$, with $X \in \operatorname{Ob} C$, factor as

$$X \longrightarrow X' \xrightarrow{u'} F$$

with u' minimal. This condition is also necessary if fiber products are representable in C.

c) Suppose that C admits a small generating subcategory (7.1), and that I, the category of representable subfunctors of F, is filtered. Then I is small.

Proof.

a) Suppose u is minimal; we prove that u is a monomorphism, that is, that for every double arrow $v, v' : Y \rightrightarrows X$ such that $uv = uv'$, one has $v = v'$. Indeed, if $X' = \operatorname{Coker}(v, v')$, then u factors as $X \xrightarrow{p} X' \xrightarrow{u'} F$, since F transforms cokernels into kernels. As p is a strict epimorphism by construction, it follows that p is an isomorphism, that is, $v = v'$. Conversely, suppose u is a monomorphism, and prove that it is minimal. Consider a factorization (8.12.3.1). Since p is a strict epimorphism, it is the cokernel of the canonical double arrow

$$v, v' : X \times_{X'} X \rightrightarrows X.$$

Since $(u'p)v = (u'p)v'$ and $u'p = u$ is a monomorphism, we have $v = v'$, hence p is an isomorphism.

b) Sufficiency: since F is left exact, $C_{/F}$ is filtered. Since we know that $F = \varinjlim_{C_{/F}} X$, we are reduced by 8.1.3(c) to proving that the full subcategory I of $C_{/F}$ formed by the subobjects of F is cofinal in $C_{/F}$. By the sufficiency assertion in a), this is exactly what is ensured by the hypothesis that every object of $C_{/F}$ is dominated by a minimal object. Necessity: since every filtered inductive limit of left exact functors is again left exact, the first condition is trivially necessary. The second then follows from the necessity assertion in a).

c) A representable subfunctor $X \hookrightarrow F$ of F is known once one knows the full subcategory $C'_{/X}$ of $C'_{/F}$, where C' is a fixed small generating subcategory of C. Use the hypothesis that I is filtered. Since $C'_{/F}$ is small, the set of its full subcategories is small, whence the conclusion.

Proposition 8.12.5. Let C be a $\mathcal{U}$-category in which finite inductive limits are representable and which admits a small generating subcategory (7.1). Let F be a presheaf on C. For F to be strictly ind-representable, it is sufficient that it satisfy the following two conditions; and these conditions are also necessary if fiber products are representable in C:

a) F is left exact.

b) Every pair (X, u), with $X \in \operatorname{Ob} C$ and $u \in F(X)$, is dominated in $C_{/F}$ by a minimal pair (8.12.3). In other words, there is a minimal pair (X', u'), with $X' \in \operatorname{Ob} C$ and $u' \in F(X')$, and a morphism $f : X \to X'$ such that $u = F(f)(u')$.

Sufficiency follows from 8.12.2 and 8.12.4(b),(c); necessity follows from 8.3.1 and 8.12.4(a).

Remark 8.12.6. When F transforms amalgamated sums in C into fiber products, one sees immediately that, for a given pair (X, u) as in b), the set of strict quotients X' of X such that

$$u \in \operatorname{Im}\bigl(F(X') \to F(X)\bigr)$$

is decreasing filtered. It follows that if the set of strict quotients of X is Artinian, then condition b) of 8.12.5 is automatically satisfied. Thus, if the preceding condition on X is satisfied for every object X of C – in which case one sometimes also says that the objects of C^{op} are *Artinian* – then 8.12.5 implies that F is strictly ind-representable if and only if F is left exact. In this case, F is strictly ind-representable as soon as it is ind-representable (8.3.1).

Corollary 8.12.7. Let C be a $\mathcal{U}$-category in which small projective limits are representable, and which admits a small cogenerating subcategory (7.9, 7.13). Then a functor $F : C \to \mathbf{Set}$ is representable if and only if F commutes with small projective limits.

Necessity is clear. For sufficiency, apply 8.12.5 to the opposite category C^{op}. To prove first that F is strictly pro-representable, it remains to prove that every pair (X, u), with $X \in \operatorname{Ob} C$ and $u \in F(X)$, is dominated by a minimal pair. But the family $(X_i)_{i \in I}$ of strict subobjects of X such that

$$u \in \operatorname{Im}\bigl(F(X_i) \to F(X)\bigr)$$

is small (7.5 in its dual form). Since F is left exact, this family is cofiltered (8.12.6), and since F commutes with small projective limits one sees that $X' = \varprojlim_i X_i$ is a smallest object of this family. Use the fact that F, being left exact, transforms monomorphisms into monomorphisms.

If $u' \in F(X')$ is the unique element whose image in $F(X)$ is u, it follows that (X', u') is a minimal pair dominating (X, u). This proves that F is pro-representable, and the conclusion follows from the following lemma.

Lemma 8.12.8.1. Let $F : C \to \mathbf{Set}$ be a functor, where C is a $\mathcal{U}$-category in which small projective limits are representable. For F to be representable, it is necessary and sufficient that it be pro-representable and commute with small projective limits.

Necessity is clear. We prove sufficiency. To say that F is representable plainly means that ${}_F\backslash C$ has an initial object; indeed, the initial objects of ${}_F\backslash C$ are precisely the isomorphisms $F \to X$, that is, the data of a representation for F. Since F is pro-representable, $({}_F\backslash C)^{\mathrm{op}}$ is filtered and equivalent to a small category. Thus

$$X = \varprojlim_{({}_F\backslash C)^{\mathrm{op}}} X$$

is representable in C, and since F commutes with the limit in question, it follows that X is an initial object of ${}_F\backslash C$. $\square$

Corollary 8.12.8. With C satisfying the hypotheses of 8.12.7, let $f : C \to C'$ be a functor from C to a $\mathcal{U}$-category C'. For f to admit a left adjoint, it is necessary and sufficient that f commute with small projective limits.

This reduces trivially to 8.12.7 by applying that statement to the composite functors of the form

$$X \longmapsto \mathrm{Hom}(Y', f(X)).$$

Examples 8.12.9. As noted in 7.13, the hypotheses on C in 8.12.7 and 8.12.8 are satisfied when C is the category of $\mathcal{U}$-sheaves of sets on a topological space $X \in \mathcal{U}$, or more generally on a $\mathcal{U}$-site (II 3.0.2). We give an instructive example, due to H. Bass, showing that the hypothesis of existence of a small cogenerating subcategory D of C is not redundant in 8.12.7. Take C to be the category of groups belonging to $\mathcal{U}$. Let J be the set of isomorphism classes of simple groups in C. For each $j \in J$, choose a simple group G_j in the class j, and let I be the filtered ordered set of $\mathcal{U}$-small subsets of J. For $i \in I$, let

$$X_i = \amalg_{j \in i} G_j.$$

The X_i then form a projective system $(X_i)_{i \in I}$ in C, with epimorphic transition morphisms, and the corresponding functor

$$F(X) = \varinjlim_i \mathrm{Hom}(X_i, X) \tag{$*$}$$

takes values in $\mathcal{U}$-sets, although the index set I plainly does not have cardinal in $\mathcal{U}$. More precisely, let us show that for every small full subcategory C_0 of C, there exists $i_0 \in I$ such that the restriction of F to C_0 is represented by X_{i_0}. This will prove both that F is $\mathcal{U}$-set-valued and that it commutes with small projective limits. To prove the assertion, it is enough to note that, for every $X \in \mathrm{Ob}\, C_0$, the cardinal of the set $J(X)$ of $j \in J$ for which there exists a nontrivial homomorphism from G_j to X is necessarily small, since such a morphism is necessarily a monomorphism, G_j being simple. Hence, if i_0 is the subset of J which is the union of the $J(X)$ for $X \in \mathrm{Ob}\, C_0$, then i_0 is small, that is, $i_0 \in I$, and it does the job. On the other hand, it is clear that F is not representable, since $\mathrm{card}\, I \notin \mathcal{U}$. From this and 8.12.7 one therefore concludes that *the category C of groups in $\mathcal{U}$ admits no small full cogenerating subcategory.* Since the object $\mathbb{Z}$ of C is nevertheless a generator, it follows from the proof of 7.12 that there exists a two-generator group G which does not embed into an injective object of the category C of groups in $\mathcal{U}$. It also seems plausible that C has no injective objects other than the unit groups.

8.13 Pro-representable functors and accessible functors

8.13.1

In this section we use some notions and results from the following paragraph, notably 9.11 and 9.13, in order to obtain a criterion of pro-representability which will be used incidentally in IV 9.16. In what follows, C denotes a $\mathcal{U}$-category satisfying condition L of 9.1(b). This condition is satisfied, for example, if small filtered inductive limits are representable in C.

Proposition 8.13.2. Let C be as above, and let

$$f : C \longrightarrow \mathbf{Set}$$

be a functor.

a) Suppose every object of C is accessible (9.3). If f is pro-representable, then f is accessible (9.2) and left exact.

b) Suppose finite projective limits are representable in C, and that C admits a cardinal filtration (9.12). If f is accessible and left exact, then f is pro-representable.

Proof.

a) The hypothesis on C means that the covariant representable functors from C to **Set** are accessible. Therefore the same is true of every small inductive limit of such functors (9.6(i)), and hence of every pro-representable functor.

b) By 8.3.3(iii), it remains to prove that $\mathrm{Ob}({}_F\backslash C)^{\mathrm{op}}$ contains a small cofinal subcategory. By hypothesis there exists a cardinal Π such that f is Π-accessible. Let (X, u), $u \in F(X)$, be an object of ${}_F\backslash C$. With the notation of 9.12(c), one then has $X = \varinjlim_i X_i$, with I filtered and large relative to Π, and with the X_i in $C' = \mathrm{Filt}^{\Pi}(C)$. Hence

$$F(X) \xleftarrow{\sim} \varinjlim_i F(X_i).$$

This shows that the small subcategory $({}_F\backslash C')^{\mathrm{op}}$ is cofinal in $({}_F\backslash C)^{\mathrm{op}}$, and completes the proof.

Corollary 8.13.3. Let C be a $\mathcal{U}$-category satisfying the following conditions:

a) finite projective limits are representable in C;

b) small filtered inductive limits are representable in C;

c) the functor Ker on the category of double arrows of C is accessible, for example it commutes with small filtered inductive limits;

d) every strict epimorphism of C is strictly universal (10.2);

e) there exists a small subcategory of C generating by strict epimorphisms (7.1).

Under these conditions, a functor $f : C \to \mathbf{Set}$ is pro-representable if and only if it is left exact and accessible (9.2).

Indeed, the hypotheses on C in 8.13.2(a) and (b) are satisfied by 9.11 and 9.13, respectively.

Corollary 8.13.4. Let $f : C \to C'$ be a functor between $\mathcal{U}$-categories satisfying conditions a) through e) of 8.13.3. For f to admit a pro-adjoint, it is necessary and sufficient that f be left exact and accessible.

Since the representable functors

$$h_{Y'} : X' \longmapsto \mathrm{Hom}(Y', X')\,, \qquad C' \to \mathbf{Set},$$

are left exact and accessible (9.11), if f has these same properties then so do their composites with f, and these composites are therefore pro-representable by 8.13.3; that is, f admits a pro-adjoint. Conversely, suppose that f admits a pro-adjoint, and let C'_1 be a small generating subcategory of C'. Choose a cardinal Π such that the objects $Y' \in \mathrm{Ob}\, C'_1$ are Π-accessible. To prove that f is Π-accessible, it is enough to prove this for its composites with the $h_{Y'}$, $Y' \in \mathrm{Ob}\, C'_1$. But by hypothesis these composites are pro-representable, hence accessible by 8.13.3, and therefore Π-accessible after increasing Π if necessary. □

Continuation anchor. The next section is Section 9, “Accessible functors, cardinal filtrations, and construction of small generating subcategories.”

Exposé I. Presheaves

Section 9, beginning through Lemma 9.13.1

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 9 from the start through Lemma 9.13.1. The notation follows the previous batches: $\mathcal{U}$ denotes the fixed universe; $\mathcal{U}$-**Set** is written as $\mathcal{U}$-**Set**; and $\operatorname{card} \operatorname{Fl} C$ denotes the cardinal of the set of arrows of a category C.

9 Accessible functors, cardinal filtrations, and construction of small generating subcategories

The present section is more technical than the other sections of this exposé. In this seminar it will be used only in IV 9 and VI 4, which themselves are not used elsewhere in the Seminar. It is therefore advisable to omit the present section on a first reading.

9.0

All categories considered in this number are assumed to be $\mathcal{U}$-categories. Except for the small indexing categories $I, J, \ldots$ that we shall use, the developments below will mainly concern "large" categories $E, F, \ldots$ that are stable under small filtered inductive limits. Most often, however, it will be enough that a slightly weaker condition hold, namely condition L of Definition 9.1 below. All cardinals considered in this number are assumed to belong to $\mathcal{U}$.

Following a suggestion of P. Deligne, we shall study, for a functor $f : E \to F$ between large categories, a condition expressing commutation of f with certain types of filtered inductive limits. This condition is remarkably stable and is satisfied by the most important functors that occur naturally. The applications we have in view for this seminar are 9.13.3 and 9.13.4, used in VI 4, and above all 9.25, which gives, in a non-trivial case, the existence of a small generating family in a category of sections of a fibred category; that result will be used in IV 9.16.

Definition 9.1.

a) Let I be a preordered set and let π be a cardinal. We say that I is *large with respect to* π if I is filtered and if every subset of I of cardinal $\leq \pi$ has an upper bound in I.

b) Let E be a category. If π is a cardinal, we say that E satisfies condition L_π if, for every small ordered set I large with respect to π, E is stable under inductive limits of type I. We say that E satisfies condition L if there exists a cardinal $\pi \in \mathcal{U}$ such that E satisfies L_π.

9.1.1

When, in Definition 9.1(a), one has $\pi \geq 2$, the second stated condition already implies that I is filtered. If π is finite, saying that I is large with respect to π simply means that I is filtered. In what follows we shall be interested almost exclusively in the case where π is infinite. Notice that if π and π' are cardinals with $\pi' \geq \pi$, then I large with respect to π' plainly implies I large with respect to π.

9.1.2

As announced in 9.0, the conditions L_π and L should be regarded as technical variants of the stronger condition of stability under small filtered inductive limits. It is clear that if π and π' are cardinals with $\pi' \geq \pi$, then condition L_π implies condition $L_{\pi'}$.

Definition 9.2. Let $f : E \to F$ be a functor. If Π is a cardinal, we say that f is Π-*preaccessible* (respectively Π-*accessible*) if E satisfies L_Π (9.1) and if, for every ordered set $I \in \mathcal{U}$ large with respect to Π, and every inductive system $(X_i)_{i\in I}$ in E of type I, the canonical morphism

$$\varinjlim f(X_i) \longrightarrow f(\varinjlim X_i)$$

is a monomorphism (respectively an isomorphism). We say that f is *preaccessible* (respectively *accessible*), relative to the universe $\mathcal{U}$, if there exists a cardinal $\Pi \in \mathcal{U}$ such that f is Π-preaccessible (respectively Π-accessible).

The category of Π-accessible (respectively accessible) functors from E to F will be denoted by $\mathrm{Hom}(E, F)_\Pi$ (respectively $\mathrm{Hom}(E, F)_{\mathrm{acc}}$).

9.2.1

Plainly, a functor which commutes with small filtered inductive limits, for example a functor admitting a right adjoint, is Π-accessible for every cardinal $\Pi \geq 2$.

Definition 9.3. Let E be a category, let X be an object of E, and let

$$h^\circ_X : E \longrightarrow (\mathcal{U}\text{-}\mathbf{Set})$$

be the covariant functor it represents. Let $\Pi \in \mathcal{U}$ be a cardinal. We say that X is a Π-preaccessible (respectively Π-accessible) object of E if the functor h°_X is Π-preaccessible (respectively Π-accessible). We say that X is preaccessible (respectively accessible) if there exists a cardinal $\Pi \in \mathcal{U}$ such that X is a Π-preaccessible (respectively Π-accessible) object of E.

9.3.1

We denote by E_Π the strictly full subcategory of E consisting of the Π-accessible objects of E.

When Definition 9.3 is applied to a category of the form $\mathrm{Hom}(C, F)$, the terminology just introduced is, *a priori*, ambiguous with the analogous terminology introduced in 9.2, when one interprets the objects of $\mathrm{Hom}(C, F)$ as functors. There does not, however, seem to be any serious risk of confusion.

Definition 9.4. Let E be a category and let $\pi \in \mathcal{U}$ be a cardinal. We say that E is a π-preaccessible (respectively π-accessible) category if there exists in E a small full subcategory C which is generating (7.1) and whose objects are π-preaccessible (respectively π-accessible) (9.3). We say that E is preaccessible (respectively accessible) if there exists a cardinal $\pi \in \mathcal{U}$ such that E is π-preaccessible (respectively π-accessible).

For important examples, see 9.11.3 below.

Proposition 9.5. Let $f : E \to F$ be a functor between categories such that F is accessible and every object of E is accessible. If f admits a left adjoint, then f is accessible.

Indeed, by hypothesis, F admits a small conservative family of representable functors $F \to (\mathcal{U}\text{-}\mathbf{Set})$ which are accessible. Thus one is immediately reduced to showing that the composite of f with each of these functors is accessible. Since f admits a left adjoint, these composites are representable functors $E \to (\mathcal{U}\text{-}\mathbf{Set})$, hence are accessible by the hypothesis on E.

Proposition 9.6. Let E and F be two categories, let $\Pi \in \mathcal{U}$ be a cardinal, and consider the full subcategory $\mathrm{Hom}(E,F)_\Pi$ of $\mathrm{Hom}(E,F)$ formed by the Π-accessible functors (9.2) from E to F.

(i) This subcategory is stable under every type of inductive limit which is representable in F.

(ii) Suppose that F is Π-accessible (9.4), and let J be a category such that $\operatorname{card} \mathrm{Fl}\, J \leq \Pi$ and such that projective limits of type J are representable in F; then projective limits of type J are representable in $\mathrm{Hom}(E,F)$. The subcategory $\mathrm{Hom}(E,F)_\Pi$ is stable under projective limits of type J.

Corollary 9.7. Let E and F be two categories, with F accessible (9.4). Then the full subcategory $\mathrm{Hom}(E,F)_{\mathrm{acc}}$ of $\mathrm{Hom}(E,F)$ formed by accessible functors is stable under every inductive or projective limit, relative to a small indexing category J, which is representable in F and hence in $\mathrm{Hom}(E,F)$.

Proof. [Proof of Proposition 9.6] Assertion (i) follows trivially from the commutation of the functor $\varinjlim$ with arbitrary inductive limits. For (ii), let $(f_j)_{j\in J}$ be a projective system of Π-accessible functors $E \to F$, and let

$$f = \varprojlim_j f_j$$

be its projective limit, computed argument by argument. We prove that f is Π-accessible; in other words, for every ordered set $I \in \mathcal{U}$ large with respect to Π, and every inductive system $(X_i)_{i\in I}$ in E, the canonical morphism

$$\varinjlim_i((\varprojlim_j f_j)(X_i)) \longrightarrow (\varprojlim_j f_j)(\varinjlim_i X_i)$$

is an isomorphism. The argument-by-argument computation of the functor $\varprojlim_j f_j$ identifies this canonical morphism with

$$\varinjlim_i \varprojlim_j f_j(X_i) \longrightarrow \varprojlim_j \varinjlim_i f_j(X_i),$$

the canonical morphism attached to the bifunctor

$$(i,j) \longmapsto f_j(X_i) : I \times J \longrightarrow F.$$

Thus Proposition 9.6 follows from the following more general assertion.

Corollary 9.8. Let F be a Π-accessible category, let I be an ordered set large with respect to Π, let J be a small category such that $\operatorname{card} \mathrm{Fl}\, J \leq \Pi$ and such that projective limits of type J are representable in F, and let $h : I \times J \to F$ be a functor. Then the canonical morphism

$$\varinjlim_i \varprojlim_j h(i,j) \longrightarrow \varprojlim_j \varinjlim_i h(i,j) \tag{9.8.1}$$

is an isomorphism. In other words, the functor

$$\varinjlim_{i} : \mathrm{Hom}(I, F) \longrightarrow F \tag{9.8.2}$$

commutes with projective limits of type J, for every small category J such that $\operatorname{card} \mathrm{Fl}\, J \le \Pi$ and such that projective limits of type J are representable in F. Equivalently, for every J as above, the functor

$$\varinjlim_{j} : \mathrm{Hom}(J, F) \longrightarrow F \tag{9.8.3}$$

is Π-accessible.

Since, by hypothesis, F admits a conservative family of covariant representable Π-accessible functors $g : F \to (\mathcal{U}\text{-}\mathbf{Set})$, one is immediately reduced to proving 9.8 in the case $F = \mathcal{U}\text{-}\mathbf{Set}$. We shall then prove the assertion in the form that (9.8.2) commutes with projective limits of type J, with $\operatorname{card} \mathrm{Fl}\, J \le \Pi$. Since I is filtered, (9.8.2) is known to commute with finite projective limits (2.8). A standard argument (cf. 2.3) therefore reduces us to proving that it commutes with products indexed by a set J of cardinal $\le \Pi$. This reduces to proving the bijectivity of (9.8.1) when J is discrete.

For injectivity, consider two elements a, b of the left-hand side. They come from $\prod_j h(i_0, j)$ for a suitable $i_0 \in I$, say from elements $\prod_j a(i_0, j)$ and $\prod_j b(i_0, j)$. Suppose that the elements $\prod_j h(\infty, j)$ of the right-hand side which they define are equal; that is, for every j there exists $i(j) \ge i_0$ such that $a(i_0, j)$ and $b(i_0, j)$ have the same image in $h(i, j)$. Since I is large with respect to $\operatorname{card} J$, there exists a common upper bound $i_1 \in I$ for all the $i(j)$. It follows that the two elements under consideration in $\prod_j h(i_0, j)$ have the same image in $\prod_j h(i_1, j)$, and therefore define the same element of the left-hand side of (9.8.1). This proves injectivity.

For surjectivity, let $\prod_j a(\infty, j)$ be an element of the right-hand side. For each $j \in J$, $a(\infty, j)$ comes from an element of $h(i(j), j)$ for a suitable $i(j) \in I$. As before, one can find a common upper bound $i \in I$ for all the $i(j)$. Then $\prod_j a(\infty, j)$ comes from an element $\prod_j a(i, j)$ of $\prod_j h(i, j)$, hence lies in the image of (9.8.1). This completes the proof.

Corollary 9.9. Let F be a Π-accessible (respectively accessible) category. Then, for every category J such that $\operatorname{card} \mathrm{Fl}\, J \le \Pi$ (respectively every small category J), and for every functor $(X_j)_{j \in J} : J \to F$ whose inductive limit in F is representable, if the X_j are Π-accessible (respectively accessible), then so is $\varprojlim X_j$.

Indeed, the functor $F \to (\mathcal{U}\text{-}\mathbf{Set})$ represented by $\varinjlim X_j$ is the projective limit of the functors represented by the X_j, and one applies Proposition 9.6(ii) to the resulting projective system of functors.

Remark 9.10. In the statements 9.6, 9.7, 9.8, and 9.9 one may everywhere replace the words "Π-accessible", "accessible" by "Π-preaccessible", "preaccessible". The proof just given proves this variant of the preceding statements as well.

Proposition 9.11. Let E be a category satisfying condition L (9.1), and let C be a small full generating subcategory (7.1). Suppose that the following condition is satisfied:

a) E is stable under kernels of double arrows, and the functor Ker on the category of double arrows of E is accessible; for example, it commutes with small filtered inductive limits.

Then every object of E is preaccessible (9.3), and *a fortiori* E is preaccessible. Suppose moreover that C is generating by strict epimorphisms (7.1), and suppose that E satisfies the following conditions:

b) E is stable under fibre products.

c) Every small strict epimorphic family in E is universally strict epimorphic (10.3); or else small coproducts are representable in E and every strict epimorphism in E is a universal strict epimorphism.

Then every object of E is accessible, and *a fortiori* E is accessible.

The fact that, under (a), every object of E is preaccessible follows from the fact that, for every object X of E, the set of strict subobjects of X is small (7.4), and from the following lemma.

Lemma 9.11.1. Under the hypotheses of 9.11(a), let $X \in \operatorname{ob} E$ and let Π be a cardinal such that E satisfies L_Π (9.1), the functor Ker in 9.11(a) is Π-accessible, and the set of strict subobjects of X has cardinal bounded by Π. Then X is Π-preaccessible.

Indeed, let I be a set large with respect to Π, let $(Y_i)_{i\in I}$ be an inductive system in E with inductive limit Y, and let

$$(u_i, v_i : X \rightrightarrows Y_i)$$

be a double arrow such that the composite double arrow $u, v : X \rightrightarrows Y$ satisfies $u = v$. We prove that there exists $j \geq i$ in I such that $u_j = v_j$. For this, consider the inductive system of double arrows $(u_j, v_j : X \rightrightarrows Y_j)_{j\geq i}$, whose inductive limit is (u, v). By hypothesis,

$$X = \operatorname{Ker}(u, v) = \varinjlim_j \operatorname{Ker}(u_j, v_j) = \varinjlim_j X_j, \qquad X_j = \operatorname{Ker}(u_j, v_j).$$

The X_j are strict subobjects of X, so there exists a subset I' of I, consisting of indices $j \geq i$, with $\operatorname{card} I' \leq \Pi$, such that every X_j is equal to one of the $X_{i'}$ for $i' \in I'$. Since I is large with respect to Π, there is an upper bound j of I' in I. Then X_j contains all the $X_{j'}$ for $j' \geq i$, hence $X_j \to X$ is an epimorphism, since the family of maps $X_{j'} \to X$ is epimorphic. It is therefore an isomorphism, since it is a strict monomorphism. Thus $u_j = v_j$, proving 9.11.1.

The second assertion of 9.11 follows from the first and from the following lemma.

Lemma 9.11.2. Under the hypotheses of 9.11(a), (b), and (c), let X be an object of E, and let Π be an infinite cardinal such that E satisfies L_Π (9.1), such that $\operatorname{card} \operatorname{ob} C_{/X} \leq \Pi$, such that for two objects $X' \to X$ and $X'' \to X$ of $C_{/X}$ the fibre product $X' \times_X X''$ is Π-preaccessible, and finally such that X is Π-preaccessible. Then X is Π-accessible.

With the notation of the proof of 9.11.1, it remains only to prove that every morphism $u : X \to Y$ comes from a morphism $u_i : X \to Y_i$. Consider the family of morphisms $Y_i \to Y$, which is a strict epimorphic family. By (b) and (c), the family

$$Y_i \times_Y X \longrightarrow X$$

is also strictly epimorphic. On the other hand, for every i, since C is generating by strict epimorphisms, the family of maps

$$T \longrightarrow Y_i \times_Y X$$

with source in C is strictly epimorphic. In the second alternative in (c), one can find such a strict epimorphism with source a small coproduct of objects of C. By transitivity of universally strict epimorphic families (II 2.5), it follows that the family of maps $T_\alpha \xrightarrow{f_\alpha} X$ with source in C which factor through one of the $Y_i \times_Y X$ is strictly epimorphic. Let J be the set of indices of this family; its cardinal is bounded by $\operatorname{card}\operatorname{ob} C_{/X} \leq \Pi$.

For each $\alpha \in J$, choose an $i = i(\alpha) \in I$ and an X-morphism

$$T_\alpha \longrightarrow Y_i \times_Y X,$$

or equivalently a morphism $v_\alpha : T_\alpha \to Y_i$ such that $p_i v_\alpha = u f_\alpha$, where $p_i : Y_i \to Y$ is the canonical morphism. Since I is large with respect to Π, one may choose $i(\alpha)$ independently of α; call it i. For every pair of indices $\alpha, \beta \in J$, consider the two composites

$$\mathrm{pr}_1\, v_\alpha, \mathrm{pr}_2\, v_\beta : T_\alpha \times_X T_\beta \rightrightarrows Y_i.$$

Their composites with $Y_i \to Y$ are equal. Since $T_\alpha \times_X T_\beta$ is Π-preaccessible by hypothesis, there exists an index $i' = i(\alpha, \beta) \geq i$ such that the composites of the arrows under consideration with $Y_i \to Y_{i'}$ are equal. The set of pairs (α, β) has cardinal $\leq \Pi^2 = \Pi$, because Π is infinite; hence one may again choose i' independently of α and β. Plainly we may suppose $i' = i$. But then, since the family $(f_\alpha : T_\alpha \to X)$ is strictly epimorphic, one can find a morphism $u_i : X \to Y_i$ such that $u_i f_\alpha = v_\alpha$. We then have $p_i u_i = u$, because for every α,

$$(p_i u_i) f_\alpha = p_i (u_i f_\alpha) = p_i v_\alpha = u f_\alpha,$$

and the family of the f_α is epimorphic. This completes the proof of 9.11.2.

Remark 9.11.3. As a special case of 9.11, one sees that every object of E is accessible in each of the following two cases. First, E is an abelian $\mathcal{U}$-category with exact filtered inductive limits and admitting a small generating subcategory; here it is the second alternative in (c) which applies. Second, E is the category of sheaves of sets on a topological space $X \in \mathcal{U}$. More generally, it is enough that E be the category of sheaves of sets on a $\mathcal{U}$-site (II 2.1), or that E be a $\mathcal{U}$-topos (IV 1.1).

Definition 9.12. Let E be a category. A *cardinal filtration* of E is an increasing filtration $(\mathrm{Filt}^\Pi(E))_{\Pi \geq \Pi_0}$ of E by strictly full subcategories $\mathrm{Filt}^\Pi(E)$, indexed by the cardinals $\Pi \in \mathcal{U}$ with $\Pi \geq \Pi_0$, where Π_0 is a fixed infinite cardinal depending on the cardinal filtration under consideration, and satisfying the following conditions:

a) For every $\Pi \geq \Pi_0$, $\mathrm{Filt}^\Pi(E)$ is equivalent to a small category.

b) E satisfies L_{Π_0} (9.3), and for every $\Pi \geq \Pi_0$, $\mathrm{Filt}^\Pi(E)$ is stable in E under filtered inductive limits indexed by ordered sets I which are large with respect to Π_0 and such that $\operatorname{card} I \leq \Pi$.

c) For every $\Pi \geq \Pi_0$ and every $X \in \operatorname{ob} E$, one can find an isomorphism

$$X \simeq \varinjlim_I X_i,$$

where $(X_i)_{i \in I}$ is an inductive system in E indexed by an ordered set I large with respect to Π (9.1), and where the X_i belong to $\mathrm{Filt}^\Pi(E)$. Moreover, if $X \in \operatorname{ob}\mathrm{Filt}^\Pi(E)$ and $\Pi' \geq \Pi$, one can take I with $\operatorname{card} I \leq (\Pi')^\Pi$.

9.12.1

It follows from (b) and (c) that every $X \in \operatorname{ob} E$ belongs to some $\operatorname{Filt}^{\Pi}(E)$ for Π sufficiently large.

Example 9.12.2. Take $E = (\mathcal{U}\text{-}\mathbf{Set})$, $\Pi_0 = \aleph_0$, and let $\operatorname{Filt}^{\Pi}(E)$ be the full subcategory of E consisting of the sets X such that $\operatorname{card} X \leq \Pi$. More generally:

Proposition 9.13. Let E be a $\mathcal{U}$-category. Suppose that E is stable under small filtered inductive limits, under sums of two objects, and under cokernels of double arrows; that E is stable under fibre products; and that epimorphic morphisms in E are universal strict epimorphisms. Let C be a small full subcategory of E which is generating by strict epimorphisms. Let Π_0 be an infinite cardinal $\geq \operatorname{card} \operatorname{Fl} C$. For every cardinal $\Pi \geq \Pi_0$, let $\operatorname{Filt}^{\Pi}(E)$ be the strictly full subcategory of E consisting of the objects X of E for which there exists a strict epimorphic family $(X_i \to X)_{i \in I}$ with target X, such that $\operatorname{card} I \leq \Pi$ and $X_i \in \operatorname{ob} C$ for every $i \in I$. Then $(\operatorname{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0}$ is a cardinal filtration of E. Moreover, for every $X \in \operatorname{ob} \operatorname{Filt}^{\Pi}(E)$, the cardinal of the set of arrows of $C_{/X}$, and *a fortiori* of the set of objects of $C_{/X}$, is bounded by Π^{Π_0}.

The last assertion is precisely 7.6.

To prove that this is a cardinal filtration, it remains to prove conditions (a), (b), and (c) of 9.12. Condition (a) follows at once from 7.5.2. For (b), suppose that

$$X = \varinjlim_{I} X_i,$$

with $\operatorname{card} I \leq \Pi$ and $X_i \in \operatorname{ob} \operatorname{Filt}^{\Pi}(E)$. The family of canonical morphisms $X_i \to X$ is therefore strictly epimorphic. By the hypothesis on the X_i, for each $i \in I$ there is a strict epimorphic family $(X_{ij} \to X_i)_{j \in J_i}$ with $\operatorname{card} J_i \leq \Pi$. Passing to the sums $\coprod_i X_i$ and $\coprod_{i,j} X_{ij}$, one sees, by transitivity of universal strict epimorphisms (II 2.5), that the family of composites

$$X_{ij} \longrightarrow X_i \longrightarrow X$$

indexed by the disjoint union of the J_i, for $i \in I$, is strictly epimorphic. Since $\operatorname{card} J \leq \Pi^2 = \Pi$, the conclusion follows. Notice that we have used only the existence of $\varinjlim_I X_i$, and not the fact that I is large with respect to Π_0, nor even that it is filtered.

It remains to prove (c). First note that for every $X \in \operatorname{ob} E$, the family of arrows $X_i \to X$ with source in $\operatorname{ob} C$ is strictly epimorphic, because C is generating by strict epimorphisms, and is plainly small. Thus there is a cardinal $\Pi \geq \Pi_0$ such that $X \in \operatorname{ob} \operatorname{Filt}^{\Pi}(E)$. It remains to prove that, if $\Pi' \geq \Pi \geq \Pi_0$ and $X \in \operatorname{ob} \operatorname{Filt}^{\Pi'}(E)$, then there is an isomorphism

$$X \simeq \varinjlim_{I} X_i,$$

where I is large with respect to Π, $\operatorname{card} I \leq (\Pi')^{\Pi}$, and the X_i lie in $\operatorname{ob} \operatorname{Filt}^{\Pi}(E)$.

Since C is generating by strict epimorphisms, one has

$$X \simeq \varinjlim_{C_{/X}} T. \tag{$*$}$$

Furthermore, by successively adjoining to C sums and cokernels of double arrows in C, and then doing the same for the category so obtained, and so on, we are reduced to the case where C

is stable under sums of two objects in E and under cokernels of double arrows in E, without destroying the hypothesis $\Pi_0 \geq \operatorname{card} \operatorname{Fl} C$. We may also suppose that, if E contains a fixed initial object $\varnothing_E$, then $\varnothing_E \in \operatorname{ob} C$. This implies that the category $C_{/X}$ is stable under sums of two objects and under cokernels of double arrows. It is then filtered: it is non-empty, since otherwise the relation $(*)$ would show that X is an initial object, so that $C_{/X}$ would contain the arrow $\varnothing_E \to X$, a contradiction.

Let I be the set, ordered by inclusion, of full subcategories i of $C_{/X}$ which are filtered and such that $\operatorname{card} \operatorname{ob} i \leq \Pi$. For every $i \in I$, let $X_i \in \operatorname{ob} E$ be the inductive limit of the composite functor

$$i \longrightarrow C_{/X} \longrightarrow E,$$

which exists by hypothesis. Then one plainly has

$$\varinjlim_{I} X_i \simeq \varinjlim_{C_{/X}} T \simeq X.$$

On the other hand, we have already noted that

$$\operatorname{card} \operatorname{ob} C_{/X} \leq (\Pi')^{\Pi_0}.$$

It follows that I is large with respect to Π and that

$$\operatorname{card} I \leq ((\Pi')^{\Pi_0})^{\Pi} = (\Pi')^{\Pi\Pi_0} = (\Pi')^{\Pi}$$

by the following lemma, whose proof is left to the reader; in applying it one takes $J = C_{/X}$ and $c = (\Pi')^{\Pi_0}$.

Lemma 9.13.1. Let J be a filtered category, and let c and Π be two cardinals such that

$$c \geq \operatorname{Sup}(\operatorname{card} \operatorname{Fl} J, \Pi)$$

and such that $\operatorname{card} \operatorname{Hom}(j, j') \leq \Pi_0$ for every pair of objects j, j' of J. Let I be the set of full filtered subcategories i of J such that $\operatorname{card} \operatorname{ob} i \leq \Pi$. Then, ordered by inclusion, I is large with respect to Π, and one has $\operatorname{card} I \leq c^{\Pi}$.

This completes the proof of Proposition 9.13.

Exposé I. Presheaves

Section 9, Remark 9.13.2 through Corollary 9.26

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 9 from Remark 9.13.2 through Corollary 9.26. The terminology follows the previous batches: "large with respect to Π" translates *grand devant* Π; "accessible" is used in the sense of 9.2–9.3; and $\operatorname{card}\operatorname{Fl} C$ denotes the cardinal of the set of arrows of a category C. A few evident typographical slips in the source have been corrected where the formulas force the correction, notably in some subscripts in 9.14, 9.21, and 9.24.

9 Accessible functors, cardinal filtrations, and construction of small generating subcategories

9.13.2

Remark 9.13.2. A slight additional effort should make it possible, in Proposition 9.13, to replace the hypothesis that E is stable under small filtered inductive limits by the hypothesis that E satisfies L (9.1), provided that E is stable under small sums, or that in E every epimorphic *family* is universally epimorphic. In the proof of c), one must then choose Π_0 so that E satisfies L_{Π_0}, and restrict to those full subcategories i of $C_{/X}$ which are not only filtered, but whose preordered set $\operatorname{ob} i$ is large with respect to Π_0. Using Section 8 and the hypothesis that E satisfies L_{Π_0}, one then finds that the X_i exist; one should conclude by means of a suitable variant of Lemma 9.13.1, which the editor has not checked.

Proposition 9.14. Let $f : E \to F$ be an accessible functor (9.2) between two categories endowed with cardinal filtrations

$$(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0} \qquad \text{and} \qquad (\mathrm{Filt}^{\Pi}(F))_{\Pi \geq \Pi_0'}.$$

Then there exists a cardinal $\Pi_1 \geq \operatorname{Sup}(\Pi_0, \Pi_0')$ such that, for every cardinal $\Pi \geq \Pi_1$, one has

$$f(\mathrm{Filt}^{\Pi}(E)) \subset \mathrm{Filt}^{\Pi^{\Pi_1}}(F). \tag{9.14.1}$$

In particular, if $\Pi = 2^c$ with $c \geq \Pi_1$, whence

$$\Pi^{\Pi_1} = 2^{c\Pi_1} = 2^c = \Pi,$$

then

$$f(\mathrm{Filt}^{\Pi}(E)) \subset \mathrm{Filt}^{\Pi}(F). \tag{9.14.2}$$

Let us immediately record the following consequence.

Corollary 9.15. Let E be a category endowed with two cardinal filtrations

$$(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0} \qquad \text{and} \qquad (\mathrm{Filt}'^{\Pi}(E))_{\Pi \geq \Pi_0'}.$$

Then there exists a cardinal $\Pi_1 \geq \mathrm{Sup}(\Pi_0, \Pi'_0)$ such that, for every cardinal $c \geq \Pi_1$, if $\Pi = 2^c$, then

$$\mathrm{Filt}^{\Pi}(E) = \mathrm{Filt}'^{\Pi}(E).$$

Proof. Proof of Proposition 9.14. Let c be a cardinal such that

$$c \geq \mathrm{Sup}(\Pi_0, \Pi'_0)$$

and such that f is c-admissible. Let also $\Pi_1 \geq c$ be such that

$$f(\mathrm{Filt}^c(E)) \subset \mathrm{Filt}^{\Pi_1}(F). \tag{$*$}$$

Such a Π_1 exists because $\mathrm{Filt}^c(E)$ is equivalent to a small category (9.12 a)); hence $f(\mathrm{Filt}^c(E))$ is also essentially small, and one may apply 9.12.1 to the objects of the latter category to find one $\mathrm{Filt}^{\Pi}(F)$ containing all of them, taking into account that the $\mathrm{Filt}^{\Pi}(F)$ are full subcategories.

Let $\Pi \geq \Pi_1$, and let $X \in \mathrm{ob}\,\mathrm{Filt}^{\Pi}(E)$. We prove that

$$f(X) \in \mathrm{Filt}^{\Pi^{\Pi_1}}(F).$$

Write

$$X = \varinjlim_I X_i,$$

where $X_i \in \mathrm{ob}\,\mathrm{Filt}^c(E)$, where I is large with respect to c, and where

$$\mathrm{card}\, I \leq \Pi^c \leq \Pi^{\Pi_1}$$

(9.12 c)). Since f is c-admissible and I is large with respect to c, one has

$$f(X) \simeq \varinjlim_I f(X_i).$$

Moreover $\mathrm{card}\, I \leq \Pi^{\Pi_1}$, and $f(X_i) \in \mathrm{ob}\,\mathrm{Filt}^{\Pi_1}(F) \subset \mathrm{ob}\,\mathrm{Filt}^{\Pi^{\Pi_1}}(F)$ by $(*)$. Therefore $f(X) \in \mathrm{Filt}^{\Pi^{\Pi_1}}(F)$ by 9.12 b). $\square$

9.15.1

The notion of a cardinal filtration (9.12) is of interest only when the objects of E are accessible. Let us point out that Proposition 9.11 implies that this condition is satisfied when, in addition to the hypotheses of Proposition 9.13, one assumes that the functor Ker on the category of double arrows of E is accessible. We also record the following.

Proposition 9.16. Let E be a category endowed with a cardinal filtration

$$(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0}.$$

Suppose that the elements of $\mathrm{Filt}^{\Pi_0}(E)$ are accessible objects of E. Then there exists a cardinal $\Pi_1 \geq \Pi_0$ in $\mathcal{U}$ such that the objects of $\mathrm{Filt}^{\Pi_0}(E)$ are Π_1-accessible. If Π_1 is chosen in this way, then for every cardinal $\Pi \geq \Pi_1$ one has, with the notation of 9.3.1,

$$E_{\Pi} \subset \mathrm{Filt}^{\Pi}(E) \subset E_{\Pi^{\Pi_0}}. \tag{9.16.1}$$

The existence of Π_1 follows immediately from the fact that $\mathrm{Filt}^{\Pi_0}(E)$ is equivalent to a small category (9.12 a)). Let $\Pi \geq \Pi_1$. If X belongs to $\mathrm{Filt}^{\Pi}(E)$, write

$$X \simeq \varinjlim_{I} X_i$$

with $\mathrm{card}\, I \leq \Pi^{\Pi_0}$ and $X_i \in \mathrm{ob}\,\mathrm{Filt}^{\Pi_0}(E)$ for every i (9.12 c)). It then follows from Corollary 9.9 that X is Π^{Π_0}-accessible; this gives the second inclusion in (9.16.1).

Conversely suppose that X is Π-accessible, and write

$$X \simeq \varinjlim_{I} X_i,$$

where I is large with respect to Π and the X_i lie in $\mathrm{ob}\,\mathrm{Filt}^{\Pi}(E)$ (9.12 c)). By the hypothesis on X, the given isomorphism $X \xrightarrow{\sim} \varinjlim_I X_i$ factors through one of the X_i; hence X is isomorphic to a direct factor of this X_i. We conclude that $X \in \mathrm{ob}\,\mathrm{Filt}^{\Pi}(E)$, and hence obtain the first inclusion in (9.16.1), by the following lemma.

Lemma 9.16.2. Every object of E which is a direct factor of an object of $\mathrm{Filt}^{\Pi}(E)$ belongs to $\mathrm{Filt}^{\Pi}(E)$.

Indeed, if X is the image of an idempotent p on an object Y of E (i.e.an endomorphism p such that $p^2 = p$), and if I is a filtered ordered set, then X is the inductive limit of the filtered inductive system $(Y_i)_{i \in I}$ defined by $Y_i = Y$ for every i, and with transition map $p : Y_i \to Y_j$ whenever $i < j$. Taking I large with respect to Π_0 and with $\mathrm{card}\, I \leq \Pi$, we see that if $Y \in \mathrm{Filt}^{\Pi}(E)$, then the same is true of X by 9.12 b).

Corollary 9.17. Under the hypotheses of Proposition 9.16, for every cardinal $c \geq \Pi_1$, if $\Pi = 2^c$, then $\mathrm{Filt}^{\Pi}(E)$ is identical with the strictly full subcategory E_Π of E formed by the Π-accessible objects.

Corollary 9.18. Let E be a category satisfying condition L_Π (9.1), where Π is a cardinal, and let C be a full subcategory of E. Denote by $\mathrm{Ind}(C)_\Pi$ the full subcategory of the category $\mathrm{Ind}(C)$ of ind-objects of C (8.2) formed by ind-objects of the form $(X_i)_{i \in I}$, where I is an ordered set large with respect to Π. Consider the canonical functor

$$(X_i)_{i \in I} \longmapsto \varinjlim_{I} X_i : \mathrm{Ind}(C)_\Pi \longrightarrow E. \tag{9.18.1}$$

a) This functor is fully faithful if and only if every object X of C is a Π-accessible object (9.3) of E.

b) Assume the hypotheses of Corollary 9.17; in particular $\Pi = 2^c$, and take $C = \mathrm{Filt}^{\Pi}(E)$. Then the functor (9.18.1) is an equivalence of categories.

Assertion a) is an immediate generalization of 8.7.5 a), and is proved in the same way. Assertion b) follows from Corollary 9.17 and from condition 9.12 c) in the definition of cardinal filtrations, which implies that the functor under consideration is essentially surjective.

Corollary 9.19. Under the hypotheses of Corollary 9.17, let $C = \mathrm{Filt}^{\Pi}(E)$, let F be a $\mathcal{U}$-category, and consider the functor

$$\mathrm{Hom}(E, F)_\Pi \longrightarrow \mathrm{Hom}(C, F) \tag{9.19.1}$$

induced by “restriction to C”, $f \mapsto f|_C$, where the source of (9.19.1) is the category of Π-accessible functors from E to F (9.2). This functor is fully faithful. If F satisfies condition L_Π (9.1), then the functor (9.19.1) is an equivalence of categories.

The first assertion is proved as in Proposition 7.8, using Corollary 9.18 b). For the second, one constructs a quasi-inverse to (9.19.1) by associating to a functor $f : C \to F$ the functor

$$(X_i)_{i \in I} \longmapsto \varinjlim_i f(X_i)$$

from $\mathrm{Ind}(E_\Pi)$ to F, and then using Corollary 9.18 b) to obtain a functor $\bar{f} : E \to F$. It remains only to show that this last functor is Π-accessible. This is proved as in the analogous assertion 8.7.3.

Corollary 9.20. Let E be a category admitting a cardinal filtration and such that every object of E is accessible (cf.Proposition 9.16), and let F be a category. Then:

a) The category $\mathrm{Hom}(E, F)_{\mathrm{acc}}$ of accessible functors from E to F (9.2) is a $\mathcal{U}$-category. (N.B. the given categories E, F are assumed to be $\mathcal{U}$-categories; see 9.0.)

b) Suppose that F is stable under small filtered inductive limits. For every full subcategory C of E equivalent to a small category, with inclusion functor $i : C \to E$, consider the corresponding functor
$$i_! : \mathrm{Hom}(C, F) \longrightarrow \mathrm{Hom}(E, F)$$
(5.1). A functor $f : E \to F$ is accessible if and only if there exists a small subcategory C of E such that f lies in the essential image of the preceding functor $i_!$.

Proof.

a) It is enough to prove that, for every cardinal Π such that E satisfies L_Π, the full subcategory $\mathrm{Hom}(E, F)_\Pi$ of $\mathrm{Hom}(E, F)_{\mathrm{acc}}$ is a $\mathcal{U}$-category. It plainly suffices to check this for cardinals of the form 2^c, with c sufficiently large. But then this follows from Corollary 9.19, since $C = \mathrm{Filt}^\Pi(E)$ is essentially small and therefore $\mathrm{Hom}(C, F)$ is plainly a $\mathcal{U}$-category.

b) By transitivity in the formation of the functors $i_!$, it suffices in the statement to restrict to subcategories C of the form $\mathrm{Filt}^\Pi(E)$, where Π is as in Corollary 9.17. One then sees easily that the composite functor

$$\mathrm{Hom}(\mathrm{Filt}^\Pi(E), F) \longrightarrow \mathrm{Hom}(E, F)_\Pi \longrightarrow \mathrm{Hom}(E, F),$$

where the first arrow is quasi-inverse to (9.19.1) and the second is the inclusion, is none other than the functor $i_!$, up to isomorphism. Thus b) follows from Corollary 9.19.

Exercise 9.20.1. This exercise uses the notions of site and topos, developed in Exposés II and IV. Let E be a $\mathcal{U}$-topos, let $\mathcal{V}$ be a universe such that $\mathcal{U} \in \mathcal{V}$, let C be a small full generating subcategory of the $\mathcal{U}$-topos E, and let $\Pi_0 \in \mathcal{U}$ be an infinite cardinal such that $\Pi_0 \geq \mathrm{card}\,\mathrm{Fl}\,C$. For every cardinal $\Pi \geq \Pi_0$, $\Pi \in \mathcal{U}$, let $\mathrm{Filt}^\Pi(E)$ be the strictly full subcategory of E formed by objects X for which there exists a strict epimorphic family $(X_i \to X)_{i \in I}$ with target X, such that $\mathrm{card}\, I \leq \Pi$ and $X_i \in \mathrm{Ob}\, C$ for every $i \in I$.

a) Show that $(\mathrm{Filt}^{\Pi}(E))_{\Pi\geq\Pi_0}$ is a cardinal filtration of the $\mathcal{U}$-category E, and that one may choose Π_0 so that, for every $\Pi \geq \Pi_0$, one has inclusions

$$E_{\Pi} \subset \mathrm{Filt}^{\Pi}(E) \subset E_{\Pi^{\Pi_0}},$$

where for every cardinal c, E_c denotes the strictly full subcategory of E formed by the c-accessible objects.

b) Choose a cardinal $\Pi \geq \Pi_0$ such that $\Pi = \Pi^{\Pi_0}$, for example a cardinal of the form 2^c with $c > \Pi_0$, and put $C = \mathrm{Filt}^{\Pi}(E)$. Show that one has an equivalence of categories

$$\mathrm{Ind}(C)_{\Pi} \xrightarrow{\sim} E$$

(with the notation $\mathrm{Ind}(C)_{\Pi}$ of Corollary 9.18).

c) Keep the notation of b), let $\mathcal{V}$ be a universe such that $\mathcal{U} \subset \mathcal{V}$, and let $\widetilde{E}_{\mathcal{V}}$ be the $\mathcal{V}$-topos of $\mathcal{V}$-sheaves on the site E endowed with its canonical topology. Denote by $\mathrm{Ind}(C,\mathcal{V})_{\Pi}$ the category of $\mathcal{V}$-Ind-objects of C indexed by preordered sets of indices which are *large with respect to* Π. Show that one has an equivalence of categories

$$\mathrm{Ind}(C,\mathcal{V})_{\Pi} \xrightarrow{\sim} \widetilde{E}_{\mathcal{V}}.$$

d) Let $c \in \mathcal{V}$ be a cardinal, and let $\mathrm{Ind}(E,\mathcal{V})'_c$ be the full subcategory of $\mathrm{Ind}(E,\mathcal{V})$ formed by $\mathcal{V}$-ind-objects of E indexed by a preordered set which is large with respect to every cardinal $< c$. Taking $c = \mathrm{card}\,\mathcal{U}$, show that one has an equivalence of categories

$$\mathrm{Ind}(E,\mathcal{V})'_c \xrightarrow{\sim} \widetilde{E}_{\mathcal{V}}.$$

9.21

This section develops the technical preliminaries for the proof of Theorem 9.22 below, which is the main result of Section 9. Let

$$p : E \longrightarrow B \tag{9.21.1}$$

be a fibred functor, where B is a small category, and where the inverse image functors

$$f^* : E_{\beta} \longrightarrow E_{\alpha}$$

associated with arrows $f : \alpha \to \beta$ of B are accessible (9.2). In particular, the fibre categories E_{α} ($\alpha \in \mathrm{ob}\,B$) satisfy condition L (9.1). Since B is small, there is a cardinal $\Pi'_0 \in \mathcal{U}$ such that all the categories E_{α} satisfy $L_{\Pi'_0}$. Let $\Pi_0 \in \mathcal{U}$ be an infinite cardinal $> \Pi'_0$, so that

$$\Pi_0 > \Pi'_0, \qquad E_{\alpha} \text{ satisfies } L_{\Pi'_0} \text{ for all } \alpha \in \mathrm{ob}\,B. \tag{9.21.2}$$

Moreover, the hypotheses just made immediately imply the existence of a cardinal $c \in \mathcal{U}$ satisfying:

$$\begin{cases} \text{a)} & c \text{ is infinite,} \\ \text{b)} & c \geq \mathrm{card}\,\mathrm{Fl}\,B, \\ \text{c)} & \text{for every arrow } f : \alpha \to \beta \text{ of } B,\ f^* : E_{\beta} \to E_{\alpha} \text{ is } c\text{-accessible.} \end{cases} \tag{9.21.3}$$

Suppose also that for each $\alpha \in \operatorname{ob} B$ one can find a full subcategory C_α of E_α satisfying:

$$\begin{cases} \text{a)} & \text{For every } X \in \operatorname{ob} C_\alpha,\ X \text{ is } c\text{-accessible in } E_\alpha. \\ \text{b)} & \text{For every } X \in \operatorname{ob} E_\alpha,\ \text{there is an isomorphism } X \simeq \varinjlim_I X_i \\ & \text{in } E_\alpha,\ \text{where the } X_i \text{ lie in } C_\alpha \text{ and } I \text{ is large with respect to } c. \end{cases} \tag{9.21.4}$$

Let Π be a cardinal such that

$$\Pi \geq \Pi_0, \qquad \text{and hence} \qquad \Pi > \Pi_0', \tag{9.21.5}$$

and for every $\alpha \in \operatorname{ob} B$ let

$$C_\alpha^\Pi \subset E_\alpha \tag{9.21.6}$$

be the full subcategory formed by objects which can be represented in the form $\varinjlim_I X_i$, where I is an ordered set large with respect to Π_0', $\operatorname{card} I \leq \Pi$, and the X_i lie in C_α. It is then clear, by 7.5.2, that C_α^Π is essentially small, i.e.equivalent to a small category.

Let

$$F = \operatorname{Hom}_B(B, E) \tag{9.21.7}$$

be the category of sections of E over B, and let

$$F^\Pi \subset F \tag{9.21.8}$$

be the strictly full subcategory of F formed by sections $X : \alpha \mapsto X(\alpha)$ such that, for every $\alpha \in \operatorname{ob} B$,

$$X(\alpha) \in C_\alpha^\Pi.$$

Since the C_α^Π are essentially small, it is clear that the same is true of F^Π. We shall show that this category is generating, and more precisely:

Lemma 9.21.9. Under the preceding hypotheses and with the preceding notation, the following hold.

(i) Every object of F is accessible (9.3). If d is a cardinal $\geq \Pi_0'$, and if $\alpha \mapsto X(\alpha)$ is an element of F such that, for every α, $X(\alpha)$ is d-accessible in E_α, then X is d-accessible in F.

(ii) Suppose that $\Pi \leq c$, or that Π_0' is finite (i.e.that the E_α are stable under small filtered inductive limits). Then every object X of F is isomorphic to an object of the form $\varinjlim_I X_i$, where the X_i lie in F^Π and I is large with respect to Π.

To prove (i), note first that by (9.21.4) and Corollary 9.9, every object of E_α is accessible. On the other hand, F satisfies condition $L_{\Pi_0'}$, by the following lemma.

Lemma 9.21.10. Let $p : E \to B$ be a fibred functor, let $F = \operatorname{Hom}_B(B, E)$, let I be a category, and let $i \mapsto X_i$ be a functor from I to F. In order that $\varinjlim_I X_i$ be representable in F, it is enough that, for every $\alpha \in \operatorname{ob} B$, $\varinjlim_I X_i(\alpha)$ be representable in the fibre category E_α. When this is the case, $\varinjlim_I X_i$ is computed "argument by argument". In particular, if the fibre categories satisfy condition $L_{\Pi_0'}$, where Π_0' is a fixed cardinal, then so does F.

We leave the easy proof of Lemma 9.21.10 to the reader, merely noting that it is convenient to use the following immediate result.

Corollary 9.21.10.1. With the hypotheses and notation of Lemma 9.21.10 for $p : E \to B$ and for I, for every $\alpha \in \operatorname{ob} B$, the inclusion functor $E_\alpha \to E$ commutes with inductive limits of type I.

Returning to the general hypotheses of Lemma 9.21.9, let X be an object of F. Since for every $\alpha \in \operatorname{ob} B$, $X(\alpha)$ is accessible in E_α, there exists a cardinal $d \in \mathcal{U}$ such that for every α, $X(\alpha)$ is d-accessible in E_α. We may choose $d \geq \Pi'_0$, so that F satisfies L_d by Lemma 9.21.10. Using Lemma 9.21.10 again to compute inductive limits $\varinjlim_I Y_i$ in F, with I large with respect to d, we immediately see that X is d-accessible. This proves (i).

We now prove Lemma 9.21.9(ii) in several steps, namely 9.21.11 through 9.21.16.

Let
$$X : \alpha \longmapsto X(\alpha)$$
be an object of F. By (9.21.4 b)), for every $\alpha \in \operatorname{ob} B$ one can find an ordered set I_α large with respect to c, and an isomorphism
$$X(\alpha) = \varinjlim_{i \in I_\alpha} X(\alpha)_i,$$
where $i \mapsto X(\alpha)_i$ is an inductive system of type I_α in E_α, with $X(\alpha)_i$ lying in C_α. Consider the product ordered set
$$I = \prod_\alpha I_\alpha.$$
It is clear that it is large with respect to c, and that the preceding inductive systems give rise to inductive systems in the categories E,
$$i \longmapsto X(\alpha)_i \in \operatorname{ob} C_\alpha, \qquad i \in \operatorname{ob} I, \tag{9.21.11}$$
indexed by the same ordered set I, and to isomorphisms
$$X(\alpha) \xrightarrow{\sim} \varinjlim_I X(\alpha)_i \quad \text{in } E_\alpha, \tag{9.21.12}$$
for all $\alpha \in \operatorname{ob} B$. In what follows we fix data (9.21.11) and (9.21.12).

Lemma 9.21.13. Under the preceding hypotheses, one can find a map $\varphi : I \to I$ such that $\varphi(i) \geq i$ for every $i \in I$, and a map
$$(i, f) \longmapsto \lambda(i, f)$$
from $I \times \operatorname{Fl} B$ to $\operatorname{Fl} E$, satisfying the following conditions.

a) For $i \in I$ and $f : \alpha \to \beta$ in $\operatorname{Fl} B$, $\lambda(i, f)$ is an f-morphism from $X(\alpha)_i$ to $X(\beta)_{\varphi(i)}$, making commutative the diagram
$$\begin{tikzcd}
X(\alpha) \arrow[r, "X(f)"] & X(\beta) \\
 & X(\beta)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[uu] \arrow[ur, "\lambda(i,f)"'] &
\end{tikzcd}$$
$$\alpha \xrightarrow[f]{} \beta.$$
Here the vertical arrows are the canonical morphisms deduced from (9.21.12).

b) For every $i \in I$ and every $f : \alpha \to \beta$ in $\mathrm{Fl}\, B$, one has commutativity in

$$\begin{tikzcd}
 & X(\beta)_{\varphi^2(i)} \\
X(\alpha)_{\varphi(i)} \arrow[ur, "\lambda(\varphi(i),f)"] & X(\beta)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[u] \arrow[ur, "\lambda(i,f)"'] & \\
\alpha \arrow[r, "f"'] & \beta.
\end{tikzcd}$$

c) For every pair of consecutive arrows

$$\alpha \xrightarrow{f} \beta \xrightarrow{g} \gamma$$

of B, one has commutativity in the following diagram:

$$\begin{tikzcd}
X(\alpha) \arrow[r, "X(f)"] & X(\beta) \arrow[r, "X(g)"] & X(\gamma) \\
 & & X(\gamma)_{\varphi^2(i)} \arrow[u] \\
 & X(\beta)_{\varphi(i)} \arrow[uu] \arrow[ur, "\lambda(\varphi(i),g)"'] & X(\gamma)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[uuu] \arrow[ur, "\lambda(i,f)"] \arrow[urr, "\lambda(i,gf)"'] & & \\
\alpha \arrow[r, "f"'] & \beta \arrow[r, "g"'] & \gamma.
\end{tikzcd}$$

Again, the vertical arrows are deduced from (9.21.12).

Notice also that the commutativity of the two upper trapezia in b) is already contained in a); thus b) is really asserting the commutativity of the lower triangle of the diagram considered.

We prove Lemma 9.21.13. Let $i \in I$, and let $f : \alpha \to \beta$ be an arrow of B. Consider the composite

$$X(\alpha)_i \longrightarrow X(\alpha) \xrightarrow{X(f)} X(\beta) \simeq \varinjlim_j X(\beta)_j.$$

It defines a morphism in E_α

$$X(\alpha)_i \longrightarrow f^*(X(\beta)) \simeq f^*\left(\varinjlim_j X(\beta)_j\right) \simeq \varinjlim_j f^*(X(\beta)_j),$$

where the last equality follows from the fact that f^* is c-accessible (9.21.3 c)) and that I is large with respect to c. By (9.21.4 a)), since $X(\alpha)_i$ lies in C_α, the morphism under consideration factors through one of the $f^*(X(\beta)_j)$; here j *a priori* depends on i and on $f \in \mathrm{Fl}\, B$. But by (9.21.3 b)), and since I is large with respect to c, we may choose j independently of f; write

$j = \varphi(i)$. Since I is filtered, we may suppose $\varphi(i) \geq i$. We thus obtain, for $i \in I$ and $f \in \operatorname{Fl} B$, a morphism

$$X(\alpha)_i \longrightarrow f^*(X(\beta)_{\varphi(i)}),$$

or equivalently an f-morphism

$$\lambda(i,f) : X(\alpha)_i \longrightarrow X(\beta)_{\varphi(i)}$$

which, by construction, makes the diagram in a) commute.

Next fix i, f, g and consider the lower triangle of the diagram in 9.21.13 c). It is not clear that this triangle commutes, but the two composites

$$X(\alpha)_i \rightrightarrows X(\gamma)_{\varphi^2(i)}$$

become equal after composition with $X(\gamma)_{\varphi^2(i)} \to X(\gamma)$, as follows from the commutativity of the two upper trapezia and of the outer trapezium; these are respectively the diagrams $D(f)$, $D(g)$, and $D(gf)$. Since $X(\alpha)_i$ is c-accessible (9.21.4 a)) and

$$X(\gamma) \simeq \varinjlim_{j \in I} X(\gamma)_j$$

with I large with respect to c, one can find an element $j \geq \varphi^2(i)$ of I such that the two arrows just considered become equal after composition with

$$X(\gamma)_{\varphi^2(i)} \longrightarrow X(\gamma)_j.$$

A priori j depends on i, f, g. But for fixed i, the set of possible pairs f, g has cardinal $\leq c^2 = c$, by (9.21.3 a) and b)); since I is large with respect to c, we may choose j independently of f and g. Write $j = \varphi'(i) \geq \varphi^2(i) \geq \varphi(i)$. Define, for every $i \in I$ and every $f : \alpha \to \beta$, $\lambda'(i,f)$ to be the composite f-morphism

$$X(\alpha)_i \longrightarrow X(\beta)_{\varphi(i)} \longrightarrow X(\beta)_{\varphi'(i)},$$

where the second arrow is the transition morphism. It is then immediate from the construction that (φ', λ') satisfies conditions 9.21.13 a) and c). Proceeding in the same way for condition b), one sees that φ' can be chosen so that this condition is also satisfied. This completes the proof of Lemma 9.21.13.

9.21.14

Let

$$J \subset I$$

be a subset of I satisfying the following conditions:

a) J is filtered;

b) for every $j \in J$, one has $\varphi(j) \in J$;

c) for every $\alpha \in \operatorname{ob} B$, the object

$$X_J(\alpha) = \varinjlim_{i \in J} X(\alpha)_i$$

is representable in E_α.

Conditions a) and c) are satisfied in particular if J is large with respect to Π'_0 (9.21.2). It then follows from Corollary 9.21.10.1 that, for every $\alpha \in \operatorname{ob} B$, $X_J(\alpha)$ is the inductive limit $\varinjlim_{i\in J} X(\alpha)_i$ in E. Now for an arrow $f : \alpha \to \beta$ of B, the arrows $\lambda(i, f)$, with i varying in J, define by 9.21.13 b) a morphism of ind-objects from $(X(\alpha)_i)_{i\in J}$ to $(X(\beta)_i)_{i\in J}$. Hence they induce a homomorphism

$$X_J(f) : X_J(\alpha) \longrightarrow X_J(\beta)$$

on inductive limits, which is visibly an f-homomorphism. Using 9.21.13 c), one finds the transitivity relations

$$X_J(gf) = X_J(g)X_J(f),$$

so that a section $X_J \in \operatorname{ob} F$ of E over B has been defined. Finally, 9.21.13 a) shows that the canonical homomorphisms

$$X_J(\alpha) \longrightarrow X(\alpha), \qquad \alpha \in \operatorname{ob} B,$$

are functorial in α; thus we have a canonical homomorphism

$$u_J : X_J \longrightarrow X.$$

Moreover, if

$$J' \supset J$$

is another subset of I satisfying conditions a), b), c) above, one obtains a canonical homomorphism

$$u_{J',J} : X_J \longrightarrow X_{J'},$$

so that the X_J form an inductive system in F, parametrized by the set, ordered by inclusion, of subsets J of I satisfying the conditions under consideration. Finally, the homomorphisms u_J above define a homomorphism from this inductive system to the constant inductive system defined by X.

9.21.15

Let K be a set of subsets J of I satisfying conditions a), b), c) of 9.21.14, and suppose that K is filtered and has union I. Then it is clear that

$$\varinjlim_{J\in K} X_J \xrightarrow{\sim} X,$$

using Corollary 9.21.10.1 to reduce the verification to an isomorphism argument by argument.

For example, take K to be the set of all subsets J of I which are large with respect to Π'_0, stable under φ, and such that $\operatorname{card} J \leq \Pi$. Then, by the definition (9.21.6) of C^{Π}_{α}, one has

$$X_J(\alpha) \in \operatorname{ob} C^{\Pi}_{\alpha}$$

for every $\alpha \in \operatorname{ob} B$, and hence $X_J \in \operatorname{ob} F^{\Pi}$. Therefore Lemma 9.21.9(ii) will be proved if we show that K is large with respect to Π (hence filtered) and has union I. For this it plainly suffices to prove that every subset S of I with $\operatorname{card} S \leq \Pi$ is contained in some $J \in K$. This follows from the preliminary hypothesis stated in Lemma 9.21.9(ii), and from the following lemma.

Lemma 9.21.16. Let I be an ordered set, let Π be an infinite cardinal, and let $\varphi : I \to I$ be a map.

a) Suppose that I is filtered. Then for every subset S of I such that $\operatorname{card} S \leq \Pi$, there exists a filtered subset $J \supset S$ of I such that $\varphi(J) \subset J$ and $\operatorname{card} J \leq \Pi$.

b) Let Π_0' be a cardinal $< \Pi$, and suppose that I is large with respect to Π. Then for every subset S of I such that $\operatorname{card} S \leq \Pi$, there exists a subset $J \supset S$ of I which is large with respect to Π_0', such that $\varphi(J) \subset J$ and $\operatorname{card} J \leq \Pi$.

We prove b); the proof of a) is analogous and simpler. Let P be the set of subsets of I of cardinal $\leq \Pi$, and let $T \mapsto i_T : P \to I$ be a map such that, for $T \in P$, i_T is an upper bound for T in I. The existence of this map is simply the assertion that I is large with respect to Π. Let A be a well-ordered set such that $\operatorname{card} A = \Pi$ and such that, for every $a \in A$, the set of $a' \leq a$ has cardinal $< \Pi$. In particular, since $\Pi_0' < \Pi$, it follows that A is large with respect to Π_0'.

Define, by transfinite recursion, maps

$$S \longmapsto S_a : P \to P \qquad (a \in A)$$

as follows:

$$S_0 = S, \qquad S_{a+1} = S_a \cup \varphi(S_a) \cup \{i_{S_a}\}, \qquad S_a = \bigcup_{a'<a} S_{a'} \quad \text{if } a \text{ is a limit ordinal.}$$

Then put, for $S \in P$,

$$J(S) = \bigcup_{a \in A} S_a.$$

It is clear that $J(S) \supset S$, that $J(S)$ is stable under φ, that $\operatorname{card} J(S) \leq \Pi$ (since $\operatorname{card} A = \Pi$), and finally that $J(S)$ is large with respect to Π_0' (using that A is large with respect to Π_0'). This proves Lemma 9.21.16 and completes the proof of Lemma 9.21.9.

We also note the following variant of Lemma 9.21.9.

Lemma 9.21.17. The notation is that of Lemma 9.21.9. Denote by F' the full subcategory

$$F' = \operatorname{Hom}_B^{\text{cart}}(B, E)$$

of $F = \operatorname{Hom}_B(B, E)$ formed by the cartesian sections of E over B. Then:

(i) Every object of F' is accessible. More precisely, let $\alpha \mapsto X(\alpha)$ be an element of F', and let d be a cardinal such that $d \geq \Pi_0'$, $d \geq c$, and such that for every $\alpha \in \operatorname{ob} B$, $X(\alpha)$ is a d-accessible object of E_α. Then X is a d-accessible object of F'.

(ii) Suppose that $\Pi \leq c$ and that the functors $f^* : E_\beta \to E_\alpha$ are Π_0'-accessible, or else that the categories E_α are stable under small filtered inductive limits and that the functors $f^* : E_\beta \to E_\alpha$ commute with these inductive limits. Then every object X of F' is isomorphic to an object of the form $\varinjlim_I X_i$, where for every $i \in I$, X_i is an object of the full subcategory

$$F'^{\Pi} = F' \cap F^{\Pi}$$

of F', and where I is large with respect to Π.

The proof being entirely analogous to that of Lemma 9.21.9, we merely indicate where modifications of that proof are necessary. First note the following.

Lemma 9.21.18. The statement of Lemma 9.21.10 remains valid when one replaces $F = \mathrm{Hom}_B(B, E)$ by $F' = \mathrm{Hom}_B^{\mathrm{cart}}(B, E)$, provided that one assumes that the inverse image functors

$$f^* : E_\beta \longrightarrow E_\alpha$$

commute with inductive limits of type I.

Thus, under the hypotheses of Lemma 9.21.17, if d is a cardinal such that $d \geq c$ and $d \geq \Pi_0'$, it follows from what precedes and from hypothesis (9.21.3 c)) that for every ordered set I large with respect to d, inductive limits of type I are representable in F' and are computed argument by argument. Lemma 9.21.17(i) follows by an immediate argument.

To prove (ii), one must supplement Lemma 9.21.13.

Lemma 9.21.19. Under the hypotheses of Lemma 9.21.13, suppose that $X \in \mathrm{ob}\, F'$, i.e.that X is a cartesian section of E over B. Then the conclusion can be strengthened as follows: there exists a function μ which assigns, to every $i \in I$ and every arrow $f : \alpha \to \beta$ of B, a morphism

$$\mu(i, f) : f^* X(\beta)_i \longrightarrow X(\alpha)_{\varphi(i)}$$

in E_α, in such a way that:

d) For every $i \in I$ and every arrow $f : \alpha \to \beta$ of B, the two following diagrams commute:

$$\begin{tikzcd}
 & f^*X(\beta)_{\varphi^2(i)} & & X(\alpha)_{\varphi^2(i)} & \\
X(\alpha)_{\varphi(i)} \arrow[ur, "\lambda(\varphi(i),f)"] & & & & f^*X(\beta)_{\varphi(i)} \arrow[ul, "\mu(\varphi(i),f)"'] \\
 & f^*X(\beta)_i \arrow[uu] \arrow[ul, "\mu(i,f)"] & & X(\alpha)_i \arrow[uu] \arrow[ur, "\lambda(i,f)"'] &
\end{tikzcd}$$

where the unlabeled arrows are the evident transition maps.

To prove this, one proceeds as in the proof of 9.21.13, by considering the composite morphism

$$f^*(X(\beta)_i) \longrightarrow f^*(X(\beta)) \xrightarrow{X(f)^{-1}} X(\alpha),$$

where, as already in the notation for condition d), an f-morphism $R \to S$ of E is identified with the corresponding morphism $R \to f^*(S)$ in E_α. Since

$$X(\alpha) = \varinjlim_j X(\alpha)_j$$

and I is large with respect to c, the preceding morphism can be factored through one of the $X(\alpha)_j$. Here j *a priori* depends on i and on f, but can be chosen independently of f; write it $\psi(i)$. Enlarging the function φ of Lemma 9.21.13 if necessary, we may suppose that $\psi = \varphi$. Proceeding as for conditions b) and c) of Lemma 9.21.13, we see that, after enlarging φ again if necessary, the two diagrams of Lemma 9.21.19 d) may be assumed commutative.

9.21.20

With φ chosen as in Lemma 9.21.19, resume the argument of 9.21.14. One must, however, assume that J satisfies, besides conditions a), b), c) stated there, the following condition:

d) For every arrow $f : \alpha \to \beta$ in B, the functor $f^* : E_\beta \to E_\alpha$ commutes with inductive limits of type J.

I claim that, under this additional condition, the section X_J of E over B is cartesian; that is, for every arrow $f : \alpha \to \beta$ of B, the morphism

$$X_J(\alpha) \longrightarrow f^* X_J(\beta) \tag{$*$}$$

is an isomorphism. Indeed, by condition d) above, the target of this morphism is identified with

$$\varinjlim_{i \in J} f^*(X(\beta)_i),$$

and the morphism itself is obtained by passing to $\varinjlim_J$ from the morphism of ind-objects in E_α

$$(X(\alpha)_i)_{i \in J} \longrightarrow (f^* X(\beta)_i)_{i \in J},$$

deduced from the morphisms

$$\lambda(i, f) : X(\alpha)_i \longrightarrow f^*(X(\beta)_{\varphi(i)})$$

for $i \in J$. But the conditions made explicit in Lemma 9.21.19 ensure that this morphism of ind-objects is in fact an isomorphism, with inverse obtained from the morphism deduced from the system of the $\mu(i, f)$, $i \in J$. This proves that $(*)$ is an isomorphism, hence that $X_J \in \operatorname{ob} F'$.

9.21.21

We now assume that the functors $f^* : E_\beta \to E_\alpha$ are Π_0'-*accessible*. Then the conditions on J considered in 9.21.20 are satisfied if J is large with respect to Π_0', stable under φ, and has $\operatorname{card} J \leq \Pi$. The proof of Lemma 9.21.17 is then completed as in 9.21.15.

Theorem 9.22. Let $p : E \to B$ be a fibred functor, where B is a category essentially equivalent to a small category, and where for every arrow $f : \alpha \to \beta$ of B, the inverse image functor

$$f^* : E_\beta \longrightarrow E_\alpha$$

is accessible (9.2). Suppose further that, for every $\alpha \in \operatorname{ob} B$, the fibre category E_α admits a cardinal filtration (9.12), and that every object of E_α is accessible (9.3). Then each of the categories

$$F = \operatorname{Hom}_B(B, E), \qquad F' = \operatorname{Hom}_B^{\mathrm{cart}}(B, E)$$

admits a small full subcategory which is generating by strict epimorphisms (7.1), and all of its objects are accessible.

It is immediate that the statement is not essentially changed when B is replaced by a full subcategory B_0 such that the inclusion functor $B_0 \to B$ is an equivalence, and E is replaced by $E_0 = E \times_B B_0$. Thus we may suppose that B is a small category.

For every $\alpha \in \operatorname{ob} B$, let

$$(\operatorname{Filt}^{\Pi}(E_\alpha))_{\Pi \geq \Pi_\alpha}$$

be a cardinal filtration of E_α. Let $c_0 \in \mathcal{U}$ be a cardinal satisfying:

$$\begin{cases} c_0 \geq \operatorname{Sup}_{\alpha \in \operatorname{ob} B} \Pi_\alpha, \qquad c_0 \geq \operatorname{card} \operatorname{Fl} B; \\ \text{for every arrow } f : \alpha \to \beta \text{ of } B,\ f^* : E_\beta \to E_\alpha \text{ is } c_0\text{-accessible;} \\ \text{for every } \alpha \in \operatorname{ob} B,\ \text{the objects of } \operatorname{Filt}^{\Pi_\alpha}(E_\alpha) \text{ are } c_0\text{-accessible.} \end{cases} \tag{9.22.1}$$

Put

$$c = 2^{c_0},$$

so that $c > c_0$, and for every $\alpha \in \operatorname{ob} B$ put

$$C_\alpha = \operatorname{Filt}^{c}(E_\alpha).$$

I claim that the preliminary hypotheses of Lemmas 9.21.9 and 9.21.17 are satisfied, taking $\Pi = \Pi_0 = c$. This is clear for (9.21.2) and (9.21.3). Condition (9.21.4 a)) follows from Corollary 9.17, and (9.21.4 b)) follows from condition 9.12 c) for a cardinal filtration. Notice also that condition 9.12 b) in the definition of a cardinal filtration implies that, for every $\alpha \in \operatorname{ob} B$, one has

$$C_\alpha^{\Pi} = C_\alpha = \operatorname{Filt}^{c}(E_\alpha),$$

where C_α^{Π} is defined in (9.21.6). Theorem 9.22 therefore follows from Lemmas 9.21.9 and 9.21.17. More precisely, we have proved:

Corollary 9.23. Under the hypotheses of Theorem 9.22, if $c_0 \in \mathcal{U}$ is a cardinal satisfying (9.22.1), and if $c = 2^{c_0}$, then the subcategory F^c of F (respectively ${F'}^c$ of F') formed by objects X such that $X(\alpha) \in \operatorname{ob} \operatorname{Filt}^{c}(E_\alpha)$ for every $\alpha \in \operatorname{ob} B$ is essentially small and generating by strict epimorphisms. More precisely, every object X of F (respectively of F') is isomorphic to an object of the form $\varinjlim_I X_i$, where the X_i lie in F^c (respectively in ${F'}^c$), and where I is an ordered set large with respect to c.

Under slightly stronger hypotheses in Theorem 9.22, one can also make Theorem 9.22 and Corollary 9.23 considerably more precise.

Corollary 9.24. Under the hypotheses of Theorem 9.22, suppose that each of the categories E_α ($\alpha \in \operatorname{ob} B$) is stable under small filtered inductive limits, and that for every $\Pi \geq \Pi_\alpha$, $\operatorname{Filt}^{\Pi}(E_\alpha)$ is stable under filtered inductive limits indexed by filtered ordered sets I such that $\operatorname{card} I \leq \Pi$; this is a slight strengthening of condition 9.12 b) for cardinal filtrations. In the case where F' is considered, suppose moreover that, for every arrow $f : \alpha \to \beta$ of B, the functor

$$f^* : E_\beta \to E_\alpha$$

commutes with small filtered inductive limits. Finally let $c_0 \in \mathcal{U}$ be a cardinal satisfying (9.22.1), and for every cardinal $\Pi \geq c = 2^{c_0}$ consider the strictly full subcategory F^{Π} (respectively ${F'}^{\Pi}$) of F (respectively of F') formed by the X such that

$$X(\alpha) \in \operatorname{Filt}^{\Pi}(E_\alpha)$$

for every $\alpha \in \operatorname{ob} B$. Then the subcategories under consideration define a cardinal filtration (9.12) of F (respectively of F').

One must verify conditions a), b), c) of 9.12. Conditions a) and b) follow from the analogous conditions for the given cardinal filtrations of the E_α, and from Lemma 9.21.10 (respectively Lemma 9.21.18, taking into account the second condition in (9.22.1)). It remains to prove c). The first part of c) follows at once from Lemma 9.21.9 (respectively Lemma 9.21.17), by taking $\Pi'_0 = 0$ and $\Pi_0 = c$.

It remains to prove that if $\Pi' \geq \Pi \geq c$ are two cardinals, then for every X in $F^{\Pi'}$ (respectively ${F'}^{\Pi'}$) one has

$$X \simeq \varinjlim_{i \in K} X_i,$$

where the X_i lie in F^{Π} (respectively in ${F'}^{\Pi}$), where K is large with respect to Π, and where $\operatorname{card} K \leq {\Pi'}^{\Pi}$. To this end, resume the proofs of Lemma 9.21.9(ii) and Lemma 9.21.17(ii). We may suppose that each I_α has cardinal $\leq \Pi'$ by condition 9.12 c) on the cardinal filtration of E_α. Hence

$$\operatorname{card} I \leq ({\Pi'}^{\Pi})^{\operatorname{card} \operatorname{ob} B} \leq ({\Pi'}^{\Pi})^c = {\Pi'}^{\Pi c} = {\Pi'}^{\Pi}.$$

The indexing set K used in 9.21.15, respectively 9.21.21, is the set of subsets J of I which are filtered (i.e.large with respect to $\Pi'_0 = 0$), stable under φ, and such that $\operatorname{card} J \leq \Pi$. It is then immediate that

$$\operatorname{card} K \leq (\operatorname{card} I)^{\Pi} \leq ({\Pi'}^{\Pi})^{\Pi} = {\Pi'}^{\Pi^2} = {\Pi'}^{\Pi},$$

which completes the proof of Corollary 9.24.

To finish, it is appropriate to give a statement freed from the somewhat technical hypotheses of Theorem 9.22, by replacing them, for the E_α, by the conjunction of the hypotheses occurring in Proposition 9.11, which ensure accessibility of the objects of the E_α, and in Proposition 9.13, which ensure the existence of a cardinal filtration in E_α.

Corollary 9.25. Let $p : E \to B$ be a fibred functor, where B is a category equivalent to a small category, and where for every arrow $f : \alpha \to \beta$ of B, the inverse image functor

$$f^* : E_\beta \to E_\alpha$$

is accessible (9.2). Suppose further that, for every $\alpha \in \operatorname{ob} B$, the fibre category E_α satisfies the following conditions:

a) E_α admits a small full subcategory generating by strict epimorphisms (7.1).

b) E_α is stable under fibre products, under kernels of double arrows, under sums of two objects, under cokernels of double arrows, and finally under small filtered inductive limits.[1]

c) The functor $\operatorname{Ker}(u, v)$ on the category of double arrows of E_α is accessible; for example, it commutes with small filtered inductive limits.

d) Every strict epimorphism of E_α (10.2) is a universal strict epimorphism.

Under these hypotheses, each of the categories

$$F = \operatorname{Hom}_B(B, E), \qquad F' = \operatorname{Hom}_B^{\text{cart}}(B, E)$$

admits a small subcategory generating by strict epimorphisms, and all of its objects are accessible (9.3).

[1]This last condition is probably unnecessary; cf.9.13.2.

Moreover, Corollary 9.24 gives:

Corollary 9.26. Under the hypotheses of 9.25, and in the case where one considers $F' = \mathrm{Hom}_B^{\mathrm{cart}}(B, E)$, suppose that the functors

$$f^* : E_\beta \to E_\alpha$$

commute with small filtered inductive limits. Then F (respectively F') admits a cardinal filtration, which can be made explicit by the procedure of Corollary 9.24.

Continuation anchor. The next part of Exposé I is Section 10, the glossary of terms used in the preceding sections.

Exposé I. Presheaves

Section 10 and Appendix II: Glossary; Universes

A. Grothendieck and J.-L. Verdier; Appendix attributed to N. Bourbaki

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 10 and the appended note "Universes." The range corresponds to source lines 7424–8921 of the Orgogozo–Laszlo TeX snapshot. The translation keeps the period terminology where it is still mathematically useful: *strict epimorphism*, *effective equivalence relation*, *direct factor*, and *base-changeable* for French *quarrable*. A few evident typographical slips in the source have been silently normalized where the formulas force the correction.

10 Glossary

For the convenience of the reader, we collect here the definitions of several terms used in the preceding sections. Throughout this section, C denotes a category.

10.1 Cartesian, cocartesian

A diagram in C

$$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \\ C \arrow[r] & D \end{tikzcd}$$

is called *cartesian* if it is commutative and if the canonical morphism from A to the fiber product $C \times_D B$ is an isomorphism. It is called *cocartesian* if the corresponding diagram in the opposite category C° is cartesian.

10.2 Epimorphism, strict epimorphism

See 10.3.

10.3 Epimorphic family, strictly epimorphic family

A family

$$f_i : A_i \longrightarrow B, \qquad i \in I,$$

of arrows with common target in C is called an *epimorphic family* if, for every object T of C, the map induced by the f_i,

$$\mathrm{Hom}_C(B,T) \longrightarrow \prod_i \mathrm{Hom}_C(A_i,T), \tag{10.3.1}$$

is injective.

The family is called *strictly epimorphic* if the image of (10.3.1) consists of those families (g_i) such that, for every object R of C, every pair of indices $i, j \in I$, and every pair of arrows $u : R \to A_i$, $v : R \to A_j$ satisfying $f_i u = f_j v$, one also has $g_i u = g_j v$. When the fiber products

$A_i \times_B A_j$ are representable for $i, j \in I$, this is equivalent to saying that the image of (10.3.1) consists of those (g_i) such that, for all i, j,

$$g_i \operatorname{pr}_1 = g_j \operatorname{pr}_2,$$

where $\operatorname{pr}_1, \operatorname{pr}_2$ are the two projections from $A_i \times_B A_j$. The family $(f_i)_{i\in I}$ is called an *effective epimorphic family* if this condition is satisfied, that is, if it is strictly epimorphic and the fiber products $A_i \times_B A_j$ are representable.

The family $(f_i)_{i\in I}$ is called *universally epimorphic* (respectively, *universally effectively epimorphic*) if the morphisms f_i are base-changeable (10.7) and if, for every arrow $B' \to B$, the family of arrows obtained from (f_i) by base change along $B' \to B$ is epimorphic (respectively, effective epimorphic).

The notion of a *universally strictly epimorphic* family will be defined in II 2.5–2.6. Note that when the morphisms f_i are base-changeable, for example if fiber products are representable in C, this notion agrees with that of a universally effectively epimorphic family, using II 2.4. In general, between the six variants of epimorphicity considered here, one has the following logical implications:

universally effective epimorphic ⇒ effective epimorphic ⇒ strictly epimorphic ⇒ epimorphic,
universally effective epimorphic ⇒ universally strictly epimorphic ⇒ strictly epimorphic,
universally strictly epimorphic ⇒ universally epimorphic ⇒ epimorphic.

A morphism $f : A \to B$ of C is called an *epimorphism* (respectively, a strict epimorphism, effective epimorphism, universal epimorphism, universally effective epimorphism, universally strict epimorphism) if the one-element family consisting of f is epimorphic (respectively, ...).

10.4 Monomorphic family, strictly monomorphic family

A family of arrows $f_i : A_i \to B$ of C is called *monomorphic* (respectively, strictly monomorphic, ...) if, as a family of arrows in the opposite category C°, it is epimorphic (respectively, strictly epimorphic, ...). An arrow $f : A \to B$ of C is called a *monomorphism* (respectively, a strict monomorphism, ...) if the one-element family consisting of f is monomorphic (respectively, strictly monomorphic, ...), that is, if f, viewed as an arrow of C°, is an epimorphism (respectively, a strict epimorphism, ...).

10.5 Monomorphism, strict monomorphism

See 10.4.

10.6 Projector, image of a projector, direct factor of an object

Let $f : X \to X$ be an endomorphism of an object of C. We say that f is a *projector* if $f^2 = f$. Then the pair (f, id_X) admits a representable cokernel X' if and only if it admits a representable kernel X''; and when this is so, there is a unique isomorphism $u : X' \to X''$ in C such that $iup = f$, where $p : X \to X'$ and $i : X'' \to X$ are the canonical morphisms. The representability of the kernel, or of the cokernel, is also equivalent to the existence of morphisms i, p and of an isomorphism u such that $iup = f$ and $pi = u^{-1}$. In particular, this property is preserved by any functor.

One then usually identifies X' and X'', and calls it the *image of the projector* f. We say that the projector f *admits an image* if $\mathrm{Ker}(f, \mathrm{id}_X)$ is representable, equivalently if $\mathrm{Coker}(f, \mathrm{id}_X)$ is representable. A subobject (respectively, a quotient) Y of X is called a *direct factor* of X if one can find a projector f in X which factors through Y, and which admits Y as image as a subobject (respectively, as a quotient) of X. By abuse of language, one sometimes says that an object of C is a *direct factor of* X if it is isomorphic to the image of a projector in X.

10.7 Base-changeable

An arrow $f : A \to B$ of C is called *base-changeable* (French: *quarrable*) if, for every arrow $B' \to B$ of C, the fiber product $A \times_B B'$ is representable in C.

10.8 Quotient, strict quotient

Let A be an object of C. Two epimorphisms $f : A \to B$, $f' : A \to B'$ with source A are called *equivalent* if there exists an isomorphism $u : B \xrightarrow{\sim} B'$ such that $uf = f'$. This gives an equivalence relation on the set of epimorphisms with source A; its classes are called the *quotients*, or *quotient objects*, of A. For every quotient of A one usually chooses an element of this class, say $f : A \to B$, and one often speaks, by abuse of language, of the quotient B of A, or of the quotient $f : A \to B$ of A. We say that B is a *strict quotient* (respectively, an effective quotient, universal quotient, ...) if the morphism $f : A \to B$ is a strict epimorphism (respectively, an effective epimorphism, universal epimorphism, ...) in the sense of 10.3.

10.9 Equivalence relations

Let F and G be two presheaves of sets on C. A diagram $p_1, p_2 : F \rightrightarrows G$ is called an *equivalence relation on* G if, for every object A of C, the map

$$(p_1(A), p_2(A)) : F(A) \longrightarrow G(A) \times G(A)$$

induces a bijection from $F(A)$ onto the graph of an equivalence relation on the set $G(A)$.

A diagram $p_1, p_2 : B \rightrightarrows C$ in C is called an equivalence relation on C if the corresponding diagram of presheaves is an equivalence relation. When finite products and fiber products are representable in C, a functor $\Phi : C \to C'$ commuting with finite products and fiber products carries equivalence relations on C to equivalence relations on $\Phi(C)$.

Let $\Pi : C \to D$ be a morphism of C such that the fiber product $C \times_D C$ is representable. Then the diagram

$$\mathrm{pr}_1, \mathrm{pr}_2 : C \times_D C \rightrightarrows C$$

is an equivalence relation on C.

10.10 Effective equivalence relation, universally effective equivalence relation

An equivalence relation $p_1, p_2 : R \rightrightarrows A$ on an object A of C is called an *effective equivalence relation* if there exists a morphism $\pi : A \to B$ such that the square

$$\begin{tikzcd} R \arrow[r, "p_2"] \arrow[d, "p_1"'] & A \arrow[d, "\pi"] \\ A \arrow[r] & B \end{tikzcd}$$

is cartesian and cocartesian. The morphism π is the cokernel of the pair (p_1, p_2), and is therefore determined up to unique isomorphism. The morphism π is then an effective epimorphism; if it is a universally effective epimorphism in the sense of 10.3, one says that the *equivalence relation* is *universally effective*.

10.11 Subobject, strict subobject

These are the notions dual to those of quotient, strict quotient, and so on; see 10.8.

References

[1] B. Mitchell, *Theory of Categories*, Academic Press, 1965.

II. Appendix: Universes (by N. Bourbaki*)

*We reproduce here, with his permission, secret papers of N. Bourbaki. The references in this text are to his learned work.

1 Definition and first properties of universes

Definition 1. A set U is called a *universe* if it satisfies the following conditions:

(U.I) if $x \in U$ and $y \in x$, then $y \in U$;

(U.II) if $x, y \in U$, then $\{x, y\} \in U$;

(U.III) if $x \in U$, then $\mathcal{P}(x) \in U$;

(U.IV) if $(x_\alpha)_{\alpha \in I}$ is a family of elements of U, and if $I \in U$, then the union $\bigcup_{\alpha \in I} x_\alpha$ belongs to U;

(U.?) if $x, y \in U$, then the pair $(x, y) \in U$.

N.B. Since, I believe, it has been decided for future editions to define the ordered pair à la Kuratowski by

$$(x, y) = \{\{x, y\}, \{x\}\},$$

condition (U.?) is useless, because it follows from (U.II).

Examples.

1) The empty set is a universe, denoted U_0.

2) Consider the nonempty finite words formed with the four symbols "$\{$", "$\}$", "$,$", and "$\emptyset$" (cf. Alg. I). Define, by induction on the length n of such a word, the notion of a *meaningful* word:

   (a) for $n = 1$, only the word $\emptyset$ is meaningful;

   (b) for a word A of length n to be meaningful, it is necessary and sufficient that there exist p distinct meaningful words $A_1, \ldots, A_p$, with $p \geq 1$ and of lengths $< n$, such that

   $$A = \{A_1, A_2, \ldots, A_p\}.$$

For example, $\emptyset$, $\{\emptyset\}$, and

$$\{\{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$$

are meaningful words. It is clear, by induction on the length, that every meaningful word denotes a term of set theory; for example, if $A = \{A_1, A_2, \ldots, A_p\}$, then A denotes the set whose elements are $A_1, A_2, \ldots, A_p$. These terms are of course *finite sets.* Let U_1 be the set of the sets thus obtained. One easily checks that U_1 satisfies conditions (U.I), (U.II), (U.III), and (U.IV) of Definition 1, though not that idiotic (U.?), which is not serious if one is willing to de-cannulate the pair. Thus U_1 is a *universe.* Notice that the elements of U_1 are finite and that U_1 is countable.

In the statements that follow, U denotes a universe.

Proposition 1. If $x \in U$ and $y \subset x$, then $y \in U$.

Indeed $\mathcal{P}(x) \in U$ by (U.III), whence $y \in \mathcal{P}(x)$ and $y \in U$ by (U.I).

Corollary. If $x \in U$, every quotient set y of x is an element of U.

Indeed y is a subset of $\mathcal{P}(x)$. Hence $y \in U$ by (U.III) and Proposition 1.

Proposition 2. If $x \in U$, then $\{x\} \in U$.

This follows from (U.II), applied with $y = x$.

Proposition 3. Every pair, triple, and quadruple of elements of U is an element of U.

For pairs this is true by the preceding N.B. or by (U.?). The case of triples follows since $(x, y, z) = ((x, y), z)$, and the case of quadruples then follows since $(x, y, z, t) = ((x, y, z), t)$.

Proposition 4. If $X, Y \in U$, then $X \times Y \in U$.

Indeed, for $x \in X$, the set $\{x\} \times Y$ is the union of the family $\{(x, y)\}_{y \in Y}$; since $\{(x, y)\} \in U$ by (U.I), (U.II), and Proposition 3, we have $\{x\} \times Y \in U$ by (U.IV). Finally, $X \times Y$ is the union of the family $(\{x\} \times Y)_{x \in X}$, so it is again an element of U by (U.IV).

Corollary 1. If $X, Y, Z, \ldots$ are elements of U, then all sets in the scale built on $X, Y, Z, \ldots$ are elements of U (cf. Chapter IV, §).

This follows from successive applications of Proposition 4 and (U.III).

Corollary 2. If $(x_\alpha)_{\alpha \in I}$ is a family of elements of U, and if $I \in U$, then the sum set $\sum_{\alpha \in I} x_\alpha$ is an element of U.

Indeed this sum set is a subset of the product $(\bigcup_{\alpha \in I} x_\alpha) \times I$, whose two factors are elements of U, by (U.IV) and the hypothesis. One then applies Propositions 4 and 1.

Proposition 5. If X and Y are elements of U, then every correspondence between X and Y, in particular every map from X to Y, is an element of U.

Indeed such a correspondence C is a triple (X, Y, Γ), where Γ is a subset of $X \times Y$, the graph of C (Chapter II, §). We have $\Gamma \in U$ by Propositions 4 and 1. Hence $C \in U$ by Proposition 3.

Proposition 6. If $X, Y \in U$, then every set Z of correspondences between X and Y, in particular every set of maps from X to Y, is an element of U.

Indeed Z is a subset of $\{X\} \times \{Y\} \times \mathcal{P}(X \times Y)$. This product is an element of U, by Proposition 2 and the corollary to Proposition 4. Therefore $Z \in U$ by Proposition 1.

Corollary. If $(x_\alpha)_{\alpha \in I}$ is a family of elements of U, and if $I \in U$, then $\prod_{\alpha \in I} x_\alpha \in U$.

Indeed this product is a set of maps from I to $\bigcup_{\alpha \in I} x_\alpha$, and this union is an element of U by (U.IV).

Proposition 7. If X is a subset of U whose cardinal is at most that of an element of U, then X is an element of U.

Let I be an element of U such that $\operatorname{card}(X) \leq \operatorname{card}(I)$. There is a surjection $i \mapsto x_i$ from a subset I' of I onto X. Then X is the union of the family $(\{x_i\})_{i \in I'}$. This union is an element of U by Proposition 1, Proposition 2, and (U.IV).

Corollary. If U is nonempty, every finite subset of U is an element of U, and U has elements of arbitrarily large finite cardinality.

On the other hand, if $U = \emptyset$, then $\emptyset$ is a finite subset of $\emptyset$ but not an element of $\emptyset$.

Indeed, if U is nonempty, Proposition 2 shows that a one-element set x_0 belongs to U. By induction on n, put $x_{n+1} = \mathcal{P}(x_n)$. Then $x_n \in U$ by (U.III), and $\mathrm{card}(x_{n+1}) = 2^{\mathrm{card}(x_n)}$, so U has elements of arbitrarily large finite cardinality.

Remark. It follows from Proposition 2 and the corollary to Proposition 7 that every nonempty universe U contains the universe U_1 of Example 2; indeed $\emptyset \in U$ by Proposition 1. Thus U_1 is the intersection of all nonempty universes. More generally:

Proposition 8. If $(U_\lambda)_{\lambda \in L}$ is a nonempty family of universes, then $U = \bigcap_{\lambda \in L} U_\lambda$ is a universe.

This follows immediately from Definition 1.

2 Universes and species of structures

Let $\mathcal{E}$ be a species of structure. To fix ideas and lighten the exposition, suppose that every structure of species $\mathcal{E}$ is defined on one underlying set. Let (X, S) be a structure of species $\mathcal{E}$, where X is the underlying set and S the structure, and let U be a universe. If $X \in U$, then all the constituent objects of the structure S on X are elements of U, by Corollary 1 to Proposition 4 of Section 1 and by (U.I); hence the structure (X, S) is an element of U.

Now suppose that $\mathcal{E}$ is a species of structure with *morphisms*. If X and X' are elements of U endowed with structures of species $\mathcal{E}$, then the set of morphisms from X to X' is again an element of U, by Proposition 6 of Section 1.

Consider the category $(U\text{-}\mathcal{E})$ defined in Section 1, no. 2, Example c). Since the objects and arrows of $(U\text{-}\mathcal{E})$ are elements of U, $(U\text{-}\mathcal{E})$ is a pair of subsets of U, endowed with a quadruple of maps. It is not, in general, an element of U. The notation $X \in (U\text{-}\mathcal{E})$ will mean that X is an element of U endowed with a structure of species $\mathcal{E}$.

The category $(U\text{-}\mathcal{E})$ is stable under many operations:

a) If $X \in (U\text{-}\mathcal{E})$, and if X' is a subset of X admitting an *induced structure*, then $X' \in (U\text{-}\mathcal{E})$. This follows from Proposition 1 of Section 1.

b) If $X \in (U\text{-}\mathcal{E})$, and if X'' is a quotient set of X admitting a *quotient structure*, then $X'' \in (U\text{-}\mathcal{E})$, by the corollary to Proposition 1 of Section 1.

c) If $X, Y \in (U\text{-}\mathcal{E})$, and if $X \times Y$ admits a *product structure*, then $X \times Y \in (U\text{-}\mathcal{E})$, by Proposition 4 of Section 1. More generally, if $(X_i)_{i \in I}$ is a family of elements of $(U\text{-}\mathcal{E})$, if one has $I \in U$, and if $\prod_{i \in I} X_i$ admits a *product structure*, then $\prod_{i \in I} X_i \in (U\text{-}\mathcal{E})$, by the corollary to Proposition 6. The analogous assertions hold for *sum structures*, by Corollary 2 to Proposition 4.

d) Let I be a preordered set, and let $(X_i, f_{ij})_{i,j \in I}$ be a *projective* (respectively, *inductive*) system of sets endowed with structures of species $\mathcal{E}$ and morphisms. Let L be the limit of this system, in the sense of Chapter III. If $X_i \in U$ for all i, if $I \in U$, and if L admits a projective (respectively, inductive) limit structure, then $L \in (U\text{-}\mathcal{E})$: indeed L is a subset of $\prod_{i \in I} X_i$ (respectively, a quotient set of $\sum_{i \in I} X_i$), and is therefore an element of U by Section 1.

Here are a few more particular examples:

* 1) Let $X, Y \in (U\text{-}\mathbf{Top})$ be two *locally compact* spaces. Then the set $\mathcal{C}(X, Y)$ of continuous maps from X to Y, endowed with the compact-open topology (General Topology, X), is an element of $(U\text{-}\mathbf{Top})$. Indeed $\mathcal{C}(X, Y) \in U$ by Proposition 6 of Section 1.

* 2) Let $X, Y \in (U\text{-}\mathbf{Ab})$. Then the group $\mathrm{Hom}_{\mathbb{Z}}(X, Y)$ is an element of $(U\text{-}\mathbf{Ab})$ by Proposition 6 of Section 1. If, moreover, $\mathbb{Z} \in U$, then the tensor product $X \otimes_{\mathbb{Z}} Y$ is an element of $(U\text{-}\mathbf{Ab})$. Indeed, this tensor product is a quotient of the free abelian group $\mathbb{Z}^{(X\times Y)}$, whose underlying set is a subset of $\mathbb{Z}^{X\times Y}$, and the latter is an element of U by Propositions 4 and 6 of Section 1.

* 3) Let $X \in (U\text{-}\mathbf{Unif})$ be a uniform space. Then its *completion* $\widehat{X}$ is an element of $(U\text{-}\mathbf{Unif})$. Indeed $\widehat{X}$ is a set of equivalence classes of Cauchy filters on X; but a filter on X is an element of $\mathcal{P}(\mathcal{P}(X))$, and hence an equivalence class of filters is an element of $\mathcal{P}(\mathcal{P}(\mathcal{P}(X)))$. Thus $\widehat{X}$ is an element of $\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathcal{P}(X))))$, and therefore an element of U by (U.III) and (U.I).

N.B. Moral: Bourbaki must take care to "canonify" his constructions. For example, in constructions in which one adjoins an element ∞, such as the Alexandroff compactification or the projective field, it will be useful to take $\infty = \emptyset$, because $\emptyset$ is an element of every nonempty universe, and to form the sum set of $\{\emptyset\}$ and the given set. Of course one should also "canonify" the two-element set used to construct sum sets; $\{\emptyset, \{\emptyset\}\}$ seems to me a good candidate, since it belongs to every nonempty universe.

3 Universes and categories

Let U be a universe and C a category. We write $C \in U$ if $\mathrm{Ob}(C) \in U$ and $\mathrm{Fl}(C) \in U$. This notation is justified by the fact that C is the sextuple consisting of $\mathrm{Ob}(C)$, $\mathrm{Fl}(C)$, and the four structural maps. Thus if $\mathrm{Ob}(C) \in U$ and $\mathrm{Fl}(C) \in U$, these four maps are elements of U, by Corollary 1 to Proposition 4 of Section 1, and so is the sextuple C, by Proposition 3 of Section 1, *cum grano salis*. This is moreover a special case of Section 2 if one considers the species of structure "category". With the notation of Section 2, the relation $C \in U$ is also written $C \in (U\text{-}\mathbf{Cat})$.

Notice that a category such as $(U\text{-}\mathbf{Set})$, $(U\text{-}\mathbf{Ord})$, $(U\text{-}\mathbf{Top})$, or $(U\text{-}\mathbf{Gr})$ is not, in general, an element of U.

Let C, D be two categories and U a universe such that $C, D \in U$. If C' is a subcategory of C, then $C' \in U$. The product category $C \times D$, the sum category of C and D, and the opposite categories C° and D° are also elements of U, by Section 2. The functor category $E = \mathrm{Hom}(C, D)$ is also an element of U: indeed, $\mathrm{Ob}(E)$ is a set of pairs of maps $\mathrm{Ob}(C) \to \mathrm{Ob}(D)$, $\mathrm{Fl}(C) \to \mathrm{Fl}(D)$, whence $\mathrm{Ob}(E) \in U$ by Section 1. As for $\mathrm{Fl}(E)$, it is a set of functorial morphisms, that is, of maps $\mathrm{Ob}(C) \to \mathrm{Fl}(D)$; hence $\mathrm{Fl}(E) \in U$ by Proposition 6 of Section 1.

Proposition 9. Let $\mathcal{E}$ be a species of structure with morphisms, and let U', U be two universes such that $U' \subset U$. Then $(U'\text{-}\mathcal{E})$ is a full subcategory of $(U\text{-}\mathcal{E})$.

Indeed, if x and y are objects of $(U'\text{-}\mathcal{E})$, then by definition they are objects of $(U\text{-}\mathcal{E})$. As for $\mathrm{Hom}(x, y)$, it is the same set in the two categories; see Section 1, no. 2, Example c).

N.B. The hypothesis that U and U' are universes is useless, so Proposition 9 would advantageously belong back in Section 1, no. 4. But the Tribe asked that it be placed here.

Remark. It may happen that the axioms of the species of structure $\mathcal{E}$ imply that the cardinals of the sets endowed with structures of species $\mathcal{E}$ are *bounded* by a fixed cardinal, say $\mathfrak{c}$; for example, the species of finite groups, finitely generated groups, finitely generated modules over a fixed ring A, or algebras of finite type over A. Suppose then that there exists an element z of U' such that $\mathfrak{c} \leq \mathrm{card}(z)$. Then, for every set x endowed with a structure of species $\mathcal{E}$, there exists an element x' of U' equipotent to x, for instance a subset of z. Endow x' with the structure transported from that of x; one obtains an element of $(U'\text{-}\mathcal{E})$. It follows that the inclusion functor from $(U'\text{-}\mathcal{E})$ into $(U\text{-}\mathcal{E})$ is then essentially surjective (Section 4, no. 2, Definition 2), and is therefore an equivalence of categories (Section 4, no. 3, Theorem 1).

4 The axiom of universes

The very agreeable stability results of Section 2 are of interest only if they can be applied to something other than the two small universes U_0 and U_1 described in Section 1. We will therefore add to the axioms of set theory the following axiom:

(A.6) *For every set x, there exists a universe U such that $x \in U$.*

This axiom implies that, if $(x_\alpha)_{\alpha \in I}$ is a *family* of sets, then there exists a universe U such that $x_\alpha \in U$ for every $\alpha \in I$. Indeed, it suffices to apply (A.6) to $x = \bigcup_{\alpha \in I} x_\alpha$, and then to apply Proposition 1 of Section 1.

In particular, given a category C, there exists a universe U such that $C \in U$ in the sense of Section 3: apply the preceding paragraph to the family $(\mathrm{Ob}(C), \mathrm{Fl}(C))$. This applies to categories of the form $(V\text{-}\mathcal{E})$, where $\mathcal{E}$ is a species of structure with morphisms and V is a set, for example a universe; in general $V \neq U$.

For example, if V is a universe, then $\mathrm{Ob}(V\text{-}\mathbf{Set}) = V$ and $\mathrm{Fl}(V\text{-}\mathbf{Set}) \subset V$, by Proposition 5 of Section 1. The universes U such that $(V\text{-}\mathbf{Set}) \in U$ are therefore those such that $V \in U$. But the relation $V \in V$ is impossible for a universe: indeed, for every subset A of V, one would have $A \in V$, by Proposition 1 of Section 1, whence $\mathrm{card}(\mathcal{P}(V)) \leq \mathrm{card}(V)$, which is impossible.

5 Universes and strongly inaccessible cardinals

Let U be a universe. By condition (U.I) of Definition 1, every element x of U is a subset of U; hence $\mathrm{card}(x) \leq \mathrm{card}(U)$. Since the cardinals of subsets of U form a well-ordered set, the cardinal

$$\mathfrak{c}(U) = \sup_{x \in U} \mathrm{card}(x) \tag{1}$$

exists. Notice that, for every cardinal $\mathfrak{c} < \mathfrak{c}(U)$, there exists an element x of U such that $\mathrm{card}(x) = \mathfrak{c}$. Indeed, by definition there exists $y \in U$ such that $\mathfrak{c} \leq \mathrm{card}(y) \leq \mathfrak{c}(U)$, and one takes for x a suitable subset of y. Conversely, if $x \in U$, then $\mathrm{card}(x) < \mathfrak{c}(U)$: indeed $\mathcal{P}(x) \in U$ by (U.III), whence $2^{\mathrm{card}(x)} \leq \mathfrak{c}(U)$. Thus the cardinal $\mathfrak{c}(U)$ has the following properties:

1) If $\mathfrak{c}$ is a cardinal $< \mathfrak{c}(U)$, then $2^{\mathfrak{c}} < \mathfrak{c}(U)$. Indeed, if x is an element of U of cardinal $\mathfrak{c}$, then $\mathcal{P}(x) \in U$ by (U.III), whence $2^{\mathfrak{c}} < \mathfrak{c}(U)$.

2) If $(\mathfrak{c}_\lambda)_{\lambda\in I}$ is a family of cardinals such that $\mathfrak{c}_\lambda < \mathfrak{c}(U)$ for every $\lambda \in I$, and if $\operatorname{card}(I) < \mathfrak{c}(U)$, then the sum cardinal $\sum_{\lambda\in I} \mathfrak{c}_\lambda$ is $< \mathfrak{c}(U)$. Indeed, let $x_\lambda \in U$ be such that $\operatorname{card}(x_\lambda) = \mathfrak{c}_\lambda$. Replacing I, if necessary, by an equipotent index set, we may suppose that $I \in U$. Then the sum set of the x_λ is an element of U, by Corollary 2 to Proposition 4 of Section 1, proving the assertion.

We make the following definition.

Definition 2. A cardinal $\mathfrak{d}$ is called *strongly inaccessible* if:

(FI.1) if $\mathfrak{c}$ is a cardinal such that $\mathfrak{c} < \mathfrak{d}$, then $2^{\mathfrak{c}} < \mathfrak{d}$;

(FI.2) if $(\mathfrak{c}_\lambda)_{\lambda\in I}$ is a family of cardinals such that $\mathfrak{c}_\lambda < \mathfrak{d}$ for all $\lambda \in I$, and if $\operatorname{card}(I) < \mathfrak{d}$, then $\sum_{\lambda\in I} \mathfrak{c}_\lambda < \mathfrak{d}$.

Examples. The cardinal 0 and the countably infinite cardinal are strongly inaccessible. No nonzero finite cardinal is strongly inaccessible. Thus we have just proved that the universe axiom (A.6) implies:

(A'.6) *Every cardinal is strictly dominated by a strongly inaccessible cardinal.*

Conversely:

Theorem 1. The assertion (A'.6) implies the axiom of universes (A.6).

Indeed, let A be a set. We must construct a universe U of which A is an element. Define by recursion a sequence $(A_n)_{n\geq 0}$ of sets by

(2) $A_0 = A$, and A_{n+1} is the union of the elements of A_n, that is, the set of elements of elements of A_n.

Put $B = \bigcup_{n=0}^{\infty} A_n$. By (A'.6), choose a strongly inaccessible cardinal $\mathfrak{c}$ such that $\operatorname{card}(B) < \mathfrak{c}$.

There exists a well-ordered set I such that $\operatorname{card}(I) = \mathfrak{c}$. Replacing I by its least segment of cardinal $\mathfrak{c}$, we may suppose that every proper segment of I has cardinal $< \mathfrak{c}$. We denote by $\mathcal{E}$ the least element of I. It follows from the hypothesis on segments that I has no largest element, and hence every element of I has a successor; for $\alpha \in I$, we denote the successor of α by $s(\alpha)$.

Now define, by transfinite recursion, a family $(B_\alpha)_{\alpha\in I}$ of sets by

$$\begin{cases} B_{\mathcal{E}} = B, \\ B_{s(\alpha)} = B_\alpha \cup \mathcal{P}(B_\alpha), \\ B_\alpha = \bigcup\limits_{\beta<\alpha} B_\beta \quad \text{if } \alpha \text{ has no predecessor.} \end{cases} \tag{3}$$

Put $U = \bigcup_{\alpha\in I} B_\alpha$. We will show that U is the required universe.

First $A \in U$, because $A = A_0$ is a subset of $B = B_{\mathcal{E}}$, and hence an element of $\mathcal{P}(B_{\mathcal{E}}) \subset B_{s(\mathcal{E})}$.

Next note that, for every $\alpha \in I$,

$$\operatorname{card}(B_\alpha) < \mathfrak{c}. \tag{4}$$

This is proved by transfinite induction on α. It is true for $\alpha = \mathcal{E}$ by hypothesis. From (4) one obtains

$$\operatorname{card}(B_{s(\alpha)}) \leq \operatorname{card}(B_\alpha) + 2^{\operatorname{card}(B_\alpha)} < \mathfrak{c} + \mathfrak{c} = \mathfrak{c},$$

using (FI.1) and the fact that $\mathfrak{c}$ is infinite. Finally, if α has no predecessor, the set I' of $\beta < \alpha$ has cardinal $< \mathfrak{c}$ by construction. Thus, if $\operatorname{card}(B_\beta) < \mathfrak{c}$ for all $\beta < \alpha$, then

$$\operatorname{card}(B_\alpha) = \operatorname{card}\Big(\bigcup_{\beta \in I'} B_\beta\Big) \leq \sum_{\beta \in I'} \operatorname{card}(B_\beta) < \mathfrak{c}$$

by (FI.2). This proves (4).

We now show that

$$\operatorname{card}(x) < \mathfrak{c} \quad \text{for every } x \in U. \tag{5}$$

It is enough to show, for every $\alpha \in I$, that $\operatorname{card}(x) < \mathfrak{c}$ for all $x \in B_\alpha$. Again proceed by transfinite induction on α. For $\alpha = \mathcal{E}$, if $x \in B_{\mathcal{E}} = B$, then $x \in A_n$ for some $n \geq 0$; hence $x \subset A_{n+1}$, so $x \subset B$ and $\operatorname{card}(x) \leq \operatorname{card}(B) < \mathfrak{c}$. The case where α has no predecessor is evident. Finally, if $x \in B_\alpha$ and $\alpha = s(\beta)$, then either $x \in B_\beta$, and the assertion follows by induction, or $x \in \mathcal{P}(B_\beta)$, whence $x \subset B_\beta$, and the assertion follows from (4).

We can now prove that U is indeed a universe:

(U.I) Let $x \in U$ and $y \in x$. We must show that $y \in U$. In other words, we must prove, for every $\alpha \in I$, the relation

$$x \in B_\alpha \text{ and } y \in x \Longrightarrow y \in B_\alpha.$$

Proceed once more by transfinite induction on α. It is true for $\alpha = \mathcal{E}$, since $x \in B_{\mathcal{E}} = B$ implies $x \in A_n$ for some n, and so, if $y \in x$, then $y \in A_{n+1}$, whence $y \in B = B_{\mathcal{E}}$. The passage to an element with no predecessor is evident. Finally pass from α to $s(\alpha)$. If $x \in B_{s(\alpha)}$ and $y \in x$, then either $x \in B_\alpha$, so $y \in B_\alpha \subset B_{s(\alpha)}$ by induction, or $x \in \mathcal{P}(B_\alpha)$, and again $y \in B_\alpha \subset B_{s(\alpha)}$.

(U.II) Let $x, y \in U$. Since the family $(B_\alpha)_{\alpha \in I}$ is increasing by (3), there exists $\alpha \in I$ such that $x, y \in B_\alpha$. Then $\{x, y\} \in \mathcal{P}(B_\alpha) \subset B_{s(\alpha)} \subset U$.

(U.III) Let $x \in U$. There exists $\alpha \in I$ such that $x \in B_\alpha$. It suffices to show, for every $\alpha \in I$, that

$$x \in B_\alpha \Longrightarrow \mathcal{P}(x) \in B_{s(s(\alpha))}.$$

Again proceed by transfinite induction on α. If $\alpha = \mathcal{E}$, then $x \in A_n$ for some n, whence every subset y of x is contained in $A_{n+1} \subset B = B_{\mathcal{E}}$. Thus $y \in \mathcal{P}(B_{\mathcal{E}}) \subset B_{s(\mathcal{E})}$ for every $y \in \mathcal{P}(x)$, and hence $\mathcal{P}(x) \subset B_{s(\mathcal{E})}$ and $\mathcal{P}(x) \in \mathcal{P}(B_{s(\mathcal{E})}) \subset B_{s(s(\mathcal{E}))}$. The passage to an element α with no predecessor is immediate, because $\beta < \alpha$ then implies $s(s(\beta)) < \alpha$. Finally pass from α to $s(\alpha)$. Let $x \in B_{s(\alpha)}$. If $x \in B_\alpha$, then $\mathcal{P}(x) \in B_{s(s(\alpha))} \subset B_{s(s(s(\alpha)))}$ by induction. If $x \in \mathcal{P}(B_\alpha)$, then $x \subset B_\alpha$, so $\mathcal{P}(x) \subset \mathcal{P}(B_\alpha)$ and $\mathcal{P}(x) \in \mathcal{P}(\mathcal{P}(B_\alpha)) \in B_{s(s(s(\alpha)))}$.

(U.IV) Let $(x_\lambda)_{\lambda \in K}$ be a family of elements of U such that $K \in U$. We must show that $x = \bigcup_{\lambda \in K} x_\lambda$ is an element of U. For every $\lambda \in K$, choose $\alpha(\lambda) \in I$ such that $x_\lambda \in B_{\alpha(\lambda)}$. We show that the set $\alpha(K)$ of the $\alpha(\lambda)$ is bounded in I. If it were not, then

$$I = \bigcup_{\lambda \in K} [\mathcal{E}, \alpha(\lambda)],$$

contradicting (FI.2), since $\operatorname{card}(K) < \mathfrak{c}$ by (5) and $\operatorname{card}([\mathcal{E}, \alpha(\lambda)]) < \mathfrak{c}$ for every $\lambda \in K$. Let β be an upper bound for $\alpha(K)$. Then $x_\lambda \in B_\beta$ for every $\lambda \in K$, because the family $(B_\alpha)_{\alpha\in I}$ is increasing. By what was proved in the verification of (U.I), each x_λ is a subset of B_β. Hence $x = \bigcup_{\lambda\in K} x_\lambda \subset B_\beta$. By (3), it follows that $x \in B_{s(\beta)}$, whence $x \in U$. □

Definition 3. Let x, y be two sets and let n be an integer ≥ 0. We say that y is a *component of order* n of x if there exists a sequence $(x_j)_{j=0,\dots,n}$ such that $x_0 = x$, $x_n = y$, and $x_{j+1} \in x_j$ for $j = 0, \dots, n-1$.

Thus x is the only component of order 0 of x. The components of order 1, respectively 2, of x are the elements of x, respectively the elements of elements of x. We say that y is a *component* of x if there exists $n \geq 0$ such that y is a component of order n of x. The relation "y is a component of x" is a preorder relation. By the selection-union scheme (Chapter II, ...), the relation "y is a component of x" is collectivizing with respect to y, so the components of x form a set.

Definition 4. Let $\mathfrak{c}$ be a cardinal. A set x is said to be of *type* $\mathfrak{c}$ (respectively, of *strict type* $\mathfrak{c}$, of *finite type*) if all components of x have cardinal $\leq \mathfrak{c}$ (respectively, $< \mathfrak{c}$, finite).

Examples. The elements of the universe U_1 of Section 1, Example 2, are all of finite type. If $\mathfrak{c}$ is a cardinal and x is of type $\mathfrak{c}$ (respectively, strict type $\mathfrak{c}$, finite type), then every component of x and every subset of x are of type $\mathfrak{c}$ (respectively, strict type $\mathfrak{c}$, finite type). Similarly, $\mathcal{P}(x)$ is of type $2^{\mathfrak{c}}$ (respectively, of strict type $2^{\mathfrak{c}}$, of finite type). If x is of type $\mathfrak{c}$ and $\mathfrak{c} \leq \mathfrak{c}'$, then x is of type $\mathfrak{c}'$.

Lemma. Let $\mathfrak{c}$ be a noncountable strongly inaccessible cardinal, and let X be a set of strict type $\mathfrak{c}$. Then there exists a cardinal $\mathfrak{d} < \mathfrak{c}$ such that X is of type $\mathfrak{d}$.

Indeed, for every integer n, let $\mathfrak{d}_n$ be the cardinal of the set X_n of components of order n of X. We have $\mathfrak{d}_0 = 1$, $\mathfrak{d}_1 = \operatorname{card}(X) < \mathfrak{c}$, and, since $X_n = \bigcup_{y\in X_{n-1}} y$, it follows by induction on n and by use of (FI.2) that $\mathfrak{d}_n < \mathfrak{c}$ for every n. Then the $\mathfrak{d}_n$ are bounded above by the cardinal $\mathfrak{d} = \sum_{j\geq 0} \mathfrak{d}_j$, which is $< \mathfrak{c}$ by (FI.2) and the noncountability of $\mathfrak{c}$. If Y is a component of order n of X, then $\operatorname{card}(Y) \leq \operatorname{card}(X_{n+1}) \leq \mathfrak{d}$. □

Proposition 10. Let U be a universe and let $\mathfrak{c}$ be a strongly inaccessible cardinal. Then the set U' of those $x \in U$ which are of strict type $\mathfrak{c}$ is a universe.

Indeed, if $x, y \in U'$ and $z \in x$, then obviously $z \in U'$ and $\{x, y\} \in U'$. If $x \in U'$, then $\operatorname{card}(x) < \mathfrak{c}$, whence $\operatorname{card}(\mathcal{P}(x)) < \mathfrak{c}$ by (FI.1). Since a component of $\mathcal{P}(x)$ is either $\mathcal{P}(x)$, or a subset of x, or a component of x, one has $\mathcal{P}(x) \in U'$. Finally, if $(x_\alpha)_{\alpha\in I}$ is a family of elements of U' such that $I \in U'$, then $\operatorname{card}(\bigcup_\alpha x_\alpha) < \mathfrak{c}$ by (FI.2); since every component of order > 0 of $\bigcup_\alpha x_\alpha$ is a component of some x_α, we do indeed have $\bigcup_\alpha x_\alpha \in U'$. □

Remark on the cardinal $\mathfrak{c}(U)$. Let U be a universe and let $\mathfrak{c}(U)$ be the cardinal defined by (1), that is,

$$\mathfrak{c}(U) = \sup_{x\in U} \operatorname{card}(x).$$

Obviously $\mathfrak{c}(U) \leq \operatorname{card}(U)$, because, recall, $x \in U$ implies $x \subset U$. But the equality $\mathfrak{c}(U) = \operatorname{card}(U)$ is not always true. For example, let $(x_\alpha)_{\alpha\in I}$ be a noncountable family of symbols with the axioms

$$\text{"}x_\alpha = \{x_\alpha\} \text{ for every } \alpha \in I\text{"} \quad \text{and} \quad \text{"}x_\alpha \neq x_\beta \text{ if } \alpha \neq \beta\text{"}.$$

In other words, x_α is a one-element set, namely itself. As in Example 2 of Section 1, form the "meaningful words" made from the symbols $\emptyset$, x_α, braces, and so on. Each denotes a *finite* set. Let U be the set of these. One easily checks that the conditions of Definition 1 are satisfied, so U is a universe. Since meaningful words are finite sequences of elements of a set of cardinal $\operatorname{card}(I)+4=\operatorname{card}(I)$, we have $\operatorname{card}(U)=\operatorname{card}(I)$; in fact $\operatorname{card}(U)\geq\operatorname{card}(I)$ would suffice, and this is evident. On the other hand, $\mathfrak{c}(U)$ is the countable cardinal, whence $\mathfrak{c}(U)<\operatorname{card}(U)$.

The strict inequality $\mathfrak{c}(U)<\operatorname{card}(U)$ is due to the fact that here we have introduced sets x_α such that $x_\alpha\in x_\alpha$. If one forbids horrors of this kind, one obtains $\operatorname{card}(U)=\mathfrak{c}(U)$ for every universe U, as well as other very pretty results "which cannot serve any purpose." This is what we shall do in the next section.

6 Artinian sets and artinian universes

Definition 5. A set x is called *artinian* if there exists no infinite sequence $(x_n)_{n\geq 0}$ such that $x_0=x$ and $x_{n+1}\in x_n$ for every $n\geq 0$.

Examples. The sets $\emptyset$, $\{\emptyset\}$, $\{\emptyset,\{\emptyset\}\}$, and more generally the elements of the universe U_1 of Section 1, are artinian.

In pictorial terms, if x is artinian and one takes an element x_1 of x, then an element x_2 of x_1, and so on, the process must stop, and one arrives at a component x_n of x which is *empty*. In other words, artinian sets are "built out of $\emptyset$." This will be made precise later (cf. (*) in the proof of Theorem 2).

Artinian sets evidently enjoy the following properties:

(AR.I) If x is artinian, every subset of x and every component of x is artinian. For x to be artinian, it is necessary and sufficient that every element of x be artinian.

(AR.II) If x and y are artinian, then $\{x,y\}$ is artinian.

(AR.III) If x is artinian, then $\mathcal{P}(x)$ is artinian.

(AR.IV) Every union of artinian sets is an artinian set.

These properties immediately show:

Proposition 11. If U is a universe, the set of artinian elements of U is a universe, necessarily artinian.

Corollary. If x is an artinian set, there exists an artinian universe U such that $x\in U$.

Indeed, by axiom (A.6), x is an element of a universe V; take for U the set of artinian elements of V.

The following proposition is still less useful than the rest of the section:

Proposition 12. Let A be an artinian set. Then:

a) for every $x\in A$, one has $x\notin x$;

b) if $x,y\in A$, one cannot have both $x\in y$ and $y\in x$;

c) for every $x \in A$, the relation "y is a component of z" between components y, z of x is an order relation;

d) for every nonempty element x of A, there exists $y \in x$ such that $x \cap y = \emptyset$.

Indeed, the negation of a), respectively of b), entails the existence of an infinite sequence $(x_n)_{n \geq 0}$ contradicting Definition 5, namely $(x, x, x, \ldots)$, respectively $(x, y, x, y, x, y, \ldots)$. The negation of c) means that there exist $x \in A$, distinct components y, z of x, and membership chains

$$y \in y_1 \in \cdots \in y_q \in z, \qquad z \in z_1 \in \cdots \in z_r \in y;$$

then, as in b), one obtains an infinite sequence $(x_n)_{n \geq 0}$ contradicting Definition 5. Finally, if d) is false, there exists a nonempty element x of A such that $x \cap y \neq \emptyset$ for every $y \in x$. Put $x_0 = x$, and choose $x_1 \in x$. Since $x \cap x_1 \neq \emptyset$, choose $x_2 \in x \cap x_1$, and so on. More formally, define by recursion an infinite sequence $(x_n)_{n \geq 1}$ of elements of x by choosing $x_1 \in x$ and $x_{n+1} \in x_n \cap x$ for $n \geq 1$. Then the sequence $(x_n)_{n \geq 0}$ contradicts Definition 5.

Remark. Let B be a set. For B to be artinian, it is necessary and sufficient that every set A of sets of components of B satisfy condition d) of Proposition 12. Necessity follows from (AR.I) and Proposition 12. Conversely, if B is not artinian, there exists an infinite sequence $(x_n)_{n \geq 0}$ with $x_0 = B$ and $x_{n+1} \in x_n$ for every $n \geq 0$. Take for A the singleton consisting of the set X of the x_n. Thus A contains a nonempty element X such that, for every element y of X, one has $y \cap X \neq \emptyset$: indeed $y = x_n$ for some n, and $x_{n+1} \in y \cap X$.

Theorem 2. Let $\mathfrak{c}$ be an infinite cardinal. Then:

a) The relation "x is an artinian set of type $\mathfrak{c}$" (respectively, of strict type $\mathfrak{c}$) is collectivizing with respect to x. The set $A_{\mathfrak{c}}$ of artinian sets of type $\mathfrak{c}$ has cardinal $2^{\mathfrak{c}}$.

b) If $\mathfrak{c}$ is strongly inaccessible, the set $U_{\mathfrak{c}}$ of artinian sets of strict type $\mathfrak{c}$ is a universe of cardinal $\mathfrak{c}$; the cardinal $\mathfrak{c}(U_{\mathfrak{c}}) = \sup_{x \in U_{\mathfrak{c}}} \operatorname{card}(x)$ is $\mathfrak{c}$.

c) If a universe U has an element of cardinal $\mathfrak{c}$, then every artinian set of type $\mathfrak{c}$ belongs to U, that is, $A_{\mathfrak{c}} \subset U$.

c') If a universe U is nonempty, then every artinian set of finite type is an element of U.

Before proving Theorem 2, let us derive a few illuminating corollaries.

Corollary 1. If a universe U is artinian, then $\operatorname{card}(U)$ is strongly inaccessible, and U is the set of artinian sets of strict type $\operatorname{card}(U)$.

Indeed put $\mathfrak{c}(U) = \sup_{x \in U} \operatorname{card}(x)$. This is a strongly inaccessible cardinal, by the beginning of Section 5. First suppose it is noncountable. Then, for every infinite cardinal $\mathfrak{c} < \mathfrak{c}(U)$, every artinian set of type $\mathfrak{c}$ is an element of U by c); hence every artinian set of strict type $\mathfrak{c}(U)$ is an element of U, by the lemma of Section 5. This last assertion remains true if $\mathfrak{c}(U)$ is countable, by c'). Conversely, by (U.I), every set of U is of strict type $\mathfrak{c}(U)$. Thus U is the universe $U_{\mathfrak{c}(U)}$ of b), whence $\mathfrak{c}(U) = \operatorname{card}(U)$ by b).

It follows from Corollary 1 that an artinian universe is uniquely determined by its cardinal, which is strongly inaccessible. Thus one has a "one-to-one correspondence" between artinian universes and strongly inaccessible cardinals. In particular:

Corollary 2. The inclusion relation $U \subset U'$ between artinian universes is a well-ordering relation.

Indeed, the relation $\mathfrak{c} \leq \mathfrak{c}'$ between cardinals is a well-ordering relation (Chapter III), and, with the notation of Theorem 2 b), the relations $\mathfrak{c} \leq \mathfrak{c}'$ and $U_{\mathfrak{c}} \subset U_{\mathfrak{c}'}$ are equivalent.

Note that Theorem 2 b) gives a second proof of Theorem 1.

We now prove Theorem 2. Given a set A, call a *chain* of A any finite sequence $(x_j)_{j=0,\ldots,n}$ such that $x_0 = A$ and $x_{j+1} \in x_j$ for $j = 0, \ldots, n-1$. The chains of A form a set, by the selection-union scheme; denote it by $G(A)$. Given a chain $X = (x_j)_{j=0,\ldots,n}$, the chains of the form $(x_i)_{i=0,\ldots,q}$, with $q \leq n$, are said to be smaller than X. This gives an ordered-set structure on $G(A)$. For A to be artinian, it is necessary and sufficient that $G(A)$ be a noetherian ordered set, in the sense of the equivalent conditions of Chapter III, §6, no. 5.

We shall show that:

> (*) *If A is artinian, then A is uniquely determined by the isomorphism class of the ordered set $G(A)$.*

Indeed, given an ordered set G and an element $g \in G$, write $S(g)$ for the set of $g' \in G$ such that $g < g'$ and such that $g \leq h \leq g'$ implies $h = g$ or $h = g'$. In other words, $S(g)$ is the set of immediate successors of g. Consider the map θ which sends a chain $X = (x_0, \ldots, x_n)$ of A to the set x_n. Then, for every $X \in G(A)$,

$$\theta(X) = \{\theta(X') \mid X' \in S(X)\}.$$

Since A is the image under θ of the least element $X_0 = (A)$ of $G(A)$, it will be enough to show that θ is uniquely determined by the isomorphism class of the ordered set $G(A)$. This follows from the following lemma.

Lemma 1. Let G be a noetherian ordered set, and let φ be a map on G such that, for every $g \in G$,

$$\varphi(g) = \{\varphi(g') \mid g' \in S(g)\}. \tag{1}$$

Then φ is uniquely determined. Moreover, if U is a universe containing an element equipotent to G, then φ takes its values in U.

The hypothesis implies that $\varphi(g) = \emptyset$ if g is a maximal element of G. Indeed, let φ and φ' be two maps satisfying (1). If $\varphi \neq \varphi'$, the set of $g \in G$ such that $\varphi(g) \neq \varphi'(g)$ is nonempty, hence has a maximal element h, since G is noetherian. Then $\varphi(g) = \varphi'(g)$ for every $g > h$, in particular for every successor $g \in S(h)$, whence $\varphi(h) = \varphi'(h)$ by (1), a contradiction. Thus $\varphi = \varphi'$.

Now let U be a universe containing an element x equipotent to G. We show that φ takes its values in U. Otherwise, let h be maximal among the $g \in G$ such that $\varphi(g) \notin U$. We have $\varphi(g') \in U$ for every $g' \in S(h)$, so

$$\varphi(h) = \{\varphi(g') \mid g' \in S(h)\}$$

is a subset of U. Since its cardinal is less than or equal to $\operatorname{card}(G)$, hence to the cardinal of an element of U, one has $\varphi(h) \in U$ by Proposition 7 of Section 1. This contradiction shows that φ takes its values in U.

We now prove Theorem 2 a). It is enough to treat the non-respective assertion, since the other follows at once. Let $\mathfrak{c}$ be an infinite cardinal. If A is a set of type $\mathfrak{c}$, then

$$\operatorname{card}(G(A)) \leq \mathfrak{c}. \tag{2}$$

Indeed, if A_n denotes the set of components of order n of A, then $\operatorname{card}(A_1) \leq \mathfrak{c}$ and $\operatorname{card}(A_{n+1}) \leq \mathfrak{c} \cdot \operatorname{card}(A_n)$. Hence $\operatorname{card}(A_n) \leq \mathfrak{c}^n$ by induction; but $\mathfrak{c}^n = \mathfrak{c}$, because $\mathfrak{c}$ is infinite (Chapter III). Therefore

$$\operatorname{card}\Big(\bigcup_{n=0}^{\infty} A_n\Big) \leq \operatorname{card}(\mathbb{N}) \cdot \mathfrak{c} = \mathfrak{c}.$$

Since $G(A)$ is a set of finite sequences of elements of $\bigcup_n A_n$, inequality (2) follows.

Now let E be a set of cardinal $\mathfrak{c}$. Giving an order structure on a subset E' of E is equivalent to giving the subset of $E \times E$ consisting of the pairs (x, y) such that $x \in E'$, $y \in E'$, and $x \leq y$. Therefore, by (2), the isomorphism classes of ordered sets $G(A)$, where A is of type $\mathfrak{c}$, form a set $\mathfrak{F}_{\mathfrak{c}}$, and

$$\operatorname{card}(\mathfrak{F}_{\mathfrak{c}}) \leq \operatorname{card}(\mathcal{P}(E \times E)) = 2^{\mathfrak{c}}.$$

Let $\mathfrak{F}'_{\mathfrak{c}}$ be the subset of $\mathfrak{F}_{\mathfrak{c}}$ formed by the classes of noetherian ordered sets having a least element. If, to each $G \in \mathfrak{F}'_{\mathfrak{c}}$, we associate the value of the function φ of Lemma 1 at the least element of G, we obtain a map θ on $\mathfrak{F}'_{\mathfrak{c}}$ whose image contains *all* artinian sets of type $\mathfrak{c}$. These therefore form a set $A_{\mathfrak{c}}$, and

$$\operatorname{card}(A_{\mathfrak{c}}) \leq \operatorname{card}(\mathfrak{F}'_{\mathfrak{c}}) \leq \operatorname{card}(\mathfrak{F}_{\mathfrak{c}}) \leq 2^{\mathfrak{c}}.$$

It remains to see that $\operatorname{card}(A_{\mathfrak{c}}) = 2^{\mathfrak{c}}$. For this it is enough to see that there exists an artinian set B of type $\mathfrak{c}$ and cardinal $\mathfrak{c}$, since the subsets of B will then be elements of $A_{\mathfrak{c}}$. This existence follows from the following lemma.

Lemma 2. For every cardinal $\mathfrak{c}$, there exists an artinian set B of type $\mathfrak{c}$ and cardinal $\mathfrak{c}$.

Proceed by transfinite induction on $\mathfrak{c}$. For $\mathfrak{c} = 0$ there is no choice: take $B = \emptyset$. If $\mathfrak{c}$ has a predecessor $\mathfrak{c}'$, let B' be an artinian set of type $\mathfrak{c}'$ and of cardinal $\mathfrak{c}'$. Then $\mathfrak{c} \leq 2^{\mathfrak{c}'}$, so there exists a subset B of $\mathcal{P}(B')$ of cardinal $\mathfrak{c}$. The elements of B are subsets of B', and hence have cardinals $\leq \mathfrak{c}' \leq \mathfrak{c}$; the components of higher order of B are components of B', and hence also have cardinals $\leq \mathfrak{c}' \leq \mathfrak{c}$. Finally, if $\mathfrak{c}$ has no predecessor, choose, for every cardinal $\mathfrak{c}_\lambda < \mathfrak{c}$, an artinian set B_λ of type $\mathfrak{c}_\lambda$ and cardinal $\mathfrak{c}_\lambda$. Then $B = \bigcup_\lambda B_\lambda$ has the required property. This proves Lemma 2 and completes the proof of part a).

We turn to b). Let $\mathfrak{c}$ be a strongly inaccessible cardinal. We already know, by a), that the artinian sets of strict type $\mathfrak{c}$ form a set $U_{\mathfrak{c}}$. The fact that $U_{\mathfrak{c}}$ is a universe follows immediately from properties (AR.I)–(AR.IV) of artinian sets, at the beginning of this section, and from cardinal estimates analogous to those of Proposition 10 of Section 5. The relation

$$\sup_{x \in U_{\mathfrak{c}}} \operatorname{card}(x) = \mathfrak{c}$$

follows from Lemma 2 applied to cardinals $< \mathfrak{c}$. Finally, to prove $\operatorname{card}(U_{\mathfrak{c}}) = \mathfrak{c}$, first suppose that $\mathfrak{c}$ is noncountable. By the lemma of Section 5, $U_{\mathfrak{c}}$ is the union

$$\bigcup_{\mathfrak{d} < \mathfrak{c}} A_{\mathfrak{d}},$$

where $A_{\mathfrak{d}}$ denotes the set of artinian sets of type $\mathfrak{d}$. But $\operatorname{card}(A_{\mathfrak{d}}) = 2^{\mathfrak{d}} < \mathfrak{c}$ by a); on the other hand, the set of cardinals $\mathfrak{d} < \mathfrak{c}$ has cardinal $\leq \mathfrak{c}$ (Chapter III). Hence

$$\operatorname{card}(U_{\mathfrak{c}}) \leq \mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c},$$

and also $\operatorname{card}(U_{\mathfrak{c}}) \geq \mathfrak{c}$, since $\operatorname{card}(U_{\mathfrak{c}}) \geq \operatorname{card}(A_{\mathfrak{d}}) = 2^{\mathfrak{d}}$ for every $\mathfrak{d} < \mathfrak{c}$.

The case $\mathfrak{c} = 0$ is trivial. It remains to treat the case where $\mathfrak{c}$ is the countably infinite cardinal. In this case $U_{\mathfrak{c}}$ is the set of artinian sets of finite type, that is, sets which are finite, as are all their components; and one uses a pretty combinatorial result.

Lemma 3 (D. König?). Consider two infinite sequences $(E_n)_{n\geq 1}$, $(f_n)_{n\geq 1}$ of finite sets E_n and maps $f_n : E_{n+1} \to E_n$. If there exists no infinite sequence $(x_n)_{n\geq 1}$ such that $x_n \in E_n$ and $f_n(x_{n+1}) = x_n$ for every n, then E_n is empty for all sufficiently large n.

In other words, if all sequences (x_n) such that $f_n(x_{n+1}) = x_n$ are finite, then their lengths are bounded. This can be expressed in terms of projective limits: a projective limit of nonempty finite sets is nonempty; see General Topology, Chapter I, 2nd ed., §9, no. 6, Proposition 8, 2°.

Argue by contradiction. If there exist nonempty E_n for arbitrarily large n, then no E_n is empty, because $E_n = \emptyset$ implies $E_{n+1} = \emptyset$, given the existence of $f_n : E_{n+1} \to E_n$. Call a finite sequence $(x_j)_{1\leq j\leq n}$ *coherent* if $f_j(x_{j+1}) = x_j$ for $j = 1, \ldots, n-1$. We prove by induction on n the existence of a coherent sequence $(a_1, \ldots, a_n)$, with $a_i \in E_i$, which, for every $q \geq n$, can be extended to a coherent sequence $(a_1, \ldots, a_n, x_{n+1}, \ldots, x_q)$ of length q. This is evident for $n = 0$, since no E_q is empty. Passing from n to $n+1$, if, for every $x \in f_n^{-1}(\{a_n\}) \subset E_{n+1}$, all coherent sequences extending $(a_1, \ldots, a_n, x)$ had lengths bounded by an integer $q(x)$, then all coherent sequences extending $(a_1, \ldots, a_n)$ would have bounded lengths, by $\sup_x q(x)$, since E_{n+1} is finite. Hence there exists $a_{n+1} \in f_n^{-1}(\{a_n\})$ such that the coherent sequence $(a_1, \ldots, a_n, a_{n+1})$ has extensions of arbitrary length. This gives an infinite sequence $(a_n)_{n\geq 1}$, contrary to the hypothesis.

It follows from Lemma 3 that if A is an artinian set of finite type, then the ordered set $G(A)$ of its chains is *finite*. Indeed, take E_n to be the set of chains

$$x_n \in x_{n-1} \in \cdots \in x_1 \in A$$

with $n+1$ terms. This is finite because the components of A of order $\leq n+1$ are finite in number and are themselves finite. Take f_n to be the map which deletes the last term:

$$(x_{n+1} \in x_n \in \cdots \in x_1 \in A) \longmapsto (x_n \in x_{n-1} \in \cdots \in x_1 \in A).$$

The set of isomorphism classes of finite ordered sets is countable: indeed, giving an order structure on a finite subset of $\mathbb{N}$ is equivalent to giving its graph, a finite subset of $\mathbb{N} \times \mathbb{N}$, and the set of finite subsets of a countable set is countable (Chapter III). It follows from (*) and Lemma 1 that the set $\mathfrak{a}$ of artinian sets of finite type is countable. It is infinite by Lemma 2, or more simply by using the sequence $(z_n)_{n\geq 0}$ defined by $z_0 = \emptyset$, $z_{n+1} = \{z_n\}$. This completes the proof of b).

Remark. We have just proved that, if A is an artinian set of finite type, then it has only finitely many components. Thus there exists an integer n such that A is of type n.

We now prove c). Let $\mathfrak{c}$ be an infinite cardinal, let U be a universe having an element of cardinal $\mathfrak{c}$, and let A be an artinian set of type $\mathfrak{c}$. The ordered set $G(A)$ has cardinal $\leq \mathfrak{c}$, by formula (2). The second assertion of Lemma 1, applied to $G(A)$, shows that the map

$$(x_0, \ldots, x_n) \rightsquigarrow x_n$$

from $G(A)$ takes its values in U. In other words, all components of A are elements of U. This proves c). The proof of c') is analogous: if A is an artinian set of finite type, then we have just seen that $G(A)$ is finite; since U is nonempty, it contains an element equipotent to $G(A)$, by the corollary to Proposition 7 of Section 1. $\square$

7 Vague metamathematical remarks

a) The axiom "*every set is artinian*" is *inoffensive*. Indeed, if one has a model M of set theory, the set M' of artinian elements of M is also a model (cf. Proposition 11).

b) The universe axiom (A.6), and the equivalent axiom (A'.6) on strongly inaccessible cardinals, is *independent* of the rest of set theory. Indeed, let $\mathfrak{c}$ be the first noncountable strongly inaccessible cardinal. The universe $U_\mathfrak{c}$ of artinian sets of strict type $\mathfrak{c}$ (Theorem 2 b)) is a model of set theory without (A.6): one calls the elements of $U_\mathfrak{c}$ "sets," the "membership relation" is the restriction to $U_\mathfrak{c}$ of the ordinary one, and so on. The "universes" of the model are therefore the ordinary universes which are elements of $U_\mathfrak{c}$. But we have seen that the only universes which are elements of $U_\mathfrak{c}$ are the two pequeños $U_0 = \emptyset$ and U_1. Thus U_1 is a "set" which is an element of no "universe." Hence we have a model of set theory in which (A.6) is false.

c) Bourbaki was too cautious in merely "presuming" that the *axiom of infinity* (A.5) is independent of the preceding axioms and schemes. It is indeed independent, because the countable universe U_1 of artinian sets of finite type is a model in which (A.5) is false and in which the preceding axioms and schemes are true.

d) It would be very interesting to prove that the universe axiom (A.6) is inoffensive. This seems difficult, and is even unprovable, says Paul Cohen.

The adjective "vague" in the title means that, in constructing models, one did not take the trouble to cannulate the symbol τ so that it does not spill out. The key to this, if one considers a model M, is to replace the ordinary

$$\tau_x(R(x))$$

by

$$\tau_x(R(x) \text{ and } x \in M).$$

One still has to check that this really transforms ordinary quantifiers into the quantifiers formerly called "typical":

$$(\forall x \in M) \qquad (\exists x \in M).$$

Exercises.

1) Let $n \geq 1$ be an integer. Show that the set of artinian sets of type n is infinite. The sets $z_0 = \emptyset$, $z_1 = \{\emptyset\}$, and $z_{q+1} = \{z_q\}$ are of type 1.

2) Call the *height* of a set A the least upper bound, finite or infinite, of the integers n such that there exists a sequence

$$x_n \in x_{n-1} \in \cdots \in x_0 = A.$$

Show that the sets of height $\leq n$ are finite and form a finite set whose cardinal p_n is calculated from $p_0 = 1$, $p_{n+1} = 2^{p_n}$. Proceed by induction on n, noting that the elements of a set A of height $\leq n$ are sets of height $\leq n-1$.

3) Let G be a noetherian ordered set having a least element g_0, and such that, for every $h \in G$, the set of $g \leq h$ is finite and totally ordered.

   a) Show that every element $h \neq g_0$ of G has a unique predecessor.

   b) Let φ be the map on G defined in Lemma 1, that is, $\varphi(g)$ is the set of the $\varphi(g')$ where g' runs over the successors of g. Put $A = \varphi(g_0)$. Show that A is an artinian set. For $g \in G$, let
   $$g_0 < g_1 < \cdots < g_n = g$$
   be the sequence of elements $\leq g$, and put
   $$f(g) = (\varphi(g_0), \varphi(g_1), \ldots, \varphi(g_n)).$$
   Show that f is an increasing surjective map from G onto the ordered set $G(A)$ of the text. Show that, if f is *injective*, it is an isomorphism from G onto $G(A)$.

   c) For $g \in G$, let M_g be the set of majorants of g. Suppose that, for every pair of distinct elements g, g' having the same predecessor p, the ordered sets M_g and $M_{g'}$ are not isomorphic. Show that the map f of b) is then an *isomorphism* from G onto $G(A)$. Hint: if $f(g) = f(g')$ with $g \neq g'$, and if
   $$g_0 < g_1 < \cdots < g_n = g, \qquad g'_0 < g'_1 < \cdots < g'_{n'} = g'$$
   are the sequences of elements $\leq g$ and $\leq g'$, show that $n = n'$, and that one may suppose that g and g' have the same predecessor p. Then consider the set of $p \in G$ such that there exist two distinct successors g, g' of p with $f(g) = f(g')$; take a maximal element q of this set and two distinct successors h, h' of q such that $f(h) = f(h')$. Observe that the restrictions of f to M_h and to $M_{h'}$ are injective, and hence, by b), are isomorphisms from M_h to $G(\varphi(h))$ and from $M_{h'}$ to $G(\varphi(h'))$. Deduce from the non-isomorphism hypothesis on M_h and $M_{h'}$ that $\varphi(h) \neq \varphi(h')$, contradicting $f(h) = f(h')$.

N.B. Exercise 3 gives very precise information on how artinian sets are "made." We already knew that such a set A is determined by the isomorphism class of the ordered set $G(A)$ (Lemma 1). We now know how to *characterize* the ordered sets G isomorphic to some $G(A)$:

α) G is noetherian and has a least element;

β) for every $g \in G$, the set of $h \leq g$ is totally ordered and finite, whence the existence and uniqueness of the predecessor of g;

γ) if g, g' are distinct elements having the same predecessor, then the set M_g of majorants of g and the set $M_{g'}$ of majorants of g' are not isomorphic.

4) Let $(A_n)_{n \geq 0}$ be the sequence of finite ordinals defined by $A_0 = \emptyset$, $A_{n+1} = A_n \cup \{A_n\}$. Show that the ordered set $G(A_n)$ has 2^n elements, and that the number of its elements of height q, in the sense of Exercise 2, is $\binom{n}{q}$.

References

[1] B. Mitchell, *Theory of Categories*, Academic Press, 1965.

Exposé II. Topologies and Sheaves

Opening; Sections 1–2.7

J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé II, from the opening overview through Remarks 2.7. The range corresponds to source lines 23–421 of the Orgogozo–Laszlo TeX snapshot. Terminology is normalized to current English usage where this is harmless: *sieve* for *crible*, *covering family*, *pretopology*, and *base-changeable* for French *quarrable*. References to Exposé I are kept in their original numerical form.

After the definition of topologies and pretopologies (no. 1) and of sheaves of sets (no. 2), no. 3 turns to the central theorem of this exposé (3.4): the existence of the sheaf associated to a presheaf. In order to cover all cases encountered in practice, the existence of this functor is proved for presheaves on a $\mathcal{U}$-site (3.0.2). Nos. 4 and 5 draw the consequences of this theorem for the behavior of inductive and projective limits in categories of sheaves. In no. 6 one defines and studies sheaves with values in arbitrary categories, and then immediately turns attention to sheaves of rings, groups, modules, etc.

1 Topologies, covering families, pretopologies

Definition 1.1. A *topology* on a category C consists, for each object X of C, of a set $J(X)$ of sieves on X, subject to the following axioms:

T 1) *Stability under base change.* For every object X of C, every sieve $R \in J(X)$, and every morphism $f : Y \to X$ with $Y \in \mathrm{Ob}(C)$, the sieve $R \times_X Y$ on Y belongs to $J(Y)$.

T 2) *Local character.* If R and R' are two sieves on X, if $R \in J(X)$, and if for every $Y \in \mathrm{Ob}(C)$ and every morphism $Y \to R$ the sieve $R' \times_X Y$ belongs to $J(Y)$, then R' belongs to $J(X)$.

T 3) For every object X of C, the maximal sieve X belongs to $J(X)$.

1.1.1

The sieves belonging to $J(X)$ will be called the *sieves covering* X, or also the *refinements of* X. From axioms (T1), (T2), and (T3), one immediately deduces that the set of sieves covering X is stable under finite intersections, and that every sieve containing a covering sieve is again a covering sieve. Thus $J(X)$, ordered by inclusion, is cofiltered (I 8).

1.1.2

Let C be a category, and let T and T' be two topologies on C. The topology T is said to be *finer than* the topology T' if, for every object X of C, every refinement of X for the topology T' is a refinement of X for the topology T. In this way one defines an order structure on the set of topologies.

1.1.3

Let $(T_i)_{i\in I}$ be a family of topologies on C. For each object X of C, the sets of sieves on X which are covering for all the topologies T_i satisfy axioms (T1), (T2), and (T3), and therefore define a topology: the *intersection topology* of the T_i, i.e. the greatest lower bound of the T_i. It is the finest of the topologies which is less fine than all the topologies T_i. It follows that the family $(T_i)_{i\in I}$ has a *least upper bound*: the intersection topology of the topologies finer than each of the T_i.

1.1.4

In particular, the assignment $J(X) =$ the set of all sieves on X is a topology finer than every topology on C; we shall call it the *discrete topology.*

There is a topology less fine than every topology on C: the topology for which $J(X) = \{X\}$ for every X of C. This topology is called the *coarse* or *chaotic* topology.

1.1.5

A category C endowed with a topology is called a *site*. The category C is called the *underlying category* of the site.

Definition 1.2. Let C be a site and let X be an object of C. A family of morphisms $(f_\alpha : X_\alpha \to X)$, $\alpha \in A$, is called *covering* if the sieve generated by the family f_α (I 4.3.3) is a sieve covering X.

1.1.6

Let C be a category. Suppose that for every object X of C we are given a set of families of morphisms with target X. Then there exists a topology T which is the least fine among the topologies for which the given families are covering, namely the intersection (1.1.3) of all topologies in question. We call this topology the *topology generated by the given sets of families of morphisms.* In general it is awkward to describe all covering sieves of this topology. The situation is more agreeable in the following case.

Definition 1.3. Let C be a category. A *pretopology* on C consists, for each object X of C, of a set $\mathrm{Cov}(X)$ of families of morphisms with target X, subject to the following axioms:

PT 0)

For every object X of C, the morphisms occurring in the families of morphisms of $\mathrm{Cov}(X)$ are base-changeable. Recall that a morphism $Y \to X$ is called base-changeable if for every morphism $Z \to X$, the fiber product $Z \times_X Y$ is representable.

PT 1)

For every object X of C, every family $(X_\alpha \to X)_{\alpha\in A}$ belonging to $\mathrm{Cov}(X)$, and every morphism $Y \to X$ with $Y \in \mathrm{Ob}(C)$, the family

$$(X_\alpha \times_X Y \to Y)_{\alpha\in A}$$

belongs to $\mathrm{Cov}(Y)$. This is stability under base change.

PT 2)

If $(X_\alpha \to X)_{\alpha \in A}$ belongs to $\mathrm{Cov}(X)$, and if for each $\alpha \in A$ the family

$$(X_{\beta_\alpha} \to X_\alpha)_{\beta_\alpha \in B_\alpha}$$

belongs to $\mathrm{Cov}(X_\alpha)$, then the family $(X_\gamma \to X)_{\gamma \in \coprod_{\alpha \in A} B_\alpha}$, where for $\gamma = (\alpha, \beta_\alpha)$ the morphism $X_\gamma \to X$ is the composite

$$X_\gamma = X_{\beta_\alpha} \to X_\alpha \to X,$$

belongs to $\mathrm{Cov}(X)$. This is stability under composition.

PT 3)

The family $X \xrightarrow{\mathrm{id}_X} X$ belongs to $\mathrm{Cov}(X)$.

1.3.1

Definition 1.1.6 allows us, for a given *pretopology* on C, to consider the *topology* on C generated by this pretopology. Note that if C is a category in which fiber products are representable, then every topology T on C can be defined by a pretopology, namely the one for which $\mathrm{Cov}(X)$ consists of *all* families covering X for the topology T.

Proposition 1.4. Let C be a category, let E be a pretopology on C, let T be the topology defined by the pretopology E (1.1.6), and let X be an object of C. Denote by $J_E(X)$ the set of sieves on X generated by the families of the pretopology, and by $J_T(X)$ the set of refinements of X for the topology T. Then $J_E(X)$ is cofinal in $J_T(X)$. In other words, for a sieve R on X to belong to $J_T(X)$, it is necessary and sufficient that there exist a sieve $R' \in J_E(X)$ such that $R' \subset R$.

Proof. For each object X of C, let $J'(X)$ be the set of sieves on X which contain a sieve of $J_E(X)$. Evidently $J'(X) \subset J_T(X)$. To show that $J'(X) = J_T(X)$, it is enough to show that the data of the $J'(X)$ define a topology on C. Now the $J'(X)$ plainly satisfy axioms (T1) and (T3). It remains to verify (T2). For this it is enough to prove that, if R' is a subsieve of $R \in J_E(X)$ such that for every $Y \to R$ the sieve $R' \times_X Y$ on Y belongs to $J'(Y)$, then R' belongs to $J'(X)$. The sieve R is generated by a family $(X_\alpha \to X)$ belonging to $\mathrm{Cov}(X)$, and for every α the sieve $R' \times_X X_\alpha$ contains a sieve generated by a family $(X_{\beta_\alpha} \to X_\alpha)$ belonging to $\mathrm{Cov}(X_\alpha)$. It follows that R' contains a sieve generated by the family $(X_{\beta_\alpha} \to X)$. Hence, by axiom (PT2), R' contains a sieve of $J_E(X)$, and therefore belongs to $J'(X)$. □

2 Sheaves of sets

Definition 2.1. Let C be a site whose underlying category is a $\mathcal{U}$-category. A presheaf F with values in $\mathcal{U}$-**Set** is called *separated* (respectively, a *sheaf*) if, for every sieve R covering an object X of C, the map

$$\mathrm{Hom}_{\widehat{C}}(X, F) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R, F)$$

is injective (respectively, bijective). The full subcategory of $\widehat{C}$ whose objects are the sheaves is called the category of sheaves of sets on C, and is most often denoted $\widetilde{C}$. When no ambiguity can result, we shall simply say the category of sheaves on C; conversely, we shall sometimes be more precise and say the category of sheaves with values in $\mathcal{U}$-**Set**, or the category of $\mathcal{U}$-sheaves.

Proposition 2.2. Let C be a $\mathcal{U}$-category, and let $\mathcal{F} = (F_i)_{i \in I}$ be a family of $\mathcal{U}$-presheaves on C. For each object X of C, denote by $J_{\mathcal{F}}(X)$ the set of sieves $R \to X$ such that, for every morphism $Y \to X$ of C with target X, the sieve $R \times_X Y$ has the following property: the map

$$\mathrm{Hom}_{\widehat{C}}(Y, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R \times_X Y, F_i)$$

is bijective (respectively, injective) for every $i \in I$. Then the sets $J_{\mathcal{F}}(X)$ define a topology on C, which is the finest topology for which each of the F_i is a sheaf (respectively, a separated presheaf).

Proof. The $J_{\mathcal{F}}(X)$ plainly satisfy axioms (T1) and (T3). It remains to show that they satisfy (T2). For this it is enough to show that:

1) if $R' \hookrightarrow R \hookrightarrow X$ are two sieves on X, with $R \in J_{\mathcal{F}}(X)$, and such that for every $Y \to R$, where Y is an object of C, the sieve $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$, then R' belongs to $J_{\mathcal{F}}(X)$;

2) if $R' \hookrightarrow R \hookrightarrow X$ are two sieves on X such that $R' \in J_{\mathcal{F}}(X)$, then R belongs to $J_{\mathcal{F}}(X)$.

In case 2), note that for every $Y \to R$, where Y is an object of C, the sieve $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$. In cases 1) and 2), the sieve R is the inductive limit of the objects Y of C lying over R (I 3.4). Inductive limits in $\widehat{C}$ are universal (I 3.3). Consequently, in cases 1) and 2), R' is the inductive limit of the $R' \times_X Y$. But in cases 1) and 2), $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$. Passing to the inductive limit over the objects Y of C lying over R, one deduces that the map

$$\mathrm{Hom}_{\widehat{C}}(R, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R', F_i)$$

is bijective (respectively, injective) for every $i \in I$. It follows that the maps

$$\mathrm{Hom}_{\widehat{C}}(X, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R', F_i), \qquad \mathrm{Hom}_{\widehat{C}}(X, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R, F_i)$$

are, in cases 1) and 2), bijections (respectively, injections). Moreover, hypotheses 1) and 2) are visibly stable under arbitrary base change $Y \to X$, with Y an object of C. Hence, for every object Y of C lying over X, and every $i \in I$, the maps

$$\begin{aligned}\mathrm{Hom}_{\widehat{C}}(Y, F_i) &\longrightarrow \mathrm{Hom}_{\widehat{C}}(R \times_X Y, F_i),\\ \mathrm{Hom}_{\widehat{C}}(Y, F_i) &\longrightarrow \mathrm{Hom}_{\widehat{C}}(R' \times_X Y, F_i)\end{aligned}$$

are in both cases bijections (respectively, injections). This shows that R' and R belong to $J_{\mathcal{F}}(X)$. □

Corollary 2.3. Let C be a $\mathcal{U}$-category, and for every object X of C let $K(X)$ be a set of sieves on X. Assume that the $K(X)$ satisfy axiom (T1) of (1.1). For a presheaf F to be a sheaf (respectively, a separated presheaf) for the topology generated (1.1.6) by the $K(X)$, it is necessary and sufficient that, for every object X of C and every sieve $R \in K(X)$, the map

$$\mathrm{Hom}_{\widehat{C}}(X, F) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R, F)$$

be bijective (respectively, injective).

Corollary 2.4. In particular, let C be a $\mathcal{U}$-category endowed with a pretopology. For a presheaf F to be a sheaf (respectively, a separated presheaf), it is necessary and sufficient that, for every object X of C and every family $(X_\alpha \to X)$ belonging to $\mathrm{Cov}(X)$, the diagram of sets

$$F(X) \longrightarrow \prod_{\alpha\in A} F(X_\alpha) \rightrightarrows \prod_{(\alpha,\beta)\in A\times A} F(X_\alpha \times_X X_\beta)$$

be exact (respectively, that the map

$$F(X) \longrightarrow \prod_{\alpha\in A} F(X_\alpha)$$

be injective).

Proof. Apply 2.3, then I 3.5 and I 2.12.

With Corollary 2.4 one recovers the definition given in [1].

Definition 2.5. Let C be a $\mathcal{U}$-category. The *canonical topology* of C is the finest topology for which the representable functors are sheaves (2.2). A sieve covering X for the canonical topology will be called a *universally strict epimorphic sieve.* A covering family for the canonical topology will be called a *universally strict epimorphic family.* When, moreover, the morphisms of the covering family are base-changeable, the family will be called *universally effective epimorphic* [2].

Remark 2.5.1. For almost all the sites that have had to be used up to now, the topology is *less fine* than the canonical topology; in other words, the representable functors on C are sheaves, i.e. the covering families of C are universally strict epimorphic. The only exception to this rule is the site $\widehat{C}$ studied in no. 5, whose topology is *finer*, and most often strictly finer, than the canonical topology.

Proposition 2.6. For a sieve $R \hookrightarrow X$ to be universally strict epimorphic, it is necessary and sufficient that, for every object $Y \to X$ of C lying over X, and for every object Z of C, the map

$$\mathrm{Hom}_C(Y, Z) \longrightarrow \varprojlim_{C/(Y\times_X R)} \mathrm{Hom}_C(\,\cdot\,, Z)$$

be a bijection.

Proof. Immediate by applying 2.2 and then I 5.3. $\square$

Remarks 2.7.

1) Proposition 2.6 gives a characterization of universally strict epimorphic sieves of a category C which is independent of the universe in which the presheaves take their values, under the sole condition that the Hom-sets $\mathrm{Hom}(X, Y)$ of C belong to this universe. Thus it allows one to define the canonical topology for any category C, without needing to specify the universes.

2) Let C be a site whose underlying category is a $\mathcal{U}$-category, let F be a $\mathcal{U}$-presheaf, and let $\mathcal{V}$ be a universe containing $\mathcal{U}$. The underlying category of C is a $\mathcal{V}$-category, and F may be regarded as a $\mathcal{V}$-presheaf. The $\mathcal{U}$-presheaf F is a sheaf (respectively, a separated presheaf) if and only if the $\mathcal{V}$-presheaf F is a sheaf (respectively, a separated presheaf). In other

words, the condition for a presheaf F to be a sheaf (respectively, a separated presheaf) does not depend, in the sense just specified, on the universe in which the presheaf F takes its values. In particular, let C be a site and let F be a $\mathcal{U}$-presheaf; one says that F is a sheaf (respectively, a separated presheaf) if there exists a universe $\mathcal{V}$ containing $\mathcal{U}$ such that the category C is a $\mathcal{V}$-category and such that F is a $\mathcal{V}$-sheaf (respectively, a separated $\mathcal{V}$-presheaf). This property does not depend on the universe $\mathcal{V}$.

Exposé II. Topologies and Sheaves

Sections 3–6.9

J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé II, Sections 3–6.9. It continues immediately after Remarks 2.7 and completes Exposé II. The range corresponds to source lines 422–2114 of the Orgogozo–Laszlo TeX snapshot. I write $\widehat{C}$ for the category of presheaves on C, and $\widetilde{C}$ for the category of sheaves on C. Terminology is normalized to current English usage: *sieve*, *covering morphism*, *bicovering morphism*, *associated sheaf* or *sheafification*, and *base-changeable* for French *quarrable*.

3 The sheaf associated to a presheaf

Definition 3.0.1. Let C be a site. A *topologically generating family* of C, or simply a *generating family* when no confusion can result, is a set G of objects of C such that every object of C is the target of a covering family of morphisms of C (1.2) whose sources are elements of G.

Definition 3.0.2. Let $\mathcal{U}$ be a universe. A $\mathcal{U}$*-site* is a site C whose underlying category is a $\mathcal{U}$-category (Exposé I, 1.1), and which has a small topologically generating family. If C is a $\mathcal{U}$-category, a $\mathcal{U}$*-topology* on C is a topology on C making C into a $\mathcal{U}$-site. A site C is said to be $\mathcal{U}$*-small*, or, by abuse of language, *small*, if the underlying category of C is small (Exposé I, 1.0).

3.0.3

It follows immediately from the definitions that every topology finer than a $\mathcal{U}$-topology is again a $\mathcal{U}$-topology, and that a small site is a $\mathcal{U}$-site.

Proposition 3.0.4. Let C be a $\mathcal{U}$-site and let G be a small topologically generating family of C. For $X \in \mathrm{Ob}(C)$, let $J_G(X)$ denote the set of covering sieves on X generated by a family of morphisms

$$(Y_\alpha \xrightarrow{u_\alpha} X)_{\alpha \in A}, \qquad Y_\alpha \in G.$$

Then:

1) the set $J_G(X)$ is small;

2) $J_G(X)$ is cofinal in the set $J(X)$ of all covering sieves on X, ordered by inclusion;

3) for every $R \in J_G(X)$, there is a small epimorphic family $(u_\alpha : Y \to R)_{\alpha \in A}$ with $Y \in G$ (Exposé I, 10.2).

Proof. For (1), put

$$A(X) = \coprod_{Y \in G} \mathrm{Hom}(Y, X).$$

The set $A(X)$ is small (Exposé I, 0), and $\mathrm{card}\, J_G(X) \le 2^{\mathrm{card}\, A(X)}$. Hence $J_G(X)$ is small.

For (2), let $R \in J(X)$. Put

$$A(R) = \coprod_{Y \in G} \mathrm{Hom}(Y, R),$$

and let R' be the sieve on X generated by the family

$$Y \xrightarrow{u} R \hookrightarrow X, \qquad u \in A(R).$$

We have $R' \subset R$, and it remains only to prove that R' is covering. By axiom (T2) of 1.1, it is enough to show that, for every morphism $Z \to R$ with $Z \in \mathrm{Ob}(C)$, the sieve $R' \times_X Z$ on Z is covering. But this sieve contains the sieve generated by the family of all morphisms $Y \to Z$ with $Y \in G$, which is covering by the hypothesis on G. Hence $R' \times_X Z$ is covering, again by axiom (T2).

For (3), let $R \in J_G(X)$. The family

$$(Y \xrightarrow{u} R), \qquad u \in A(R) = \coprod_{Y \in G} \mathrm{Hom}(Y, R),$$

is epimorphic by construction. For every $Y \in \mathrm{Ob}(C)$, $\mathrm{Hom}(Y, R)$ is contained in $\mathrm{Hom}(Y, X)$, which is small. Hence $A(R)$ is small. □

3.0.5

Let C be a $\mathcal{U}$-site, let $\mathcal{V}$ be a universe such that $C \in \mathcal{V}$ and $\mathcal{U} \subset \mathcal{V}$, and let G be a $\mathcal{U}$-small topologically generating family of C. The category $\widehat{C}$ of presheaves of $\mathcal{U}$-sets on C is a $\mathcal{V}$-category (Exposé I, 1.1.1). Let X be an object of C. The set $J(X)$ of covering sieves on X, ordered by inclusion, is $\mathcal{V}$-small and cofiltered (1.1.1). For every $\mathcal{U}$-presheaf F, the inductive limit

$$\varinjlim_{R \in J(X)} \mathrm{Hom}_{\widehat{C}}(R, F)$$

is therefore representable by an element of $\mathcal{V}$ (Exposé I, 2.4.1). Moreover, by 3.0.4(3), for every $R \in J_G(X)$, the set $\mathrm{Hom}_{\widehat{C}}(R, F)$ is $\mathcal{U}$-small; since $J_G(X)$ is a $\mathcal{U}$-small cofinal subset of $J(X)$ (loc. cit.), it follows from Exposé I, 2.4.2 that

$$\varinjlim_{R \in J(X)} \mathrm{Hom}_{\widehat{C}}(R, F)$$

is $\mathcal{U}$-small. We choose, for each F and X, an element of $\mathcal{U}$ representing this inductive limit, and put

$$LF(X) = \varinjlim_{R \in J(X)} \mathrm{Hom}_{\widehat{C}}(R, F).$$

If $g : Y \to X$ is a morphism of C, the base-change functor $g^* : J(X) \to J(Y)$ defines a map

$$LF(g) : LF(X) \longrightarrow LF(Y),$$

making $X \mapsto LF(X)$ into a presheaf on C.

Since the morphism $\mathrm{id}_X : X \to X$ is an element of $J(X)$, for every object X of C one has a map

$$\ell(F)(X) : F(X) \longrightarrow LF(X),$$

and these maps plainly define a morphism of presheaves

$$\ell(F) : F \longrightarrow LF.$$

It is also clear that $F \mapsto LF$ is functorial in F, and that the morphisms $\ell(F)$ define a natural transformation

$$\ell : \mathrm{Id} \longrightarrow L.$$

Finally, let $R \hookrightarrow X$ be a refinement of X. The definition of $LF(X)$ and Exposé I, 1.4 give a map

$$Z_R : \mathrm{Hom}_{\widehat{C}}(R, F) \longrightarrow \mathrm{Hom}_{\widehat{C}}(X, LF).$$

For every morphism $g : Y \to X$ of C, the definition of LF shows that the square

$$\begin{array}{ccc} \mathrm{Hom}_{\widehat{C}}(R, F) & \xrightarrow{Z_R} & \mathrm{Hom}_{\widehat{C}}(X, LF) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_{\widehat{C}}(R \times_X Y, F) & \xrightarrow{Z_{R\times_X Y}} & \mathrm{Hom}_{\widehat{C}}(Y, LF) \end{array} \tag{*}$$

is commutative, where the vertical arrows are the evident ones.

We now copy the standard argument of Demazure.

Lemma 3.1.

1) For every refinement $i_R : R \hookrightarrow X$ and every $u : R \to F$, the diagram

$$\begin{array}{ccc} R & \xrightarrow{i_R} & X \\ u \downarrow & & \downarrow Z_R(u) \\ F & \xrightarrow{\ell(F)} & LF \end{array} \tag{**}$$

is commutative.

2) For every morphism $v : X \to LF$, there exists a refinement R of X and a morphism $u : R \to F$ such that $Z_R(u) = v$.

3) Let Y be an object of C, and let $u, v : Y \rightrightarrows F$ be two morphisms such that $\ell(F)\circ u = \ell(F)\circ v$. Then the equalizer, or kernel, of the pair (u, v) is a refinement of Y.

4) Let R and R' be two refinements of X, and let $u : R \to F$ and $u' : R' \to F$ be two morphisms. Then $Z_R(u) = Z_{R'}(u')$ if and only if u and u' coincide on some refinement

$$R'' \hookrightarrow R \times_X R'.$$

Proof. Only assertion (1) is nontrivial. We must prove

$$Z_R(u) \circ i_R = \ell(F) \circ u.$$

By Exposé I, 3.4 it is enough to prove equality after composition with every morphism $g : Y \to R$, where Y is an object of C. Let $f = i_R \circ g$, and consider $R \times_X Y$. The canonical monomorphism $R \times_X Y \to Y$ has a section, hence is an isomorphism by 1.3(3). By definition of $\ell(F)$, the morphism $Z_{R\times_X Y}(u \circ g')$ is $\ell(F) \circ u \circ g$. On the other hand, the commutativity of $(*)$ gives

$$Z_{R\times_X Y}(u \circ g') = Z_R(u) \circ f.$$

Thus $\ell(F) \circ u \circ g = Z_R(u) \circ i_R \circ g$, as required. □

Proposition 3.2.

1) The functor L is left exact (Exposé I, 2.3.2).

2) For every presheaf F, LF is a separated presheaf.

3) A presheaf F is separated if and only if the morphism $\ell(F) : F \to LF$ is a monomorphism. In that case LF is a sheaf.

4) The following properties are equivalent:

   (i) $\ell(F) : F \to LF$ is an isomorphism;

   (ii) F is a sheaf.

Proof. For (1), it is enough, by Exposé I, 3.1, to prove that for every object X of C, the functor $F \mapsto LF(X)$ commutes with finite projective limits. But for every $R \hookrightarrow X$ in $J(X)$, the functor

$$F \longmapsto \mathrm{Hom}_{\widehat{C}}(R, F)$$

commutes with projective limits, by the definition of projective limit; and the filtered inductive limit $\varinjlim_{J(X)}$ commutes with finite projective limits because $J(X)$ is a cofiltered set (Exposé I, 2.7).

For (2), let X be an object of C, and let $f, g : X \rightrightarrows LF$ be two morphisms which coincide on a refinement $R \hookrightarrow X$. By Lemma 3.1(2), there is a covering sieve $R' \hookrightarrow X$, which we may suppose contained in R, and two morphisms $u, v : R' \rightrightarrows F$ such that $Z_{R'}(u) = f$ and $Z_{R'}(v) = g$. By Lemma 3.1(1), $\ell(F) \circ u = \ell(F) \circ v$. Hence, by Lemma 3.1(4), u and v coincide on a refinement $R'' \hookrightarrow R'$. If w denotes the restriction of u to R'', then

$$Z_{R''}(w) = Z_{R'}(u) = Z_{R'}(v),$$

and consequently $f = g$. Thus LF is separated.

For (3), if F is separated, then $\ell(F)$ is a monomorphism, because a filtered inductive limit of monomorphisms is a monomorphism. Conversely, if $\ell(F)$ is a monomorphism, then F is a subpresheaf of a separated presheaf, hence is separated. We now prove that LF is then a sheaf. Let $i : R \hookrightarrow X$ be a covering sieve of an object X of C, and let $u : R \to LF$ be a morphism. It is enough to prove that u factors through X. Put

$$R' = F \times_{LF} R, \qquad v = Z_{R'}(\mathrm{pr}_1).$$

To prove $u = v \circ i$, it is enough, by Exposé I, 3.4, to test after every morphism $m : Y \to R$ with Y an object of C. Put $R'' = Y \times_R R'$, with projections p_1, p_2. The projection $R'' \to Y$ is a monomorphism making R'' into a covering sieve of Y by Lemma 3.1(2). We have

$$v \circ i \circ p_2 = \ell(F) \circ \mathrm{pr}_1$$

by Lemma 3.1(1), and hence

$$v \circ i \circ p_2 = u \circ m \circ p_2.$$

Pulling back along p_1 gives

$$v \circ i \circ m \circ p_2 = u \circ m \circ p_2.$$

Since LF is separated, $v \circ i \circ m = u \circ m$. This proves the sheaf condition.

Assertion (4) is clear from (3). $\square$

Remark 3.3. Let $J'(X)$ be a cofinal subset of $J(X)$. Then

$$LF(X) = \varinjlim_{R \in J'(X)} \mathrm{Hom}_{\widehat{C}}(R, F).$$

In particular, if the topology of C is defined by a pretopology $X \mapsto \mathrm{Cov}(X)$ (1.1.3), then the functor L can be described by means of the covering families of $\mathrm{Cov}(X)$ (Exposé I, 2.12 and 3.5). Making the formulae explicit, one recovers the usual construction.

Theorem 3.4. Let C be a $\mathcal{U}$-site. The inclusion functor

$$i : \widetilde{C} \hookrightarrow \widehat{C}$$

from sheaves to presheaves has a left adjoint

$$\underline{a} : \widehat{C} \longrightarrow \widetilde{C},$$

which is left exact (Exposé I, 2.3.2). The functor $i \circ \underline{a}$ is canonically isomorphic to $L \circ L$ (cf. 3.0.5). For every presheaf F, the adjunction morphism

$$F \longrightarrow i\,\underline{a}(F)$$

is, through the preceding isomorphism, the composite

$$F \xrightarrow{\ell(F)} LF \xrightarrow{\ell(LF)} L(LF).$$

Definition 3.5. The sheaf $\underline{a}F$ is called the *sheaf associated to the presheaf* F, or the *sheafification* of F.

Theorem 3.4 follows immediately from Proposition 3.2.

Proposition 3.6. Let C be a $\mathcal{U}$-site and let $\mathcal{V} \supset \mathcal{U}$ be a universe. Denote by

$$\widehat{C}_{\mathcal{U}}, \quad \widetilde{C}_{\mathcal{U}}$$

the categories of $\mathcal{U}$-presheaves and $\mathcal{U}$-sheaves, and by

$$\widehat{C}_{\mathcal{V}}, \quad \widetilde{C}_{\mathcal{V}}$$

the categories of $\mathcal{V}$-presheaves and $\mathcal{V}$-sheaves. Let

$$\underline{a}_{\mathcal{U}} : \widehat{C}_{\mathcal{U}} \to \widetilde{C}_{\mathcal{U}}, \qquad \underline{a}_{\mathcal{V}} : \widehat{C}_{\mathcal{V}} \to \widetilde{C}_{\mathcal{V}}$$

be the corresponding associated-sheaf functors. Then the diagram

$$\begin{array}{ccc} \widehat{C}_{\mathcal{U}} & \xrightarrow{\underline{a}_{\mathcal{U}}} & \widetilde{C}_{\mathcal{U}} \\ \downarrow & & \downarrow \\ \widehat{C}_{\mathcal{V}} & \xrightarrow{\underline{a}_{\mathcal{V}}} & \widetilde{C}_{\mathcal{V}} \end{array}$$

where the vertical functors are the canonical inclusions, commutes up to canonical isomorphism.

Proof. This follows from the construction of the functors $\underline{a}_{\mathcal{U}}$ and $\underline{a}_{\mathcal{V}}$ (3.4 and 3.0.5). $\square$

4 Exactness properties of the category of sheaves

The exactness properties of the category of sheaves are deduced from those of the category of presheaves by means of Theorem 3.4. This section makes that philosophy explicit in a number of typical statements which are among the most useful.

Theorem 4.1. Let C be a $\mathcal{U}$-site, let $\widetilde{C}$ be the category of sheaves, let

$$\underline{a} : \widehat{C} \longrightarrow \widetilde{C}$$

be the associated-sheaf functor, and let

$$i : \widetilde{C} \longrightarrow \widehat{C}$$

be the inclusion functor.

1) The functor $\underline{a}$ commutes with inductive limits and is exact.

2) $\mathcal{U}$-inductive limits in $\widetilde{C}$ are representable. For every small category I and every functor $E : I \to \widetilde{C}$, the canonical morphism

$$\varinjlim_I E \longrightarrow \underline{a}\left(\varinjlim_I iE\right)$$

is an isomorphism.

3) $\mathcal{U}$-projective limits in $\widetilde{C}$ are representable. For every object X of C, the functor $F \mapsto F(X)$ on $\widetilde{C}$ commutes with projective limits; equivalently, the inclusion $i : \widetilde{C} \to \widehat{C}$ commutes with projective limits.

Proof. These properties follow essentially from Theorem 3.4 and Exposé I, 2.11. □

Thus, in the category of sheaves, products indexed by an element of $\mathcal{U}$, fiber products, sums indexed by an element of $\mathcal{U}$, amalgamated sums, kernels, cokernels, images, and coimages are representable.

Corollary 4.1.1. Let C be a $\mathcal{U}$-site and let F be a sheaf of sets on C. The canonical homomorphism

$$\varinjlim_{C/F} \underline{a}X \longrightarrow F$$

is an isomorphism.

Proof. This follows from Exposé I, 3.4 and the fact that $\underline{a}$ commutes with inductive limits. □

Proposition 4.2. Every morphism of $\widetilde{C}$ which is both an epimorphism and a monomorphism is an isomorphism.

Proof. Let $u : G \to H$ be a morphism of $\widetilde{C}$ which is both an epimorphism and a monomorphism. First note that u is a monomorphism of presheaves (4.1(1)). Form the amalgamated sum

$H \amalg_G H = K$ and the two canonical morphisms $i_1, i_2 : H \rightrightarrows K$ in the category of presheaves. Since u is a monomorphism of presheaves, the square

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ u \downarrow & & \downarrow i_2 \\ H & \xrightarrow{i_1} & K \end{array}$$

is immediately verified to be both cartesian and cocartesian (Exposé I, 10.1). Applying the associated-sheaf functor, we get a cartesian and cocartesian square in the category of sheaves:

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ u \downarrow & & \downarrow \underline{a}i_2 \\ H & \xrightarrow{\underline{a}i_1} & \underline{a}K. \end{array} \tag{*}$$

Since u is an epimorphism of sheaves, $\underline{a}i_1$ is an isomorphism; since the square $(*)$ is cartesian, u is an isomorphism. □

Proposition 4.3.

1) Representable inductive limits in $\widetilde{C}$ are universal (Exposé I, 2.5).

2) Every epimorphic family of morphisms is an effective universal epimorphic family (2.6).

3) Every equivalence relation is effective and universal (Exposé I, 10.6).

4) $\mathcal{U}$-filtered inductive limits commute with finite projective limits (Exposé I, 2.6).

Proof. Assertions (1)–(4) hold in the category of sets and hence in the category of presheaves of sets (Exposé I, 3.1). Assertions (1) and (4) then follow at once from the corresponding assertions for presheaves and from Theorem 4.1.

We prove (3). Let $p_1, p_2 : R \rightrightarrows X$ be an equivalence relation in $\widetilde{C}$. It is also an equivalence relation of presheaves (4.1.1). Hence there exists a morphism of presheaves $u : X \to Y$ such that

$$\begin{array}{ccc} R & \xrightarrow{p_2} & X \\ p_1 \downarrow & & \downarrow u \\ X & \xrightarrow{u} & Y \end{array}$$

is both cartesian and cocartesian in the category of presheaves. Applying the associated-sheaf functor, one obtains a cartesian and cocartesian square in the category of sheaves (4.1(1)). Thus $R \rightrightarrows X$ is an effective equivalence relation. Since every equivalence relation is effective by the preceding argument, every equivalence relation is universally effective.

We prove (2). Let $(u_i : X_i \to X)_{i \in I}$ be an epimorphic family in $\widetilde{C}$. By 2.6 and Exposé I, 2.12, it is enough to prove:

a) for every morphism of sheaves $v : Y \to X$, the family

$$\mathrm{pr}_{2,i} : X_i \times_X Y \to Y, \qquad i \in I,$$

is epimorphic in $\widetilde{C}$;

b) for every sheaf Z, the diagram of sets

$$\operatorname{Hom}(X,Z) \longrightarrow \prod_{i\in I}\operatorname{Hom}(X_i,Z) \rightrightarrows \prod_{(i,k)\in I\times I}\operatorname{Hom}(X_i\times_X X_k, Z)$$

is exact.

Let X' be the union of the images of the morphisms u_i, in the sense of presheaves, and let $j : X' \hookrightarrow X$ be the canonical injection. For every i, the morphism $u_i : X_i \to X$ factors as $X_i \xrightarrow{u_i'} X' \xrightarrow{j} X$, and the family $u_i' : X_i \to X'$ is epimorphic in $\widehat{C}$. Since j is a monomorphism, the morphism $\underline{a}j : \underline{a}X' \to X$ is a monomorphism of sheaves (4.1). Since $u_i = \underline{a}j\,\underline{a}u_i'$ for each i, $\underline{a}j$ is also an epimorphism of sheaves. Hence $\underline{a}j$ is an isomorphism by Proposition 4.2.

Let $v : Y \to X$ be a morphism of sheaves. After base change we obtain a morphism $j_v : X' \times_X Y \to Y$ and morphisms $u'_{i,v} : X_i \times_X Y \to X' \times_X Y$. Since $\underline{a}$ commutes with fiber products, $\underline{a}j_v$ is an isomorphism. Since epimorphic families of presheaves remain epimorphic after base change, the family $u'_{i,v}$ is epimorphic in $\widehat{C}$. As $\underline{a}$ commutes with inductive limits, the family

$$\underline{a}u'_{i,v} : X_i \times_X Y \to \underline{a}(X' \times_X Y)$$

is epimorphic in $\widetilde{C}$. This proves (a).

For (b), let Z be a sheaf. Since $\underline{a}j$ is an isomorphism, the map

$$\operatorname{Hom}(X,Z) \longrightarrow \operatorname{Hom}(X',Z)$$

is a bijection. Since the u_i' form an epimorphic family in $\widehat{C}$, the diagram

$$\operatorname{Hom}(X',Z) \longrightarrow \prod_{i\in I}\operatorname{Hom}(X_i,Z) \rightrightarrows \prod_{(i,k)\in I\times I}\operatorname{Hom}(X_i\times_{X'} X_k, Z)$$

is exact. Finally, since X' is a subpresheaf of X, the presheaf $X_i \times_{X'} X_k$ is canonically isomorphic to $X_i \times_X X_k$. This gives (b). $\square$

Remarks 4.3.1.

1) Proposition 4.3(2) formally gives a second proof of Proposition 4.2. In any category, a morphism which is both an effective universal epimorphism and a monomorphism is plainly an isomorphism.

2) From the preceding propositions it follows that every morphism in the category of sheaves factors uniquely as an effective epimorphism followed by an effective monomorphism. For non-additive categories, this property is the natural analogue of axiom (AB2) for abelian categories, namely coim $\simeq$ im [Tohoku].

4.4.0

Let C be a $\mathcal{U}$-site. The functor $h : C \to \widehat{C}$ (Exposé I, 1.3.1), composed with the associated-sheaf functor, gives a functor

$$\epsilon_C : C \longrightarrow \widetilde{C},$$

called the *canonical functor* from C to $\widetilde{C}$. It will be used constantly below. The functor ϵ_C commutes with finite projective limits. When the topology of C is less fine than the canonical topology, ϵ_C is fully faithful and commutes with projective limits; in that case it is, moreover, defined even if C need not have a small topologically generating family. We shall not study in detail the behavior of ϵ_C with respect to inductive limits (cf. SGA 3, IV). We shall, however, need the following propositions.

Theorem 4.4. Let C be a $\mathcal{U}$-site and let $(u_i : X_i \to X)_{i\in I}$ be a family of morphisms of C with target X. The following conditions are equivalent:[1]

(i) The family
$$(\epsilon_C u_i : \epsilon_C X_i \to \epsilon_C X)_{i\in I}$$
is an epimorphic family in $\widetilde{C}$ (Exposé I, 10.2).

(ii) The family $(u_i : X_i \to X)_{i\in I}$ is a covering family of C (1.2).

Proof. (ii) $\Rightarrow$ (i). Let $R \hookrightarrow X$ be the sieve generated by the u_i (Exposé I, 4.3.3), and let $u_i' : X_i \to R$ be the morphisms induced by the u_i. The family u_i' is epimorphic in $\widehat{C}$, and R is a covering sieve on X. Hence, for every sheaf F, the map
$$\mathrm{Hom}(X, F) \longrightarrow \mathrm{Hom}(R, F)$$
is bijective, while
$$\mathrm{Hom}(R, F) \longrightarrow \prod_i \mathrm{Hom}(X_i, F)$$
is injective. Thus
$$\mathrm{Hom}(X, F) \longrightarrow \prod_i \mathrm{Hom}(X_i, F)$$
is injective, and consequently
$$\mathrm{Hom}(\underline{a}X, F) \longrightarrow \prod_i \mathrm{Hom}(\underline{a}X_i, F)$$
is injective. This is exactly (i).

(i) $\Rightarrow$ (ii). With the notation just introduced, let $J_R : R \hookrightarrow X$ be the canonical inclusion of the sieve generated by the u_i. From (i) it follows that the morphism of sheaves
$$\underline{a}J_R : \underline{a}R \longrightarrow \underline{a}X$$
is both an epimorphism and a monomorphism. Hence it is an isomorphism by Proposition 4.2. With the notation of Section 3, we have a commutative diagram
$$\begin{array}{ccccc} X & \xrightarrow{\ell(X)} & LX & \xrightarrow{\ell(LX)} & \underline{a}X \\ \uparrow J_R & & \uparrow LJ_R & & \uparrow \wr \\ R & \xrightarrow{\ell(R)} & LR & \xrightarrow{\ell(LR)} & \underline{a}R. \end{array}$$

[1]When the site C need not have a small topologically generating family and when the topology of C is less fine than the canonical topology, one has (ii) $\Rightarrow$ (i); see the proof of 4.4.

By Lemma 3.1(2), there exists a covering sieve $J_{S_1} : S_1 \to X$ and a morphism $u_1 : S_1 \to LR$ such that

$$\ell(LX) \circ \ell(X) \circ J_{S_1} = \underline{a} J_R \circ \ell(LR) \circ u_1 = \ell(LX) \circ LJ_R \circ u_1. \tag{4.6.1}$$

Put $m = \ell(X) \circ J_{S_1}$ and $n = LJ_R \circ u_1$. These are morphisms $S_1 \to LX$. Let $S_2 \hookrightarrow S_1$ be the equalizer of the pair (m, n). For every object Y of C and every morphism $\alpha : Y \to S_1$, equation (4.6.1) gives

$$\ell(LX)m\alpha = \ell(LX)n\alpha.$$

By Lemma 3.1(3), the equalizer $S_2 \times_{S_1} Y$ of $m\alpha$ and $n\alpha$ is a covering sieve on Y. Hence, by axiom (T2), $J_{S_2} : S_2 \hookrightarrow X$ is a covering sieve on X. Thus we have a morphism $u_2 : S_2 \to LR$ and a commutative diagram

$$\begin{array}{ccc} S_2 & \xrightarrow{J_{S_2}} & X \\ {\scriptstyle u_2}\downarrow & & \downarrow{\scriptstyle \ell(X)} \\ LR & \xrightarrow{LJ_R} & LX \end{array} \tag{4.6.2}$$

with $R \hookrightarrow X$ inserted in the evident way through J_R.

Let Y be an object of C and let $\beta : Y \to S_2$ be a morphism. By Lemma 3.1(1), there exists a covering sieve $J_{Q_1} : Q_1 \hookrightarrow Y$ and a morphism $v_1 : Q_1 \to R$ such that

$$u_2 \beta J_{Q_1} = \ell(R) v_1.$$

Put

$$f = J_{S_2} \beta J_{Q_1}, \qquad g = J_R v_1.$$

Let Q_2 be the equalizer of the pair $f, g : Q_1 \rightrightarrows X$, and let $J_{Q_2} : Q_2 \hookrightarrow Y$ be the canonical inclusion. For every object Z of C and every morphism $\gamma : Z \to Q_1$, the commutativity of (4.6.2) gives

$$\ell(X) f \gamma = LJ_R\, u_2 \beta J_{Q_1} \gamma = LJ_R\, \ell(R) v_1 \gamma = \ell(X) J_R v_1 \gamma = \ell(X) g \gamma.$$

By Lemma 3.1(3), the equalizer $Q_2 \times_{Q_1} Z$ of $f\gamma$ and $g\gamma$ is a covering sieve on Z. By axiom (T2), $J_{Q_2} : Q_2 \hookrightarrow Y$ is a covering sieve on Y. Hence, for every morphism $\beta : Y \to S_2$, there exists a covering sieve $Q_2 \hookrightarrow Y$ and a morphism $v_2 : Q_2 \to R$ making

$$\begin{array}{cccc} Q_2 & \xrightarrow{v_2} & R & \\ \downarrow & & \downarrow J_R & \\ Y & \xrightarrow{\beta} & S_2 & \xrightarrow{J_{S_2}} X \end{array}$$

commutative.

Now let $S_2 \cap R$ denote the fiber product of S_2 and R over X. The sieve

$$(S_2 \cap R) \times_{S_2} Y$$

on Y contains Q_2; hence it is covering. Axiom (T2) implies that $S_2 \cap R$ is a covering sieve on X. Since R contains $S_2 \cap R$, R is itself a covering sieve on X. $\square$

Corollary 4.4.4. Let T be the topology of the $\mathcal{U}$-site C. Let T' (respectively T'') be the finest topology on C among those for which the objects of $\widetilde{C}$ are sheaves (respectively separated presheaves) (2.2). Then

$$T = T' = T''.$$

Indeed, trivially $T \le T' \le T''$, while Theorem 4.4 and Proposition 2.2 imply $T'' \le T$. $\square$

Definition 4.5. An *initial object* of a category A is an object ϕ_A representing the empty inductive limit, i.e. such that for every $X \in \mathrm{Ob}(A)$ there is exactly one arrow $\phi_A \to X$. An object ϕ_A is called *strict initial* if it is initial and every morphism with target ϕ_A is an isomorphism.

Let $(S_i)_{i\in I}$ be a family of objects of a category A. Suppose that the sum $S = \coprod_{i\in I} S_i$ is representable. The sum S is said to be *disjoint* if the structural morphisms $S_i \to S$ are base-changeable monomorphisms and, for every pair i, j with $i \neq j$, the fiber product $S_i \times_S S_j$ is an initial object of A. The sum is said to be *universally disjoint* if it is disjoint and remains a disjoint sum after every base change $T \to S$. It follows that, for every $i \neq j$, the objects $S_i \times_S S_j$ are strict initial objects of A.

Example 4.5.1. In the category of sets, direct sums are disjoint and universal. Hence the same is true in every category of presheaves of sets (Exposé I, 3.1), and therefore in every category of sheaves of sets on a $\mathcal{U}$-site (4.1). In particular, the initial object of $\widetilde{C}$ is strict.

Proposition 4.6. Let C be a $\mathcal{U}$-category and let $(s_i : X_i \to X)_{i\in I}$ be a family of morphisms of C. For every $\mathcal{U}$-topology $\mathcal{T}$ on C (3.0.2), let $\widetilde{C}_{\mathcal{T}}$ be the corresponding category of sheaves, and let

$$\epsilon_{\mathcal{T}} : C \to \widetilde{C}_{\mathcal{T}}$$

be the corresponding canonical functor.

1) Let $\mathcal{T}$ be a $\mathcal{U}$-topology such that

$$\coprod_{i\in I} \epsilon_{\mathcal{T}}(X_i) \xrightarrow{(\epsilon_{\mathcal{T}}(s_i))} \epsilon_{\mathcal{T}}(X)$$

is an isomorphism. Then, for every topology $\mathcal{T}'$ finer than $\mathcal{T}$, the morphism

$$\coprod_{i\in I} \epsilon_{\mathcal{T}'}(X_i) \xrightarrow{(\epsilon_{\mathcal{T}'}(s_i))} \epsilon_{\mathcal{T}'}(X)$$

is an isomorphism.

2) Let $\mathcal{T}$ be a $\mathcal{U}$-topology on C. The following properties (i) and (ii) are equivalent:

(i) The following three conditions hold:

(a) The family $(s_i : X_i \to X)_{i\in I}$ is covering for $\mathcal{T}$.

(b) For every $i \in I$, the diagonal morphism of presheaves

$$\Delta_i : X_i \hookrightarrow X_i \times_X X_i$$

is transformed by the associated-sheaf functor for $\mathcal{T}$ into an isomorphism; this is automatically the case if the s_i are monomorphisms.

(c) For every pair $i \neq j$, the presheaf $X_i \times_X X_j$ is transformed by the associated-sheaf functor for $\mathcal{T}$ into the initial object of $\widetilde{C}_{\mathcal{T}}$.

(ii) $\epsilon_{\mathcal{T}}(X)$ is the sum of the $\epsilon_{\mathcal{T}}(X_i)$.

Proof. For (1), a sheaf for $\mathcal{T}'$ is a sheaf for $\mathcal{T}$. Hence, for every sheaf F for $\mathcal{T}'$, the morphism

$$\mathrm{Hom}_{\widehat{C}}(X, F) \longrightarrow \prod_{i \in I} \mathrm{Hom}_{\widehat{C}}(X_i, F)$$

is an isomorphism. This implies the assertion.

For (2), by definition property (ii) holds if and only if the morphism of presheaves

$$\Phi = (h(s_i))_{i \in I} : \coprod_i h(X_i) \longrightarrow h(X)$$

is transformed by the associated-sheaf functor into an isomorphism; equivalently, by Proposition 4.2, $\underline{a}(\Phi)$ is both an epimorphism and a monomorphism of sheaves. By Theorem 4.4, $\underline{a}(\Phi)$ is an epimorphism if and only if (a) holds. The diagonal

$$\Delta : \coprod_i h(X_i) \longrightarrow \left(\coprod_i h(X_i)\right) \times_{h(X)} \left(\coprod_i h(X_i)\right)$$

is the direct sum of morphisms $\Delta_{i,j}$, indexed by $I \times I$, defined as follows:

- if $i = j$, $\Delta_{i,i}$ is the diagonal

$$h(X_i) \to h(X_i) \times_{h(X)} h(X_i);$$

- if $i \neq j$, $\Delta_{i,j}$ is the morphism

$$\phi_{\widehat{C}} \to h(X_i) \times_{h(X)} h(X_j),$$

  where $\phi_{\widehat{C}}$ denotes the initial object of $\widehat{C}$.

The morphism $\underline{a}(\Phi)$ is a monomorphism if and only if $\underline{a}(\Delta)$ is an isomorphism; and by Theorem 4.1, $\underline{a}(\Delta)$ is an isomorphism if and only if each $\underline{a}(\Delta_{i,j})$ is an isomorphism, i.e. if and only if (b) and (c) hold. ☐

Corollary 4.6.1. Let C be a $\mathcal{U}$-category and let X be an object of C.

1) Let $\mathcal{T}$ be a $\mathcal{U}$-topology on C. The object X of C is transformed by the associated-sheaf functor for $\mathcal{T}$ into the initial object of $\widetilde{C}_{\mathcal{T}}$ if and only if the empty sieve covers X.

2) When X is a strict initial object of C, the sheaf associated to X, for every $\mathcal{U}$-topology finer than the canonical topology, is an initial object.

Proof. For (1), take the indexing set I in Proposition 4.6(2) to be empty. For (2), by (1) and Proposition 4.6(1), it is enough to prove that the empty sieve covers X for the canonical topology; equivalently, by 2.6, it is enough to prove that every object $Y \to X$ over X is initial, which follows from the definition of strict initial object. ☐

Corollary 4.6.2. Let C be a $\mathcal{U}$-site, and let $(s_i : X_i \to X)_{i \in I}$ be a family of base-changeable morphisms in C, all with target X, satisfying:

α the family s_i is covering;

β each $s_i : X_i \to X$ is a monomorphism;

γ for each pair $i \neq j$, the object $X_i \times_X X_j$ is covered by the empty sieve, i.e. for every sheaf F on C, the set $F(X_i \times_X X_j)$ is a singleton.

Then $\epsilon_C(X)$ is the sum of the $\epsilon_C(X_i)$, $i \in I$.

Proof. This follows immediately from Proposition 4.6(2) and Corollary 4.6.1(1). □

Corollary 4.6.3. Let C be a $\mathcal{U}$-category and let $(s_i : X_i \to X)_{i \in I}$ be a family of base-changeable morphisms with the same target. Suppose that the canonical topology of C is a $\mathcal{U}$-topology. The following conditions are equivalent:

(i) There exists a $\mathcal{U}$-topology $\mathcal{T}$ on C, less fine than the canonical topology, such that $\epsilon_{\mathcal{T}}(X)$ is the sum of the $\epsilon_{\mathcal{T}}(X_i)$, $i \in I$.

(ii) The object X is the disjoint universal sum, in C, of the X_i (4.5).

Proof. (ii) $\Rightarrow$ (i). Since X is the universal sum of the X_i, the family s_i is covering for the canonical topology of C (2.6). Thus condition α of Corollary 4.6.2 holds for the canonical topology. Condition β is immediate, and condition γ follows from Proposition 4.6(1),(2). Hence (i) holds.

(i) $\Rightarrow$ (ii). Let $\mathcal{T}$ be a topology on C, less fine than the canonical topology, such that $\epsilon_{\mathcal{T}}(X)$ is the sum of the $\epsilon_{\mathcal{T}}(X_i)$. For every sheaf F for $\mathcal{T}$, the canonical map

$$\mathrm{Hom}(X, F) \to \prod_i \mathrm{Hom}(X_i, F)$$

is a bijection. Since every representable presheaf is a sheaf, it follows at once that X is the sum in C of the X_i. Applying the same argument when I is empty, we see that if an object of C is transformed into the initial object of $\widetilde{C}_{\mathcal{T}}$, then it is an initial object of C. The functor $\epsilon_{\mathcal{T}} : C \to \widetilde{C}_{\mathcal{T}}$ is fully faithful and commutes with finite projective limits. Hence condition (b) of Corollary 4.6.2 implies that the s_i are monomorphisms in C, and condition (c), by what has just been proved, implies that $X_i \times_X X_j$ is an initial object of C for $i \neq j$. Thus X is a disjoint sum of the X_i. Since $\epsilon_{\mathcal{T}}$ commutes with fiber products, this property is stable under every base change $Y \to X$, and therefore X is the disjoint universal sum of the X_i. □

Corollary 4.6.4. Let C be a $\mathcal{U}$-category and let X be an object of C. Suppose that the canonical topology on C is a $\mathcal{U}$-topology. The following properties are equivalent:

(i) There exists on C a topology less fine than the canonical topology such that $\epsilon_{\mathcal{T}}(X)$ is an initial object of $\widetilde{C}_{\mathcal{T}}$.

(ii) X is a strict initial object of C.

Proof. Take I to be empty in Corollary 4.6.3. □

Proposition 4.7. Let C be a $\mathcal{U}$-category, let $\widetilde{C}$ be the category of sheaves on C for the canonical topology, let

$$\epsilon_C : C \to \widetilde{C}$$

be the canonical functor (4.4.0), and let $R \rightrightarrows X$ be an equivalence relation in C admitting a cokernel Y in C. Let $\pi : X \to Y$ be the canonical morphism and suppose it is base-changeable. Consider:

(i) $\epsilon_C(Y)$ is the quotient of the equivalence relation

$$\epsilon_C(R) \rightrightarrows \epsilon_C(X).$$

(ii) The equivalence relation $R \rightrightarrows X$ is effective and universal (cf. no. 7).

Then (ii) implies (i). If C has a $\mathcal{U}$-topology less fine than the canonical topology, then (i) implies (ii).

Proof. (ii) $\Rightarrow$ (i). The canonical morphism $R \to X \times_Y X$ is an isomorphism. Let $S \to Y$ be the sieve generated by $\pi : X \to Y$. For every presheaf F, the diagram

$$\mathrm{Hom}(S, F) \longrightarrow \mathrm{Hom}(X, F) \rightrightarrows \mathrm{Hom}(R, F)$$

is exact (Exposé I, 2.12). It is therefore enough to show that, for every sheaf F for the canonical topology, the map

$$\mathrm{Hom}(Y, F) \longrightarrow \mathrm{Hom}(S, F)$$

is a bijection; equivalently, that S is covering for the canonical topology, or that the sieve generated by $\pi : X \to Y$ is covering. This follows from Definition 2.5.

(i) $\Rightarrow$ (ii). The functor $\epsilon_C : C \to \widetilde{C}$ commutes with finite projective limits and is fully faithful. Since the canonical morphism

$$\epsilon_C(R) \to \epsilon_C(X) \times_{\epsilon_C(Y)} \epsilon_C(X)$$

is an isomorphism (4.3), the canonical morphism $R \to X \times_Y X$ is an isomorphism. The morphism $\epsilon_C(X) \to \epsilon_C(Y)$ is an epimorphism; hence, by Theorem 4.4, $\pi : X \to Y$ is a covering morphism for the canonical topology. □

Proposition 4.8. Let C be a $\mathcal{U}$-site. The category of sheaves on C has the following properties:

a) finite projective limits are representable;

b) direct sums indexed by an element of $\mathcal{U}$ are representable, and are disjoint and universal (4.5);

c) equivalence relations are effective and universal.

These facts were proved in Theorem 4.1(2),(3), Proposition 4.3(3), and Example 4.5.1.

These properties have been singled out because they will later make it possible to characterize the $\mathcal{U}$-categories equivalent to categories of sheaves on categories belonging to $\mathcal{U}$ (Exposé IV, 1.2).

Remark. Recall that property (a) is equivalent to the following: there exists a final object, and fiber products are representable (Exposé I, 2.3.1).

4.9

Let C be a site whose underlying category is a $\mathcal{U}$-category. Then the category of $\mathcal{U}$-sheaves on C satisfies the conditions considered in Exposé I, 7.3, either (i) or (ii); hence, by loc. cit., the various variants considered in Exposé I, 7.1 for the notion of a *generating family* in C coincide. With this understood:

Proposition 4.10. With the preceding notation, consider the canonical functor

$$\epsilon : C \to \widetilde{C}$$

(4.4.0), and let $G \subset \mathrm{Ob}(C)$ be a topologically generating family in C (3.0.1). Then the family $(\epsilon(X))_{X\in G}$ of objects of $\widetilde{C}$ is a generating family. In particular, the family $(\epsilon(X))_{X\in \mathrm{Ob}(C)}$ is generating.

We must prove that every morphism $u : F \to F'$ in $\widetilde{C}$ such that

$$\mathrm{Hom}(\epsilon(X), F) \longrightarrow \mathrm{Hom}(\epsilon(X), F') \tag{4.10.1}$$

is a bijection for every $X \in G$, is an isomorphism. The preceding map is identified with

$$u(X) : F(X) \longrightarrow F'(X), \tag{4.10.2}$$

and it remains to show that, if this map is bijective for every $X \in G$, then it is bijective for every $X \in \mathrm{Ob}(C)$. Let $(X_i \to X)_{i\in I}$ be a covering family of X by objects of G. For every pair (i, j) in $I \times I$, consider the set of all morphisms

$$X_{ijk} \to X_i \times_X X_j$$

in $\widehat{C}$, with $X_{ijk} \in G$, where k ranges over an index set $I_{i,j}$. We obtain a morphism of exact diagrams of sets

$$\begin{array}{ccccc}
F(X) & \to & \prod\limits_i F(X_i) & \rightrightarrows & \prod\limits_{i,j,k} F(X_{ijk}) \\
\downarrow & & \downarrow \wr & & \downarrow \wr \\
F'(X) & \to & \prod\limits_i F'(X_i) & \rightrightarrows & \prod\limits_{i,j,k} F'(X_{ijk}),
\end{array}$$

where the last two vertical arrows are bijections by hypothesis. Hence the first vertical arrow is a bijection as well. □

Corollary 4.11. Suppose that C is a $\mathcal{U}$-site (3.0.2). Then $\widetilde{C}$ is a $\mathcal{U}$-category and has a $\mathcal{U}$-small generating family.

Indeed, by hypothesis one may take in Proposition 4.10 a small generating family G, which proves the existence of a small generating family. On the other hand, if $F, F' \in \mathrm{Ob}(\widetilde{C})$, a homomorphism $F \to F'$ is known once the maps (4.10.1), or equivalently (4.10.2), are known for every $X \in G$ (Exposé I, 7.1.1). Hence the map

$$\mathrm{Hom}(F, F') \longrightarrow \prod_{X\in G} \mathrm{Hom}(F(X), F'(X))$$

is injective. Since the right-hand side is $\mathcal{U}$-small, the left-hand side is $\mathcal{U}$-small, which proves that $\widetilde{C}$ is a $\mathcal{U}$-category.

Corollary 4.12. Let C be a $\mathcal{U}$-site. Then for every object X of $\widetilde{C}$, both the set of subobjects of X and the set of quotient objects of X are $\mathcal{U}$-small.

This follows from Exposé I, 7.4 and 7.5, respectively, which apply by Corollary 4.11.

5 Extension of a topology from C to $\widehat{C}$

Proposition 5.1. Let C be a $\mathcal{U}$-site and let $f : H \to K$ be a morphism of $\widehat{C}$. The following conditions are equivalent:

(i) For every $X \to K$, with $X \in \mathrm{Ob}(C)$, the corresponding morphism

$$H \times_K X \to X$$

has as image a covering sieve of X.

(ii) The morphism of associated sheaves

$$\underline{a}(f) : \underline{a}H \to \underline{a}K$$

is an epimorphism in $\widetilde{C}$.

(ii bis) For every sheaf F on C, the map

$$\mathrm{Hom}(K, F) \to \mathrm{Hom}(H, F)$$

deduced from f is injective.

Proof. It is clear that (ii) is equivalent to (ii bis).

We prove (i) $\Rightarrow$ (ii). By Theorem 4.4, (i) means that

$$\underline{a}(H \times_K X) \to \underline{a}(X)$$

is an epimorphism. Since

$$\underline{a}(H \times_K X) \simeq \underline{a}(H) \times_{\underline{a}(K)} \underline{a}(X),$$

because $\underline{a}$ commutes with finite projective limits (4.1), the morphism in question is obtained from

$$\underline{a}(f) : \underline{a}(H) \to \underline{a}(K)$$

by base change. As epimorphisms in $\widetilde{C}$ are universal, this shows that (ii) implies (i) after base change; conversely, since the family of all $X \to K$ is epimorphic and the corresponding family $\underline{a}(X) \to \underline{a}(K)$ is epimorphic in $\widetilde{C}$ (4.1), to verify that $\underline{a}(f)$ is an epimorphism it is enough to check it after every such base change. This proves (i) $\Rightarrow$ (ii), and the equivalence follows. $\square$

Definition 5.2.

1) A morphism $f : H \to K$ satisfying the equivalent conditions of Proposition 5.1 is called a *covering morphism*. A family of morphisms $(f_i : H_i \to K)_{i \in I}$ with common target is called *covering* if the corresponding morphism

$$f : \coprod_i H_i \to K$$

is covering.

2) A morphism $f : H \to K$ is called *bicovering* if it is covering and if its diagonal morphism

$$H \to H \times_K H$$

is covering. A family $(f_i : H_i \to K)_{i \in I}$ with common target is called *bicovering* if the corresponding morphism $\coprod_i H_i \to K$ is bicovering.

By condition (i) of Proposition 5.1, to say that a family $(f_i : H_i \to K)_{i \in I}$ is covering means that, for every morphism $X \to K$, with $X \in \mathrm{Ob}(C)$, the family

$$H_i \times_K X \to X, \qquad i \in I,$$

has as image a covering sieve of X; or equivalently, by condition (ii), that the family of morphisms

$$\underline{a}(f_i) : \underline{a}H_i \to \underline{a}K$$

in $\widetilde{C}$ is epimorphic, taking into account that $\underline{a}$ commutes with direct sums (4.1).[2]

Proposition 5.3. Let C be a $\mathcal{U}$-site and let $f : H \to K$ be a morphism of $\widehat{C}$. The following conditions are equivalent:

(i) f is bicovering (5.2).

(i bis) f is covering, and for every object X of C and every pair of morphisms $u, v : X \rightrightarrows H$ such that $fu = fv$, the equalizer of (u, v) is a covering sieve of X.

(ii) $\underline{a}(f) : \underline{a}H \to \underline{a}K$ is an isomorphism in $\widetilde{C}$.

(ii bis) For every sheaf F on C, the map

$$\mathrm{Hom}(K, F) \to \mathrm{Hom}(H, F)$$

is a bijection.

Proof. The equivalence (i) $\Leftrightarrow$ (i bis) follows from condition (i) of Proposition 5.1, applied to the diagonal morphism $H \to H \times_K H$. The equivalence (ii) $\Leftrightarrow$ (ii bis) is trivial.

(i) $\Rightarrow$ (ii). The morphism

$$\underline{a}(f) : \underline{a}H \to \underline{a}K$$

is an epimorphism by Proposition 5.1. Since $\underline{a}$ commutes with fiber products (4.1), the diagonal

$$\underline{a}H \to \underline{a}H \times_{\underline{a}K} \underline{a}H$$

is an epimorphism by Proposition 5.1. A diagonal morphism is always a monomorphism; hence it is an isomorphism by Proposition 4.2. Thus $\underline{a}(f)$ is a monomorphism. It is therefore an isomorphism by Proposition 4.2.

(ii) $\Rightarrow$ (i). Since $\underline{a}$ commutes with fiber products, both f and the diagonal $H \to H \times_K H$ are transformed by $\underline{a}$ into isomorphisms. In particular they are transformed into epimorphisms. Hence they are covering. $\square$

[2] The property of being covering (respectively bicovering), for a morphism or family of morphisms of $\widehat{C}$, is stable under base change.

5.3.1

It follows from Proposition 5.3 and from the fact that $\underline{a}$ commutes with direct sums that a family

$$(f_i : H_i \to K)_{i \in I}$$

of morphisms of $\widehat{C}$ is bicovering if and only if the f_i induce, on associated sheaves, an isomorphism

$$\coprod_i \underline{a} H_i \xrightarrow{\sim} \underline{a} K,$$

or equivalently if and only if, for every sheaf F, the map

$$\mathrm{Hom}(K, F) \longrightarrow \prod_i \mathrm{Hom}(H_i, F)$$

is bijective.

Proposition 5.4. Let C be a $\mathcal{U}$-site. There exists on $\widehat{C}$ a topology, clearly unique, such that a family $H_i \to K$ of arrows of $\widehat{C}$ with common target is covering for this topology if and only if it is covering in the sense of Definition 5.2. It is also the least fine topology on $\widehat{C}$ among the topologies T satisfying:

a) T is finer than the canonical topology of $\widehat{C}$, i.e. every epimorphic family in $\widehat{C}$ is covering for T;

b) every covering family in C is covering in $\widehat{C}$.

Proof. We only give indications. It is left to the reader to show that the families covering in the sense of Definition 5.2 are the covering families of a topology T_C on $\widehat{C}$; one uses Theorem 4.1. The covering families for the canonical topology on $\widehat{C}$ are the epimorphic families of $\widehat{C}$ (2.6 and Exposé I, 3.1). Since $\underline{a}$ commutes with inductive limits (4.1), the topology T_C is finer than the canonical topology of $\widehat{C}$. Moreover, the covering families of C are covering families for T_C (5.1).

Let T' denote the least fine topology on $\widehat{C}$ having properties (a) and (b). Then T_C is finer than T'. Let

$$(f_i : H_i \to K)_{i \in I}$$

be a T_C-covering family. We show that it is a covering family for T'. Let $s_i : H_i \to H = \coprod_i H_i$ be the canonical monomorphisms, and let $f = (f_i) : H \to K$ be the morphism defined by the f_i. The family s_i is covering for T'. Thus, by axiom (T2), to prove that (f_i) is covering for T', it is enough to prove that the morphism $f : H \to K$ is covering for T'.

There exists an epimorphic family in $\widehat{C}$

$$u_\lambda : X_\lambda \to K, \qquad \lambda \in \Lambda, \quad X_\lambda \in \mathrm{Ob}(C)$$

(Exposé I, 3.4). By axiom (T2), to prove that $f : H \to K$ is covering for T', it is enough to prove that, for every λ, the projection

$$H \times_K X_\lambda \to X_\lambda$$

is covering for T'. Let

$$v_j : Y_j \to H \times_K X_\lambda, \qquad j \in J, \quad Y_j \in \mathrm{Ob}(C),$$

be an epimorphic family in $\widehat{C}$. The family $(\mathrm{pr}_2 \circ v_j)_{j\in J}$ is covering for T_C. Hence it is a covering family in C by Proposition 5.1, and therefore is covering for T'. Consequently the sieve generated by $H \times_K X_\lambda \to X_\lambda$ contains a T'-covering sieve; it is therefore covering for T'. This proves the proposition. □

Remark 5.4.1. The proof of Proposition 5.4 actually shows that every topology T' on $\widehat{C}$ which is finer than the canonical topology of $\widehat{C}$, is the least fine of the topologies T on $\widehat{C}$ satisfying:

a) T is finer than the canonical topology of $\widehat{C}$;

b) every T'-covering family of the form

$$u_i : X_i \to X, \qquad i \in I,$$

where X and the X_i are objects of C, is covering for T.

In particular, every topology T' on $\widehat{C}$ finer than the canonical topology is uniquely determined by the families $u_i : X_i \to X$, $i \in I$, with X_i and X objects of C, which are covering for T'.

Remark 5.4.2. One can easily show that, for every topology T' on $\widehat{C}$ finer than the canonical topology, the set of families of morphisms

$$(X_i \to X)_{i\in I}$$

with common target in C, which are covering for T', is the set of covering families of a topology on C. Thus Proposition 5.4 and Remark 5.4.1 establish a one-to-one correspondence between topologies on C and topologies on $\widehat{C}$ finer than the canonical topology.

5.5.0

Let C be a small category. Let $\mathcal{C}$ be the set of strictly full subcategories of $\widehat{C}$ (strictly full means that every object isomorphic to an object of the subcategory is itself an object of the subcategory) whose inclusion functor admits a left adjoint commuting with finite projective limits. Also let $\mathcal{T}$ denote the set of topologies on C. Theorem 3.4 defines a map

$$\Phi : \mathcal{T} \longrightarrow \mathcal{C}.$$

We now define a map in the other direction. For this, to every element

$$i' : C' \hookrightarrow \widehat{C}, \qquad \underline{a}' \dashv i',$$

with $\underline{a}'$ left adjoint to i', we associate a topology T_e on C. For every object X of C, $J_e(X)$ is defined to be the set of subobjects of X whose inclusion morphism is transformed by $\underline{a}'$ into an isomorphism. Using the hypotheses on $\underline{a}'$, one immediately checks that this defines a topology T_e on C. We have therefore defined a map

$$\Psi : \mathcal{C} \longrightarrow \mathcal{T}.$$

We then have the following result, due to J. Giraud.

Theorem 5.5. The map Φ is a bijection, and Ψ is its inverse.

Proof. The composite $\Psi \circ \Phi$ is the identity. Indeed, this follows immediately from Proposition 5.1.

The composite $\Phi \circ \Psi$ is the identity. Let

$$i' : C' \hookrightarrow \widehat{C}, \qquad \underline{a}' \dashv i',$$

be an element of $\mathcal{C}$, let T_e be the topology corresponding to it under Ψ, and let C_e be the category of sheaves for T_e. Using the definition of the topology T_e and the definition of bicovering morphisms (5.2), one proves the equivalence of the following assertions:

(i) a morphism u of $\widehat{C}$ is bicovering for the topology T_e;

(ii) the morphism u of $\widehat{C}$ is transformed by $\underline{a}'$ into an isomorphism.

It is clear that C' is a full subcategory of C_e. It remains only to prove that every sheaf F for the topology T_e is an object of C'. By the equivalence above, the morphism

$$F \longrightarrow i'\underline{a}'(F)$$

is bicovering; since its source and target are sheaves, Proposition 5.3(ii) implies that it is an isomorphism. □

6 Sheaves with values in a category

6.0

Let C and D be two categories. A contravariant functor

$$F : C^\circ \to D$$

is called a *presheaf on C with values in D*.

Definition 6.1. Let C be a site and let D be a category. A presheaf $F : C^\circ \to D$ is called a *sheaf on C with values in D*, or more briefly a *D-valued sheaf*, if for every object S of D, the presheaf of sets

$$X \longmapsto \mathrm{Hom}_D(S, F(X)), \qquad X \in \mathrm{Ob}(C),$$

is a sheaf. The full subcategory of $\mathrm{Hom}(C^\circ, D)$ consisting of sheaves on C with values in D will be denoted

$$\mathrm{Hom}^\sim(C^\circ, D).$$

Remarks 6.2.

1) Condition 6.1 means that for every object X of C, every covering sieve R of X, and every object S of D, one has an isomorphism

$$\mathrm{Hom}_D(S, F(X)) \xrightarrow{\sim} \varprojlim_{Y \to R} \mathrm{Hom}_D(S, F(Y))$$

(Exposé I, 3.5 and 2.1). When D is the category of $\mathcal{U}$-sets, this recovers the definition of sheaves of sets.

2) Let D' be a $\mathcal{U}$-category and let $G : D \to D'$ be a functor commuting with projective limits. By composition, G sends sheaves with values in D to sheaves with values in D'.

6.3.0

Let γ be a species of algebraic structure defined by finite projective limits (Exposé I, 2.9), and let $\mathcal{U}$-γ be the category of γ-sets which belong to $\mathcal{U}$. Denote by

$$\mathrm{For} : \mathcal{U}\text{-}\gamma \to \mathcal{U}\text{-}\mathbf{Set}$$

the forgetful functor to the underlying set. The functor For commutes with projective limits and therefore, by the preceding remark, defines by composition functors

$$\widehat{\mathrm{For}} : \mathrm{Hom}(C^\circ, \mathcal{U}\text{-}\gamma) \longrightarrow \widehat{C},$$

and

$$\widetilde{\mathrm{For}} : \mathrm{Hom}^\sim(C^\circ, \mathcal{U}\text{-}\gamma) \longrightarrow \widetilde{C}.$$

The latter is called the *underlying sheaf of sets* functor. In fact, it factors canonically through the category γ-$\widetilde{C}$ of γ-objects of $\widetilde{C}$. Denoting the resulting functor again by

$$\widetilde{\mathrm{For}} : \mathrm{Hom}^\sim(C^\circ, \mathcal{U}\text{-}\gamma) \longrightarrow \gamma\text{-}\widetilde{C},$$

one may state:

Proposition 6.3.1. The functor $\widetilde{\mathrm{For}}$ establishes equivalences of categories between:

1) sheaves on C with values in $\mathcal{U}$-γ;

2) γ-objects of the category of sheaves of sets;

3) γ-objects of the category of presheaves of sets whose underlying presheaf is a sheaf.

Proof. The proof uses essentially Exposé I, 3.2 and the fact that a finite projective limit of sheaves, computed in the category of presheaves, is again a sheaf (4.1). □

6.3.2

The preceding proposition justifies the abuse of language by which one identifies a sheaf with values in $\mathcal{U}$-γ with the corresponding γ-object in $\widetilde{C}$. We shall henceforth use this abuse of language systematically.

We shall use the classical terminology: *sheaves of groups*, *sheaves of commutative groups*, which we shall most often call *abelian sheaves*, *sheaves of rings*, *sheaves of A-modules*, and so on.

6.3.3

We denote by

$$\widetilde{C}_\gamma \quad \text{(respectively } \widehat{C}_\gamma\text{)}$$

the category of γ-objects of $\widetilde{C}$ (respectively of $\widehat{C}$). When γ is the species of structure "A-module", we also write $\widetilde{C}_A$, or simply $\widetilde{C}_{\mathrm{ab}}$ when $A = \mathbb{Z}$.

Assume that C is a $\mathcal{U}$-site. The associated-sheaf functor is left exact (4.1), and therefore:

Proposition 6.4. The inclusion functor

$$i_\gamma : \widetilde{C}_\gamma \to \widehat{C}_\gamma$$

has a left adjoint

$$a_\gamma : \widehat{C}_\gamma \to \widetilde{C}_\gamma$$

which is left exact; in other words, there is an isomorphism

$$\mathrm{Hom}_{\widetilde{C}_\gamma}(a_\gamma(\,\cdot\,),\,\cdot\,) \xrightarrow{\sim} \mathrm{Hom}_{\widehat{C}_\gamma}(\,\cdot\,, i_\gamma(\,\cdot\,)).$$

Let X be an object of $\widehat{C}_\gamma$. The underlying sheaf of sets of $a_\gamma(X)$ is canonically isomorphic to the sheaf of sets associated to the underlying presheaf of sets of X. The morphism of underlying presheaves of sets associated to the adjunction morphism

$$\mathrm{id} \to i_\gamma a_\gamma$$

is identified with the adjunction morphism applied to the underlying presheaf of sets.

Proposition 6.5. Suppose that the forgetful functor

$$\mathrm{For} : \gamma\text{-}\mathcal{U} \to \mathcal{U}\text{-}\mathbf{Set}$$

has a left adjoint[3]

$$\mathrm{Lib} : \mathcal{U}\text{-}\mathbf{Set} \to \gamma\text{-}\mathcal{U},$$

i.e.

$$\mathrm{Hom}_{\mathcal{U}\text{-}\mathbf{Set}}(\,\cdot\,, \mathrm{For}(\,\cdot\,)) \xrightarrow{\sim} \mathrm{Hom}_{\gamma\text{-}\mathcal{U}}(\mathrm{Lib}(\,\cdot\,),\,\cdot\,).$$

Then the functor

$$\mathrm{For}^{\sim} : \widetilde{C}_\gamma \to \widetilde{C}$$

(respectively

$$\mathrm{For}^{\wedge} : \widehat{C}_\gamma \to \widehat{C})$$

has a left adjoint

$$\mathrm{Lib}^{\sim} \quad (\text{respectively } \mathrm{Lib}^{\wedge}),$$

and there is a canonical isomorphism

$$\mathrm{Lib}^{\sim} = a_\gamma \, \mathrm{Lib}^{\wedge}\, i.$$

Proof. The proof is formal and is left to the reader. □

Corollary 6.6. Let $(X_i)_{i\in I}$ be a generating family of $\widetilde{C}$ (4.9). Then the family

$$\mathrm{Lib}^{\sim}(X_i)$$

is a generating family of $\widetilde{C}_\gamma$.

Proof. The proof is formal once one has observed that the functor $\mathrm{For}^{\sim}$ commutes with projective limits and is conservative: $\mathrm{For}^{\sim}(u)$ is an isomorphism if and only if u is an isomorphism. □

[3] One proves that such a left adjoint always exists in the usual examples: free group, free abelian group, free A-module, and so on.

Proposition 6.7. Let A be a $\mathcal{U}$-sheaf of rings, or a small ring. Let

$$\widetilde{C}_A \quad \text{(respectively } \widehat{C}_A\text{)}$$

be the category of sheaves (respectively presheaves) of unital A-modules (6.3.3) on a $\mathcal{U}$-site C. Then $\widetilde{C}_A$ is a $\mathcal{U}$-abelian category satisfying the axioms (AB5), "existence of left exact filtered inductive limits", and (AB3)*, "existence of infinite products", of TOHOKU. It has a family of generators indexed by an element of $\mathcal{U}$.

Proof. It is clear that $\widehat{C}_A$ is an abelian category satisfying axioms (AB3), (AB4), and (AB5) (Exposé I, 2.8 and 3.3). From Proposition 6.4 it follows that $\widetilde{C}_A$ is an additive category in which kernels and cokernels are representable.

More explicitly, let i_A and a_A be the inclusion and associated-sheaf functors, and let $u : X \to Y$ be a morphism of $\widetilde{C}_A$. The functor a_A is left exact and commutes with inductive limits. The functor i_A commutes with projective limits. Hence the functor

$$i_A a_A : \widehat{C}_A \to \widehat{C}_A$$

is additive and left exact. We get canonical isomorphisms

$$i_A(\mathrm{Ker}(u)) \xrightarrow{\sim} \mathrm{Ker}(i_A(u)), \tag{*}$$

and

$$\mathrm{coker}(u) \xrightarrow{\sim} a_A\ \mathrm{coker}(i_A(u)). \tag{**}$$

It follows that there is a canonical isomorphism

$$\mathrm{coim}(u) \xrightarrow{\sim} a_A\ \mathrm{coim}(i_A(u)). \tag{***}$$

We prove that the canonical morphism $\mathrm{coim}(u) \to \mathrm{im}(u)$ is an isomorphism. By $(*)$, it is enough to prove exactness of the sequence

$$0 \longrightarrow i_A(\mathrm{coim}(u)) \longrightarrow i_A(Y) \longrightarrow i_A(\mathrm{coker}(u)).$$

Using the isomorphisms $(**)$ and $(***)$, and observing that $a_A i_A(Y)$ is isomorphic to Y, this sequence is identified with

$$0 \longrightarrow i_A a_A(\mathrm{coim}(i_A(u))) \longrightarrow i_A a_A i_A(Y) \longrightarrow i_A a_A(\mathrm{coker}(i_A(u))),$$

which is simply the image under the left exact functor $i_A a_A$ of the exact sequence

$$0 \longrightarrow \mathrm{coim}(i_A(u)) \longrightarrow i_A(Y) \longrightarrow \mathrm{coker}(i_A(u)).$$

The category $\widetilde{C}_A$ is therefore abelian.

The functor

$$a_A : \widehat{C}_A \to \widetilde{C}_A$$

is left exact and commutes with arbitrary inductive limits. Hence it is additive, exact, and commutes with inductive limits. Since $\widehat{C}_A$ satisfies axiom (AB5), so does $\widetilde{C}_A$. Finally, it is clear that in $\widetilde{C}_A$, products indexed by an element of $\mathcal{U}$ are representable; hence axiom (AB3)* holds. This completes the proof of the first assertion.

For every ring A which is an element of $\mathcal{U}$, the functor

$$\mathrm{For} : \mathcal{U}\text{-}A\text{-}\mathrm{Mod} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

has a left adjoint X_A, the free A-module generated by X. It remains only to apply Proposition 4.10 and Corollary 6.6. $\square$

Notation 6.8. Let H be a sheaf of sets. The sheaf of A-modules associated to the presheaf (cf. notation 6.5)

$$X \longmapsto \mathrm{Lib}_A\, H(X), \qquad X \in \mathrm{Ob}(C),$$

is denoted A_H. When $H = \underline{a}(X)$, the sheaf associated to the presheaf represented by X, one sometimes simply writes A_X, by abuse of notation. The proof of Proposition 6.7 shows that the family of sheaves of A-modules A_X, $X \in \mathrm{Ob}(C)$, is a generating family, indexed by an element of $\mathcal{U}$, for the category of sheaves of A-modules.

Remark 6.9. By TOHOKU, Proposition 6.7 shows that when A is an element of $\mathcal{U}$, the category $\widetilde{C}_A$ has enough injectives. It is known, moreover, that infinite products are not necessarily exact in $\widetilde{C}_A$, and therefore that the category $\widetilde{C}_A$ does not in general have enough projectives [ROOS].

References

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[4] J.-E. ROOS, *C. R.*

Exposé III. Functoriality of Categories of Sheaves

Complete draft translation

J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé III. The range corresponds to source lines 23–1551 of the Orgogozo–Laszlo TeX snapshot. Terminology is normalized to current English usage where this is harmless: *continuous functor*, *cocontinuous functor*, *induced topology*, *comparison lemma*, *sieve*, *bicovering morphism*, *associated sheaf*, and *base-changeable* for French *quarrable*. Numbering follows the original exposé. Cross-references to other exposés are kept numerically rather than converted into active links.

In Exposé I, 5, the behavior of categories of presheaves with respect to functors between the categories of arguments was studied. In this exposé this study is extended to the case of sites and categories of sheaves (nos. 1 and 2). After introducing the induced topology (no. 3), we turn in no. 4 to the comparison lemma, which will play an important role in Exposé IV. In no. 5, for the patient reader, certain commutative diagrams related to localized categories of the type C/X are studied. In Theorem I, 5.1 smallness hypotheses are imposed on the categories under consideration. These hypotheses are not, in general, satisfied by the sites encountered in practice ($\mathcal{U}$-sites). This forces a few contortions (4.2, 4.3, 4.4).

1 Continuous functors

Definition 1.1. Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a functor between the underlying categories. We say that u is *continuous* if, for every sheaf of sets F on C', the presheaf

$$X \longmapsto F(u(X))$$

on C is a sheaf.

This notion of continuity depends a priori on the universe $\mathcal{U}$ for which the two sites are $\mathcal{U}$-sites. Proposition 1.5 shows that it does not depend on it.

1.1.1

Let

$$i : \widetilde{C} \to \widehat{C} \qquad \text{(respectively } i' : \widetilde{C'} \to \widehat{C'}\text{)}$$

denote the canonical inclusion functor from sheaves to presheaves. By Definition 1.1, the functor u is continuous if and only if there exists a functor

$$u_s : \widetilde{C'} \to \widetilde{C}$$

such that

$$iu_s = \widehat{u}^* i', \qquad \widehat{u}^* F = F \circ u$$

(Exposé I, 5.0).

Proposition 1.2. Let C be a small site, let C' be a $\mathcal{U}$-site, and let $u : C \to C'$ be a functor between the underlying categories. The following properties are equivalent:

i) The functor u is continuous.

ii) For every object X of C and every sieve $R \hookrightarrow X$ covering X, the morphism

$$u_! R \hookrightarrow u(X)$$

is bicovering in $\widehat{C'}$ (Exposé II, 5.2 and Exposé I, 5.1).[1]

iii) For every bicovering family $H_i \to K$, $i \in I$, in $\widehat{C}$, the family

$$u_! H_i \to u_! K, \qquad i \in I,$$

is bicovering in $\widehat{C'}$.

iv) There exists a functor

$$u^s : \widetilde{C} \to \widetilde{C'}$$

commuting with inductive limits and "extending u", that is, such that the diagram of canonical functors

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd}$$

commutes up to isomorphism.

Moreover, when C is no longer necessarily a small site but is a $\mathcal{U}$-site, one still has i) $\Leftrightarrow$ iv), and the functor u^s in iv) is necessarily left adjoint to $u_s : \widetilde{C'} \to \widetilde{C}$, and is therefore determined up to a unique isomorphism.

Proof. The proof of i) $\Leftrightarrow$ iv) when C is a $\mathcal{U}$-site will be given in 4.2. Suppose that C is small. Clearly iii) $\Rightarrow$ ii).

i) $\Rightarrow$ iii). Let $H_i \to K$, $i \in I$, be a bicovering family of $\widehat{C}$. For every sheaf F on C', the presheaf u^*F (Exposé I, 5.0) is a sheaf on C. Hence (Exposé II, 5.3) the canonical map

$$\mathrm{Hom}(K, u^*F) \longrightarrow \prod_i \mathrm{Hom}(H_i, u^*F)$$

is bijective. By adjunction (Exposé I, 5.1), it follows that the canonical map

$$\mathrm{Hom}(u_! K, F) \longrightarrow \prod_i \mathrm{Hom}(u_! H_i, F)$$

is bijective. Therefore (Exposé II, 5.3) the family $u_! H_i \to u_! K$, $i \in I$, is bicovering.

ii) $\Rightarrow$ i). Let X be an object of C, let $R \hookrightarrow X$ be a covering sieve, and let F be a sheaf on C'. By Exposé II, 5.3, the map

$$\mathrm{Hom}(u_! X, F) \longrightarrow \mathrm{Hom}(u_! R, F)$$

is bijective. By adjunction (Exposé I, 5.1), the map

$$\mathrm{Hom}(X, u^*F) \longrightarrow \mathrm{Hom}(R, u^*F)$$

[1] "Covering" instead of "bicovering" was not necessarily sufficient; see Example 1.9.3 below.

is bijective. Therefore u^*F is a sheaf.

i) $\Rightarrow$ iv). Use the usual notation $\underline{a}$ and i (respectively $\underline{a}'$ and i') for the associated-sheaf functor and the inclusion into presheaves. For every sheaf G on C, put

$$u^sG = \underline{a}'u_!iG.$$

For every sheaf F on C', one has the sequence of natural isomorphisms

$$\begin{aligned}\operatorname{Hom}(G, u_sF) &\simeq \operatorname{Hom}(iG, iu_sF) \simeq \operatorname{Hom}(iG, u^*i'F)\\ &\simeq \operatorname{Hom}(u_!iG, i'F) \simeq \operatorname{Hom}(\underline{a}'u_!iG, F).\end{aligned}$$

The first isomorphism comes from the full faithfulness of i, the second from the definition of u_s, and the third and fourth from adjunction. Thus u^s is left adjoint to u_s. In particular u^s commutes with inductive limits. For every presheaf K on C and every sheaf F on C', one has natural isomorphisms

$$\operatorname{Hom}(u^s\underline{a}K, F) \simeq \operatorname{Hom}(\underline{a}K, u_sF) \simeq \operatorname{Hom}(K, iu_sF) \simeq \operatorname{Hom}(K, u^*i'F) \simeq \operatorname{Hom}(u_!K, i'F) \simeq \operatorname{Hom}(\underline{a}'u_!K, F),$$

whence a functorial isomorphism

$$u^s\underline{a}K \simeq \underline{a}'u_!K.$$

In particular, using this isomorphism for representable K, and taking Exposé I, 5.4(3) into account, one obtains a diagram, commutative up to isomorphism:

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'}. \end{tikzcd}$$

iv) $\Rightarrow$ ii). Let $h : C \to \widehat{C}$ (respectively $h' : C' \to \widehat{C'}$) be the functor which sends an object to the presheaf it represents. Consider the diagram

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "h"'] & C' \arrow[d, "h'"] \\ \widehat{C} \arrow[r, "u_!"] \arrow[d, "\underline{a}"'] & \widehat{C'} \arrow[d, "\underline{a}'"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'}. \end{tikzcd}$$

The upper square is commutative (Remark 1.4(3)), and $\underline{a}h = \epsilon_C$, $\underline{a}'h' = \epsilon_{C'}$. For every object K of $\widehat{C}$, there is a canonical isomorphism (Exposé I, 3.4)

$$\varinjlim_{(h(X)\to K)\in \operatorname{Ob}(C/K)} h(X) \xrightarrow{\sim} K.$$

Since $u_!$, u^s, $\underline{a}$, and $\underline{a}'$ commute with inductive limits, this gives isomorphisms

$$\varinjlim_{(h(X)\to K)} u^s\underline{a}h(X) \xrightarrow{\sim} u^s\underline{a}K, \qquad \varinjlim_{(h(X)\to K)} \underline{a}'u_!h(X) \xrightarrow{\sim} \underline{a}'u_!K.$$

We have $u_! h = h' u$, and, by hypothesis, an isomorphism $\underline{a}' h' u \simeq u^s \underline{a} h$. Hence there is an isomorphism

$$u^s \underline{a} K \simeq \underline{a}' u_! K,$$

which is immediately checked to be functorial in K. Thus the diagram

$$\begin{tikzcd} \widehat{C} \arrow[r, "u_!"] \arrow[d, "\underline{a}"'] & \widehat{C'} \arrow[d, "\underline{a}'"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd}$$

commutes up to isomorphism. Since $\underline{a}i$ is isomorphic to the identity functor, we have $u^s \simeq \underline{a}' u_! i$, whence the uniqueness of u^s. Let $v : H \to K$ be a bicovering morphism of $\widehat{C}$. Then $\underline{a}(v)$ is an isomorphism (Exposé II, 5.3), and therefore $\underline{a}' u_!(v)$, isomorphic to $u^s \underline{a}(v)$, is an isomorphism. Thus $u_!(v)$ is sent by $\underline{a}'$ to an isomorphism; hence $u_!(v)$ is bicovering (Exposé II, 5.3), which proves ii).

The supplementary assertion, in the case where C is small, has been proved along the way. □

1.2.1

The functor $u^s : \widetilde{C} \to \widetilde{C'}$ of Proposition 1.2(iv) can often be interpreted as an "inverse image" functor for a "morphism of topoi" $\widetilde{C'} \to \widetilde{C}$; cf. Exposé IV, 4.9. Its properties are summarized in the following proposition.

Proposition 1.3. Let $u : C \to C'$ be a continuous functor from a small site C to a $\mathcal{U}$-site C'.

1) The functor u^s is left adjoint to u_s.

2) There is a canonical isomorphism $u^s \simeq \underline{a}' u_! i$.

3) There is a canonical isomorphism $u^s \underline{a} \simeq \underline{a}' u_!$.

4) The functor u^s commutes with inductive limits.

5) When $u_!$ is left exact (cf. Exposé I, 5.4(4)), so is u^s. More generally, u^s commutes with all types of finite projective limits with which $u_!$ commutes.

Proof. Assertions (1)–(4) were proved in the course of the proof of Proposition 1.2. Assertion (5) follows from the isomorphism in (2), using the fact that i commutes with projective limits and that $\underline{a}'$ commutes with finite projective limits (Exposé II, 4.1). □

Remark 1.4. Combining Exposé I, 5.4(3) and Proposition 1.3(3), one obtains a diagram

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "h"'] & C' \arrow[d, "h'"] \\ \widehat{C} \arrow[r, "u_!"] \arrow[d, "\underline{a}"'] & \widehat{C'} \arrow[d, "\underline{a}'"] \\ \widetilde{C} \arrow[r] & \widetilde{C'}, \end{tikzcd}$$

in which the upper square is commutative and the lower square is commutative up to isomorphism. But the functors $\underline{a}$, $\underline{a}'$, $u_!$, and u^s are defined only up to isomorphism. The reader may check that the functors $\underline{a}$, $\underline{a}'$, and u^s can be chosen so that:

1) the composite functors $\underline{a}h$ and $\underline{a}'h'$ are injective on sets of objects;

2) the diagram
$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\underline{a}h"'] & C' \arrow[d, "\underline{a}'h'"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd}$$
commutes.

More precisely, one can choose $\underline{a}$ and $\underline{a}'$ so as to satisfy condition (1). Once $\underline{a}$ and $\underline{a}'$ have been chosen, one can choose u^s so as to satisfy condition (2).

Proposition 1.5. Let $\mathcal{U} \subset \mathcal{V}$ be two universes, let C and C' be two $\mathcal{U}$-sites, and let $u : C \to C'$ be a functor between the underlying categories. Then u is continuous relative to $\mathcal{U}$ if and only if it is continuous relative to $\mathcal{V}$. Moreover, when u is continuous, let $u^s_{\mathcal{U}}$ (respectively $u^s_{\mathcal{V}}$) denote the functor between categories of $\mathcal{U}$-sheaves (respectively $\mathcal{V}$-sheaves) introduced in Proposition 1.2(iv). Then the diagram

$$\begin{tikzcd} \widetilde{C}_{\mathcal{U}} \arrow[r, "u^s_{\mathcal{U}}"] \arrow[d] & \widetilde{C'}_{\mathcal{U}} \arrow[d] \\ \widetilde{C}_{\mathcal{V}} \arrow[r, "u^s_{\mathcal{V}}"] & \widetilde{C'}_{\mathcal{V}} \end{tikzcd} \tag{1.5.1}$$

where the vertical functors are the canonical inclusion functors, is commutative up to canonical isomorphism.

Proof. We give the proof only in the case where C is $\mathcal{U}$-small. The general case is proved in 4.3. It is clear that if u^* carries every $\mathcal{V}$-sheaf on C' to a $\mathcal{V}$-sheaf on C, then it carries every $\mathcal{U}$-sheaf on C' to a $\mathcal{U}$-sheaf on C.

Suppose conversely that u^* carries every $\mathcal{U}$-sheaf on C' to a $\mathcal{U}$-sheaf on C. Denote by

$$\widehat{C}_{\mathcal{U}} \hookrightarrow \widehat{C}_{\mathcal{V}}, \qquad \widehat{C'}_{\mathcal{U}} \hookrightarrow \widehat{C'}_{\mathcal{V}}$$

the categories of $\mathcal{U}$-presheaves and $\mathcal{V}$-presheaves, and by $u_{\mathcal{U}!}$ and $u_{\mathcal{V}!}$ the "inverse image" functors for $\mathcal{U}$-presheaves and $\mathcal{V}$-presheaves respectively. There is a commutative diagram (cf. the explicit construction of $u_!$ in Exposé I, 5.1)

$$\begin{tikzcd} \widehat{C}_{\mathcal{U}} \arrow[r, "u_{\mathcal{U}!}"] \arrow[d, hook] & \widehat{C'}_{\mathcal{U}} \arrow[d, hook] \\ \widehat{C}_{\mathcal{V}} \arrow[r, "u_{\mathcal{V}!}"] & \widehat{C'}_{\mathcal{V}}. \end{tikzcd} \tag{*}$$

The inclusion functors from $\mathcal{U}$-presheaves to $\mathcal{V}$-presheaves are fully faithful and commute with projective limits. It follows from Exposé II, 5.1(i) that a bicovering morphism of $\mathcal{U}$-presheaves is a bicovering morphism of $\mathcal{V}$-presheaves. Let $R \hookrightarrow X$ be a sieve covering an object X of C. By Proposition 1.2(ii), $u_{\mathcal{U}!}R \to u_{\mathcal{U}!}X$ is bicovering. Using the commutativity of $(*)$ and the preceding observation, it follows that $u_{\mathcal{V}!}R \to u_{\mathcal{V}!}X$ is bicovering. Hence (Proposition 1.2) u^* carries $\mathcal{V}$-sheaves on C' to $\mathcal{V}$-sheaves on C. The commutativity of (1.5.1) follows from the commutativity of $(*)$, from Exposé II, 3.6, and from Proposition 1.3(2). □

Proposition 1.6. Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a functor between the underlying categories. Consider the conditions:

i) u is continuous (cf. Definition 1.1 and Proposition 1.2).

ii) For every covering family $(X_i \to X)_{i\in I}$ of C, the family $(uX_i \to uX)_{i\in I}$ of C' is covering.

There is an implication i) ⇒ ii).

Assume that the topology of C can be defined by a pretopology T (Exposé II, 1.3) and that u commutes with fiber products.[2] Then i) and ii) are equivalent, and are equivalent to the following condition:

iii) The functor u carries the covering families of T to covering families of C'.

In particular, if fiber products are representable in C and if u commutes with fiber products, then i) and ii) are equivalent.

Proof. Let us prove i) ⇒ ii). Let $(X_i \to X)_{i\in I}$ be a covering family of C. For every sheaf F on C', u^*F is a sheaf. Hence the map

$$\mathrm{Hom}(X, u^*F) \longrightarrow \prod_i \mathrm{Hom}(X_i, u^*F)$$

is injective (Exposé II, 5.1). Thus

$$\mathrm{Hom}(u_!X, F) \longrightarrow \prod_i \mathrm{Hom}(u_!X_i, F)$$

is injective. Hence the family $u_!X_i \to u_!X$, $i \in I$, is covering in C' (Exposé II, 5.1).

Every covering family of the pretopology T is a covering family of C, whence ii) ⇒ iii). Let us prove iii) ⇒ i). Let X be an object of C, and let $(X_i \to X)_{i\in I}$ be a covering family in $\mathrm{Cov}(X)$ (Exposé II, 1.3). The morphisms $X_i \to X$ are base-changeable and u commutes with fiber products. Thus the fiber products $u(X_i) \times_{u(X)} u(X_j)$ are representable and canonically isomorphic to $u(X_i \times_X X_j)$. Let F be a sheaf on C'. Then the diagram of sets

$$F(u(X)) \longrightarrow \prod_i F(u(X_i)) \rightrightarrows \prod_{i,j} F(u(X_i \times_X X_j))$$

is exact (Exposé II, 2.1; Exposé I, 3.5 and 2.12). Hence the diagram

$$u^*F(X) \longrightarrow \prod_i u^*F(X_i) \rightrightarrows \prod_{i,j} u^*F(X_i \times_X X_j)$$

[2] In fact, it is enough that u commute with the fiber products occurring in base changes of morphisms coming from covering families for T.

is exact. Thus u^*F is a sheaf (Exposé II, 2.4), and i) follows. The last assertion follows from the preceding argument and from the fact that, when fiber products are representable in C, all covering families of C[3] define a pretopology on C whose associated topology is the topology of C. $\square$

Proposition 1.7. Let C be a small site, let C' be a $\mathcal{U}$-site, and let $u : C \to C'$ be a continuous functor. Let γ be an algebraic structure defined by finite projective limits, such that in the category of γ-objects of $\mathcal{U}$-**Set**, $\mathcal{U}$-inductive limits are representable.[4] We use the notation of Proposition II, 6.4. The functor u_s commutes with projective limits and therefore defines a functor

$$u_s^\gamma : \widetilde{C}_\gamma \to \widetilde{C'}_\gamma.$$

The functor u_s^γ admits a left adjoint u_γ^s, with the following properties:

1) There is a canonical isomorphism

$$u_\gamma^s \xrightarrow{\sim} \underline{a}'_\gamma u_{\gamma!} i_\gamma,$$

and u_γ^s commutes with all finite projective limits with which $u_{\gamma!}$ commutes.

2) There is a canonical isomorphism

$$u_\gamma^s \underline{a}_\gamma \xrightarrow{\sim} \underline{a}'_\gamma u'_{\gamma!}.$$

3) The functor u_γ^s commutes with inductive limits.

4) If u^s is left exact, then the diagram (notation of Exposé II, 6.3.0)

$$\begin{tikzcd} \widetilde{C}_\gamma \arrow[r, "u_\gamma^s"] \arrow[d, "j^\sim"'] & \widetilde{C'}_\gamma \arrow[d, "j'^\sim"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd}$$

commutes up to isomorphism, and u_γ^s is exact.

5) Suppose that $j^\sim$ (respectively $j'^\sim$) has a left adjoint $\mathrm{Lib}^\sim$ (respectively $\mathrm{Lib}'^\sim$) (Exposé II, 6.5). There is a canonical isomorphism

$$u_\gamma^s \,\mathrm{Lib}^\sim \xrightarrow{\sim} \mathrm{Lib}'^\sim u^s.$$

Moreover, when C is not necessarily a small site but is a $\mathcal{U}$-site, the functor u_s^γ admits a left adjoint u_γ^s. Finally, if $\mathcal{V}$ is a universe containing $\mathcal{U}$, and if $u^s_{\gamma\mathcal{U}}$ (respectively $u^s_{\gamma\mathcal{V}}$) denotes the left adjoint of u_s relative to $\mathcal{U}$ (respectively $\mathcal{V}$), the following diagram, analogous to (1.5.1),

$$\begin{tikzcd} \widetilde{C}_{\gamma\mathcal{U}} \arrow[r, "u^s_{\gamma\mathcal{U}}"] \arrow[d, hook] & \widetilde{C'}_{\gamma\mathcal{U}} \arrow[d, hook] \\ \widetilde{C}_{\gamma\mathcal{V}} \arrow[r, "u^s_{\gamma\mathcal{V}}"] & \widetilde{C'}_{\gamma\mathcal{V}} \end{tikzcd} \tag{1.7.1}$$

[3] Indexed by sets belonging to a suitable universe.

[4] In fact one can show that this condition is always satisfied.

commutes up to canonical isomorphism.

Proof. The proof in the case where C is a small site is left to the reader. It will be completed in 4.3 in the case where C is a $\mathcal{U}$-site. $\square$

Notation 1.8. From now on we shall use the notation u_s for the functor u_s^γ. This notation causes no confusion, because the functor u_s defined on sheaves of sets commutes with finite projective limits.

Similarly, when u^s is left exact, we shall use the notation u^s for the functor u_γ^s. This abuse of notation is justified by Proposition 1.7(4).

Example 1.9.1. Let X be a small topological space. Let $\mathrm{Open}(X)$ denote the category of open subsets of X, whose objects are the open subsets of X and whose morphisms are inclusions, endowed with the canonical topology. A family $(U_i \subset U)_{i\in I}$ is covering if and only if the open subsets U_i cover U (Exposé II, 2.6). Thus sheaves of sets on $\mathrm{Open}(X)$ are sheaves of sets in the sense of Godement [TF]. Let $f : X \to Y$ be a continuous map. It gives a functor

$$f^{-1} : \mathrm{Open}(Y) \to \mathrm{Open}(X),$$

which sends an open set U of Y to its inverse image $f^{-1}(U)$. The functor f^{-1} is continuous (Proposition 1.6). The functor

$$f_s^{-1} : \widetilde{\mathrm{Open}(X)} \to \widetilde{\mathrm{Open}(Y)}$$

is the direct image functor for sheaves in the sense of [TF]. The functor

$$(f^{-1})^s : \widetilde{\mathrm{Open}(Y)} \to \widetilde{\mathrm{Open}(X)}$$

is the inverse image functor for sheaves in the sense of [TF]. The functor $(f^{-1})^s$ commutes with finite projective limits, by Proposition 1.3(5).

Example 1.9.2. Let X be a topological space and let $Y \hookrightarrow X$ be an open subspace. Every open subset of Y is an open subset of X, hence there is a functor

$$\Phi : \mathrm{Open}(Y) \to \mathrm{Open}(X).$$

The functor Φ is continuous (Proposition 1.6). The functor $\Phi_s : \widetilde{\mathrm{Open}(X)} \to \widetilde{\mathrm{Open}(Y)}$ is the "restriction to Y" functor. The functor

$$\Phi^s_{\mathbf{Ab}} : \widetilde{\mathrm{Open}(Y)}_{\mathbf{Ab}} \to \widetilde{\mathrm{Open}(X)}_{\mathbf{Ab}}$$

(Exposé II, 6.3.3; Exposé I, 1.7) is the "extension by zero outside Y" functor introduced in [TF]. The functor

$$\Phi^s : \widetilde{\mathrm{Open}(Y)} \to \widetilde{\mathrm{Open}(X)}$$

(Proposition 1.3) is analogous to extension by zero: for every sheaf H on Y and every point $y \in Y$, the stalk of $\Phi^s H$ at y is canonically isomorphic to the stalk of H at y; for every point $x \in X$ with $x \notin Y$, the stalk of $\Phi^s H$ at x is empty. The functor Φ^s is called the "extension by the empty set outside Y" functor. It commutes with fiber products and with finite products over a nonempty set of indices, but it does not carry the final object of $\widetilde{\mathrm{Open}(Y)}$ to the final object of $\widetilde{\mathrm{Open}(X)}$.

Example 1.9.3. Let C and C' be two small sites, and let $u : C \to C'$ be a functor between the underlying categories which carries every covering family of C to a covering family of C'. The functor u is not continuous in general (cf. however Proposition 1.6). Here is a counterexample. Let S be an infinite set belonging to $\mathcal{U}$, and let C be the full subcategory of the category of sets whose objects are the finite nonempty subsets of S. Endow C with the canonical topology. Notice that the covering families of C are the surjective families of maps, and that covering families are nonempty. Notice also that, up to isomorphism, there are only two constant sheaves on C: the sheaf with value the empty set and the sheaf with value a one-point set. Put $C = C'$, and let $u : C \to C'$ be a constant functor. The functor u carries covering families of C to covering families of C', but u is not continuous. Indeed, since the representable presheaves of C' are sheaves, for every object X of C' there exists a sheaf F on C' such that the cardinality of $F(X)$ is at least 2; consequently the presheaf u^*F is not a sheaf.

2 Cocontinuous functors

Definition 2.1. Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a functor between the underlying categories. We say that u is *cocontinuous* if it has the following property:

(COC) For every object Y of C, and every sieve $R \hookrightarrow u(Y)$ covering $u(Y)$, the sieve of Y generated by the arrows $Z \to Y$ such that $u(Z) \to u(Y)$ factors through R, covers Y.

Proposition 2.2. Let C be a small site, let C' be a $\mathcal{U}$-site, and let $u : C \to C'$ be a functor between the underlying categories. Let

$$\widehat{u}_* : \widehat{C} \to \widehat{C'}$$

denote the functor right adjoint to $F \mapsto F \circ u = \widehat{u}^*F$ (Exposé I, 5.1). The functor u is cocontinuous if and only if, for every sheaf G on C, $\widehat{u}_*G$ is a sheaf on C'.

Proof. Sufficiency. Suppose that for every sheaf G on C, the presheaf $\widehat{u}_*G$ is a sheaf. Let Y be an object of C, and let $R \hookrightarrow u(Y)$ be a covering sieve. We have an isomorphism

$$\mathrm{Hom}(u(Y), u_*G) \xrightarrow{\sim} \mathrm{Hom}(R, u_*G),$$

whence, by adjunction (Exposé I, 5.1), an isomorphism

$$\mathrm{Hom}(\widehat{u}^*u(Y), G) \xrightarrow{\sim} \mathrm{Hom}(\widehat{u}^*R, G).$$

It follows (Exposé II, 5.3) that the morphism

$$\widehat{u}^*R \to \widehat{u}^*u(Y)$$

is bicovering. Since $\widehat{u}^*$ commutes with projective limits, this morphism is a monomorphism, and therefore is a covering monomorphism. On the other hand, there is a canonical morphism

$$Y \xrightarrow{p} \widehat{u}^*u(Y)$$

deduced from the adjunction morphism $\mathrm{id}_C \to \widehat{u}^* u_!$ (Exposé I, 5.4(3)). By base change along p, we obtain a covering sieve of Y:

$$\widehat{u}^* R \times_{\widehat{u}^* u(Y)} Y \to Y.$$

The arrows $Z \to Y$ which factor through this sieve are precisely those described in property (COC).

Necessity. Let S be an object of C' and let $R \hookrightarrow S$ be a sieve covering S. We must prove that, for every sheaf G on C, the map

$$\mathrm{Hom}(S, u_* G) \xrightarrow{\sim} \mathrm{Hom}(R, u_* G)$$

is an isomorphism. Using Proposition II, 5.3 and the adjunction isomorphisms, it is enough to prove that the monomorphism

$$\widehat{u}^* R \to \widehat{u}^* S$$

is covering. For this, it is enough to prove (Exposé II, 5.1) that, for every base change $Y \to \widehat{u}^* S$, where Y is an object of C, the sieve

$$\widehat{u}^* R \times_{\widehat{u}^* S} Y$$

is covering. By the properties of adjoint functors and Exposé I, 5.4(3), the morphism $Y \to \widehat{u}^* S$ factors as

$$Y \xrightarrow{p} \widehat{u}^* u(Y) \to \widehat{u}^* S.$$

Thus the sieve $\widehat{u}^* R \times_{\widehat{u}^* S} Y \hookrightarrow Y$ is obtained by base change along p from the monomorphism

$$\widehat{u}^* R \times_{\widehat{u}^* S} \widehat{u}^* u(Y) \hookrightarrow \widehat{u}^* u(Y).$$

Since $\widehat{u}^*$ commutes with projective limits, this monomorphism is the image under $\widehat{u}^*$ of the covering sieve

$$R \times_S u(Y) \hookrightarrow u(Y).$$

The hypothesis (COC) therefore implies that

$$\widehat{u}^* R \times_{\widehat{u}^* S} Y \hookrightarrow Y$$

is covering. $\square$

Proposition 2.3. Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a cocontinuous functor. Let

$$u^* : \widetilde{C'} \to \widetilde{C}$$

be the functor $u^* = \underline{a}\widehat{u}^* i'$, where $\underline{a}$ denotes the associated-sheaf functor for $\widehat{C}$, $\widehat{u}^*$ the functor $F \mapsto F \circ u$, and i' the inclusion functor $\widetilde{C'} \to \widehat{C'}$.

1) The functor u^* commutes with inductive limits and is exact.

2) There is a canonical isomorphism

$$u^* \underline{a}' \simeq \underline{a}\widehat{u}^*,$$

where $\underline{a}'$ denotes the associated-sheaf functor for $\widehat{C'}$.

3) The functor u^* has a right adjoint $u_* : \widetilde{C} \to \widetilde{C'}$. When C is small, the diagram

$$\begin{tikzcd} \widetilde{C} \arrow[r, "u_*"] \arrow[d, hook, "i"'] & \widetilde{C'} \arrow[d, hook, "i'"] \\ \widehat{C} \arrow[r, "\widehat{u}_*"] & \widehat{C'} \end{tikzcd} \tag{2.3.1}$$

in which the lower functor is right adjoint to $\widehat{u}^*$ (Exposé I, 5.1), is commutative up to canonical isomorphism.

4) Let $\mathcal{V} \supset \mathcal{U}$ be a universe. Let

$$u_{*\mathcal{U}} : \widetilde{C}_{\mathcal{U}} \to \widetilde{C'}_{\mathcal{U}} \quad (\text{respectively } u_{*\mathcal{V}} : \widetilde{C}_{\mathcal{V}} \to \widetilde{C'}_{\mathcal{V}})$$

denote the right adjoint relative to the universe $\mathcal{U}$ (respectively $\mathcal{V}$) whose existence is asserted in (3). Then the diagram

$$\begin{tikzcd} \widetilde{C}_{\mathcal{U}} \arrow[r, "u_{*\mathcal{U}}"] \arrow[d, hook] & \widetilde{C'}_{\mathcal{U}} \arrow[d, hook] \\ \widetilde{C}_{\mathcal{V}} \arrow[r, "u_{*\mathcal{V}}"] & \widetilde{C'}_{\mathcal{V}} \end{tikzcd} \tag{2.3.2}$$

commutes up to isomorphism.

Proof. Assertions (1) and (2) follow immediately from Exposé II, 3.4. Assertion (3), when C is small, follows from Proposition 2.2. When C is a $\mathcal{U}$-site, it will be proved in 4.4. Assertion (4), when C is small, follows from the analogous commutativity assertion when the topologies on C and C' are chaotic, and from Exposé I, 3.5. When the topologies on C and C' are chaotic, the commutativity of (2.3.2) is immediate from the explicit description of u_* (Exposé I, 5.1). □

2.4

We shall not develop here the considerations concerning γ-objects of categories of sheaves. It will be enough for us to observe that the functors u^* and u_* introduced in this number always commute with finite projective limits, unlike what happened in the preceding number for the functor u^s. Consequently they extend naturally to functors defined on γ-objects; these extensions are adjoint to one another and commute with the "underlying sheaf of sets" functors. We shall make the abuses of notation indicated in 1.8, writing u^* and u_* for the extensions to γ-objects.

Proposition 2.5. Let C and C' be two $\mathcal{U}$-sites and let

$$C \underset{u}{\overset{v}{\rightleftarrows}} C'$$

be a pair of functors, where v is left adjoint to u. The following properties are equivalent:

i) The functor u is continuous.

ii) The functor v is cocontinuous.

Moreover, under these equivalent conditions, there are canonical isomorphisms $v_* \simeq u_s$ and $v^* \simeq u^s$. In particular, u^s commutes with finite projective limits.

Proof. The property of a functor being continuous or cocontinuous does not depend on the universe $\mathcal{U}$ for which C and C' are $\mathcal{U}$-sites; this is immediate for cocontinuous functors, and follows from Proposition 1.5 for continuous functors. Thus, after enlarging the universe if necessary, we may suppose that C and C' are small. The proposition then follows from Proposition 2.2 and Exposé I, 5.6. □

Proposition 2.6. Let $u : C \to C'$ be a continuous and cocontinuous functor between two $\mathcal{U}$-sites. Then the functor

$$u^* : \widetilde{C'} \to \widetilde{C}$$

(Proposition 2.3) commutes with $\mathcal{U}$-inductive and projective limits. Let

$$u_! : \widetilde{C} \to \widetilde{C'}, \qquad u_* : \widetilde{C} \to \widetilde{C'}$$

denote the left and right adjoints of u^*, respectively. The functor $u_!$ is fully faithful if and only if u_* is fully faithful. When u is fully faithful, $u_!$ is fully faithful; the converse holds when the topologies of C and C' are less fine than the canonical topology.

The functor u^* has a right adjoint u_* (Proposition 2.3) and a left adjoint $u_!$ (Proposition 1.2). Hence u^* commutes with inductive and projective limits (Exposé I, 2.11). The fact that $u_!$ is fully faithful if and only if u_* is fully faithful is a general property of adjoint functors (Exposé I, 5.7(1)). When u is fully faithful, the functor

$$\widehat{u}_* : \widehat{C}_{\mathcal{V}} \to \widehat{C'}_{\mathcal{V}},$$

relative to $\mathcal{V}$-presheaves for a sufficiently large universe $\mathcal{V}$, is fully faithful (Exposé I, 5.7). Therefore $u_* : \widetilde{C} \to \widetilde{C'}$ is fully faithful by the commutativity of (2.3.1) and (2.3.2), and hence $u_!$ is fully faithful by what precedes. The converse follows from the existence of the commutative diagram in Proposition 1.2(iv), taking into account the fact that the functors ϵ_C and $\epsilon_{C'}$ are fully faithful when the topologies are less fine than the canonical topology. □

Example 2.7. With the notation of 1.9.2, the functor

$$\Phi : \mathrm{Open}(Y) \to \mathrm{Open}(X)$$

is continuous (1.9.2) and cocontinuous (Proposition 2.2). Moreover, let $i : Y \to X$ be the canonical injection and let $i^{-1} : \mathrm{Open}(X) \to \mathrm{Open}(Y)$ be the functor it defines (1.9.1). The functor i^{-1} is right adjoint to Φ. Thus there is a sequence of three adjoint functors

$$\Phi^s, \quad \Phi_s = \Phi^*, \quad \Phi_*,$$

and canonical isomorphisms

$$\Phi^* \simeq (i^{-1})^s, \qquad \Phi_* \simeq i^{-1}_s.$$

3 Induced topology

3.1

Let C' be a site, let C be a category, and let $u : C \to C'$ be a functor. For every universe $\mathcal{U}$ such that C' is a $\mathcal{U}$-site and C is a $\mathcal{U}$-small category, denote by $\mathcal{T}_{\mathcal{U}}$ the finest among the topologies T on C which make u continuous (Definition 1.1). Such a topology exists by Exposé II, 2.2. The topology $\mathcal{T}_{\mathcal{U}}$ does not depend on $\mathcal{U}$. Indeed, if $\mathcal{V} \supset \mathcal{U}$ is a universe, one has $\mathcal{T}_{\mathcal{U}} = \mathcal{T}_{\mathcal{V}}$ (Definition 1.1 and Proposition 1.5). The topology $\mathcal{T}_{\mathcal{U}}$ is called the *topology induced on C by the topology of C' by means of the functor u.*[5]

Proposition 3.2. Let C be a small category, let C' be a $\mathcal{U}$-site, let $u : C \to C'$ be a functor, and let $\mathcal{T}$ be the topology on C induced by u. Let X be an object of C and let $R \to X$ be a sieve on X. The following properties are equivalent:

i) The sieve $R \hookrightarrow X$ is covering for $\mathcal{T}$.

ii) For every base change $Y \to X$, where Y is an object of C, the morphism

$$u_!(R \times_X Y) \to u(Y)$$

is bicovering in $\widehat{C'}$ (Exposé II, 5.2).

Proof. i) $\Rightarrow$ ii) follows from axiom (T1) for topologies and from Proposition 1.2.

ii) $\Rightarrow$ i). For every sheaf F on C', the map

$$\operatorname{Hom}(Y, u^*F) \to \operatorname{Hom}(R \times_X Y, u^*F)$$

is isomorphic, by adjunction (Exposé I, 5.1), to the map

$$\operatorname{Hom}(u(Y), F) \to \operatorname{Hom}(u_!(R \times_X Y), F),$$

which is bijective (Exposé II, 5.3). Hence, by Exposé II, 2.2, the sieve $R \hookrightarrow X$ is covering for $\mathcal{T}$. $\square$

Corollary 3.3. Let C' be a site, let C be a category, let $u : C \to C'$ be a functor, and let $\mathcal{T}$ be the topology on C induced by the topology of C'. Let $(X_i \to X)_{i \in I}$ be a family of base-changeable morphisms of C, and suppose that u commutes with fiber products.[6] The following properties are equivalent:

i) The family $(X_i \to X)_{i \in I}$ is covering for $\mathcal{T}$.

ii) The family $(u(X_i) \to u(X))_{i \in I}$ is covering.

Proof. i) $\Rightarrow$ ii) follows from Proposition 1.6.

ii) $\Rightarrow$ i). Let $\mathcal{U}$ be a universe such that C is $\mathcal{U}$-small and C' is a $\mathcal{U}$-site. Let $R \hookrightarrow X$ be the sieve generated by the $X_i \to X$. The presheaf R is the cokernel of the pair of arrows (Exposé I, 2.12 and 3.5)

$$\coprod_{i,j} X_i \times_X X_j \rightrightarrows \coprod_i X_i,$$

[5] When no confusion can result, this topology is called simply the *topology induced on C by the topology of C'*.

[6] In fact, it is enough that u commute with fiber products of the form $X_i \times_X X'$.

where the coproduct is taken in $\widehat{C}$. Since $u_!$ commutes with inductive limits (Exposé I, 5.4), the presheaf $u_!R$ is the cokernel of

$$\coprod_{i,j} u(X_i \times_X X_j) \rightrightarrows \coprod_i u(X_i).$$

Since u commutes with fiber products, $u_!R$ is the cokernel of

$$\coprod_{i,j} u(X_i) \times_{u(X)} u(X_j) \rightrightarrows \coprod_i u(X_i),$$

and hence $u_!R \to u(X)$ is the sieve on $u(X)$ generated by the $u(X_i) \to u(X)$, $i \in I$. Using once again that u commutes with fiber products, one shows that, for every base change $Y \to X$ with Y an object of C, the morphism

$$u_!(R \times_X Y) \to u(Y)$$

is the sieve on $u(Y)$ generated by the morphisms

$$u(X_i) \times_{u(X)} u(Y) \to u(Y), \qquad i \in I.$$

Since $(u(X_i) \to u(X))_{i\in I}$ is covering, the family $u(X_i) \times_{u(X)} u(Y) \to u(Y)$, $i \in I$, is covering. Thus $u_!(R \times_X Y) \to u(Y)$ is a covering sieve, and therefore by Proposition 3.2 the sieve $R \to X$ is covering for $\mathcal{T}$. □

Corollary 3.4. Let C' be a $\mathcal{U}$-site, let C be a full subcategory of C', and let $u : C \to C'$ be the inclusion functor. Suppose that fiber products are representable in C and that u commutes with fiber products. The following conditions are equivalent:

- i) a) For every object X of C, every covering family $(Y_j \to X)_{j\in J}$ of C' is dominated by a covering family $(X_i \to X)_{i\in I}$, where the X_i are objects of C.
  - b) There exists a small set G of objects of C such that every object of C is the target of a covering family in C' of morphisms whose sources belong to G.
- ii) The topology induced by the topology of C' is a $\mathcal{U}$-topology (Exposé I, 3.0.2), and u is continuous and cocontinuous for this topology.

Proof. This follows immediately from Corollary 3.3 and Definition 2.1. □

Proposition 3.5. Let C be a $\mathcal{U}$-site and let $\epsilon_C : C \to \widetilde{C}$ be the canonical functor (Exposé II, 4.4.0). Endow $\widetilde{C}$ with the canonical topology. Then the topology of the site C is the topology induced by the topology of $\widetilde{C}$.

Proof. The covering families of $\widetilde{C}$ for the canonical topology are the universal effective epimorphic families (Exposé II, 2.5), that is (Exposé II, 4.3), the epimorphic families. Let T be the topology of the site C, and let $\mathcal{T}_{\mathcal{U}}$ be the finest topology T' on C such that, for every sheaf F on $\widetilde{C}$, $F \circ \epsilon_C$ is a sheaf for T'. It is enough to show that $T = \mathcal{T}_{\mathcal{U}}$; then $\mathcal{T}_{\mathcal{U}}$ is a $\mathcal{U}$-topology, and hence $\mathcal{T}_{\mathcal{U}}$ is the induced topology.

A) The topology T is finer than $\mathcal{T}_{\mathcal{U}}$. Let $(X_i \to X)_{i\in I}$ be a covering family for $\mathcal{T}_{\mathcal{U}}$. Then for every sheaf F on $\widetilde{C}$, the map

$$F(\epsilon_C(X)) \to \prod_i F(\epsilon_C(X_i))$$

is injective. Hence (Exposé II, 5.2) the family $(\epsilon_C X_i \to \epsilon_C X)_{i\in I}$ is covering for the canonical topology of $\widetilde{C}$. Therefore (Exposé II, 5.2) $(X_i \to X)_{i\in I}$ is covering for T.

B) The topology T is less fine than $\mathcal{T}_{\mathcal{U}}$. It is enough to show that $\epsilon_C : C \to \widetilde{C}$ is continuous (Definition 1.1). But it will be proved in Exposé IV, no. 1, that every sheaf on $\widetilde{C}$ is representable; hence, for every sheaf F on $\widetilde{C}$, $F \circ \epsilon_C$ is a sheaf on C. We shall not use Proposition 3.5 before Exposé IV, 1.

We record two results which will be used in Exposé VI, 7. □

Proposition 3.6. Let $(C_i)_{i\in I}$ be a family of sites, let C be a category, and for each $i \in I$, let $u_i : C_i \to C$ be a functor. Let $\mathcal{U}$ be a universe such that the categories C_i and C are $\mathcal{U}$-small. There exists a topology $\mathcal{T}_{\mathcal{U}}$ on C which is the least fine among the topologies for which the u_i are continuous. The topology $\mathcal{T}_{\mathcal{U}}$ does not depend on the universe $\mathcal{U}$ for which the categories considered are small.

The last assertion of Proposition 3.6 follows from Proposition 1.5.

Let T be a topology on C. The functors u_i are continuous if and only if, for every $i \in I$ and every covering sieve $R \hookrightarrow X$ of an object X of C_i, the morphism

$$u_{i!}(R) \to u_i(X)$$

of $\widehat{C}$ is bicovering (Proposition 1.2(ii)). Thus Proposition 3.6 is a consequence of the following lemma.

Lemma 3.6.1. Let C be a small category and let $(u_i : F_i \to G_i)_{i\in I}$ be a family of arrows of $\widehat{C}$. Then there exists on C a least fine topology among those which make the morphisms u_i covering (respectively bicovering) (Exposé II, 5.2).

To say that $u : F \to G$ is covering for a given topology T means that, for every arrow $X \to G$, with $X \in \operatorname{Ob} C$, the corresponding arrow $F \times_G X \to X$ is covering; equivalently, the family of arrows $X' \to X$ of C which factor through that arrow is covering. The existence of a least fine topology among those for which the $u_i : F_i \to G_i$ are covering follows from Exposé I, 1.1.6, proving the non-parenthesized assertion of Lemma 3.6.1. The parenthesized assertion follows by recalling that a morphism $u : F \to G$ is bicovering if and only if the morphisms $u : F \to G$ and $\Delta_u : F \to F \times_G F$ are covering.

Proposition 3.7. Let $(C_i)_{i\in I}$ be a family of sites, let C be a category, and for each $i \in I$, let $u_i : C_i \to C$ be a functor. Let $\mathcal{U}$ be a universe such that the categories C_i and C are $\mathcal{U}$-small. There exists a topology $\mathfrak{T}_{\mathcal{U}}$ which is the finest topology for which the u_i are cocontinuous. The topology $\mathfrak{T}_{\mathcal{U}}$ does not depend on the universe $\mathcal{U}$ for which the categories considered are small.

Let $\mathcal{U}$ be a universe for which the categories considered are small. The functors u_i are cocontinuous for a topology $\mathfrak{T}$ on C if and only if, for every $i \in I$ and every sheaf F on C_i, the presheaf $\widehat{u}_{i*}F$ is a sheaf for $\mathfrak{T}$ (Proposition 2.2). It follows that $\mathfrak{T}_{\mathcal{U}}$ is the finest topology for which the presheaves $\widehat{u}_{i*}F$, $i \in I$, $F \in \operatorname{Ob}(C_i)$, are sheaves (Exposé II, 2.2). The last assertion follows from Proposition 2.2.

4 The comparison lemma

Theorem 4.1 (comparison lemma). Let C be a small category, let C' be a site whose underlying category is a $\mathcal{U}$-category, and let $u : C \to C'$ be a fully faithful functor. Endow C with the topology induced by u (3.1). Consider the properties:

i) Every object of C' can be covered by objects coming from C.

ii) The functor $F \mapsto F \circ u$ induces an equivalence from the category of sheaves on C' to the category of sheaves on C.

One always has i) $\Rightarrow$ ii). When C' is a $\mathcal{U}$-site and the topology of C' is less fine than the canonical topology, one has ii) $\Rightarrow$ i).

4.1.1

We first prove i) $\Rightarrow$ ii). The proof is in two steps.

First step. For every presheaf H of $\widehat{C'}$, the adjunction morphism

$$\varphi : u_! u^* H \to H$$

(Exposé I, 5.1) is bicovering (Exposé II, 5.3), and the functor $u_s : \widetilde{C'} \to \widetilde{C}$ (1.1.1) is fully faithful.

Let C/H be the small category whose objects are pairs consisting of an object Y of C and a morphism $u(Y) \to H$, and whose morphisms are the commutative diagrams

$$\begin{tikzcd} u(Y) \arrow[rr, "u(m)"] \arrow[dr] & & u(Y') \arrow[dl] \\ & H & \end{tikzcd} .$$

We have

$$u^* H = \varinjlim_{Y \in \mathrm{Ob}(C/H)} Y$$

(Exposé I, 3.4), and hence (Exposé I, 5.4)

$$u_! u^* H = \varinjlim_{Y \in \mathrm{Ob}(C/H)} u(Y).$$

The adjunction morphism is the evident morphism coming from this description of $u_! u^* H$ as an inductive limit. It has the following property:

$(*)$ For every object Y of C, every morphism $m : u(Y) \to H$ factors uniquely as the canonical morphism $\alpha(m) : u(Y) \to u_! u^* H$ followed by φ.

It follows immediately from (i) that φ is covering (Exposé II, 5.1). Let us prove that it is bicovering. Let $p, q : Z \rightrightarrows u_! u^* H$ be two morphisms from an object Z of C' such that $\varphi p = \varphi q$. For every object Y of C and every morphism $n : u(Y) \to Z$, one has $\varphi p n = \varphi q n$. Property $(*)$ then implies $pn = qn$, and since the $u(Y)$ cover Z, the kernel of (p, q) is a covering sieve of Z. Thus φ is bicovering (Exposé II, 5.3). For every sheaf H on C', $u^* H$ is a sheaf on C, denoted $u_s H$ (1.1.1). Moreover

$$u^s u_s H = \underline{a}' u_! u_s H$$

(Proposition 1.3), and the adjunction morphism $u^s u_s H \to H$ is obtained by applying the associated-sheaf functor to $\varphi : u_! u^* H \to H$. Therefore (Exposé II, 5.3) the adjunction morphism $u^s u_s H \to H$ is an isomorphism. Hence $u_s : \widetilde{C'} \to \widetilde{C}$ is fully faithful.

Second step. The functor u is cocontinuous and the functor $u^s : \widetilde{C} \to \widetilde{C'}$ (Proposition 1.2) is fully faithful. Therefore $u_s : \widetilde{C'} \to \widetilde{C}$ is an equivalence.

Let Y be an object of C, and let $i : R \hookrightarrow u(Y)$ be a covering sieve. Since u^* commutes with projective limits (Exposé I, 5.5), the morphism

$$u^*(i) : u^*(R) \hookrightarrow u^*u(Y)$$

is a monomorphism. Since u is fully faithful, $u^*u(Y) \simeq Y$, and hence this gives a sieve $u^*(R) \to Y$ on Y. To show that u is cocontinuous, it is enough to show that $u^*(i) : u^*(R) \hookrightarrow Y$ is a covering sieve for the induced topology on C (Definition 2.1). For this it is enough to prove (Proposition 3.2) that, for every base change $m : X \to Y$, the morphism

$$u_!(u^*(R) \times_Y X) \to u(X)$$

is bicovering. Since u^* commutes with projective limits, one has

$$u^*(R) \times_Y X = u^*(R \times_{u(Y)} u(X)),$$

and $R \times_{u(Y)} u(X)$ is a covering sieve of $u(X)$. It is therefore enough to prove that, for every object Y of C and every covering sieve $i : R \hookrightarrow u(Y)$, the morphism in $\widehat{C'}$

$$u_!u^*(R) \xrightarrow{u_!u^*(i)} u(Y)$$

is bicovering. But this morphism factors as the adjunction morphism $u_!u^*(R) \to R$, which is bicovering by the first step, followed by the monomorphism $R \hookrightarrow u(Y)$, which is covering. Hence it is bicovering (Exposé II, 5.3). This proves that u is cocontinuous. Since u is continuous and fully faithful, Proposition 2.6 (used in the case where C is small) implies that u^s (denoted $u_!$ in Proposition 2.6) is fully faithful. Since u^s and u_s are adjoint to one another and both are fully faithful, they are quasi-inverse functors; hence u_s is an equivalence.

4.1.2

We now prove ii) $\Rightarrow$ i). The objects Y of C form a generating family of $\widetilde{C}$ (Exposé II, 4.10). Therefore, for every object X of C', there exists an epimorphic family in $\widetilde{C}$

$$v_i : Y_i \to u_sX, \qquad i \in I.$$

We have $u_su(Y_i) = Y_i$, and, since u_s is an equivalence of categories, $v_i = u_s(w_i)$, where the $w_i : u(Y_i) \to X$ form an epimorphic family of $\widetilde{C'}$. It follows from Exposé II, 5.1, that the family $w_i : u(Y_i) \to X$, $i \in I$, is covering.

4.1 End of the proof of Proposition 1.2

Let G be a full subcategory of C whose objects form a small topologically generating family of C (Exposé II, 3.0.1). Endow G with the topology induced by the inclusion functor $i : G \to C$ (3.1). The functor $(u \circ i)_s = i_s \circ u_s$ (1.1.1) has a left adjoint by the first part of the proof of Proposition 1.2. Since i_s is an equivalence of categories (Theorem 4.1), the functor u_s has a left adjoint u^s, and there is a functorial isomorphism

$$u^s \simeq (u \circ i)^s \circ i_s.$$

Let us prove that the diagram

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd} \tag{4.2.1}$$

commutes up to isomorphism. For every sheaf H on C' and every $X \in \operatorname{Ob} C$,

$$\operatorname{Hom}_{\widetilde{C'}}(\epsilon_{C'} uX, H) \simeq u_s H(X).$$

Moreover,

$$\operatorname{Hom}_{\widetilde{C'}}(u^s \epsilon_C X, H) \simeq \operatorname{Hom}_{\widetilde{C}}(\epsilon_C X, u_s H) \simeq u_s H(X),$$

by adjunction and by the definition of ϵ_C. Hence $\epsilon_{C'} uX \simeq u^s \epsilon_C X$ for every $X \in \operatorname{Ob} C$. This proves i) $\Rightarrow$ iv). Let us prove iv) $\Rightarrow$ i). There is a commutative diagram

$$\begin{tikzcd} G \arrow[r, "i"] \arrow[d, "\epsilon_G"'] & C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"'] \\ \widetilde{G} \arrow[r, "i^s"] & \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'}, \end{tikzcd} \tag{4.2.2}$$

and therefore, by the first part of the proof, the functor $u \circ i : G \to C'$ is continuous. From Theorem 4.1 and Definition 1.1 it follows immediately that u is continuous, hence iv) $\Rightarrow$ i). One also obtains the uniqueness of u^s, knowing from the first part of the proof the uniqueness when C is small.

4.2 End of the proof of Propositions 1.5 and 1.7

Let $u : C \to C'$ be a functor between two $\mathcal{U}$-sites, and let G be a full subcategory of C whose set of objects is a small topologically generating family of C (Exposé II, 3.0.1). Endow G with the topology induced by the inclusion functor $i : G \to C$. It follows immediately from Theorem 4.1 and Definition 1.1 that u is continuous if and only if $u \circ i$ is continuous. From this observation and from the first part of the proof of Proposition 1.5 the general case follows. This observation also reduces the proof of Proposition 1.7 to the case where C is small.

4.3 End of the proof of Proposition 2.3

Let $u : C \to C'$ be a functor between two $\mathcal{U}$-sites, and let G be a full subcategory of C whose set of objects is a small topologically generating family of C (Exposé II, 3.0.1), endowed with the topology induced by the inclusion functor $i : G \to C$. It follows from the proof of Theorem 4.1 (second step in 4.1.1) that i is cocontinuous. Therefore, when u is cocontinuous, the functor $u \circ i : G \to C'$ is cocontinuous. Moreover,

$$i^* : \widetilde{C} \to \widetilde{G}$$

is an equivalence of categories (Theorem 4.1). Thus the functor $u^* : \widetilde{C'} \to \widetilde{C}$ has a right adjoint. This proves assertion (3). To prove assertion (4), it is enough to observe that

$$(u \circ i)_* \simeq u_* \circ i_*$$

and that i_* is an equivalence (Theorem 4.1). The commutativity of (2.3.2) follows from the commutativity of these diagrams when C is small.

5 Localization

5.1

Let now C be a $\mathcal{U}$-site and let X be an object of $\widehat{C}$. Unless explicitly stated otherwise, the category C/X will be endowed with the topology $\mathcal{T}$ induced by the functor

$$j_X : C/X \to C$$

(3.1). The notation C/X will denote the category C/X endowed with the topology $\mathcal{T}$. Proposition I, 5.11 shows that the functor $j_{X!}$ commutes with fiber products and therefore carries every monomorphism to a monomorphism. In particular, for every object $(Y \to X)$ of C/X, the functor $j_{X!}$ gives a bijective correspondence between the sieves, in the category C/X, of the object $(Y \to X)$, and the sieves, in C, of the object Y.

Proposition 5.2. Let C be a $\mathcal{U}$-site, let X be an object of $\widehat{C}$, and let $j_X : C/X \to C$ be the corresponding continuous functor.

1) A sieve R of an object $(Z \to X)$ is covering in C/X if and only if the sieve $j_{X!}(R) \hookrightarrow Z$ is covering in C.

2) The functor $j_X : C/X \to C$ is cocontinuous and continuous.

3) Let $m : Y \to X$ be a morphism of $\widehat{C}$. Then there is a commutative diagram

$$\begin{tikzcd} C/Y \arrow[rr, "j_m"] \arrow[dr, "j_Y"'] & & C/X \arrow[dl, "j_X"] \\ & C & \end{tikzcd} .$$

The topology induced by j_Y on C/Y is equal to the topology induced by j_m on C/Y.

4) The topology induced by $j_X : C/X \to C$ is a $\mathcal{U}$-topology (Exposé II, 3.0.2).

Proof.

1) If the sieve R of $(Z \to X)$ is covering, the sieve $j_{X!}(R) \hookrightarrow Z$ is covering (Proposition 1.6). Conversely, if $j_{X!}(R) \to Z$ is covering in C, the same is true for every sieve obtained by base change in C/X. Hence R is covering (Proposition 3.2).

2) This follows immediately from (1) by applying Definition 2.1.

3) This follows immediately from the description of covering sieves given in (1).

4) Let $(G_i)_{i\in I}$ be a small topologically generating family of C. One checks immediately that the small family of pairs

$$(u : G_i \to X), \qquad u \in \coprod_{i\in I} \mathrm{Hom}_{\widehat{C}}(G_i, X),$$

is topologically generating in C/X.

□

Terminology and notation 5.3. By the preceding proposition, the functor j_X is both continuous and cocontinuous. It therefore defines a sequence of three adjoint functors (4.3.2) between categories of sheaves of sets (Propositions 1.3 and 2.3):

$$j_X^s : (C/X)^\sim \to \widetilde{C}, \qquad j_X^* = j_{X,s} : \widetilde{C} \to (C/X)^\sim, \qquad j_{X*} : (C/X)^\sim \to \widetilde{C}.$$

In the special situation of Proposition 5.2, we shall use the following terminology and notation:

1) The functor j_{X*} will be called the *direct image functor*.

2) The functor $j_{X,s} = j_X^*$ will be denoted j_X^* and called the *restriction functor to C/X*.

3) The functor j_X^s on sheaves of sets will be called the "extension by the empty set to the category C" functor and will be denoted $j_{X!}$ (cf. 2.9.2).

Thus there is a sequence of three adjoint functors between $(C/X)^\sim$ and $\widetilde{C}$:

$$j_{X!}, \quad j_X^*, \quad j_{X*}.$$

Proposition 5.4. The functor $j_{X!} : (C/X)^\sim \to \widetilde{C}$ factors through the category $\widetilde{C}/\underline{a}X$, where $\underline{a}$ is the associated-sheaf functor:

$$(C/X)^\sim \xrightarrow{e_X^\sim} \widetilde{C}/\underline{a}X \to \widetilde{C}.$$

The functor

$$e_X^\sim : (C/X)^\sim \to \widetilde{C}/\underline{a}X$$

is an equivalence of categories. The restriction functor to C/X, composed with the equivalence $e_X^\sim$, is isomorphic to the "base change by $\underline{a}X \to \epsilon$" functor

$$F \longmapsto (F \times \underline{a}X \xrightarrow{\mathrm{pr}_2} \underline{a}X),$$

where ϵ is the final object of $\widetilde{C}$.

Proof. The image under $j_{X!}$ of the final object of $(C/X)^\sim$ is the object $\underline{a}X$, whence the factorization. To show that $e_X^\sim$ is an equivalence, we restrict ourselves to a few indications. By Exposé I, 5.11, a presheaf on C/X is defined by a presheaf F on C endowed with a morphism $F \to X$. One then proves that the following two properties are equivalent:

i) The presheaf on C/X defined by $F \to X$ is a sheaf.

ii) The following diagram is cartesian, where i and $\underline{a}$ denote the inclusion into presheaves and the associated-sheaf functor:

$$\begin{tikzcd} F \arrow[r] \arrow[d] & i\underline{a}F \arrow[d] \\ X \arrow[r] & i\underline{a}X. \end{tikzcd}$$

It follows immediately that $e^{\sim}_X$ is an equivalence. The last assertion is trivial. □

Proposition 5.5.

1) Let C be a $\mathcal{U}$-site and let X be an object of $\widehat{C}$. The following diagram of categories and functors is commutative up to canonical isomorphisms:

$$\begin{tikzcd} C/X \arrow[r, "h_X"] \arrow[d] & \widehat{(C/X)} \arrow[r, "\underline{a}_X"] \arrow[d, "\widehat{e}_X"] & (C/X)^{\sim} \arrow[rr, "i_X"] \arrow[d, "e^{\sim}_X"] & & \widehat{(C/X)} \arrow[d, "\widehat{e}_X/X"] \\ C/X \arrow[r, "h/X"] & \widehat{C}/X \arrow[r, "\underline{a}/X"] & \widetilde{C}/\underline{a}X \arrow[r, "i/\underline{a}X"] & \widehat{C}/i\underline{a}X \arrow[r, "\pi"] & \widehat{C}/X. \end{tikzcd}$$

The notation $\underline{a}_X$ and i_X denotes the associated-sheaf functor and the inclusion into presheaves for the site C/X. The notation $\underline{a}/X$ denotes the natural extension of $\underline{a}$, the associated-sheaf functor for the site C, to the category of arrows with target X; similarly for $i/\underline{a}X$. Finally, π denotes the base-change functor by the canonical arrow $X \to i\underline{a}X$.

2) Suppose in addition that $m : Y \to X$ is a morphism of $\widehat{C}$. The following diagram is commutative up to canonical isomorphism:

$$\begin{tikzcd} \widetilde{C}/\underline{a}X/\underline{a}Y \arrow[d, equal] & (C/X)^{\sim}/\underline{a}_X Y \arrow[l, "f"'] & \\ & & (C/X/Y)^{\sim} \arrow[ul, "g"'] \arrow[dl, "e^{\sim}_m"] \\ \widetilde{C}/\underline{a}Y & (C/Y)^{\sim} \arrow[l, "e^{\sim}_Y"'] & . \end{tikzcd}$$

The arrow f is the natural extension of the equivalence $e^{\sim}_X$ to the category of arrows with target $\underline{a}_X Y$, and therefore is an equivalence of categories. The arrow g is none other than the equivalence of Proposition 5.4 applied to the situation $(C/X)/[m]$.

3) Let

$$j_{\underline{a}X!} : \widetilde{C}/\underline{a}X \to \widetilde{C}$$

denote the functor "forget $\underline{a}X$", and let

$$j^*_{\underline{a}X} : \widetilde{C} \to \widetilde{C}/\underline{a}X$$

denote the functor “product with $\underline{a}X$”. The following diagrams are commutative up to canonical isomorphism:

$$
\begin{tikzcd}
(C/X)^{\sim} \arrow[r, "j_{X!}"] \arrow[d] & \widetilde{C} \arrow[d, equal] \\
\widetilde{C}/\underline{a}X \arrow[r, "j_{\underline{a}X!}"] & \widetilde{C}
\end{tikzcd}
\qquad
\begin{tikzcd}
\widetilde{C} \arrow[r, "j_X^*"] \arrow[d, equal] & (C/X)^{\sim} \arrow[d, "e_X^{\sim}"] \\
\widetilde{C} \arrow[r, "j_{\underline{a}X}^*"] & \widetilde{C}/\underline{a}X.
\end{tikzcd}
$$

Proof. The proof of these assertions is immediate from the definition of the equivalences $e_X^{\sim}$ (Proposition 5.4) and $\widehat{e}_X$ (Exposé I, 5.11). $\square$

References

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Exposé IV. Topoi

Draft translation, Sections 0–6.10

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Translator's status note. This is a working English translation of SGA 4, Exposé IV, from the introduction through Section 6.10. The range corresponds to source lines 1–4120 of the Orgogozo–Laszlo TeX snapshot. Terminology is normalized to current mathematical English where this is harmless: *topos/topoi*, *geometric morphism* for French *morphisme de topos*, *direct image*, *inverse image*, *sober space*, *soberification*, *classifying topos*, *big site*, *big topos*, *cocontinuous functor*, *morphism of sites*, *fiber functor*, *constant object*, *section functor*. Numbering follows the source.

0. Introduction

0.1

We have seen in Exposé II various exactness properties of categories of the form

$$\widetilde{\mathcal{C}} = \text{the category of sheaves of sets on } \mathcal{C},$$

where $\mathcal{C}$ is a small site. These properties may be expressed by saying that, in many respects, these categories, which we shall call *topoi*, inherit the familiar properties of the category (**Set**) of small sets. On the other hand, experience has shown that one should often regard various situations in mathematics *mainly as a technical means for constructing the corresponding categories of sheaves* of sets, i.e. the corresponding *topoi*. It appears that all the genuinely important notions attached to a site–for instance its cohomological invariants, studied in Exposé V, various other "topological" invariants, such as the homotopy invariants recently studied by M. Artin and B. Mazur, and the notions studied in J. Giraud's book on non-commutative cohomology–are in fact expressed directly in terms of the associated topos.

From this viewpoint, it is appropriate to regard two sites as essentially equivalent when their associated topoi are equivalent categories, and to regard the datum of a site, at least in the practically most important case where its topology is no finer than the canonical topology, as equivalent to the datum of a topos E, namely the associated topos of sheaves of sets on the site, together with a generating family of objects of E (cf. Exposé II, 4.9 and 1.2.1 below). This viewpoint is analogous to the one in which one attaches a group to a system of generators and a system of relations among these generators, and focuses on the structure of the group rather than on the generators and relations by which it was produced, these being regarded as auxiliary data of the situation. Moreover, the comparison lemma, Exposé III, 4.1, gives numerous examples of pairs of sites $\mathcal{C}, \mathcal{C}'$ which are not isomorphic, and not even equivalent as categories, but give rise to equivalent topoi; hence $\mathcal{C}$ and $\mathcal{C}'$ should be regarded as essentially equivalent.

0.2

In this exposé we give a characterization of topoi by simple exactness properties, due to J. Giraud. We study the natural notion of morphism of topoi, inspired by the notion of a continuous map from one topological space to another, and we develop in the setting of topoi some familiar constructions from ordinary sheaf theory, such as sheaves $\mathcal{H}om$, sheaf tensor products, and supports. Finally, following M. Artin, we show how a topos can be reconstructed from an "open" part of it, the complementary "closed" part, and a certain left exact functor relating them, which, up to minor qualifications, may in fact be chosen arbitrarily.

0.3

This gives a gluing procedure for topoi which, when applied to topoi coming from ordinary topological spaces, will generally produce a topos no longer of that same type. This is a first indication of the remarkable stability of the notion of topos under various natural constructions, a stability lacking in the notion of topological space, from which the notion of topos is inspired. As a second remarkable example, let us also mention the classifying topos associated with a group object of a topos (cf. [1] or 5.9 below), inspired by the classical notion of classifying space of a topological group, and the notion of a "modular topos" associated with various moduli problems in algebraic geometry or analytic geometry [10], [13].

Other topoi, such as the *étale topos* of a scheme, studied systematically in this seminar from Exposé VII onward, or the *crystalline topos* of a relative scheme [6], arise naturally when one wants to develop usable cohomology theories for abstract algebraic varieties, and more generally for schemes, replacing the classical Betti cohomology of algebraic varieties over the complex numbers.

0.4

Thus one may say that the notion of topos, a natural derivative of the *sheaf-theoretic point of view* in topology, is itself a substantial enlargement of the notion of topological space.[1] It embraces many situations which previously were not regarded as belonging to topological intuition. The characteristic feature of such situations is that one has a notion of "localization"; this is formalized precisely by the notion of site, and ultimately by that of topos, via the topos associated with the site. As the term "topos" itself is meant to suggest, it seems reasonable and legitimate to the authors of this seminar to regard topology as the study of *topoi*, and not merely of topological spaces.

0.5

It has seemed useful to include in this general exposé on topoi a fairly large number of examples, many of which have only distant connections with the original aim of this seminar, namely the study of étale cohomology. The hurried reader, interested exclusively in étale cohomology, may of course omit these examples without loss, and likewise most of the present exposé, referring back to it only as needed.

1 Definition and characterization of topoi

Definition 1.1. A $\mathcal{U}$-*topos*, or simply a topos when no confusion can arise, is a category E such that there exists a site $C \in \mathcal{U}$ for which E is equivalent to the category $\widetilde{C}$ of $\mathcal{U}$-sheaves of sets on C.

[1] Cf. [9], or 4.1 and 4.2 below, for the precise relation between topoi and topological spaces.

1.1.1

Let E be a $\mathcal{U}$-topos. We shall always regard E as endowed with its canonical topology (Exposé II, 2.5), making it a site, and even, by 1.1.2(d) below, a $\mathcal{U}$-site (Exposé II, 3.0.2). Unless explicitly stated otherwise, we shall consider no topology on E other than the one just described.

1.1.2

We saw in Exposé II, 4.8 and 4.11, that a $\mathcal{U}$-topos E is a $\mathcal{U}$-category satisfying the following conditions:

a) finite projective limits are representable in E;

b) coproducts indexed by an element of $\mathcal{U}$ are representable in E, and are disjoint and universal;

c) equivalence relations in E are effective and universal;

d) E admits a generating family indexed by an element of $\mathcal{U}$.

In fact, we shall see that these intrinsic properties *characterize* $\mathcal{U}$-topoi.

Theorem 1.2 (J. Giraud). Let E be a $\mathcal{U}$-category. The following properties are equivalent:

i) E is a $\mathcal{U}$-topos.

ii) E satisfies conditions a), b), c), and d) of 1.1.2.

iii) The $\mathcal{U}$-sheaves on E for the canonical topology are representable, and E has a small generating family, namely condition 1.1.2(d).

iv) There exist a category $C \in \mathcal{U}$ and a fully faithful functor $i : E \to \widehat{C}$, where $\widehat{C}$ denotes the category of $\mathcal{U}$-presheaves on C, which admits a left adjoint a that is left exact.

i′ There exists a site $C \in \mathcal{U}$, in which projective limits are representable and whose topology is no finer than its canonical topology, such that E is equivalent to the category $\widetilde{C}$ of $\mathcal{U}$-sheaves of sets on C.

Moreover:

Corollary 1.2.1. Let E be a $\mathcal{U}$-topos, and let C be a full subcategory of E, endowed with the topology induced from that of E. Consider the functor

$$E \longrightarrow \widetilde{C}$$

which sends $X \in \operatorname{Ob} E$ to the restriction to C of the functor represented by X. This functor is an equivalence of categories if and only if $\operatorname{Ob} C$ is a generating family of E.

The equivalence i) $\Leftrightarrow$ iv) follows at once from Exposé II, 5.5, and we have already recalled that i) $\Rightarrow$ ii). Since i′ $\Rightarrow$ i is trivial, it remains to prove ii) $\Rightarrow$ iii) and iii) $\Rightarrow$ i′, which is done in 1.2.4 and 1.2.3 below. The corollary is then obtained by noting that the functor under consideration factors as

$$E \longrightarrow \widetilde{E} \longrightarrow \widetilde{C}.$$

Since $E \to \widetilde{E}$ is an equivalence by (iii), the question reduces to determining when $\widetilde{E} \to \widetilde{C}$ is an equivalence. The comparison lemma of Exposé III, 4.1, then gives the conclusion; cf. the proof in 1.2.3 below.

1.2.3. Proof of iii) $\Rightarrow$ i')

Let $\mathfrak{X} = (X_i)_{i\in I}$, $I \in \mathcal{U}$, be a small generating family of E. Since E is a $\mathcal{U}$-category, the set of isomorphism classes of finite diagrams in E whose objects are among the X_i is $\mathcal{U}$-small. Hence the smallest set $\mathfrak{X}'$ of objects of E, containing the finite projective limits of objects of $\mathfrak{X}$, is a countable union of small sets and is therefore small.[2] Replacing $\mathfrak{X}$ successively by $\mathfrak{X}^{(n+1)} = (\mathfrak{X}^{(n)})'$ and then by $\bigcup_n \mathfrak{X}^{(n)}$, we may assume, after enlarging the generating family, that $\mathfrak{X}$ is stable under finite projective limits.

Let $\mathcal{V}$ be a universe containing $\mathcal{U}$ such that E is $\mathcal{V}$-small. For each object H of E, write

$$I(H) = \coprod_{i\in I} \mathrm{Hom}(X_i, H).$$

Let H be an object of E, and let $H' \hookrightarrow H$ be the subsheaf of H, for the canonical topology, which is the "union" of the images of the morphisms $u : X_i \to H$, $(u,i) \in I(H)$. Since H' is a subsheaf of a $\mathcal{U}$-sheaf, H' is a $\mathcal{U}$-sheaf. It is therefore representable. Since $\mathfrak{X}$ is generating and, for every $i \in I$, the map $\mathrm{Hom}(X_i, H') \to \mathrm{Hom}(X_i, H)$ is bijective, the monomorphism $H' \hookrightarrow H$ is an isomorphism. Thus the family $(u : X_i \to H)$, $(u,i) \in I(H)$, is covering for the canonical topology of E. Hence every object of E can be covered, for the canonical topology of E, by objects of $\mathfrak{X}$.

Let C be the full subcategory of E defined by the objects of $\mathfrak{X}$, and let $u : C \hookrightarrow E$ be the inclusion. The topology induced on C by the canonical topology of E is no finer than the canonical topology of C; when $\mathcal{U}$ contains an element of infinite cardinality, finite projective limits are representable in C. It follows from Exposé III, 5.1, that the functor $F \mapsto F \circ u$ is an equivalence from E to the category of $\mathcal{U}$-sheaves on C for the induced topology. $\square$

1.2.4. Proof of ii) $\Rightarrow$ iii)

The proof has four steps. Let $\mathcal{V}$ be a universe containing $\mathcal{U}$, such that E is an element of $\mathcal{V}$. Let $\widetilde{E}$ be the category of $\mathcal{V}$-sheaves on E for the canonical topology, and let $J_E : E \to \widetilde{E}$ be the canonical functor.

1.2.4.1. Let $(g_i : G_i \to H)$, $i \in I \in \mathcal{U}$, be an epimorphic family in $\widetilde{E}$. If the G_i and the fiber products $G_i \times_H G_j$ are representable, then the sheaf H is representable.

Indeed, the coproduct $\coprod_{i\in I} G_i$ is representable in $\widetilde{E}$, by a representable sheaf G. The same holds for $K = \coprod_{(i,j)\in I\times I}(G_i \times_H G_j)$. Moreover, the diagram $K \rightrightarrows G \to H$ is exact, and K is the fiber square of G over H. The latter property holds in the category of presheaves, hence in the category of sheaves. It follows from condition c) and Exposé II, 4.7, that H is representable.

1.2.4.2. Let X_α, $\alpha \in A \in \mathcal{U}$, be a generating family of E. For every sheaf H, let

$$I(H) = \coprod_{\alpha\in A} \mathrm{Hom}_E(X_\alpha, H).$$

The family $(u : X_\alpha \to H, (u,\alpha) \in I(H))$ is epimorphic in $\widetilde{E}$. When H is a $\mathcal{U}$-sheaf, $I(H)$ is an element of $\mathcal{U}$.

[2] We are reasoning here with classes of objects of E up to isomorphism.

Indeed, every sheaf H is the target of an epimorphic family of morphisms whose sources are representable sheaves; therefore it suffices to prove the first assertion when H is representable. Let G be the image of the family $(u : X_\alpha \to H, (u, \alpha) \in I(H))$. The morphism $G \to H$ is a monomorphism. We are then in the situation of the first step, because $X_\alpha \times_G X_\beta$ is isomorphic to $X_\alpha \times_H X_\beta$, which is representable, and because $I(H)$ is an element of $\mathcal{U}$. Hence G is representable. Since the X_α form a generating family, $G \to H$ is an isomorphism. The final assertion is immediate from the definition of $\mathcal{U}$-sheaves.

1.2.4.3. Every subsheaf of a representable sheaf is representable.

Indeed, let $G \to H$ be a subsheaf of a representable sheaf. Then G is a $\mathcal{U}$-sheaf. The family $(u : X_\alpha \to G, (u, \alpha) \in I(G))$ is therefore epimorphic in $\widetilde{E}$ and indexed by an element of $\mathcal{U}$. Moreover, the fiber products $X_\alpha \times_G X_\beta$ are isomorphic to the fiber products $X_\alpha \times_H X_\beta$, which are representable. Thus we are in the situation of the first step, and G is representable.

1.2.4.4. Every $\mathcal{U}$-sheaf is representable.

Indeed, by the first and second steps, it suffices to show that the fiber products $X_\alpha \times_H X_\beta$ are representable. But these fiber products are subobjects of the products $X_\alpha \times X_\beta$. The conclusion follows from the third step. This completes the proof of Theorem 1.2.

Remark 1.3. Of course, for a given $\mathcal{U}$-topos E, there is generally no preferred way to represent it, up to equivalence, as $\widetilde{C}$ with C a *small* site; or, what is essentially the same by 1.2.1 when one restricts to C whose topology is no finer than the canonical topology, there is no preferred *small* generating family in E. Once one drops the requirement that C be small, however, there is, by 1.2(iii), a completely canonical choice of a $\mathcal{U}$-site C such that E is equivalent to $\widetilde{C}$, namely E itself. This is one technical reason why, in practice, it is inconvenient to work only with small sites: indeed, the most important sites of all, namely $\mathcal{U}$-topoi, are not small. Moreover, the sites generating topoi which arise in many questions in algebraic geometry, and even in topology (cf. 2.5), are also not small; for example, the étale site of a scheme.

Proposition 1.4. Let E be a $\mathcal{U}$-topos, E' a category, not necessarily a $\mathcal{U}$-category, and F a presheaf on E with values in E'. Then F is a sheaf with values in E' if and only if F transforms $\mathcal{U}$-inductive limits in E into projective limits in E'.

Let $\mathcal{V}$ be a universe containing $\mathcal{U}$ and such that E' is a $\mathcal{V}$-category. By composing F with the functors $\mathrm{Hom}(X', -) : E' \to \mathcal{V}\text{-}\mathbf{Set}$ defined by the objects X' of E', we reduce to the case $E' = \mathcal{V}\text{-}\mathbf{Set}$. Suppose first that F transforms $\mathcal{U}$-inductive limits into projective limits. Then it follows immediately from the definitions that F is a sheaf, since a covering family $X_i \to X$ in E, by definition of the canonical topology on E, allows X to be regarded as the inductive limit of the usual diagram $X_i \times_X X_j \rightrightarrows X_i$.

Conversely, suppose that F is a sheaf, and prove that it transforms $\mathcal{U}$-inductive limits into projective limits. If $\mathcal{V} \subset \mathcal{U}$, we may assume $\mathcal{V} = \mathcal{U}$, and it suffices to apply criterion 1.2(iii). In the general case a little more work is needed. Let C be a full subcategory of E generated by a generating family indexed by an element of $\mathcal{U}$, and let $u : C \to E$ be the inclusion. Endow C with the topology induced by the canonical topology of E. Let $\widetilde{C}_{\mathcal{U}}$ and $\widetilde{C}_{\mathcal{V}}$ denote the categories of sheaves on C, and let $\widetilde{E}_{\mathcal{V}}$ denote the category of $\mathcal{V}$-sheaves on E for the canonical topology. Let $J_E : E \to \widetilde{E}_{\mathcal{V}}$ be the canonical functor, and let

$$i_{\mathcal{U},\mathcal{V}} : \widetilde{C}_{\mathcal{U}} \longrightarrow \widetilde{C}_{\mathcal{V}}$$

be the inclusion. The diagram

$$\begin{tikzcd}
\widetilde{C}_{\mathcal{U}} \arrow[r, "i_{\mathcal{U},\mathcal{V}}"] & \widetilde{C}_{\mathcal{V}} \\
E \arrow[u] \arrow[r, "J_E"] & \widetilde{E}_{\mathcal{V}} \arrow[u]
\end{tikzcd}$$

where the vertical arrows are induced by $F \mapsto F \circ u$, is commutative. By the explicit construction of the associated-sheaf functor and by Exposé II, 4.1, the functor $i_{\mathcal{U},\mathcal{V}}$ commutes with $\mathcal{U}$-inductive limits. Moreover, by Exposé III, 5.1, and Corollary 1.5, the vertical arrows in the diagram are equivalences of categories. Hence $J_E : E \to \widetilde{E}_{\mathcal{V}}$ commutes with $\mathcal{U}$-inductive limits. For every object X of E, one has

$$F(X) \simeq \mathrm{Hom}_{\widetilde{E}_{\mathcal{V}}}(J_E(X), F).$$

Therefore F transforms $\mathcal{U}$-inductive limits of E into projective limits.

Corollary 1.5. Let E be a $\mathcal{U}$-topos, E' a $\mathcal{U}$-category, and $f : E \to E'$ a functor. Then f admits a right adjoint if and only if f commutes with $\mathcal{U}$-inductive limits.

This is an immediate consequence of 1.4 in the case $E' = \mathcal{U}$-**Set**.

Corollary 1.6. Let E, E' be two $\mathcal{U}$-topoi and let $f : E \to E'$ be a functor. The following conditions are equivalent:

i) f commutes with $\mathcal{U}$-inductive limits;

ii) f admits a right adjoint;

iii) f is continuous in the sense of Exposé III, 1.1.

The equivalence i) $\Leftrightarrow$ ii) was proved in 1.5. To prove i) $\Leftrightarrow$ iii), apply the definition of continuous functors, choosing a universe $\mathcal{V}$ such that $\mathcal{U} \in \mathcal{V}$, so that E and E' are $\mathcal{V}$-small sites. One must express that, for every $\mathcal{V}$-sheaf F on E', the composite $F \circ f$ is a sheaf on E, that is, by 1.4, that it transforms $\mathcal{U}$-inductive limits into projective limits. This follows from (i), since by 1.4 the functor F itself transforms $\mathcal{U}$-inductive limits into projective limits. It is also necessary, as is seen by taking F representable.

Corollary 1.7. With the notation of 1.6, f is the inverse-image functor u^* of a geometric morphism $u : E' \to E$ if and only if f is left exact and sends epimorphic families to epimorphic families.

Necessity is trivial from the definition. Sufficiency follows from Exposé III, 1.6, which implies that f is continuous under the stated conditions, hence commutes with $\mathcal{U}$-inductive limits by 1.6.

Remark 1.8. With the notation of 1.5, one can show that f admits a left adjoint if and only if it commutes with $\mathcal{U}$-projective limits. Equivalently, a covariant functor $E \to (\mathcal{U}$-**Set**$)$ is representable if and only if it commutes with $\mathcal{U}$-projective limits.[3] We indicate only the principle of the proof, which has two steps:

[3] A more general statement is found in Exposé I, 8.12.8 and 8.12.9; the sketch below corresponds to the proof given there.

a) Standard arguments show that, if F commutes with projective limits, it is pro-representable by a strict projective system $(T_i)_{i \in I}$, where I is a filtered ordered set, not necessarily small. One may assume that, if $i > j$, then $T_i \to T_j$ is not an isomorphism; under this hypothesis, F is representable if and only if I is small, which in fact implies that the projective system is essentially constant.

b) To prove that I is small, knowing that for every object X of E the set $F(X) = \varinjlim_i \mathrm{Hom}(T_i, X)$ is small, it suffices to have a *small cogenerating family* $(X_j)_{j \in J}$, that is, a generating family for the opposite category E°. One shows that such a small cogenerating family always exists in a $\mathcal{U}$-topos E.

c) To prove this last point, standard arguments show that every object X of E admits a monomorphism into an "injective" object; then, for every generating family (L_α) of E, if one embeds each L_α in this way into an injective object I_α, the family (I_α) is cogenerating.

2 Examples of topoi

2.0

In this section we have gathered a fairly large number of typical examples of topoi which the reader will already have met elsewhere, and which are meant to ease access to the "yoga" of topoi. For other examples, drawn from algebraic geometry, of topologies on sites giving rise to topoi, the reader may consult SGA 3, IV, 6, and, for the étale topos, Exposé VII of the present seminar. Since later references to this section will mostly concern notation or terminology, we leave the verification of the statements accompanying these examples as exercises for the reader's general instruction. All the examples in this section will be made more precise in Section 4, where we examine their functorial dependence on the data, and in later sections as illustrations of the general notions concerning topoi.

2.1. Topos associated with a topological space

Let X be a small topological space, and let $\mathrm{Open}(X)$ be the category of open subsets of X, endowed with the canonical topology. We denote by $\mathrm{Top}(X)$ the topos of $\mathcal{U}$-sheaves on $\mathrm{Open}(X)$. This topos is equivalent to the category of spaces *étalés* over X, by associating to such a space X' over X the sheaf

$$U \longmapsto \Gamma(X'/U) = \mathrm{Hom}_X(U, X')$$

on $\mathrm{Open}(X)$. By abuse of language we shall sometimes denote by the same letter X the topos $\mathrm{Top}(X)$ defined by the topological space X.

It is clearly this example which served chiefly as a guide and intuitive support for the development of the theory of topoi. One should nevertheless be careful that the topoi deduced from topological spaces are of a very special nature, notably because they have been described by sites $C = \mathrm{Open}(X)$ in which *all morphisms are monomorphisms*; the underlying category is therefore just a preordered set. It follows in particular that the sheaves represented by the objects of C are subsheaves of the final sheaf, and hence that *subsheaves of the final sheaf form a generating family* for the topos in question. This property is not shared by most of the topoi arising naturally in algebraic geometry or algebra, cf. the examples below. Up to minor qualifications, it characterizes topoi of the form $\mathrm{Top}(X)$ (cf. 7.1.9 below).

One checks easily that the map $\mathrm{Open}(X) \to \mathrm{Top}(X)$, which sends an open set of X to the sheaf it represents, is a bijection from $\mathrm{Open}(X)$ onto the set of subobjects of the final object of $\mathrm{Top}(X)$, and that this bijection is even an isomorphism for the natural order structures. This suggests that it should be possible to reconstruct the topological space X, up to homeomorphism, from $\mathrm{Top}(X)$ up to equivalence. We shall see below (4.2) that this is indeed so, subject to a slight restriction on X.

2.2. The punctual or final topos, and the empty or initial topos

When X is a topological space reduced to a single point, the functor

$$F \longmapsto F(X) : \mathrm{Top}(X) \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

is an equivalence of categories. In particular, the category $\mathcal{U}$-**Set** is a $\mathcal{U}$-topos. We saw in Exposé II that this $\mathcal{U}$-topos is typical from the point of view of exactness properties, in that the verification of many properties of general topoi, especially exactness properties, reduces to this particular topos. The interpretation given here in terms of the one-point topological space justifies the abuse of language by which a topos equivalent to $\mathcal{U}$-**Set** is called a *punctual topos*, although as a category it is not at all equivalent to the one-object category. This terminology corresponds to the correct geometric intuition for the role played by these topoi. Also by abuse of language, a punctual topos is sometimes called a *final topos*; we shall say “the final topos” for $\mathcal{U}$-**Set**.

When X is the empty topological space, so that $\mathrm{Open}(X)$ is the one-object category, a presheaf F on $\mathrm{Open}(X)$ is a sheaf if and only if its value at the unique object $\varnothing$ of $\mathrm{Open}(X)$ is a singleton. It follows that $\mathrm{Top}(\varnothing)$ is isomorphic to the category of singleton $\mathcal{U}$-sets, a category equivalent to the one-object category. Thus the one-object category, and any $\mathcal{U}$-category equivalent to it, is a $\mathcal{U}$-topos. It is sometimes called, by abuse of language, the *empty topos* or *initial topos*; note that it is not equivalent to the empty category.

2.3. Topos associated with a space with operators

Let X be a topological space and let G be a discrete group acting on X by homeomorphisms. In [Toh., 5.1] the category of G-*sheaves on* X was defined; we shall also say sheaves on (X, G). These are sheaves of sets on X endowed with actions of G compatible with the action of G on X. Giraud’s criterion 1.2(iii) immediately shows that this category is a topos, denoted $\mathrm{Top}(X, G)$. When G is the trivial group, this gives the example of 2.1; when X is a point, this gives the topos of sets with a left action of G, or G-*sets*, also called the *classifying topos of the discrete group* G, and denoted B_G. One checks easily that the only subobjects of the final object e of B_G are e and the empty sheaf $\varnothing$. In particular, if G is not the trivial group, subobjects of the final object of B_G do not form a generating family for B_G. Thus B_G is then not equivalent to a topos of the form $\mathrm{Top}(X)$ considered in 2.1.

The notion of G-sheaf was introduced in loc. cit. to develop the cohomological theory of abelian G-sheaves. Interpreting these as the abelian sheaves of the topos (X, G), that theory is included in the theory of Exposé V, developed for general topoi.

More generally, one may wish to attach an appropriate topos to a topological space X endowed with a topological group G, not necessarily discrete, of automorphisms, giving rise to an adequate cohomological theory. The same question arises in the contexts of differentiable manifolds, real or complex analytic manifolds or spaces, and schemes. This is indeed possible; cf. 2.5 below.

2.4. Classifying topos of a group object

Let E be a topos and let G be a group object of E. Let (E, G) be the category of objects of E on which G acts. Giraud's criterion immediately shows that this is a topos. It is called the *classifying topos* of the group object G, and is denoted B_G. When E is the punctual topos, i.e. when G is an ordinary group, one recovers the classifying topos of 2.3.

The terminology is justified by the fact that B_G plays a universal role in the classification of torsors, or principal homogeneous bundles, under G, and more generally under $G_{E'} = f^*(G)$, where E' is a topos "over E", i.e. endowed with a morphism $f : E' \to E$ (cf. 3.1 below). This role, made explicit in [3, Chap. V] or in 5.9 below, shows that B_G plays in the context of topoi the same role as the classical classifying spaces of topological groups in the homotopy theory of topological spaces. The latter may be regarded (cf. 2.5) as a weakened version of the former, obtained by retaining only the "homotopy type" of the classifying topos, in a suitable sense which need not be made precise here.

2.5. The big site and big topos of a topological space. Classifying topos of a topological group

Let $\mathcal{U}$-**Top**, or simply (**Top**), be the category of topological spaces belonging to $\mathcal{U}$. Finite projective limits are representable in (**Top**). Consider on (**Top**) the pretopology for which $\mathrm{Cov}(X)$ is the set of surjective families of open immersions $u_i : X_i \to X$. We shall regard (**Top**) as a site via the topology generated by this pretopology. For every object X of (**Top**), consider the category

$$(\mathbf{Top})_{/X}$$

of objects of (**Top**) over X, i.e. topological spaces over X, as a site by means of the topology induced from that of (**Top**) by the forgetful functor $(\mathbf{Top})_{/X} \to (\mathbf{Top})$. This site is called the *big site* associated with X. It is not an element of $\mathcal{U}$; nor is it a $\mathcal{U}$-site in the sense of Exposé II, 3.0.2, so precautions are needed when applying the usual results to it. To avoid this inconvenience, one may choose a universe $\mathcal{V}$ such that $\mathcal{U} \in \mathcal{V}$, so that $(\mathbf{Top})_{/X}$ becomes a $\mathcal{V}$-site, and work with the associated $\mathcal{V}$-topos, which may be denoted $\mathrm{TOP}(X)$ and will be called the *big topos of* X. If one is reluctant to enlarge $\mathcal{U}$, one may choose a cardinal c bounding the cardinalities of X and of all topological spaces one expects to use in the arguments–most often $\sup(\mathrm{card}\, X, \mathrm{card}\, \mathbb{R})$ will suffice–and replace $(\mathbf{Top})_{/X}$ by the subcategory $(\mathbf{Top})'_{/X}$ formed by those X' over X with $\mathrm{card}\, X' \leq c$, endowed with the induced topology, and denote the sheaf topos on that site by $\mathrm{TOP}(X)$. To fix ideas, suppose the first definition has been adopted.

The advantage of the big topos of X over the small one is that the site defining it contains $(\mathbf{Top})_{/X}$ as a full subcategory. Since the topology of this site is manifestly no finer than the canonical topology, the canonical functor from $(\mathbf{Top})_{/X}$ to $\mathrm{TOP}(X)$, sending a space X' over X to the sheaf it represents, is *fully faithful*. Thus *a space X' over X is known up to X-isomorphism once the sheaf it defines is known*; the notion of sheaf on the big site of X may therefore be regarded as a *generalization* of the notion of topological space over X, in terms of which all constructions of sheaf theory acquire meaning for topological spaces over X.

Thus, when G is a group object of the category $(\mathbf{Top})_{/X}$ of topological spaces over X, one can associate to it the classifying topos B_G (2.4), hence classifying cohomology groups, classifying homotopy groups, and so on, defined as the corresponding invariants of the $\mathcal{V}$-topos B_G. In particular, when X is a point, G is an ordinary topological group. Under the usual local hypotheses ensuring that the singular cohomology of the Cartesian powers G^n agrees with sheaf cohomology, say for constant coefficients–for example when G is locally contractible–the cohomology of the classifying topos of G is canonically isomorphic to that of the classifying space of G in the sense of topologists.

The introduction of classifying topoi through big sites has, compared with classifying spaces, the advantage of giving a richer theory, since it provides useful cohomological invariants for coefficients more general than constant or locally constant coefficients. Moreover, the definition considered here adapts in the evident way to the usual other contexts: differentiable manifolds, real or complex analytic manifolds or spaces, and schemes. This point of view makes it possible, in particular, to connect the traditional and the "arithmetic" approaches to characteristic classes by considering the classical groups as coming from schemes defined over the ring of integers; cf. Section 7 for indications in this direction. Similarly, J. Giraud's general results [3] on the classification of extensions of group objects, developed in the very general and flexible setting of topoi, may, through big topoi, specialize to results on the classification of extensions of topological groups, or of real or complex Lie groups, which seem to have been known to topologists essentially only in the case of extensions with abelian kernel [11].

2.6. Topoi of the form $\widehat{C}$

Let C be a small category. The category $\widehat{C}$ of $\mathcal{U}$-presheaves on C is plainly a $\mathcal{U}$-topos, since it is of the form $\widetilde{C}$ when C is endowed with the chaotic topology. We shall give below some details on the relations between C and $\widehat{C}$. Let us note here only that a topos E equivalent to one of the form $\widehat{C}$ is rather special, since *it admits a small generating family* (X_i) *formed of connected projective objects*, i.e. objects X such that the functor $Y \mapsto \mathrm{Hom}(X, Y)$ sends epimorphisms to epimorphisms and coproducts to coproducts. It suffices to take in $\widehat{C}$ the generating family formed by the functors represented by the objects $X \in \mathrm{Ob}\, C$. Note also that if, in a topos E, one has a covering family $X_i \to X$ with the X_i projective and connected, then every other covering family of X is refined by this one. Hence in a topos E of the form $\widehat{C}$, every object X admits a covering family refining all the others. A topos of the form $\mathrm{Top}(X)$, with X a topological space whose points are closed, has this property only when X is discrete.

When the category C has a single object, C is identified with a monoid G. A presheaf on C is then a set on which G acts on the *right*, since it is a functor $G^\circ \to \mathbf{Set}$, and $\widehat{C}$ is the topos of sets with a right monoid of operators. Taking 2.3 into account, this may also be denoted B_{G°: it is the *topos of sets with monoid of operators* G°, where G° is the opposite monoid. When G is a group, using the isomorphism $g \mapsto g^{-1}$ from G to G°, one recovers the classifying topos B_G of 2.3.

2.7. Classifying topos of a pro-group

2.7.1

Let $\mathcal{G} = (G_i)_{i \in I}$ be a projective system of groups, with $G_i, I \in \mathcal{U}$. Suppose the projective system is *strict*, i.e. the transition morphisms $G_j \to G_i$ are surjective. If E is a set, an *action of* $\mathcal{G}$ *on*

E (say on the left) is the following structure: a family $(E_i)_{i\in I}$ of subsets of E, with union E; and, for every $i \in I$, an action of the group G_i on the set E_i. These data are required to satisfy the following condition: for $j \geq i$, E_i is the subset of E_j formed by the elements fixed under the kernel of $G_j \to G_i$. The $\mathcal{U}$-sets endowed with an action of $\mathcal{G}$ form a category in the evident way. Giraud's criterion immediately shows that this category is a $\mathcal{U}$-topos. It is denoted $B_{\mathcal{G}}$ and called the *classifying topos of* $\mathcal{G}$. When I has an initial object i_0, and $G = G_{i_0}$, this recovers the classifying topos of 2.3.

2.7.2

Another important example is the case where the groups G_i are finite, so that

$$G = \varprojlim G_i$$

is a compact totally disconnected topological group, or *profinite group*. An action of $\mathcal{G}$ on E is then the same as a continuous action of G on E, or equivalently an action such that the stabilizer of every point of E is an open subgroup of G. The classifying topos $B_{\mathcal{G}}$ will also be denoted B_G, where of course G is considered with its profinite topology.

2.7.3

Using the remarks of 2.6, it is easy to check that the topos $B_{\mathcal{G}}$ defined by a strict projective system $\mathcal{G} = (G_i)_{i\in I}$ of groups is equivalent to a topos of the form $\widehat{C}$ only if this projective system is essentially constant. In the case of a profinite group, this means that the group is actually finite.

2.7.4

The following geometric interpretation of the classifying topos B_G of a discrete group G is useful for giving the correct geometric intuition for these topoi; cf. also 4.5, 5.8, 5.9, and 7.2 below. Let X be a connected, locally connected, and locally simply connected topological space, let x be a point of X, and let G be its fundamental group at x. It is known that, up to isomorphism, every discrete group G can be obtained in this way. Galois theory of coverings of X gives an equivalence between the category B_G of G-sets and the category of *étale coverings* of X, that is, of spaces X' over X which are locally X-isomorphic to X-spaces of the form $X \times I$, where I is a discrete space. This latter category may also be interpreted as the category of locally constant sheaves on X, i.e. the category of locally constant objects of the topos $\mathrm{Top}(X)$.

When G is a profinite group, there is an analogous geometric interpretation of B_G as the category of X-schemes which are sums of finite étale coverings of a connected scheme X, endowed with a geometric point x and an isomorphism $G \simeq \pi_1(X, x)$. It is again known that every profinite group can occur as the fundamental group of a suitable connected scheme, the spectrum of a field if desired. Finally, pro-groups which are not necessarily profinite and not necessarily essentially constant also occur in the classification of coverings of connected and locally connected spaces which are not locally simply connected, or in the classification of étale coverings, not necessarily finite or ind-finite, of non-normal connected schemes. For this last case, cf. SGA 3, X, 6.

Exercise 2.7.5. Define, for a topos E, the notions of connectedness, local connectedness, simple connectedness, and local simple connectedness. Define the notions of constant and locally constant object of E (cf. IX, 2.0). Let $f : E' \to E$ be a geometric morphism, with E' simply connected (for example E' the punctual topos), and E connected and locally connected. Define a strict pro-group

$$\pi_1(E, f) = (G_i)_{i \in I} = \mathcal{G},$$

called the *fundamental pro-group of E at f*, and an equivalence of categories from $B_{\mathcal{G}}$ to the category of locally constant objects of E. Show that when E is locally simply connected, $\pi_1(E, f)$ is essentially constant and therefore identifies with an ordinary discrete group $\pi_1(E, f)$, called the *fundamental group of E at f*. Show that every strict pro-group $\mathcal{G}$ is isomorphic, as a pro-group, to the fundamental pro-group of a suitable connected and locally connected topos E, for a suitable f, with E' the punctual topos: take $E = B_{\mathcal{G}}$, and $f : E' \to E$ defined by the forgetful functor $f^* : E \to E' = \mathbf{Set}$. When $\mathcal{G}$ is essentially constant, i.e. isomorphic as a pro-group to an ordinary discrete group, prove that one may take E above to be locally simply connected, again taking $E = B_G$.

2.8. Example of a false topos

Let $\mathcal{G} = (G_i)_{i \in I}$ be a strict pro-group, where I is a filtered ordered set and where $i > j$ implies that $G_i \to G_j$ is not an isomorphism. Suppose that $\mathrm{card}(I) \notin \mathcal{U}$. Consider the category of sets $E \in \mathcal{U}$ on which $\mathcal{G}$ acts on the left, in the sense of 2.7.1. This is a $\mathcal{U}$-category, and, as in 2.7.1, it satisfies conditions a), b), c) of 1.1.2. However, it is not a $\mathcal{U}$-topos, since it has no generating family that is $\mathcal{U}$-small. Likewise, it is not a $\mathcal{V}$-topos for any universe $\mathcal{V}$.

3 Geometric morphisms of topoi

Definition 3.1. Let E and E' be two $\mathcal{U}$-topoi. A *geometric morphism* from E to E', or sometimes, by abuse of language, a continuous map from E to E', is a triple $u = (u_*, u^*, \varphi)$, formed of functors

$$u_* : E \longrightarrow E', \qquad u^* : E' \longrightarrow E,$$

and an adjunction isomorphism of bifunctors, for $X' \in \mathrm{Ob}\, E'$ and $Y \in \mathrm{Ob}\, E$,

$$\varphi : \mathrm{Hom}_E(u^* X', Y) \xrightarrow{\sim} \mathrm{Hom}_{E'}(X', u_* Y),$$

where u^* is moreover required to be left exact, i.e. to commute with finite projective limits. The functor u_* is called the direct-image functor of the geometric morphism u, the functor u^* is called the inverse-image functor of u, and φ is called the adjunction isomorphism for u.

3.1.1

Unless explicitly stated otherwise, for a geometric morphism $u : E \to E'$ we shall denote the corresponding direct-image and inverse-image functors by u_* and u^*.[4] Since u_* is right adjoint to u^*, and u^* is left adjoint to u_* by the adjunction isomorphism φ, either of the two functors u_*, u^* determines the other up to unique isomorphism, by the usual formalism of adjoint functors. In practice, depending on the case, it may be more convenient to define a geometric morphism

[4] Sometimes one should also write u^{-1} instead of u^*, cf. 13.2.3.

$u : E \to E'$ either by giving $u^* : E' \to E$, or by giving $u_* : E \to E'$. In the first case one must simply verify that the given functor u^* admits a right adjoint and is left exact. In the second, one must verify that the given functor u_* admits a left adjoint which is left exact. In either case, from the partial datum and from a *choice* of adjoint functor and adjunction isomorphism, one obtains a geometric morphism $u : E \to E'$, which is unique up to unique isomorphism in terms of the datum u^*, respectively u_*, in an evident sense made entirely explicit below in 3.2.1.

3.1.2

If $u : E \to E'$ is a geometric morphism, it follows from the properties of adjoint functors that the direct image $u_* : E \to E'$ commutes with projective limits, while $u^* : E' \to E$ commutes with inductive limits.[5] Since u^* is also assumed left exact, it follows in particular that u^* is *exact*. Thus, in the pair (u_*, u^*), it is the *inverse-image functor* u^* which has the strongest exactness properties.

These properties ensure that, for any kind of algebraic structure Σ whose data can be described in terms of arrows between underlying sets and sets obtained from them by repeated application of finite projective limits and arbitrary inductive limits, and for any "object of E' endowed with a structure of kind Σ", its image under u^* is endowed with the same structures. Rather than undertake the unattractive task of giving a precise formal meaning to this statement and justifying it formally, we advise the reader to spell it out and verify it for structures such as group, ring, module over a ring, comodule over a co-ring, bialgebra over a ring, and torsor under a group. In these examples, the first three kinds of structure are defined exclusively in terms of finite projective limits, while the others implicitly involve constructions using inductive limits as well. Moreover, u^* "commutes" with all the usual functorial operations made in terms of such structures, more precisely with all operations expressible in terms of inductive limits and finite projective limits: construction of free objects, such as free groups or free modules generated by an object, tensor products, cf. Section 12 below, and so on.

The direct image $u_* : E \to E'$, which commutes with projective limits, consequently respects every algebraic structure on an object, or a family of objects, of E which is definable solely in terms of projective limits, such as group, ring, or module structures among the examples just mentioned. On the other hand, u_* is generally not right exact, i.e. it does not generally commute with finite inductive limits, and it does not in general even send epimorphisms to epimorphisms. This failure of exactness of u_* is precisely the source of cohomological phenomena, to be studied from the standpoint of commutative homological algebra in the following exposé. Consequently, u_* does not generally extend to objects such as comodules, bialgebras, or torsors under a group, and does not in general commute with operations such as the free module generated by an object, tensor products of modules, and so on.

3.1.3

In practice, when one has a functor $f : E \to F$ from one $\mathcal{U}$-topos to another, one should make explicit all its exactness properties, including the possible existence of left or right adjoints, in order to understand the "geometric nature" of f. Such an understanding is usually an indispensable guide to the correct geometric intuition. Thus, if f commutes with arbitrary

[5]Moreover, by 1.5 and 1.8, for a given functor $u_* : E \to E'$, respectively $u^* : E' \to E$, this functor admits a left, respectively right, adjoint if and only if it commutes with $\mathcal{U}$-projective, respectively $\mathcal{U}$-inductive, limits.

inductive limits and finite projective limits, it should be written

$$f = u^*,$$

where

$$u : F \longrightarrow E$$

is a geometric morphism; in other words, f should be interpreted as an inverse-image functor for a continuous map of topoi. If f commutes with arbitrary projective limits, so that it has a left adjoint, and if that left adjoint, which a priori commutes with arbitrary inductive limits, also commutes with finite projective limits, then f should be written

$$f = v_*,$$

where

$$v : E \longrightarrow F$$

is a geometric morphism. In some cases f may satisfy both of these conditions; cf. 4.10 for an example. In that case one should introduce simultaneously the geometric morphisms

$$u : F \longrightarrow E, \qquad v : E \longrightarrow F,$$

which give rise to a sequence of three adjoint functors

$$v^*, \qquad v_* = u^* = f, \qquad u_*.$$

One must then carefully distinguish the geometric morphisms u and v, under penalty of losing the geometric intuition of the situation.

In this connection, note that when between two topoi E, F one has a sequence of three adjoint functors

$$e, \quad f, \quad g \qquad (e, g : F \rightrightarrows E,\ f : E \to F),$$

then f commutes with arbitrary inductive and projective limits, and hence can always be written as u^*, where $u : F \to E$ is a geometric morphism. Then g is u_*. On the other hand, of course, e can be written as v^*, and then f as v_*, only if e also commutes with finite projective limits. This will certainly be the case if e is itself the right adjoint of a fourth functor d. Without such a condition, one often writes

$$e = u_!$$

for the left adjoint of an inverse-image functor $f = u^*$, when such a left adjoint exists; this notation is suggested by the example of an open immersion $u : X \to Y$ of topological spaces. Similarly, if e is of the form v^*, i.e. if f is of the form v_*, one sometimes writes $g = v^!$ for the right adjoint of a direct-image functor v_*, when such an adjoint exists.[6]

[6] Cf. below, 14.4, in the case of abelian sheaves.

3.2

Let E, E' be two $\mathcal{U}$-topoi, and let

$$u = (u_*, u^*, \varphi), \qquad v = (v_*, v^*, \Psi) : E \rightrightarrows E'$$

be two geometric morphisms from E to E'. A *morphism from u to v* is a morphism from u_* to v_* in the category $\mathcal{H}om(E, E')$ of functors from E to E'. Morphisms of geometric morphisms compose in the evident way, thereby defining a category, in fact a $\mathcal{U}$-category, denoted

$$\mathcal{H}om_{\text{top}}(E, E'),$$

called the *category of geometric morphisms*, or of continuous maps, from E to E'. There is an evident functor

$$u \longmapsto u_* : \mathcal{H}om_{\text{top}}(E, E') \longrightarrow \mathcal{H}om(E, E'),$$

which is fully faithful by the definition of arrows in the first category, though it is not injective on objects.

3.2.1

If u and v are as above, the theory of adjoint functors gives a canonical bijection

$$\operatorname{Hom}(u_*, v_*) \xrightarrow{\sim} \operatorname{Hom}(v^*, u^*).$$

In particular, one obtains a contravariant functor on $\mathcal{H}om_{\text{top}}(E, E')$:

$$u \longmapsto u^* : \mathcal{H}om_{\text{top}}(E, E')^\circ \longrightarrow \mathcal{H}om(E, E').$$

In accordance with the geometric intuition, according to which the direct-image functor u_* goes in the same direction as the continuous map giving rise to it, the direction of arrows between geometric morphisms should therefore be defined in terms of direct-image functors, not in terms of inverse-image functors, even though the latter, as we have seen, have the characteristic exactness properties of the notion of geometric morphism.

3.2.2

Having defined the category $\mathcal{H}om_{\text{top}}(E, E')$ of morphisms from the topos E to the topos E', the notion of *isomorphism* between two morphisms u, v from E to E' is also defined. In practice, one usually does not distinguish essentially between two isomorphic geometric morphisms, at least when a *canonical* isomorphism between them is available, just as one often does not distinguish between two objects of a category when one has chosen a canonical isomorphism between them. In this connection, let us point out that when one deals with diagrams of geometric morphisms and commutativity questions for such diagrams, a notion meaningful by 3.3, one usually means commutativity only up to "canonical" isomorphism. By abuse of language, such diagrams are then treated as genuinely commutative.

One might try to justify this abuse by introducing the set $\operatorname{Hom}_{\text{top}}(E, E')/\operatorname{Isom}$ of morphisms from E to E' up to isomorphism, and by calling such an isomorphism class a morphism, instead of following Definition 3.1. But this meets the same serious drawbacks that arise whenever one tries to identify two isomorphic objects of a category without having a canonical isomorphism between them. Experience shows that such a viewpoint is impracticable, and that one must retain the fine notion of 3.1, together with the notion of morphism between geometric morphisms, even if this sometimes forces one to fight with compatibilities among canonical isomorphisms.[7]

3.3.1

Let E, E', E'' be three $\mathcal{U}$-topoi, and consider geometric morphisms

$$u : E \longrightarrow E', \qquad v : E' \longrightarrow E''.$$

The theory of adjoint functors gives an adjunction isomorphism between the composite functors $v_* u_*$ and $u^* v^*$, in terms of the adjunction isomorphisms for the pairs (u_*, u^*) and (v_*, v^*). On the other hand, $u^* v^*$ is left exact as a composite of left exact functors. Thus one obtains a morphism from E to E'', called the *composite of the morphisms u and v*, and denoted vu:

$$vu : E \longrightarrow E''.$$

One then verifies trivially that composition of morphisms is associative, and that for every $\mathcal{U}$-topos E there is a morphism from E to itself which is a two-sided unit for composition, namely $(\mathrm{id}_E, \mathrm{id}_E, \varphi)$, where φ is the evident adjunction isomorphism from id_E to itself. If $\mathcal{V}$ is a universe such that $\mathcal{U} \in \mathcal{V}$, this defines a category

$$(\mathcal{V}\text{-}\mathcal{U}\text{- Top}),$$

whose objects are the $\mathcal{U}$-topoi belonging to $\mathcal{V}$, whose arrows are the morphisms between such $\mathcal{U}$-topoi, and whose composition is the one just described.

3.3.2

In fact, the composition map

$$\mathrm{Hom}_{\mathrm{top}}(E, E') \times \mathrm{Hom}_{\mathrm{top}}(E', E'') \longrightarrow \mathrm{Hom}_{\mathrm{top}}(E, E'')$$

is the map on objects induced by a "composition functor for morphisms"

$$\mathcal{H}om_{\mathrm{top}}(E, E') \times \mathcal{H}om_{\mathrm{top}}(E', E'') \longrightarrow \mathcal{H}om_{\mathrm{top}}(E, E''),$$

whose effect on arrows is the usual convolution product for morphisms between functors, here direct-image functors. These composition functors satisfy a strict associativity property for four topoi E, E', E'', E''', refining the associativity of composition of geometric morphisms. In language which is becoming familiar, one may also say that $\mathcal{U}$-topoi are the objects, or 0-arrows, of a *2-category*, whose 1-arrows are geometric morphisms and whose 2-arrows are morphisms of geometric morphisms.

It is this fact–that $\mathcal{U}$-topoi belonging to a universe $\mathcal{V}$ form a 2-category, and not merely an ordinary category as ordinary topological spaces do–which, from the technical point of view, is the most important difference between the theory of topoi and that of topological spaces. It is the source of certain technical complications already mentioned, but also of essentially new facts compared with traditional topology.

[7] For examples of such battles, apparently victorious, we refer to M. Hakim's book on relative schemes [9].

3.4

The fact that $\mathcal{U}$-topoi belonging to a universe $\mathcal{V}$ form a 2-category makes it possible, in particular, to define the notion of *equivalence of two $\mathcal{U}$-topoi* E, E': we say that E and E' are *equivalent* if there exist geometric morphisms $u : E \to E'$ and $v : E' \to E$ such that the composites uv and vu are isomorphic to the identity morphisms of E and of E', respectively. Then u and v are said to be *quasi-inverse equivalences* of each other.

One sees at once that a geometric morphism $u : E \to E'$ is an equivalence if and only if u_* is an equivalence, or equivalently if u^* is an equivalence. Use the fact that a functor $f : E \to E'$ between two topoi which is an equivalence is both of the form u_* and of the form v^*, for geometric morphisms u and v. Also, E and E' are equivalent in the sense of the preceding paragraph if and only if they are equivalent as categories, i.e. as objects of the 2-category ($\mathcal{V}$-**Cat**). As expected, this shows that the notions of equivalence introduced here do not depend on the choice of the universe $\mathcal{V}$ in 3.3.1.

3.4.1

In practice, one usually does not distinguish essentially between equivalent $\mathcal{U}$-topoi, just as one often does not distinguish between equivalent categories, provided however that one has an explicit equivalence between them, or at least an equivalence defined up to unique isomorphism. This notion of equivalence of topoi replaces here the traditional notion of homeomorphism between topological spaces. See the example in 4.2 below for the precise relation between these two notions.

4 Examples of geometric morphisms of topoi

We return here to the examples of Section 2, using the notion of geometric morphism. The comments of 2.0 apply equally to this section. The universe $\mathcal{U}$ will generally be left implicit.

4.1. The topos $\mathrm{Top}(X)$ for variable topological space X

4.1.1

Let

$$f : X \longrightarrow Y$$

be a continuous map of topological spaces. We shall associate to it canonically a geometric morphism

$$\mathrm{Top}(f) \text{ or } f : \mathrm{Top}(X) \longrightarrow \mathrm{Top}(Y),$$

with the notation of 2.1. When $\mathrm{Top}(X)$ is defined as $\widetilde{\mathrm{Open}(X)}$, the most convenient description of $\mathrm{Top}(f)$ is by the direct-image functor for sheaves

$$f_* : \mathrm{Top}(X) \longrightarrow \mathrm{Top}(Y),$$

defined by the formula

$$f_*(F) = F \circ f^{-1},$$

where

$$f^{-1} : \mathrm{Open}(Y) \longrightarrow \mathrm{Open}(X)$$

is the evident functor $U \mapsto f^{-1}(U)$. We have already noted that this functor is continuous and left exact, and thus defines the above functor f_*, which has a right adjoint f^* that is left exact (Exposé III, 1.9.1). Strictly speaking, the geometric morphism $\mathrm{Top}(f)$ depends on the choice of the right adjoint f^* of f_*, and is therefore defined only up to canonical isomorphism. We shall henceforth refrain from explicitly signaling phenomena of this sort.

When one adopts the viewpoint of *étalé spaces* to define $\mathrm{Top}(X)$, it is the inverse-image functor

$$f^* : \mathrm{Top}(Y) \longrightarrow \mathrm{Top}(X)$$

which is most convenient for defining $\mathrm{Top}(f)$, by simply putting

$$f^*(Y') = X \times_Y Y'$$

for every étale space Y' over Y. It is clear that the fiber product is an étale space over X, and that the functor f^* thus obtained is left exact and commutes with arbitrary inductive limits, hence defines a geometric morphism $\mathrm{Top}(f)$. For the compatibility of the two definitions we refer to [TF].

For composable continuous maps

$$X \xrightarrow{f} Y \xrightarrow{g} Z,$$

there are transitivity isomorphisms

$$\mathrm{Top}(gf) \simeq \mathrm{Top}(g)\,\mathrm{Top}(f)$$

of geometric morphisms. For three composable continuous maps f, g, h, these *transitivity isomorphisms* satisfy a compatibility relation which we do not write out here, and which is precisely the one considered in SGA 1, VI, 7.4(B). This may be expressed by saying that, for X varying in the category (**Top**),

$$X \longmapsto \mathrm{Top}(X)$$

is a "pseudo-functor"

$$(\mathcal{U}\text{-}\mathbf{Top}) \longrightarrow (\mathcal{V}\text{-}\mathcal{U}\text{- Top}), \tag{4.1.1.1}$$

or, in the terminology of 2-categories, that one has a non-strict functor of 2-categories. In practice we shall usually allow the abuse of language that identifies $\mathrm{Top}(gf)$ and $\mathrm{Top}(g)\,\mathrm{Top}(f)$, reasoning as though (4.1.1.1) were a genuine functor of ordinary categories. We shall allow the analogous abuses in the other examples below.

4.1.2

The preceding considerations extend immediately to topoi associated with topological spaces with groups of operators (2.3). If $f = (f^{\mathrm{sp}}, f^{\mathrm{gr}})$,

$$f : (X, G) \longrightarrow (Y, H)$$

is a morphism of spaces with operators, where

$$f^{\mathrm{sp}} : X \to Y, \qquad f^{\mathrm{gr}} : G \to H$$

are respectively a continuous map and a group homomorphism, compatible in the evident sense, then one associates to it a geometric morphism

$$\mathrm{Top}(f) \text{ or } f : \mathrm{Top}(X, G) \longrightarrow \mathrm{Top}(Y, H),$$

whose definition is left to the reader. When G and H are the trivial groups, this gives the definition of 4.1.1. When instead X and Y are points, the inverse-image functor f^* is the "restriction of the group of operators"

$$f^* : B_H \longrightarrow B_G,$$

which sends an H-set to the G-set it defines via $f : G \to H$. We shall encounter this example again in other forms in 4.5 and 4.6.1.

4.1.3

Given a continuous map $f : X \to Y$ of topological spaces, one also associates to it a morphism of the corresponding big topoi (2.5),

$$\mathrm{TOP}(f) \text{ or } f : \mathrm{TOP}(X) \longrightarrow \mathrm{TOP}(Y),$$

most conveniently defined by the inverse-image functor

$$f^* : \mathrm{TOP}(Y) \longrightarrow \mathrm{TOP}(X),$$

which is just the *restriction functor*. This morphism $\mathrm{TOP}(f)$ is a special case of the so-called "inclusion" morphism for an induced topos, which will be studied in Section 5. Thus, with the same reservations as in 4.1.1, the topos $\mathrm{TOP}(X)$ may be regarded as a functor in X, for X varying in (**Top**).

4.2. Faithfulness properties of $X \mapsto \mathrm{Top}(X)$

We shall specify to what extent a topological space X can be reconstructed from the topos $\mathrm{Top}(X)$. For this purpose, for two spaces X and Y, one must describe the category of morphisms from $\mathrm{Top}(X)$ to $\mathrm{Top}(Y)$, in order to state the faithfulness properties of the "functor" $X \mapsto \mathrm{Top}(X)$. We merely state the results obtained, referring the reader to [9] for details. The reader who wants to verify them directly may consult Exercise 7.8.

4.2.1

Recall that a topological space X is called *sober* if every irreducible closed subset of X has exactly one generic point. Almost all spaces used in practice are sober; this is true in particular for Hausdorff spaces, more generally for spaces all of whose points are closed, and for the underlying space of a scheme. With any topological space X one associates a sober topological space X_{sob} and a continuous map

$$\varphi : X \longrightarrow X_{\mathrm{sob}} \qquad (4.2.1.1)$$

which is *universal* for continuous maps from X to sober spaces. Equivalently, one constructs a left adjoint $X \mapsto X_{\mathrm{sob}}$ to the inclusion functor from sober spaces to all topological spaces. Explicitly, the points of X_{sob} are the irreducible closed subsets of X, and the open subsets are those of the form U', where U is an open subset of X and $U' \subset X_{\mathrm{sob}}$ is the set of irreducible closed subsets of X meeting U. The map (4.2.1.1) sends $x \in X$ to the closure of $\{x\}$. The space X is sober if and only if the preceding map is bijective, hence a homeomorphism.

The functor

$$\varphi^{-1} : \mathrm{Open}(X_{\mathrm{sob}}) \longrightarrow \mathrm{Open}(X)$$

induced by φ is an isomorphism. It follows that the geometric morphism

$$\mathrm{Top}(\varphi) : \mathrm{Top}(X) \longrightarrow \mathrm{Top}(X_{\mathrm{sob}})$$

defined by φ is also an isomorphism. This explains a priori why X_{sob} must enter into the question of reconstructing X from $\mathrm{Top}(X)$: the latter depends only on X_{sob}, up to isomorphism, so the question can have an affirmative answer only if X is sober. We shall specify below (7.1) how X_{sob} can in fact be reconstructed in terms of $\mathrm{Top}(X)$, by interpreting its points as "points" of the topos $\mathrm{Top}(X)$, or equivalently as fiber functors.

4.2.2

On every topological space one should introduce the order relation $\leq$ defined by

$$x \leq y \quad \Longleftrightarrow \quad \overline{\{x\}} \subset \overline{\{y\}} \quad \Longleftrightarrow \quad x \in \overline{\{y\}},$$

which is also expressed by saying that x is a *specialization* of y, or that y is a *generalization* of x. For a space of the form X_{sob}, this is just the inclusion relation among irreducible closed subsets of X.

This being so, on the set of maps from a space X to a space Y one introduces the so-called specialization order induced from that of Y, namely

$$f \leq g \quad \Longleftrightarrow \quad f(x) \leq g(x) \text{ for all } x \in X.$$

With these conventions one has the following result.

4.2.3

Let X, Y be topological spaces, with Y sober, and let $f, g : X \to Y$ be continuous maps. Then there is at most one morphism from $\mathrm{Top}(f)$ to $\mathrm{Top}(g)$, and such a morphism exists if and only if g is a specialization of f. Finally, every geometric morphism $\mathrm{Top}(X) \to \mathrm{Top}(Y)$ is isomorphic to a morphism of the form $\mathrm{Top}(f)$, where $f : X \to Y$ is a continuous map, uniquely determined by the first assertion.

If $\mathrm{cat}(I)$ denotes the category associated with an ordered set I, defined by declaring that for $i \geq j$ there is exactly one arrow from i to j, the preceding result may be summarized by saying that there is a canonical equivalence of categories

$$\mathrm{cat}(\mathrm{Hom}_{\mathbf{Top}}(X,Y)) \xrightarrow{\approx} \mathcal{H}om_{\mathrm{top}}(\mathrm{Top}(X), \mathrm{Top}(Y)).$$

Formally, these results imply:

Corollary 4.2.4.

a) Let $f : X \to Y$ be a continuous map. Then $\mathrm{Top}(f) : \mathrm{Top}(X) \to \mathrm{Top}(Y)$ is an equivalence of topoi if and only if $f_{\mathrm{sob}} : X_{\mathrm{sob}} \to Y_{\mathrm{sob}}$ is a homeomorphism. Thus, when X and Y are sober, this is equivalent to f being a homeomorphism.

b) Let X, Y be topological spaces. The topoi $\mathrm{Top}(X)$ and $\mathrm{Top}(Y)$ are equivalent if and only if X_{sob} and Y_{sob} are homeomorphic. Thus, if X and Y are sober, this is equivalent to X and Y being homeomorphic.

4.3. Morphisms to the final topos: constant objects of a topos, section functors

Let P, for "point", denote the standard final topos, i.e. $P = \mathbf{Set}$ (2.2). Let E be any topos. We shall see that *up to unique isomorphism, there exists a unique geometric morphism*

$$f : E \longrightarrow P,$$

more precisely that $\mathcal{H}om_{\mathrm{top}}(E, P)$ is equivalent to the one-object category. Thus, between any two objects of this category there exists a unique arrow, and it is an isomorphism. This justifies, to some extent, the name "final topos".

For this, recall from 3.2.1 that $\mathcal{H}om_{\mathrm{top}}(E, P)$ is equivalent to the full subcategory of $\mathcal{H}om(P, E)^{\circ}$ formed by the functors

$$f^* : P = \mathbf{Set} \longrightarrow E$$

which commute with inductive limits and are left exact. Let e be a one-point set. Then every set X can be written canonically as the sum of X copies of e. It follows that the category of functors $g : \mathbf{Set} \to E$ which commute with inductive limits is equivalent to E, by associating to g the object $g(e)$ of E. Conversely, g is reconstructed from $T = g(e)$, up to unique isomorphism, by

$$g(I) = T \times I,$$

where $T \times I = \coprod_{i \in I} T_i$ with $T_i = T$ for all $i \in I$. The functor in I thus defined by T indeed commutes with inductive limits, since it is plainly left adjoint to the functor $X \mapsto \mathrm{Hom}(T, X)$ from E to $\mathbf{Set}$. For g to be left exact, it is plainly necessary that $g(e) = T$ be a final object of E, since e is a final object of $\mathbf{Set}$. It follows easily from the fact that coproducts in E are universal that this condition is also sufficient. Thus the category of inverse-image functors f^* is equivalent to the category of final objects of E, and this is itself plainly equivalent to the final category.

The preceding discussion shows that choosing a morphism $f : E \to P$ is essentially the same as choosing a final object e_E of E. In terms of this object, one has canonical isomorphisms of functors

$$f^*(I) \simeq e_E \times I = \text{the sum of } I \text{ copies of } e_E \qquad (I \in \mathrm{Ob}\,\mathbf{Set}), \tag{4.3.2}$$

and

$$f_*(X) = \mathrm{Hom}(e_E, X) \qquad (X \in \mathrm{Ob}\, E). \tag{4.3.3}$$

4.3.4

The preceding two functors will play an important role below. For a set I, the object $f^*(I) = e_E \times I$ of (4.3.2) is called the *constant object of value* I in E, or, when E is realized as a category $\widetilde{C}$ by means of a site C, the *constant sheaf of value I on C*. It will often be denoted I_E, or I_C when E is defined by the site C. The fact that $I \mapsto I_E$ is the inverse-image functor of a geometric morphism specifies its exactness properties; in particular, it respects all usual kinds of algebraic structure, sending a group to a group object of E, and so on (3.1.2). For example, if G

is a group object of E, we shall say that it is a *constant group object*, or, as the case may be, a *constant sheaf of groups*, if it is isomorphic to a group of the form $G_{0,E}$, where G_0 is an ordinary group. The same terminology applies to every other kind of "algebraic" structure, in the sense made more or less precise in 3.1.2.

4.3.5

One should be careful that the functor $I \mapsto I_E$ is not necessarily fully faithful, nor even faithful: take for E the empty topos. Hence a constant object of E does not generally determine, up to unique isomorphism, the set I from which it comes. The topos E is said to be 0-*acyclic*, or *connected and non-empty*, if the functor $I \mapsto I_E$ is fully faithful. By the general properties of adjoint functors, this is equivalent to saying that the adjunction morphism

$$I \longrightarrow f_* f^*(I) = f_*(I_E) = \mathrm{Hom}(e_E, I_E)$$

is an isomorphism, so that the functor (4.3.3) recovers the "value" of a constant object of E. One checks easily that this holds if and only if e_E is not the initial object $\varnothing_E$ of E, i.e. E is not an empty topos, which expresses the faithfulness of $I \mapsto I_E$,[8] and e_E is a connected object of E, i.e. is not a sum of two objects of E which are not empty, meaning not initial objects of E.

4.3.6

The functor (4.3.3) is also often called the *section functor* and denoted Γ_E, $\Gamma(E, -)$, or simply Γ:

$$\mathrm{Hom}(e_E, X) = \Gamma_E(X) = \Gamma(E, X) = \Gamma(X). \tag{4.3.6.1}$$

It commutes with arbitrary projective limits, but is not generally right exact; its derived functors on abelian group objects will be studied in the next exposé.

4.4. Morphisms from the empty topos

Let $\varnothing_{\mathrm{top}}$ be an empty topos, corresponding to a category of sheaves Φ equivalent to the final category (2.2). Let E be a topos. The category of functors from E to Φ is evidently equivalent to the one-object category, and every such functor commutes with inductive and projective limits. Thus it may be regarded as an inverse-image functor for a geometric morphism $\varnothing_{\mathrm{top}} \to E$. It follows that $\mathcal{H}om_{\mathrm{top}}(\varnothing_{\mathrm{top}}, E)$ is equivalent to the one-object category, and in particular that *there exists, up to unique isomorphism, a unique geometric morphism*

$$\varnothing_{\mathrm{top}} \longrightarrow E. \tag{4.4.1}$$

This justifies, to some extent, the terminology "initial topos" introduced in 2.2.

One may also determine the geometric morphisms

$$E \longrightarrow \varnothing_{\mathrm{top}}. \tag{4.4.2}$$

One checks at once that such a morphism exists if and only if the initial object of E is also final, i.e. if and only if E itself is an empty topos, and that in this case $\mathcal{H}om_{\mathrm{top}}(E, \varnothing_{\mathrm{top}})$ is again equivalent to the one-object category. The unique morphism (4.4.2), modulo isomorphism, is then an equivalence of topoi.

[8] Or equivalently the fact that this functor is conservative.

4.5. The classifying topos B_G for variable group object G

4.5.1

Let E be a topos and let
$$f : G \longrightarrow H$$
be a morphism of group objects in E. It induces a "restriction of the group of operators" functor
$$f^* : B_H \longrightarrow B_G,$$
with the notation of 2.4. It is trivial that this functor commutes with inductive limits and projective limits; a fortiori it may be interpreted as the inverse-image functor associated with a geometric morphism
$$B_f \text{ or } f : B_G \longrightarrow B_H.$$
The corresponding direct-image functor
$$f_* : B_G \longrightarrow B_H$$
is described easily by the formula
$$f_*(X) = \mathcal{H}om_G(H_s, X),$$
where X is an object of E with a left action of G, where H_s is H endowed with the left action of G induced by f, and where $\mathcal{H}om_G$ denotes the evident subobject of the $\mathcal{H}om$-object defined below in 10.2. The group object H acts on the left on $\mathcal{H}om_G(H_s, X)$ through the right actions of H on H_s by right translations.

Since the inverse-image functor f^* commutes with arbitrary projective limits, it is itself the right adjoint of a functor
$$f_! : B_G \longrightarrow B_H,$$
so that there is a sequence of three adjoint functors as in 3.1.3:
$$f_!, \quad f^*, \quad f_*.$$
The functor $f_!$ is described by the formula
$$f_!(X) = H \overset{G}{\times} X,$$
where the right-hand side denotes the contracted product, deduced from the action of G on X on the left and on H on the right via right translations and f, defined as the quotient of $H \times X$ by the action of G given by
$$g \circ (h, x) = (hg^{-1}, gx).$$

The functor $f_!$, being a left adjoint, plainly commutes with inductive limits, but it is generally not left exact, and hence cannot in turn be regarded as an inverse-image functor for a geometric morphism $B_H \to B_G$. In fact, it is easy to check that it can be left exact only if $f : G \to H$ is an isomorphism. Similarly, the functor f_*, which commutes with projective limits, is generally not right exact, and a fortiori does not generally admit a right adjoint. At least when E is the punctual topos, i.e. when G and H are ordinary groups, f_* is right exact only if f is an isomorphism.

When there is a second homomorphism of group objects $g : H \to K$, one obtains as in 4.1.1 a transitivity property up to canonical isomorphism. Thus, subject to the usual reservation, the classifying topos B_G may be regarded as depending *functorially* on the group object G. We leave to the reader the generalization of this functorial behavior to the case where both G and the topos E vary simultaneously.

4.5.2. The topos $B_{\mathcal{G}}$ for a variable pro-group $\mathcal{G}$

We leave to the reader the task of specifying the covariant character of the topos $B_{\mathcal{G}}$ (2.7) with respect to $\mathcal{G}$, by imitating the exposition given in 4.5.1. One should be careful, however, that in the case of a morphism $f : \mathcal{G} \to \mathcal{H}$ of pro-groups which are not essentially constant, the corresponding geometric morphism $B_{\mathcal{G}} \to B_{\mathcal{H}}$ does not generally allow the definition of a functor $f_!$, whose right adjoint would be the restriction functor f^* for the pro-group of operators. Assuming, for simplicity, that $\mathcal{G}$ and $\mathcal{H}$ are defined by profinite groups G and H, one sees easily that $f_!$ exists if and only if the image of $f : G \to H$ has finite index in H, and in that case $f_!$ is given by the same formula as in 4.5.

4.6. The topos $\widehat{C}$ for variable category C

4.6.1

Let

$$f : C \longrightarrow C'$$

be a functor from a category $C \in \mathcal{U}$ to another category C'. It gives a functor

$$f^* : \widehat{C'} \longrightarrow \widehat{C}, \qquad f^*(F) = F \circ f.$$

It is trivial that this functor commutes with projective and inductive limits; a fortiori it may be regarded as the inverse-image functor for a geometric morphism

$$\widehat{f} \text{ or } f : \widehat{C} \longrightarrow \widehat{C'}.$$

The corresponding direct-image functor

$$f_* : \widehat{C} \longrightarrow \widehat{C'}$$

is just the functor also denoted f_* in Exposé I, 5.1. Moreover, as expected from the fact that f^* also commutes with arbitrary projective limits, f^* also admits a left adjoint

$$f_! : \widehat{C} \longrightarrow \widehat{C'},$$

which was also denoted $f_!$ in Exposé I, 5.1. Thus one obtains a sequence of three adjoint functors

$$f_!, \quad f^*, \quad f_*,$$

the first being an extension of $f : C \to C'$ to categories of presheaves, for the usual embeddings of C, C' into $\widehat{C}, \widehat{C'}$. Note that $f_!$ is generally not left exact, nor is f_* generally right exact; this removes any ambiguity about the variance direction of the topos $\widehat{C}$ associated with the variable category C: it is *covariant* in C, subject to the usual reservation concerning transitivity isomorphisms. When the sets of objects of C and C' are each a singleton, so that C and C' identify with monoids G and G', the corresponding topoi are the classifying topoi B_{G° and $B_{G'^\circ}$, and one recovers the variance for variable G already encountered from other viewpoints in 4.1.2 and 4.5.

4.6.2

The 2-functorial dependence of the topos $\widehat{C}$ on C may be made precise by introducing, for two categories $C, C' \in \mathcal{U}$, a canonical functor

$$\mathcal{H}om(C, C') \longrightarrow \mathcal{H}om_{\mathrm{top}}(\widehat{C}, \widehat{C'}). \tag{4.6.2.1}$$

It remains to define this functor on arrows. For this, note that $f \mapsto f^*$ identifies, up to equivalence of categories, the target with a full subcategory of $\mathcal{H}om(\widehat{C'}, \widehat{C})^{\circ}$. If $f, g : C \to C'$ are two functors, any morphism $u : f \to g$ of functors defines a morphism $F \circ g \to F \circ f$ of functors in $F \in \mathrm{Ob}\,\widehat{C'}$, because F is a contravariant functor. This is a morphism $g^* \to f^*$, hence defines a morphism $\widehat{f} \to \widehat{g}$, as asserted.

When C is the one-object category, $\widehat{C}$ is the punctual topos P, and (4.6.2.1) is interpreted as a natural functor

$$C' \longrightarrow \mathcal{H}om_{\mathrm{top}}(P, \widehat{C'}) \overset{\mathrm{def}}{=} \mathrm{Pt}(\widehat{C'}), \tag{4.6.2.2}$$

from C' to the *category of points* of $\widehat{C'}$, which will be studied in Section 6. This functor is not necessarily an equivalence of categories (7.3.3); a fortiori (4.6.2.1) is not necessarily an equivalence of categories.

On the other hand, the functor (4.6.2.1) is always *fully faithful*. To see this, note that if $f, g : E \to E'$ are two geometric morphisms such that $f_!$ and $g_!$ are defined (3.1.3), then the theory of adjoint functors gives canonical isomorphisms

$$\mathrm{Hom}(f_!, g_!) \simeq \mathrm{Hom}(g^*, f^*) \simeq \mathrm{Hom}(f_*, g_*) \overset{\mathrm{def}}{=} \mathrm{Hom}(f, g).$$

Applying this to functors of the form $\widehat{f}, \widehat{g}$ associated with $f, g : C \to C'$, one obtains the result, because the natural extension map $\mathrm{Hom}(f, g) \to \mathrm{Hom}(f_!, g_!)$ is bijective (Exposé I, 7.8).

4.6.3

One may ask when the functor (4.6.2.1) is an equivalence of categories, i.e. when it is essentially surjective. This is a special case of determining all geometric morphisms $\widehat{C} \to \widehat{C'}$. More generally, if C is a category in $\mathcal{U}$ and E is a topos, one may try to determine the geometric morphisms

$$f : \widehat{C} \longrightarrow E.$$

The category of these morphisms is equivalent to the opposite of the category of functors $f^* : E \to \widehat{C}$ commuting with arbitrary inductive limits and finite projective limits. Interpreting functors $E \to \widehat{C}$ as functors $E \times C^{\circ} \to \mathcal{U}\text{-}\mathbf{Set} = \mathbf{Set}$, or equivalently as functors $F : C^{\circ} \to \mathcal{H}om(E, \mathbf{Set})$, the exactness condition in question is expressed by requiring that, for every object X of C, the functor $F(X) : E \to \mathbf{Set}$ commute with arbitrary inductive limits and finite projective limits; as we shall say in Section 6, $F(X)$ is a *fiber functor* on E. Equivalently, $F(X)$ is the inverse-image functor of a geometric morphism $P \to E$. Thus one finally obtains a canonical equivalence of categories

$$\mathcal{H}om_{\mathrm{top}}(\widehat{C}, E) \xrightarrow{\approx} \mathcal{H}om(C, \mathrm{Pt}(E)), \tag{4.6.3.1}$$

where, for brevity, $\mathrm{Pt}(E) = \mathcal{H}om_{\mathrm{top}}(P, E)$ denotes the category of points of the topos E.

When E is of the form $\widehat{C'}$, it follows at once from the definitions that the composite of (4.6.2.1) and the equivalence (4.6.3.1) is the functor

$$\mathcal{H}om(C, C') \longrightarrow \mathcal{H}om(C, \mathrm{Pt}(\widehat{C'})) \tag{4.6.3.2}$$

defined by $F \mapsto i \circ F$, where $i : C' \to \mathrm{Pt}(\widehat{C'})$ is the canonical embedding (4.6.2.2). It follows immediately that *(4.6.2.1) is an equivalence if and only if C is empty or (4.6.2.2) is essentially surjective*, and hence an equivalence of categories. This last condition on C' is satisfied in some interesting cases, notably when C' is the one-object category defined by a group G (7.2.5).

Remark 4.6.4. The fact that a topos $\widehat{C}$ has been attached to an arbitrary category C suggests that a category C has invariants of topological nature, such as cohomology and homotopy groups, just as a topos does. The cohomology groups of $\widehat{C}$ with coefficients in an abelian group object F, in the general sense studied in Exposé V, are just the values of the derived functors $\varprojlim^{(n)}$ of $\varprojlim$, already familiar to topologists. J.-L. Verdier and, independently, D. G. Quillen verified that, when one restricts to constant coefficients, or more generally to locally constant coefficients, these cohomology groups identify with the cohomology groups of the semi-simplicial set $\mathrm{Ner}(C)$ canonically associated with C [5, no. 212, Prop. 4.1], and also that, up to isomorphism in the homotopy category of [2], every semi-simplicial set can be obtained from a category C. With a suitable notion of homotopy type for topoi, which we do not specify here, one may say that the semi-simplicial homotopy types of topologists are precisely the homotopy types of topoi of the special form $\widehat{C}$, and more generally of topoi E in which every object admits a cover refining all others (2.6).[9] In the absence of this condition on E, its homotopy type can at best be expressed by a suitable projective system of semi-simplicial sets [1].

4.7. The topos $\widetilde{C}$ for variable site C (cocontinuous functors)

Let

$$f : C \longrightarrow C'$$

be a cocontinuous functor between sites in $\mathcal{U}$, i.e. a functor such that the functor $\widehat{f}_* : \widehat{C} \to \widehat{C'}$ sends $\widetilde{C}$ into $\widetilde{C'}$, or equivalently induces a functor

$$\widetilde{f}_* : \widetilde{C} \longrightarrow \widetilde{C'} \tag{4.7.1}$$

making the diagram

$$\begin{tikzcd} \widetilde{C} \arrow[r, "\widetilde{f}_*"] \arrow[d, "i"'] & \widetilde{C'} \arrow[d, "i'"] \\ \widehat{C} \arrow[r, "\widehat{f}_*"] & \widehat{C'} \end{tikzcd} \tag{4.7.2}$$

commute, where i, i' are the inclusion functors. We then saw in Exposé III, 2.3, that $\widetilde{f}_*$ has a left adjoint $\widetilde{f}^*$, and that this left adjoint is left exact. In other words, $\widetilde{f}_*$ is the direct-image functor associated with a geometric morphism

$$\widetilde{f} \text{ or } f : \widetilde{C} \longrightarrow \widetilde{C'},$$

[9] And still more generally of topoi which are "locally ∞-connected" in an evident sense which we leave to the reader to specify.

whose inverse-image functor is, of course, $\widetilde{f}^*$. Taking left adjoints of the functors involved, the commutative diagram (4.7.2) also gives a diagram commutative up to isomorphism

$$\begin{tikzcd} \widetilde{C} & \widetilde{C'} \arrow[l, "\widetilde{f}^*"'] \\ \widehat{C} \arrow[u, "\underline{a}"] & \widehat{C'} \arrow[l, "\widehat{f}^*"] \arrow[u, "\underline{a}'"'] \end{tikzcd} \tag{4.7.3}$$

where $\underline{a}$ and $\underline{a}'$ are the associated-sheaf functors. This immediately recovers the formula

$$\widetilde{f}^* = \underline{a}\,\widehat{f}^* i'.$$

The transitivity property for geometric morphisms $\widehat{f} : \widehat{C} \to \widehat{C'}$ implies the same property for the geometric morphisms $\widetilde{f} : \widetilde{C} \to \widetilde{C'}$ associated with cocontinuous functors. Thus the topos $\widetilde{C}$ varies functorially, in a *covariant* way, with C, when cocontinuous functors are taken as the “morphisms” of sites.

In all cases encountered so far, the cocontinuous functor f used is also continuous, that is, it extends to a functor

$$\widetilde{f}_! : \widetilde{C} \longrightarrow \widetilde{C'}$$

commuting with inductive limits and left adjoint to $\widetilde{f}^*$, so that there is a sequence of three adjoint functors

$$\widetilde{f}_!, \quad \widetilde{f}^*, \quad \widetilde{f}_*.$$

One should note that $\widetilde{f}_!$ is generally not left exact, nor is $\widetilde{f}_*$ generally right exact; this removes any ambiguity about the variance direction of the topos $\widetilde{C}$ when C varies by functors assumed only to be cocontinuous, or even continuous and cocontinuous.

Remark 4.7.4. Given a geometric morphism

$$F : E \longrightarrow E',$$

there exists a functor $F_!$ left adjoint to F^* if and only if one can realize E and E', up to equivalence, as $\widetilde{C}$ and $\widetilde{C'}$ for two sites $C, C' \in \mathcal{U}$, and find a *continuous and cocontinuous* functor $f : C \to C'$ such that F identifies with $\widetilde{f}$. This is plainly sufficient. For necessity, it suffices to take for C and C' small full generating subcategories of E and E', respectively, such that

$$F_!(\operatorname{Ob} C) \subset \operatorname{Ob} C',$$

endow them with the topologies induced from those of E and E', and take f to be the functor induced by $F_!$ (cf. 1.2.1). Recall (1.8) that existence of $F_!$ also means that F^* commutes with projective limits, or equivalently here, since this functor is left exact, that it commutes with products.

4.7.5

If one asks which geometric morphisms $F : E \to E'$ can be realized by a cocontinuous, not necessarily continuous, functor of sites, one obtains similarly the following answer: the set of objects X of E such that the functor

$$X' \longmapsto \mathrm{Hom}(X, F^*(X'))$$

from E' to **Set** commutes with projective limits, or equivalently with products, must be a *generating family of E*.[10]

4.8. The geometric morphism $\widetilde{C} \to \widehat{C}$ for a site C

Let C be a small site, with associated topoi $\widetilde{C}$ and $\widehat{C}$, the latter independent of the topology put on C. In Exposé II, 3.4, the associated-sheaf functor

$$\underline{a} : \widehat{C} \longrightarrow \widetilde{C}$$

was defined and shown to be left exact and to commute with inductive limits. It is therefore the inverse-image functor associated with a geometric morphism

$$p : \widetilde{C} \longrightarrow \widehat{C}, \qquad p^* = \underline{a},$$

whose direct-image functor is the canonical inclusion

$$i = p_* : \widetilde{C} \longrightarrow \widehat{C}.$$

It is well known that this latter functor is generally not right exact–its derived functors on abelian objects give rise to the presheaves of cohomology $\mathcal{H}^n(F)$ of Exposé V, 2–and that $\underline{a}$ generally does not commute with arbitrary projective limits. This removes any ambiguity about the direction of the "natural" geometric morphism between $\widehat{C}$ and $\widetilde{C}$.

When one has a cocontinuous functor

$$f : C \longrightarrow C'$$

of sites, one deduces a diagram of geometric morphisms

$$\begin{tikzcd}
\widetilde{C} \arrow[r, "\widetilde{f}"] \arrow[d, "p"'] & \widetilde{C'} \arrow[d, "p'"] \\
\widehat{C} \arrow[r, "\widehat{f}"] & \widehat{C'}
\end{tikzcd}$$

commutative up to canonical isomorphism. This is precisely the content of the commutativity of the diagram of functors (4.7.2). Thus the geometric morphism $p : \widetilde{C} \to \widehat{C}$ is functorial in C, when C varies by *cocontinuous* functors between sites.

[10]Editorial note in the source: this statement does not appear correct. The expressed condition is equivalent to F^* commuting with arbitrary products, or equivalently to F^* having a left adjoint $F_!$. For any site C, the identity functor $F = \mathrm{id} : C \to C$ is cocontinuous when the codomain is given the trivial topology. Then $F^* : \widehat{C} \to \widetilde{C}$ is the associated-sheaf functor, and it does not in general commute with arbitrary products.

4.9. Effect of a continuous functor of sites. Morphisms of sites

4.9.1

Let

$$f : C \longrightarrow C'$$

be a continuous functor from C to C', i.e. such that the functor $\widehat{f}^* : \widehat{C'} \to \widehat{C}$ sends $\widetilde{C'}$ into $\widetilde{C}$, and hence induces a functor

$$f_s : \widetilde{C'} \longrightarrow \widetilde{C}.$$

We saw in Exposé III, 1.2, that f_s has a left adjoint

$$f^s : \widetilde{C} \longrightarrow \widetilde{C'},$$

which moreover extends f in an evident sense. One should be careful that in general f^s is not left exact, even if f is also cocontinuous, nor does f_s commute with inductive limits. Thus the datum of f does not, without further hypotheses, describe a geometric morphism in either direction between $\widetilde{C}$ and $\widetilde{C'}$. The case where f is cocontinuous, i.e. where f_s commutes with inductive limits and can be regarded as an inverse-image functor for a geometric morphism $\widetilde{f} : \widetilde{C} \to \widetilde{C'}$, was examined in 4.7. We now examine the case where f^s is left exact, hence may be regarded as the inverse-image functor for a geometric morphism in the *opposite* direction:

$$\mathrm{Top}(f) = g : \widetilde{C'} \longrightarrow \widetilde{C}. \tag{4.9.1.1}$$

Notice that *we have deliberately avoided denoting this morphism by $\widetilde{f}$ or f, to prevent confusion with the situation of 4.7, following the general recommendations of 3.1.3.* One sometimes says that the functor $f : C \to C'$ is a *morphism of sites from C' to C*–note: *not from C to C'*–if it is continuous and if the functor f^s is left exact. Equivalently, there exists a geometric morphism (4.9.1.1) whose corresponding inverse-image functor

$$g^* : \widetilde{C} \longrightarrow \widetilde{C'}$$

extends the functor f, i.e. makes the diagram of functors

$$\begin{tikzcd} C \arrow[r, "f"] \arrow[d, "\epsilon"'] & C' \arrow[d, "\epsilon'"] \\ \widetilde{C} \arrow[r, "g^*"] & \widetilde{C'} \end{tikzcd}$$

commute up to isomorphism, where ϵ, ϵ' are the canonical functors of Exposé II, 4.4.0.

4.9.2

In practice, one recognizes that a continuous functor $f : C \to C'$ is a morphism of sites from C' to C by the fact that finite projective limits are representable in C and that f commutes with them (Exposé III, 1.3(5)). Under the stated condition on C, almost always satisfied in practice, and assuming moreover that the topology of C' is no finer than the canonical topology, which is also almost always satisfied, this sufficient condition–that f be left exact–is also necessary.

4.9.3

In the spirit of the preceding discussion, if C and C' are two $\mathcal{U}$-sites, one should define the category of morphisms of sites $\mathcal{M}or_{\text{site}}(C', C)$ from C to C' as the full subcategory of the *opposite* category $\mathcal{H}om(C, C')^\circ$ of the category of functors from C to C', formed by the functors which are willing to be morphisms of sites, from C' to C. In this way one obtains a canonical functor, defined up to unique isomorphism,

$$\mathcal{M}or_{\text{site}}(C', C) \longrightarrow \mathcal{H}om_{\text{top}}(\widetilde{C'}, \widetilde{C}).$$

Proposition 4.9.4. Let E be a $\mathcal{U}$-topos and C a $\mathcal{U}$-site. The functor $f \mapsto f^*|_C = f^* \circ \epsilon_C$, which associates with every geometric morphism $f : E \to \widetilde{C}$ the restriction to C of its inverse-image functor $f^* : \widetilde{C} \to E$, induces an equivalence of categories

$$\mathcal{H}om_{\text{top}}(E, \widetilde{C}) \xrightarrow{\approx} \mathcal{M}or_{\text{site}}(E, C) \hookrightarrow \mathcal{H}om(C, E)^\circ.$$

When finite projective limits are representable in C, the corresponding fully faithful functor

$$\mathcal{H}om_{\text{top}}(E, \widetilde{C}) \longrightarrow \mathcal{H}om(C, E)^\circ$$

has as essential image the set of functors $g : C \to E$ which are left exact and continuous, or equivalently left exact and send covering families to covering families.

The last assertion follows from the first by 4.9.2. On the other hand, Exposé III, 1.2(iv), together with Theorem 1.2(iii) above, implies that $G \mapsto G \circ \epsilon$ induces an equivalence between the category of continuous functors from $\widetilde{C}$ to E and the category of continuous functors from C to E; a quasi-inverse is given by $g \mapsto g^s$, with the notation of loc. cit.. By definition, g is a morphism of sites if and only if g^s is the inverse-image functor associated with a geometric morphism $E \to \widetilde{C}$. The conclusion follows from 3.2.1, and the "or equivalently" from Exposé III, 1.6.

The main point to retain from 4.9.4 is that, when finite projective limits are representable in C, giving a geometric morphism $f : E \to \widetilde{C}$ is "the same as" giving a functor $g : C \to E$ which is left exact and sends covering families to covering families.

4.9.5

Using 4.9.1 and 4.9.3, one sees as usual that for a variable $\mathcal{U}$-site C, using the notion of morphism of sites and of morphism of morphisms of sites just described, the topos $\widetilde{C}$ depends functorially, or more exactly 2-functorially, on the site C. Notice that, with the terminology introduced, $\widetilde{C}$ depends *covariantly* on the site C.

4.9.6

It is immediate that, contrary to what happens for cocontinuous functors (cf. 4.7.4), *every* geometric morphism $F : E \to E'$ can be realized by a morphism of sites $f : C \to C'$, that is, by a continuous functor $f : C' \to C$ satisfying the above condition. It suffices to choose small full generating subcategories C and C' of E and E', respectively, endowed with the topologies induced by those of E and E', such that

$$F^*(\mathrm{Ob}\, C') \subset \mathrm{Ob}\, C,$$

and to take f to be the functor induced by F^*. This explains why most geometric morphisms of topoi encountered in practice are in fact described by morphisms of sites, rather than by cocontinuous functors as in 4.7.

4.10. Relations between the small and big topoi associated with a topological space X

We resume the notation of 2.5. In particular, $\mathcal{V}$ is a universe such that $\mathcal{U} \in \mathcal{V}$. We depart from the convention of 2.1 by denoting by $\mathrm{Top}(X)$ the topos of $\mathcal{V}$-sheaves, not $\mathcal{U}$-sheaves, on X. Thus we shall reason with $\mathcal{V}$-topoi rather than $\mathcal{U}$-topoi. It would be possible to keep $\mathcal{U}$-topoi by adopting the appropriate convention in the definition of $\mathrm{TOP}(X)$, so that it is a $\mathcal{U}$-topos; cf. 2.5. With this understood, we shall define two geometric morphisms

$$\begin{cases} f : \mathrm{Top}(X) \longrightarrow \mathrm{TOP}(X), \\ g : \mathrm{TOP}(X) \longrightarrow \mathrm{Top}(X), \qquad gf \simeq \mathrm{id}_{\mathrm{Top}(X)}, \end{cases} \tag{4.10.1}$$

giving rise to a sequence of three adjoint functors

$$g^* = f_!, \qquad g_* = f^*, \qquad g^! = f_*, \tag{4.10.1.1}$$

with the notation of 3.1.3. We shall define the first two functors in this sequence successively; the third is then defined as the right adjoint of the second.

4.10.2

The functor

$$g_* = f^* : \mathrm{TOP}(X) \longrightarrow \mathrm{Top}(X)$$

is simply the *restriction functor* from $(\mathbf{Top})_{/X}$ to the site $\mathrm{Open}(X)$ of open subsets of X. It indeed sends sheaves to sheaves, as follows immediately from the definition. Denoting sheaves on the big site of X by underlined letters, we denote, for such a sheaf $\underline{F}$, by $\underline{F}_X$ its restriction to the site $\mathrm{Open}(X)$. Thus we obtain a functor

$$\mathrm{Res} : \underline{F} \longmapsto \underline{F}_X : \mathrm{TOP}(X) \longrightarrow \mathrm{Top}(X). \tag{4.10.2.1}$$

It is clear that this functor commutes with $\mathcal{V}$-projective limits, since these are computed argument by argument; one could also invoke the existence of the left adjoint constructed in 4.10.4 below. I claim that it also commutes with $\mathcal{V}$-inductive limits. To see this, we give a very convenient interpretation of big sheaves on X, i.e. objects of $\mathrm{TOP}(X)$, in terms of ordinary sheaves, meaning $\mathcal{V}$-sheaves, on topological spaces X' over X.

4.10.3

For a big sheaf $\underline{F}$ on X, and for every space X' over X, with $X' \in \mathcal{U}$ understood, one defines in the evident way, as in 4.10.2, a sheaf $\underline{F}_{X'}$, the restriction of $\underline{F}$ to X'. If

$$u : X'' \longrightarrow X'$$

is a morphism in $(\mathbf{Top})_{/X}$, one defines in the evident way a morphism $u_*(\underline{F}_{X''}) \to \underline{F}_{X'}$, or equivalently a *transition morphism*

$$\varphi_u : u^*(\underline{F}_{X'}) \longrightarrow \underline{F}_{X''}. \tag{4.10.3.1}$$

As u varies, these morphisms satisfy an evident transitivity condition for a composite

$$X''' \xrightarrow{u} X'' \xrightarrow{v} X'$$

of morphisms in $(\mathbf{Top})_{/X}$, which we leave the reader to spell out. This gives a natural functor from the category $\mathrm{TOP}(X)$ of big sheaves on X to the category of systems

$$(\underline{F}_{X'}) \quad (X' \in \mathrm{Ob}(\mathbf{Top})_{/X}), \qquad (\varphi_u) \quad (u \in \mathrm{Fl}(\mathbf{Top})_{/X}),$$

satisfying the transitivity condition just described and, moreover, such that for every morphism $u : X'' \to X'$ which is an *open immersion*, or more generally an étale map, the transition morphism φ_u is an isomorphism. Since the functors $u^* : \mathrm{Top}(X') \to \mathrm{Top}(X'')$ used in the description of the φ_u commute with $\mathcal{V}$-inductive limits, it follows at once that, in the preceding description of objects of $\mathrm{TOP}(X)$ in terms of small sheaves $F_{X'}$, $\mathcal{V}$-inductive limits are computed argument by argument. In other words, the functors $\underline{F} \mapsto \underline{F}_{X'}$ commute with $\mathcal{V}$-inductive limits.

In particular, this is true of the functor $\underline{F} \mapsto \underline{F}_X$ considered in 4.10.2. Hence it has a right adjoint (1.5). It remains to construct a left adjoint for it, whose existence also follows a priori from 1.8, and to verify that this left adjoint is left exact. This will complete the definition of the announced geometric morphisms (4.10.1) in terms of the functors (4.10.1.1).

4.10.4

The functor

$$g^* = f_! : \mathrm{Top}(X) \longrightarrow \mathrm{TOP}(X)$$

is obtained by associating with every small sheaf F on X the system of its inverse images $(F_{X'})$, for $X' \in \mathrm{Ob}(\mathbf{Top})_{/X}$, by the structural morphisms $X' \to X$; the transition morphism φ_u for $u : X'' \to X'$ is the transitivity isomorphism for inverse images of sheaves (4.1.1). One sees at once that this gives a fully faithful functor

$$\mathrm{Prol} : \mathrm{Top}(X) \longrightarrow \mathrm{TOP}(X),$$

whose essential image consists of the big sheaves $\mathcal{F}$ on X for which all the transition morphisms φ_u of (4.10.3.1) are isomorphisms. We leave to the reader the definition of an adjunction isomorphism between this canonical prolongation functor and the restriction functor of 4.10.2, proving that the latter is right adjoint to the former. This is immediate in terms of the description 4.10.3 of the category $\mathrm{TOP}(X)$.

Remarks 4.10.5.

a) The construction of 4.10.4 shows at the same time that $g^* = f_!$ is fully faithful, or equivalently that $g_* = f^*$ is a localization functor, or again that $g^! = f_*$ is fully faithful. The big sheaves on X which belong to the essential image of $g^* = f_! = \mathrm{Prol}$ deserve the name of big *étale sheaves* on X, since they form a category equivalent to that of ordinary sheaves on X, or equivalently to that of étale spaces over X. It is also easy to see that if X' is a topological space over X, then the big sheaf on X represented by X' is étale in the preceding sense if and only if X' is an étale space over X.

b) The full faithfulness of f_* may also be expressed by saying that the adjunction morphism $f^*f_* \to \mathrm{id}_{\mathrm{Top}(X)}$ is an isomorphism, i.e. that $g_*f_* \simeq \mathrm{id}$, i.e. that $gf \simeq \mathrm{id}$. Thus g makes $\mathrm{TOP}(X)$ a topos over $\mathrm{Top}(X)$, admitting a section f over $\mathrm{Top}(X)$.

c) The fact that $g^* = f_!$ is fully faithful, exact, and commutes with $\mathcal{V}$-inductive limits partially justifies the very convenient viewpoint, due to J. Giraud, that in practically all questions of sheaf theory on X, it is harmless to replace ordinary, or small, sheaves by the associated big sheaves. This is especially true for cohomological questions, since g_* is exact, so the functors $R^i g_*$ vanish for $i > 0$. Therefore, for every big sheaf $\mathcal{F}$ on X, there are canonical isomorphisms
$$H^i(\mathrm{TOP}(X), \mathcal{F}) \simeq H^i(\mathrm{Top}(X), \mathcal{F}_X) = H^i(X, F).$$
Applying this to a sheaf of the form $\mathrm{Prol}(F)$, where F is a small sheaf on X, and using $\mathrm{Prol}(F)_X \simeq F$, one obtains a canonical isomorphism
$$H^i(X, F) = H^i(\mathrm{TOP}(X), \mathrm{Prol}(F)).$$
Thus *the cohomological invariants of X, computed via the small or the big topos of X, are essentially identical.* The same result also holds in non-commutative cohomology.

Exercise 4.10.6. Let S be a $\mathcal{U}$-site whose topology is no finer than the canonical topology, and let $\mathcal{M}$ be a subset of $\mathrm{Fl}\, S$ satisfying the following conditions:

a) The morphisms in $\mathcal{M}$ are base-changeable, and $\mathcal{M}$ is stable under base change.

b) $\mathcal{M}$ contains the identity arrows and is stable under composition.

c) If a morphism $u : X \to Y$ admits a covering family $Y_i \to Y$ such that the morphisms $X \times_Y Y_i \to Y_i$ belong to $\mathcal{M}$, then u belongs to $\mathcal{M}$.

d) For every $X \in \mathrm{Ob}\, S$, every covering family of X is refined by a covering family $(f_i : X_i \to X)$ with $f_i \in \mathcal{M}$.

For every $X \in \mathrm{Ob}\, S$, consider the site $S(X)$, the "small site of X", whose underlying category is the full subcategory of $S_{/X}$ formed by the objects whose structural morphism is in $\mathcal{M}$, endowed with the topology induced from that of S.

1) For every arrow $u : X \to Y$ of S, show that base change by u from $S(Y)$ to $S(X)$ is a continuous functor, giving a functor commuting with inductive limits
$$S(u)^* : S(Y)^\sim \longrightarrow S(X)^\sim.$$

2) Define an equivalence between the topos $S^\sim$ and the category of systems
$$(F_X) \quad (X \in \mathrm{Ob}\, S), \qquad (\varphi_u) \quad (u \in \mathrm{Fl}\, S)$$
formed of objects $F_X \in \mathrm{Ob}\, S(X)^\sim$, and, for every arrow $u : X \to Y$ in S, a morphism $\varphi_u : S(u)^*(F_Y) \to F_X$, subject to a transitivity condition for a composite $v \circ u$ of arrows of S, and to the condition that $u \in \mathcal{M}$ implies that φ_u is an isomorphism.

3) Define the restriction and prolongation functors
$$\mathrm{Res}_X : (S_{/X})^\sim \longrightarrow S(X)^\sim, \qquad \mathrm{Prol}_X : S(X)^\sim \longrightarrow (S_{/X})^\sim.$$
Show that Res_X commutes with small inductive and projective limits and that Prol_X is fully faithful, its essential image consisting of the sheaves F such that φ_u is an isomorphism for every arrow u of $S_{/X}$.

4) Define an adjunction morphism making Res_X right adjoint to Prol_X. Deduce that there exists a geometric morphism
$$f : S(X)^\sim \longrightarrow (S_{/X})^\sim$$
which makes $S(X)^\sim$ a subtopos of $(S_{/X})^\sim$, and such that
$$f_! = \mathrm{Prol}_X, \qquad f^* = \mathrm{Res}_X\,.$$
Show that Prol_X sends abelian sheaves to abelian sheaves.

5) Show that if $S(X)$ has fiber products, then Prol_X is exact. Deduce that there then exists a geometric morphism
$$g : (S_{/X})^\sim \longrightarrow S(X)^\sim$$
which is a left retraction of f, i.e. such that
$$g^* = \mathrm{Prol}_X, \qquad g_* = \mathrm{Res}_X\,.$$
For an example where $S(X)$ does not have fiber products, take S to be the category of schemes with the étale topology, take $\mathcal{M}$ to be the class of smooth morphisms, and take X to be a noetherian scheme of dimension > 0.

6) Show that for every abelian sheaf F on $(S_{/X})^\sim$ there is an isomorphism
$$H^q(X, F) \simeq H^q(X, \mathrm{Res}_X F) \qquad \forall q.$$
Show, using for instance hypercoverings, that for every abelian sheaf G on $S(X)^\sim$ there is an isomorphism
$$H^q(X, G) \simeq H^q(X, \mathrm{Prol}_X G) \qquad \forall q.$$

7) Buy the expositor a chocolate medal.

5 Induced topoi

5.1

Let E be a topos and let X be an object of E. Then the category $E_{/X}$ of objects of E over X is a topos. This follows immediately, for instance, from Giraud's criterion, 1.2(ii). Equally, using 1.2.1, one may write E as a category of sheaves $C^\sim$, where C is a generating subcategory of E, chosen so that $X \in \mathrm{Ob}\, C$. Then
$$E_{/X} = C^\sim_{/X}$$
is equivalent to $(C_{/X})^\sim$, by Exposé III, 5.4, and hence is a topos.

The topos $E_{/X}$ is called the *topos induced on the object* X of E.

5.2

We now define a canonical geometric morphism

$$j_X : E_{/X} \longrightarrow E, \tag{5.2.1}$$

called the *inclusion morphism* of the induced topos $E_{/X}$ into the ambient topos E, or better, in view of 5.7, the *localization morphism of E at X*. This morphism corresponds to a sequence of three adjoint functors

$$j_{X!} \quad j_X^* \quad j_{X*}, \tag{5.2.2}$$

which may be described as follows.

First, $j_{X!} : E_{/X} \to E$ is the functor which forgets the structural arrow to X. Second,

$$j_X^* : E \longrightarrow E_{/X}$$

is given by

$$j_X^*(Z) = (X \times Z, \mathrm{pr}_1),$$

where $\mathrm{pr}_1 : X \times Z \to X$ is the first projection. It is also the base-change functor for the morphism $X \to e_E$, where e_E is the final object of E. It is immediate from the definition of base change that j_X^* is right adjoint to $j_{X!}$. In particular it commutes with projective limits. It also commutes with inductive limits, by the special exactness properties of topoi (Exposé II, 4).

It follows from the latter fact, by 1.6, that j_X^* admits a right adjoint

$$j_{X*} : E_{/X} \longrightarrow E.$$

For an object X' over X, the object $j_{X*}(X')$ is also sometimes denoted by one of

$$\prod_{X/e_E} (X'/X), \qquad \mathcal{H}om_{X/e_E}(X, X'), \qquad \mathrm{Res}_{X/e_E}(X'/X),$$

the last notation being the "Weil restriction"; one of these notations is no doubt already familiar to the learned reader of our modest work.

5.3

One should note that the functor $j_{X!}$ commutes with fiber products and sends monomorphisms to monomorphisms, but is not in general left exact. It is left exact only when X is a final object of E, since $X = j_{X!}(X, \mathrm{id}_X)$ and (X, id_X) is final in $E_{/X}$. Equivalently, this is the case precisely when j_X is in fact an equivalence of topoi, and hence when all three functors in (5.2.2) are equivalences.

Similarly, j_{X*} is not in general right exact and need not send epimorphisms to epimorphisms. These observations remove any possible ambiguity about the direction of the geometric morphism relating $E_{/X}$ to E.

5.4

The functor j_X^* is often called the *localization functor*, or the *restriction functor*. The latter terminology is justified as follows. Identify objects of E, respectively of $E_{/X}$, with sheaves on E, respectively on $E_{/X}$, by 1.2(iii). Under this identification, the adjunction between $j_{X!}$ and j_X^*

says that $j_X^*(F)$ is the restriction to $E_{/X}$ of the sheaf F on E, where "restriction" is understood in the generalized sense: composition with the "inclusion" functor $j_{X!} : E_{/X} \to E$. In familiar notation we shall therefore also write, interchangeably,

$$j_X^*(F) = F|X = F_X = \text{ the restriction of } F \text{ to } X. \tag{5.4.1}$$

Similarly, when $E = C^\sim$, with C a small site, and $X = \varepsilon(S)$ for $S \in \operatorname{Ob} C$, so that the equivalence recalled in 5.1

$$E_{/X} = C^\sim_{/\varepsilon(S)} \simeq (C_{/S})^\sim$$

holds, where $C_{/S}$ carries the topology induced from that of C, the functor j_X^* is simply the restriction of a variable sheaf F on C to the category $C_{/S}$.

These considerations already indicate the important role played by induced topoi and localization morphisms in all questions where one reasons by localization on E, and hence in practically every question involving sites or topoi.

5.5

Let

$$f : X \longrightarrow Y$$

be an arrow of E, so that X, more precisely (X, f), may be regarded as an object of $E_{/Y}$. There is an evident canonical isomorphism

$$(E_{/Y})_{/X} \xrightarrow{\sim} E_{/X}. \tag{5.5.1}$$

This is the transitivity of induced topoi. Applying the construction of 5.1 to $E_{/Y}$ instead of E, one obtains a canonical geometric morphism

$$\operatorname{loc}(f), \text{ or simply } f : E_{/X} \longrightarrow E_{/Y}, \tag{5.5.2}$$

also called the *localization morphism associated with* f. When Y is the final object of E, this recovers (5.2.1). The localization morphism for f is associated with three adjoint functors

$$f_! \quad f^* \quad f_*, \tag{5.5.3}$$

which are interpreted respectively as the forgetful functor, the restriction functor, and a functor denoted, according to preference and with X' as argument, by

$$\prod_{X/Y} (X'/X), \qquad \mathcal{H}om_{X/Y}(X, X'), \qquad \operatorname{Res}_{X/Y}(X'/X).$$

5.6

The transitivity of forgetful functors implies that, for two composable morphisms

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

in E, there is a canonical isomorphism of geometric morphisms

$$\operatorname{loc}(gf) \simeq \operatorname{loc}(g)\operatorname{loc}(f) : E_{/X} \longrightarrow E_{/Y} \longrightarrow E_{/Z}.$$

For three composable morphisms, the usual compatibility relation holds for these transitivity isomorphisms. Thus, with the usual abuse of language, one may regard

$$X \longmapsto E_{/X}$$

as a covariant functor from E to the category of topoi. More precisely, it is a 2-functor, not necessarily strict, from E to the 2-category of topoi. Before continuing the general theory of induced topoi in 5.10, we give some instructive examples.

5.7

Let X be a topological space, hence giving a topos $\mathrm{Top}(X)$. Let X' be an object of $\mathrm{Top}(X)$, which it is convenient to interpret as an *étale space* over X, say $p : X' \to X$. The category $\mathrm{Top}(X)_{/X'}$ identifies with the category of étale spaces X'' over X, endowed with an X-morphism $X'' \to X'$. Such a morphism makes X'' an étale space over X', and in this way one obtains a canonical equivalence of topoi

$$\mathrm{Top}(X)_{/X'} \simeq \mathrm{Top}(X').$$

The localization morphism is therefore a geometric morphism $\mathrm{Top}(X') \to \mathrm{Top}(X)$, and one immediately checks that this is precisely the morphism

$$\mathrm{Top}(p) : \mathrm{Top}(X') \longrightarrow \mathrm{Top}(X)$$

associated with the structural continuous map $p : X' \to X$. Intuitively, the localization morphism is just the translation, into topos language, of the étale-space morphism $p : X' \to X$. This both justifies the term "localization morphism" and calls for caution with the term "inclusion morphism", which seems appropriate mainly when p is an open immersion. It is therefore prudent, in 5.1, to reserve "inclusion morphism" for the case where $X \to e_E$ is a monomorphism, i.e. where X identifies with a subobject of the final object of E.

5.8

Let E be a topos, let G be a group object of E, let $H \subset G$ be a subgroup, and put

$$X = G/H.$$

We regard this quotient homogeneous space as an object of E with a left action of G, i.e. as an object of the classifying topos B_G. We determine the induced topos

$$(B_G)_{/X} \qquad (X = G/H)$$

by defining an equivalence of topoi

$$c : B_H \xrightarrow{\sim} (B_G)_{/X} \tag{5.8.1}$$

such that the diagram

$$\begin{tikzcd} B_H \arrow[r, hook, "c"] \arrow[rr, bend right, "B_i"'] & (B_G)_{/X} \arrow[r, "j_X"] & B_G \end{tikzcd} \tag{5.8.2}$$

commutes up to canonical isomorphism. Here j_X is the localization morphism in B_G, and

$$B_i : B_H \longrightarrow B_G$$

is the morphism induced by the inclusion $i : H \to G$.

To define (5.8.1), we only describe the inverse-image and direct-image functors c^* and c_*, leaving to the reader the verification that they are quasi-inverse equivalences and give rise to the commutative diagram (5.8.2). Let e be the subobject of E underlying X, namely the image of the section of X induced, over a chosen final object e_E, by the unit section of G. If X' is an object of B_G over X, define

$$c^*(X') = X' \times_X e.$$

This object is stable under the left action of H, obtained by restricting the given action of G on X'. This gives a functor

$$c^* : (B_G)_{/X} \longrightarrow B_H.$$

Conversely, if X' is an object of B_H, the morphism $X' \to e_H$, where e_H is the final object of B_H, gives after applying $i_!$ a morphism in B_G

$$i_!(X') \longrightarrow i_!(e_H) \simeq X.$$

Thus $i_!(X')$ is an object of $(B_G)_{/X}$, and the desired functor is

$$c_* : X' \longmapsto (i_!(X') \to X) : B_H \longrightarrow (B_G)_{/X}.$$

5.8.3

It follows from (5.8.2) that if G is a group object of a topos and H is a subgroup, then the functor which restricts the operators from G to H may be interpreted as a localization functor. In particular, taking H to be the unit subgroup, the functor which forgets the action of G is a localization functor.

More generally, if (X, G) is an object of E with an action of G, i.e. an object of B_G, then the functor

$$(B_G)_{/(X,G)} \longrightarrow E_{/X}$$

which forgets the action of G may be interpreted as a localization functor, relative to a suitable object of $(B_G)_{/(X,G)}$ covering the final object. One takes

$$E_{G,(X,G)} = E_G \times (X, G),$$

where $E_G = G_s = G$ carries the left-translation action of G. This gives a cartesian diagram

$$\begin{tikzcd} E_G \arrow[d] & Z = E_{G,(X,G)} \arrow[l] \arrow[d] \\ e_G & (X, G) \arrow[l] \end{tikzcd} \tag{5.8.3.1}$$

and, by transitivity of induced topoi, the induced topos over Z identifies with $E_{/X}$. The resulting diagram of topoi

$$\begin{tikzcd} (B_G)_{/E_G} \simeq E \arrow[d, "j"'] & (B_G)_{/Z} \simeq E_{/X} \arrow[l, "i"'] \arrow[d, "j'"] \\ B_G & (B_G)_{/(X,G)} \arrow[l, "f"] \end{tikzcd} \tag{5.8.3.2}$$

commutes up to canonical isomorphism. The inverse-image functors associated with the vertical arrows are the functors forgetting the action of G, while i^* identifies with the localization functor $E \to E_{/X}$. This diagram is in fact 2-cartesian in the sense of Proposition 5.11 below.

5.8.4

The reader may extend the preceding considerations to a pro-group $\mathcal{G} = (G_i)$, in the special case where H is defined as the kernel of $\mathcal{G} \to G_{i_0}$. For profinite groups this means that one restricts to open subgroups H of G, equivalently closed subgroups of finite index.

Exercise 5.9. Let E be a topos, let G be a group object of E, let B_G be the classifying topos of G, and let

$$\pi : B_G \longrightarrow E$$

be the morphism induced by the homomorphism from G to the unit group of E, using $B_e \simeq E$.

a) Let $E_G = G_s$ be the object of B_G whose underlying object of E is G, with G acting by left translations. Regard $\pi^*(G)$ as a group object of B_G, namely the object G of E with the trivial action of G. Show that right translations of G on G_s define a morphism in B_G

$$E_G \times \pi^*(G) \longrightarrow E_G.$$

Show that this makes E_G an object of B_G with a right action of the B_G-group $\pi^*(G)$, and that this action makes E_G a torsor under $\pi^*(G)$.

b) If $q : E' \to E$ is a topos over E, and B is any topos over E, define the category $\mathcal{H}om_{\mathrm{top},E}(E', B)$ of geometric morphisms $E' \to B$ compatible with the structural morphisms to E, up to a specified isomorphism. For $B = B_G$, show that the rule $f \mapsto f^*(E_G)$ defines an equivalence of categories

$$\mathcal{H}om_{\mathrm{top},E}(E', B_G) \longrightarrow \mathrm{Tors}(E', q^*(G)),$$

where the right-hand side is the category of right $q^*(G)$-torsors on E'.

c) Let $H \subset G$ be a subgroup. The quotient

$$X = E_G/\pi^*(H)$$

is the object G/H with its usual left action of G. Passing to induced topoi in the diagram $E_G \to X \leftarrow e_G$, show that the resulting diagram is equivalent to the diagram attached by 4.5 to the group homomorphisms $e \to H \to G$. Thus the classifying topos B_H is to be interpreted intuitively as a homogeneous space over B_G, with group $\pi^*(G)$, associated with the universal torsor E_G over B_G.

d) Revisit the example of 5.7 when X' is an étale covering of X, locally trivial in the sense of 2.7.4, and interpret it in terms of the present exercise.

5.10. Inverse images of induced topoi

Let

$$f : E' \longrightarrow E$$

be a morphism of $\mathcal{U}$-topoi, let X be an object of E, and put $X' = f^*(X)$. We define a diagram of geometric morphisms

$$\begin{tikzcd} E_{/X} \arrow[d, "j_X"'] & E'_{/X'} \arrow[l, "f_{/X}"'] \arrow[d, "j_{X'}"] \\ E & E' \arrow[l, "f"'] \end{tikzcd} \tag{5.10.1}$$

where j_X and $j_{X'}$ are the localization morphisms and where $f_{/X}$ is defined below. The diagram is commutative up to canonical isomorphism.

We define $f_{/X}$ by its inverse-image functor

$$(f_{/X})^* : E_{/X} \longrightarrow E'_{/X'}, \tag{5.10.2}$$

namely the functor induced by f^* in the evident sense. Since the forgetful functors $j_{X!}$ and $j_{X'!}$ commute with inductive limits and fiber products and are conservative, it follows at once that $(f_{/X})^*$ commutes with inductive limits and fiber products, just as f^* does. It also sends the final object X to the final object X'. Hence it is left exact and is associated to a geometric morphism

$$f_{/X} : E'_{/X'} \longrightarrow E_{/X}, \tag{5.10.3}$$

defined up to unique isomorphism. The diagram obtained from (5.10.1) by passing to inverse-image functors is commutative up to canonical isomorphism by construction and because f^* commutes with products $X \times Y$. Thus (5.10.1) itself is commutative, with the unique isomorphism inducing the canonical isomorphism on inverse-image functors.

5.10.4

When $f : E' \to E$ is associated with a continuous map $Y' \to Y$ of topological spaces, so that X is an étale space over Y and $X' = X \times_Y Y'$, the identifications $\mathrm{Top}(Y)_{/X} \simeq \mathrm{Top}(X)$ and $\mathrm{Top}(Y')_{/X'} \simeq \mathrm{Top}(X')$ identify $f_{/X}$ with the morphism $\mathrm{Top}(X') \to \mathrm{Top}(X)$ induced by the canonical continuous map $X' \to X$. Thus $E'_{/X'}$ plays the role of the fiber product of $E_{/X}$ and E' over E. The following result makes this intuition precise.

Proposition 5.11. With the notation of 5.10, the diagram (5.10.1), together with the compatibility isomorphism

$$\alpha : f \circ j_{X'} \xrightarrow{\sim} j_X \circ f_{/X},$$

is 2-cartesian. More explicitly, for every $\mathcal{U}$-topos F, associating to a geometric morphism $g : F \to E'_{/X'}$ the triple

$$(f_{/X}g,\ j_{X'}g,\ \alpha_g)$$

defines an equivalence from $\mathcal{H}om_{\mathrm{top}}(F, E'_{/X'})$ to the category of triples (g_1, g_2, β), where $g_1 : F \to E_{/X}$, $g_2 : F \to E'$, and $\beta : f \circ g_2 \xrightarrow{\sim} j_X \circ g_1$ is an isomorphism of geometric morphisms to E.

5.11.1

We record the elementary categorical calculation behind the preceding statement. Suppose given categories E, E', an object X of E such that $A \mapsto A \times X$ is defined, and a left exact functor

$$\varphi : E_{/X} \longrightarrow E'. \tag{*}$$

Since $E_{/X}$ has final object (X, id_X), so does E', with final object $e' \simeq \varphi(X)$. Choose e'. To φ we associate the pair

$$(\Psi, u), \qquad \Psi = \varphi \circ j : E \to E', \qquad u \in \mathrm{Hom}(e', \Psi(X)), \tag{**}$$

where $j(A) = A \times X$. The element u is obtained by applying φ to the diagonal section $\delta_X : X \to X \times X$ over the final object X of $E_{/X}$:

$$u = \varphi(\delta_X) : e' = \varphi(X) \longrightarrow \varphi(X \times X) = \Psi(X).$$

Conversely, the pair (Ψ, u) recovers φ up to unique isomorphism. For an object $p : X' \to X$ of $E_{/X}$, the graph morphism $\Gamma_p = (\mathrm{id}_{X'}, p)$ gives a cartesian square in $E_{/X}$. Applying φ yields the canonical formula

$$\varphi(X', p) \simeq \Psi(X') \times_{\Psi(X)} e'. \tag{5.11.1.1}$$

This is functorial in (X', p). The functor Ψ is left exact, and more generally commutes with any type of projective limits with which φ commutes.

Conversely, when finite projective limits are representable in E and E', a pair (Ψ, u) with Ψ left exact defines φ by (5.11.1.1). This φ is left exact and has the corresponding limit-commutation properties. Moreover one recovers Ψ by the canonical isomorphism

$$\Psi(X') \simeq \varphi(j(X')), \tag{5.11.1.2}$$

and under this isomorphism $u : e' \to \Psi(X)$ identifies with $\varphi(\delta_X)$.

Lemma 5.11.2. Let E and E' be categories in which finite projective limits are representable, let X be an object of E, and let $\mathrm{Lex}(E, E')$, respectively $\mathrm{Lex}(E_{/X}, E')$, denote the category of left exact functors. Then $\mathrm{Lex}(E_{/X}, E')$ is equivalent to the category whose objects are pairs (Ψ, u), with $\Psi \in \mathrm{Ob}\,\mathrm{Lex}(E, E')$ and $u : e' \to \Psi(X)$, where e' is the final object of E'. The construction and a quasi-inverse are as described above. If φ and (Ψ, u) correspond, then φ commutes with a given type of projective limits if and only if Ψ does. If in E and E' base-change functors commute with a given type of inductive limits, then φ commutes with inductive limits of that type if and only if Ψ does.

We leave to the reader the explicit categorical structure on the category of pairs (Ψ, u). If $\Gamma : \mathrm{Lex}(E, E') \to \mathbf{Set}$ is given by $\Gamma(\Psi) = \mathrm{Hom}(e', \Psi(X))$, then u may be viewed as an element of $\Gamma(\Psi)$, which explains the notation $\mathrm{Lex}(E, E')_{/\Gamma}$. The remaining functorial details and the assertions on inductive limits follow at once from the explicit formulas (5.11.1.1) and (5.11.1.2).

Proposition 5.12. Let E and E' be $\mathcal{U}$-topoi and let X be an object of E. On $\mathcal{H}om_{\mathrm{top}}(E', E)$ consider the contravariant functor

$$\gamma : f \longmapsto \Gamma(E', f^*(X)) = \mathrm{Hom}(e', f^*(X)).$$

Then there is an equivalence of categories

$$\mathcal{H}om_{\mathrm{top}}(E', E_{/X}) \xrightarrow{\sim} \mathcal{H}om_{\mathrm{top}}(E', E)_{/\gamma}. \tag{5.12.1}$$

It sends a geometric morphism $h : E' \to E_{/X}$ to the pair (f, u), where $f = j_X h : E' \to E$, and where

$$u : e' = h^*(X, \mathrm{id}_X) \longrightarrow f^*(X) = h^*(j_X^* X) = h^*(X \times X)$$

is induced by the diagonal $\delta_X : X \to X \times X$.

5.12.2

Using 5.12, the proof of Proposition 5.11 becomes almost tautological. To give a geometric morphism $g : F \to E'_{/X'}$ amounts to giving a morphism $g_2 : F \to E'$ together with a section of $g_2^*(X')$. But $g_2^*(X') = g_2^* f^*(X) = (fg_2)^*(X)$, so such a section is equivalently a lifting of $fg_2 : F \to E$ to a morphism $g_1 : F \to E_{/X}$. This is exactly the data of a triple (g_1, g_2, α) as in 5.11.

Remark 5.13. For any diagram of topoi

$$\begin{tikzcd} E_1 \arrow[d] & \\ E & E_2 \arrow[l] \end{tikzcd}$$

there exists a 2-fiber product in the 2-category of topoi, independent up to equivalence of the chosen enlargement of universes. For other natural examples of fiber products of topoi, besides Proposition 5.11, see the final chapter of Giraud's book.

Exercise 5.14.

a) If F and G are topoi over a topos E, define $\mathcal{H}om_{\mathrm{top},E}(F, G)$ to be the category of pairs (f, α), where $f : F \to G$ is a geometric morphism and $\alpha : p \xrightarrow{\sim} qf$ is an isomorphism of the structural morphisms to E. Define composition functors $\mathcal{H}om_{\mathrm{top},E}(F, G) \times \mathcal{H}om_{\mathrm{top},E}(G, H) \to \mathcal{H}om_{\mathrm{top},E}(F, H)$ and associativity isomorphisms.

b) If X, Y are objects of E, define a natural functor from the discrete category $\mathrm{Hom}(X, Y)$ to $\mathcal{H}om_{\mathrm{top},E}(E_{/X}, E_{/Y})$, and check compatibility with composition in E and the compositions of part a).

c) Prove that the functor of b) is an equivalence. In particular, an object X of a topos E is recovered, up to unique isomorphism, from the induced topos $E_{/X}$ considered as a topos over E, up to E-equivalence. This result is due to P. Deligne; the case where E is the punctual topos was previously treated by Mme Hakim.

6 Points of a topos and fiber functors

6.0

Let P be the standard punctual $\mathcal{U}$-topos

$$P = (\mathcal{U}\text{-}\mathbf{Set}).$$

Because the symbol P evokes a geometric object of the nature of a point, whereas ($\mathcal{U}$-**Set**) evokes a large category, to be interpreted as the category of sheaves on P, it is often preferable, according to context, to use one notation or the other exclusively, although strictly speaking they denote one and the same object.

Definition 6.1. Let E be a $\mathcal{U}$-topos. A *point* of E is a geometric morphism $P \to E$ from the punctual topos to E. The *category of points* of E, denoted $\mathrm{Pt}(E)$ or $\mathrm{Pt}(E)$, is the category $\mathcal{H}om_{\mathrm{top}}(P, E)$. If $p : P \to E$ is a point of E, with inverse-image functor $p^* : E \to (\mathcal{U}\text{-}\mathbf{Set})$, and if F is an object of E, the set $p^*(F)$ is called the *fiber of F at p* and is denoted F_p.

6.1.1

If F is a group object, ring object, and so on, in E, then F_p is respectively a group, a ring, and so on.

6.1.2

By 3.2.1, the category of points of E is equivalent, via $p \mapsto p^*$, to the opposite of the category of functors

$$\varphi : E \longrightarrow (\mathcal{U}\text{-}\mathbf{Set})$$

which commute with $\mathcal{U}$-inductive limits and are left exact. Thus giving a point of E is essentially the same thing as giving such a functor φ.

Definition 6.2. A *fiber functor* of the $\mathcal{U}$-topos E is a functor

$$\varphi : E \longrightarrow (\mathcal{U}\text{-}\mathbf{Set})$$

which commutes with $\mathcal{U}$-inductive limits and is left exact. The *category of fiber functors* on E, denoted $\mathrm{Fib}(E)$, is the full subcategory of $\mathcal{H}om(E, (\mathcal{U}\text{-}\mathbf{Set}))$ formed by the fiber functors.

6.2.1

The category $\mathrm{Fib}(E)$ of fiber functors on E is equivalent to the opposite of the category $\mathrm{Pt}(E)$ of points of E, via the functor $p \mapsto p^*$:

$$\mathrm{Pt}(E) \xrightarrow{\sim} \mathrm{Fib}(E)^{\circ}. \tag{6.2.1.1}$$

Thus a functor $\varphi : E \to (\mathcal{U}\text{-}\mathbf{Set})$ is a fiber functor if and only if it is the fiber functor $F \mapsto F_p$ associated with a point p of E. Equivalently, by 1.7, φ is left exact and sends covering families to surjective families. This last condition may also be expressed by saying that φ commutes with $\mathcal{U}$-small sums and sends epimorphisms to epimorphisms.

6.3

Let $C \in \mathcal{U}$ be a site. A *point of the site* C is a point of the topos $C^{\sim}$, and the *category of points of the site* C is

$$\mathrm{Pt}(C) = \mathrm{Pt}(C^{\sim}).$$

Consider the functor

$$\varphi \longmapsto \varphi|C = \varphi \circ \varepsilon_C : \mathrm{Fib}(C^{\sim}) \longrightarrow \mathcal{H}om(C, (\mathcal{U}\text{-}\mathbf{Set})). \tag{6.3.1}$$

It is fully faithful, with essential image the full subcategory of site morphisms $(\mathcal{U}\text{-}\mathbf{Set}) \to C$ in the sense of 4.9.4; denote this subcategory also by $\mathrm{Fib}(C)$. A functor $\Psi : C \to (\mathcal{U}\text{-}\mathbf{Set})$ is called a *fiber functor on the site* C if it belongs to this essential image. Such a functor is continuous, hence sends covering families to covering families, i.e. to surjective families, and is left exact. When finite projective limits are representable in C, as in nearly all practical cases, these two properties characterize the fiber functors on C: they are precisely the left exact functors $C \to (\mathcal{U}\text{-}\mathbf{Set})$ which send covering families to surjective families.

The functor $\varphi \mapsto \varphi|C$ is an equivalence

$$\mathrm{Fib}(C^{\sim}) \xrightarrow{\sim} \mathrm{Fib}(C),$$

so it is essentially the same thing to give a fiber functor on the topos $C^{\sim}$, or a point of it, as to give a fiber functor on the site C. If Ψ is a fiber functor on C, and φ is the corresponding fiber functor on $C^{\sim}$, then φ is recovered by

$$\varphi(F) \simeq \varinjlim_{C_{/F}} \Psi(X), \tag{6.3.2}$$

where $C_{/F}$ is the category of objects X of C equipped with a morphism $X \to F$ in $\widehat{C}$, equivalently with an element of $F(X)$. This follows immediately from the canonical formula

$$F \simeq \varinjlim_{C_{/F}} \varepsilon_C(X).$$

More generally, for a presheaf G on C, one has a canonical isomorphism

$$\varphi(aG) \simeq \varinjlim_{C_{/G}} \Psi(X), \tag{6.3.3}$$

where aG denotes the sheaf associated with G.

6.4.0

We refer to Exposé I, 6.1 for the notion of a conservative family of functors

$$\varphi_i : E \longrightarrow E_i \qquad (i \in I).$$

When E and the E_i are topoi and the φ_i are inverse-image functors f_i^* of geometric morphisms $f_i : E_i \to E$, the exactness properties required in Exposé I, 6.2, 6.3 and 6.4 are satisfied, apart from the harmless finiteness restriction there indicated. In this setting it is equivalent to say that $(f_i^*)_{i \in I}$ is conservative, conservative for monomorphisms, conservative for epimorphisms, or faithful.

When the family (f_i^*) is conservative we shall also say that the family of geometric morphisms (f_i) is conservative. In particular:

Definition 6.4.1. Let E be a $\mathcal{U}$-topos and let $(p_i : P \to E)_{i\in I}$ be a family of points of E. This family is *conservative* if the associated family of fiber functors $F \mapsto F_{p_i}$ from E to ($\mathcal{U}$-**Set**) is conservative. The topos E is said to have *enough points* if it admits a conservative family of points, equivalently if the family of all its points is conservative.

6.4.2

All topoi used up to this point have enough points. It is nevertheless possible, deliberately, to construct topoi which do not have enough points; such a topos is necessarily non-empty in the geometric sense of 2.2. A topos with enough points admits a conservative family of points indexed by a $\mathcal{U}$-small set (Proposition 6.5). However, the set of isomorphism classes of points of a $\mathcal{U}$-topos need not itself be $\mathcal{U}$-small, although it is so in many practical cases. Finally, for an important existence theorem of P. Deligne giving enough points and covering all cases occurring in algebraic geometry, see Exposé VI, 9.

6.4.3

If $(p_i)_{i\in I}$ is a conservative family of points of E, then two arrows $u, v : F \rightrightarrows G$ in E are equal if and only if $u_{p_i} = v_{p_i}$ for all i. A morphism $u : F \to G$ is a monomorphism, respectively an epimorphism, if and only if each map $u_{p_i} : F_{p_i} \to G_{p_i}$ is injective, respectively surjective. An object F is initial, respectively final, if and only if every F_{p_i} is empty, respectively a singleton. Finally, F is a subobject of the final object e_E if and only if each F_{p_i} has at most one element.

Proposition 6.5.

a) Let C be a $\mathcal{U}$-site and let $(\varphi_i)_{i\in I}$ be a family of fiber functors on C. The corresponding family of points of $E = C^\sim$ is conservative if and only if, for every family $(X_j \to X)_{j\in J}$ in C, the following implication holds: if for every $i \in I$ the family $(\varphi_i(X_j) \to \varphi_i(X))_{j\in J}$ is surjective, then $(X_j \to X)_{j\in J}$ is covering.

b) If a $\mathcal{U}$-topos E has enough points, then it admits a $\mathcal{U}$-small conservative family of points.

Assertion b) is a special case of Exposé I, 7.7. For a), apply that criterion to the generating family of E consisting of the $\varepsilon(X)$, $X \in \operatorname{Ob} C$. Recall that $(X_j \to X)$ is covering if and only if $(\varepsilon(X_j) \to \varepsilon(X))$ is epimorphic. If the family of points is conservative, this is equivalent to surjectivity on all fibers, i.e. to surjectivity of all the families $(\varphi_i(X_j) \to \varphi_i(X))$. This proves necessity. Conversely, the same criterion reduces sufficiency to checking that a monomorphism $F \to \varepsilon(X)$ which is an isomorphism on all fibers is an isomorphism. Choose a covering family $\varepsilon(X_j) \to F$, refined so that the composites $\varepsilon(X_j) \to \varepsilon(X)$ come from morphisms $X_j \to X$. The fiberwise hypothesis and the assumption of a) imply that $(X_j \to X)$ is covering. Hence $(\varepsilon(X_j) \to \varepsilon(X))$ is epimorphic, and therefore $F \to \varepsilon(X)$ is epimorphic.

Corollary 6.5.1. Let C be a category. A topology on C making C into a $\mathcal{U}$-site with enough fiber functors, equivalently such that the associated topos has enough points, is completely determined by the full subcategory $\Phi \subset \mathcal{H}om(C, \mathbf{Set})$ formed by the fiber functors on C.

Indeed, Proposition 6.5(a) describes the covering families in terms of the family of fiber functors. Below, in 6.8.5, we shall see that Φ is contained in the category of pro-representable functors on C, so that $\Phi^\circ \simeq \mathrm{Pt}(C^\sim)$ identifies, up to equivalence, with a full subcategory of $\mathrm{Pro}(C)$. Compare this corollary with Giraud's theorem, Exposé II, 5.5, which establishes a bijection between topologies on C and certain full subcategories of $\mathcal{H}om(C^\circ, \mathbf{Set})$.

Exercise 6.5.2. Let C be a category equivalent to a $\mathcal{U}$-small category, and let P be a subcategory of $\mathrm{Pro}(C)$. For $p \in P$, write $X \mapsto X_p$ for the functor pro-represented by p. A family $(X_i \to X)_{i\in I}$ in C is called P-covering if for every $p \in P$ the family $(X_{ip} \to X_p)_{i\in I}$ is surjective. Show that there exists a topology T_P on C for which the covering families are precisely the P-covering families; that T_P is the finest topology on C for which all functors $X \mapsto X_p$, $p \in P$, are fiber functors; and that T_P gives C enough fiber functors. For any topology T on C, show that among the topologies finer than T with enough fiber functors there is a least one, namely T_P, where P is the strictly full subcategory of $\mathrm{Pro}(C)$ equivalent to $\mathrm{Pt}(C^\sim)$ described in 6.8.5.

Problem 6.5.3. Characterize the strictly full subcategories P of $\mathrm{Pro}(C)$, stable under filtered projective limits and satisfying the appropriate further conditions, which arise from a topology on C as the essential image of $\mathrm{Pt}(C^\sim)$ in $\mathrm{Pro}(C)$.

6.6

Let

$$f : E \longrightarrow E'$$

be a morphism of $\mathcal{U}$-topoi. It induces a canonical functor

$$\mathrm{Pt}(f) = (p \longmapsto f \circ p) : \mathrm{Pt}(E) \longrightarrow \mathrm{Pt}(E').$$

For two composable geometric morphisms

$$E \xrightarrow{f} E' \xrightarrow{g} E'',$$

one has, tautologically,

$$\mathrm{Pt}(gf) = \mathrm{Pt}(g) \circ \mathrm{Pt}(f).$$

Thus the dependence of $\mathrm{Pt}(E)$ on E is genuinely functorial: if $\mathcal{V}$ is a universe with $\mathcal{U} \in \mathcal{V}$, one obtains an honest functor from the category of $\mathcal{U}$-topoi which are elements of $\mathcal{V}$ to the category of categories which are elements of $\mathcal{V}$.

6.7

Let E be a $\mathcal{U}$-topos. Every object F of E defines a contravariant functor

$$p \longmapsto F_p = p^*(F) : \mathrm{Pt}(E)^\circ \longrightarrow (\mathcal{U}\text{-}\mathbf{Set}),$$

and hence, for variable F, a canonical functor

$$E \longrightarrow \widehat{\mathrm{Pt}(E)} = \mathcal{H}om(\mathrm{Pt}(E)^\circ, (\mathcal{U}\text{-}\mathbf{Set})). \tag{6.7.1}$$

By definition, this functor is faithful if and only if E has enough points. It has a certain formal analogy with a Fourier transform.

Let X be an object of E, and let $\widehat{X}$ be the corresponding presheaf on $C = \mathrm{Pt}(E)$. The category

$$C_{/\widehat{X}} = \mathrm{Pt}(E)_{/\widehat{X}}$$

is the category of pairs (p, ξ), where p is a point of E and $\xi \in X_p$. There is a canonical equivalence of categories

$$\mathrm{Pt}(E_{/X}) \xrightarrow{\sim} \mathrm{Pt}(E)_{/\widehat{X}}. \tag{6.7.2}$$

Composed with $\mathrm{Pt}(E)_{/\widehat{X}} \to \mathrm{Pt}(E)$, it is just the functor on points induced by the localization morphism $j_X : E_{/X} \to E$. This is the special case of Proposition 5.12 obtained by taking $E' = P = (\mathbf{Set})$.

Corollary 6.7.3. Let E be a topos. If E has enough points, then so does every induced topos $E_{/X}$. Conversely, if $(X_i)_{i \in I}$ is a family of objects of E covering the final object, and if each $E_{/X_i}$ has enough points, then E has enough points.

For the first assertion, let $f : X' \to X''$ be a morphism in $E_{/X}$ which is not an isomorphism. Since E has enough points, there is a point p of E such that $f_p : X'_p \to X''_p$ is not an isomorphism. Then for some $q \in X_p$, the map induced on the fibers over q is not an isomorphism. Under (6.7.2), this element q is a point of $E_{/X}$, and the map just described is precisely the map on fibers at that point.

Conversely, suppose the X_i cover the final object of E, and that each $E_{/X_i}$ has enough points. If $f : X' \to X''$ in E is not an isomorphism, then for some i, its restriction $f_i : X'_i \to X''_i$ over X_i is not an isomorphism. A point p_i of $E_{/X_i}$ detects this failure. Its image p in E then detects the failure of f to be an isomorphism. Thus E has enough points.

Remark 6.7.4. The same argument shows that if (p_α) is a conservative family of points of E, then the family of points of $E_{/X}$ lying above one of the p_α is conservative. Likewise, if the X_i cover the final object of E, and P_i is a conservative set of points of $E_{/X_i}$, then the set of their images in E under the localization morphisms $E_{/X_i} \to E$ is conservative.

6.8

Let E be a topos and let p be a point of E. A *neighborhood of p in E* is a pair (X, u), where $X \in \mathrm{Ob}\, E$ and $u \in X_p$. By (6.7.2), one may also interpret u as a lifting of p to a point of the induced topos $E_{/X}$. If φ_p is the fiber functor associated with p, a neighborhood of p is also an object of the category $E_{/\varphi_p}$. This category of pairs (X, u) is called the *category of neighborhoods* of p and is denoted $\mathrm{Neigh}(p)$. Since φ_p is left exact and finite projective limits are representable in E, finite projective limits are representable in $\mathrm{Neigh}(p)$, and the canonical functor $\mathrm{Neigh}(p) \to E$ commutes with them. Hence $\mathrm{Neigh}(p)^\circ$ is filtered. Moreover, for every $F \in \mathrm{Ob}\, E$, there is a canonical isomorphism

$$F_p = \varphi_p(F) \simeq \varinjlim_{\mathrm{Neigh}(p)^\circ} F(X). \tag{6.8.1}$$

In other words, the fiber functor at p is the filtered inductive limit of the functors represented in E by the neighborhoods of p. It is in fact ind-representable, equivalently representable by a pro-object $(X_i)_{i \in I}$ of E, with I a filtered ordered set in $\mathcal{U}$. To see this, choose a small full generating subcategory $C \subset E$. The full subcategory of $\mathrm{Neigh}(p)$ formed by the neighborhoods (X, u) with $X \in C$ is cofinal: every neighborhood (X, u) is refined by one whose object lies in C, using a covering family from C and the surjectivity on fibers.

6.8.2

Let C be a $\mathcal{U}$-site and let p be a point of C, i.e. of $C^\sim$. By abuse of notation write again φ_p for its restriction to C, and write $X_p = \varphi_p(X) = \varepsilon(X)_p$. A *neighborhood of p in the site C* is a pair (X, u), with $X \in \operatorname{Ob} C$ and $u \in X_p$. These neighborhoods form the category $C_{/\varphi_p}$, also denoted $\operatorname{Neigh}_C(p)$. Then $\operatorname{Neigh}_C(p)^\circ$ is filtered, admits a cofinal $\mathcal{U}$-small full subcategory, and for every sheaf F on C there is a functorial isomorphism

$$F_p \simeq \varinjlim_{\operatorname{Neigh}_C(p)^\circ} F(X). \tag{6.8.3}$$

Indeed, if $C' \subset C$ is a topologically generating full subcategory, then $\operatorname{Neigh}_{C'}(p)^\circ$ is cofinal in $\operatorname{Neigh}_C(p)^\circ$. Thus one reduces to the case $C \in \mathcal{U}$. Then $\widehat{C}$ is a topos and the associated-sheaf functor $a : \widehat{C} \to C^\sim$ allows one to apply (6.8.1) to $\widehat{C}$, with generating full subcategory C. This gives, more generally, for every presheaf P on C,

$$(aP)_p \simeq \varinjlim_{\operatorname{Neigh}_C(p)^\circ} P(X). \tag{6.8.4}$$

The general large-site case follows after enlargement of universes and the compatibility of associated sheaves with such enlargement.

6.8.5

To each point p of $C^\sim$ one has therefore associated a pro-object of the site C, defined by the canonical functor $\operatorname{Neigh}_C(p) \to C$. For variable p this gives a functor

$$\operatorname{Pt}(C^\sim) \longrightarrow \operatorname{Pro}(C). \tag{6.8.5.1}$$

It is fully faithful. Indeed, via associated fiber functors it is the composite $\operatorname{Fib}(C) \to \operatorname{Fib}(C^\sim) \to \operatorname{Pro}(C)^\circ$. The first functor is an equivalence by 4.9.4, recalled for fiber functors in 6.3, and the second is fully faithful by the general formalism of pro-objects.

6.8.6

If C is $\mathcal{U}$-small and carries the chaotic topology, so that $C^\sim = \widehat{C}$, then the preceding functor is an equivalence

$$\operatorname{Pt}(\widehat{C}) \xrightarrow{\sim} \operatorname{Pro}(C). \tag{6.8.6.1}$$

Essential surjectivity follows because the functors $P \mapsto P(X)$ on $\widehat{C}$ are fiber functors, and a filtered inductive limit of fiber functors on a topos is again a fiber functor. Under the embedding $C \subset \operatorname{Pro}(C)$, the induced functor $C \to \operatorname{Pt}(\widehat{C})$ is the canonical one already considered in 4.6.2.2; a quasi-inverse to (6.8.6.1) is the canonical extension of that functor to pro-objects.

6.8.7

Let $(X_i)_{i \in I}$ be a pro-object of the $\mathcal{U}$-site C. It defines on $C^\sim$ the functor

$$\varphi(F) = \varinjlim_I F(X_i). \tag{6.8.7.1}$$

In view of 6.8.5 it is natural to ask when this is a fiber functor. The following conditions are equivalent:

(i) φ is a fiber functor on $C^{\sim}$.

(ii) Its restriction to C is a fiber functor on the site C.

(iii) For every covering family $(Y_\alpha \to Y)$ in C, every $i_0 \in I$, and every morphism $X_{i_0} \to Y$, there exist $i \geq i_0$, an index α, and a morphism $X_i \to Y_\alpha$ making the evident square

$$\begin{tikzcd} X_i \arrow[r] \arrow[d] & Y_\alpha \arrow[d] \\ X_{i_0} \arrow[r] & Y \end{tikzcd}$$

commute.

Condition (iii) simply expresses that $\varphi|C$ sends covering families to covering families. The implications (i)⇒(ii)⇒(iii) are immediate. Conversely, φ is manifestly left exact; by 4.9.4 it remains to show that it sends covering families of sheaves to covering families. Given a covering family $F_\alpha \to F$, an index i_0, and an element $x_0 \in F(X_{i_0})$, choose a covering family $Z_\gamma \to X_{i_0}$ over which x_0 lifts to some $F_\alpha(Z_\gamma)$. By (iii), for some $i \geq i_0$ there is an X_{i_0}-morphism $X_i \to Z_\gamma$, so the image of x_0 in $F(X_i)$ lifts to some $F_\alpha(X_i)$. This proves the claim.

Exercise 6.9. Let C be a site in $\mathcal{U}$, and let φ be a fiber functor on C, pro-represented by a pro-object $(X_i)_{i\in I}$ of $\mathrm{Pro}(C)$. For every index i, every object Y, every morphism $u : X_i \to Y$, and every covering sieve R on Y, choose an index $j \geq i$ such that $X_j \to X_i \xrightarrow{u} Y$ factors through R. For fixed i, let $J(i)$ be the set formed by i and all such j; define $J(I')$ similarly for subsets $I' \subset I$. For a finite subset I', choose an upper bound $m(I')$, with $m(\{i\}) = i$; let $M(I')$ be the union of these upper bounds over all finite subsets of I'. Let $P(I')$ be the union of the iterates $(MJ)^n(I')$.

a) Show that $P(I')$ is filtered for the induced order, and that I is the increasing filtered union of the $P(I')$ as I' ranges over finite subsets of I.

b) Show that there is a cardinal $c \in \mathcal{U}$, depending only on $a = \operatorname{card}\mathrm{Fl}(C)$, such that $\operatorname{card} P(I') \leq c$ for every finite I'. If a is infinite, one may take $c = 2^a$.

c) Using 6.8.7, show that for every I', the pro-object $(X_i)_{i\in P(I')}$ pro-represents a fiber functor $\varphi_{I'}$, and that φ is the filtered inductive limit of the $\varphi_{I'}$ for finite I'.

d) Deduce that there is a small full subcategory Φ of $\mathrm{Fib}(C)$ such that every object of $\mathrm{Fib}(C)$ is a filtered inductive limit of objects of Φ. If a is infinite, Φ may be chosen of cardinal $\leq 2^{2^a}$; if a is finite, one may of course take $\Phi = C$.

Exercise 6.10. Let C be a $\mathcal{U}$-site, let D be a category in which small filtered inductive limits are representable, and let $\mathrm{Sh}(C, D)$ be the category of sheaves on C with values in D. Extending (6.8.3), define a fiber functor

$$F \longmapsto F_p : \mathrm{Sh}(C, D) \longrightarrow D. \tag{6.10.1}$$

If finite projective limits are representable in D and commute with filtered inductive limits, then this functor is left exact.

Continuation anchor. The next untranslated point is SGA 4, Exposé IV, Section 7, “Examples of fiber functors and points of topoi” (French source line 4121 of `04/04.tex`).